OFFLINE REFERENCE

Strength Coach documentation

Design record, rules, readiness, training evidence, operations and roadmap for the local-first strength coaching app.

Generated 2026-10-23 · commit 7899a02 · 42 documents

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Evidence-Based Strength Coach

A local-first strength-coaching tool built around a **fixed methodology**. It ingests training and readiness logs, critiques the most recent session against explicit programme rules, prescribes the next session, and explains both with citations to the exact rules used.

New here? Start with the user guide.

Design principle: Python writes the what, the LLM writes the why. All arithmetic, cleaning, calibration and prescription generation is deterministic Python. The LLM layer is small, tool-bound and auditable — and the system still emits a safe, valid prescription if the LLM call fails entirely.

Status

v0 complete; v0.1 complete; v0.2 complete; v0.3 complete; v0.4 next. The full slice runs: import → readiness → prescribe a day → critique the last session → review it → calibrate. Days are prescribable through rolling session types (squat, hinge, bench+squat, bench+hinge); athlete state, constraints, suitability, e1RM overrides, symptoms and the exercise catalogue are editable in the CLI and local web UI; the LLM can narrate or propose within the rules; VBT targets attach where the data supports them.

Delivered in v0.2/v0.3: readiness score and overrides, wearable/HRV/RHR baselines, injury signal, continuity calibration, cardio capture, measurements, session RPE/length, recent stress, per-lift readiness, e1RM calibration/decay, rolling session types, readiness volume valve, weekly volume ledger/MRV cap, exercise selection with A/B/C variant comparability, and the local web UI with import, state, Programme, Exercises, Terminal and calibration reports. v0.3 is complete. v0.4 covers cardio prescription/model plus the deeper labelled calibration set moved from v0.3. Prospective validation remains a future gate, not an active candidate; review follow-ups are in ROADMAP §4.20. v0.5 is video. See ROADMAP.md.

What runs today:

- **ingest** — both exports, cleaned under the documented sentinel policy, quarantined rather than dropped, reconstructed into sessions with a canonical session type;
- **store** — SQLite, append-only, numbered migrations, a replayable `run` / `run_event` trace;
- **rules** — the constitution loaded as hash-bearing rules, one predicate per `check.fn`, verdicts with citable evidence, and `unknown` where an input does not exist;
- **prescribe** — a deterministic session of one of the four canonical types, whose loads are solved from a recent e1RM (or a manual override), rounded down to real increments, and validated for executability;
- **critique** — six measured rubric items against a stored session, two explicitly unmeasurable rather than guessed;
- **eval** — a backtest that replays sessions through the rubric and compares them with hand-made marks, per item and per session.

The CLI that ties these together is `strength-coach`; see the user guide.

Quickstart

```
python3 -m venv .venv
.venv/bin/pip install -e ".[dev]"
source .venv/bin/activate           # so `strength-coach` is on PATH

.venv/bin/python -m pytest -m "not integration and not golden"   # fast iteration (~2 min)
.venv/bin/python -m pytest tests/test_prescribe.py --lf             # targeted / last failed
.venv/bin/python -m pytest            # full, parallel, ~4-5 min; -n 0 to debug serially
.venv/bin/python tools/profile_data.py    # profile your own export
.venv/bin/python tools/baseline.py        # strength baseline, latest date in the log
.venv/bin/python tools/llm_check.py       # show the LLM configuration (see below)
.venv/bin/python tools/import_export.py   # load data/raw into data/coach.db
.venv/bin/python tools/eval_prepare.py --session <date>   # labelling sheet
.venv/bin/python tools/backtest.py        # rubric vs hand marks
./run-web.sh                              # local web UI at http://127.0.0.1:8000
```

The tools read `data/raw/training_log.csv` and `data/raw/readiness.csv`. See `data/README.md` for the expected schema.

Using it

```
python -m strength_coach.cli import                    # data/raw -> data/coach.db
python -m strength_coach.cli run --date 2026-10-20     # critique the last session, prescribe
python -m strength_coach.cli review --session 2026-10-15  # mark it; calibration updates
python -m strength_coach.cli calibrate                 # agreement report so far
```

`run` is the v0 slice: it critiques the most recent stored session and prescribes the requested date, then prints every item with the rule id or stored set that justifies its load. Re-running the same date writes nothing unless you pass `--force`.

`review` is the calibration loop. It shows the session beside the rubric's scores, asks for your mark on each item, stores them append-only, and immediately prints running agreement. Do it after any session you want to judge: the report gets sharper with every review, and an item is flagged for revisiting only once it has five marks and still disagrees by more than 0.25. Nothing is retuned automatically — a rubric change is a decision, not a side effect.

Privacy

This is a public repository and it contains **no training data and no personal metrics**.

- `data/` is git-ignored (the raw exports are a complete personal training history).
- `private/` is git-ignored (documents derived from a real log).
- The code, schemas, rules and tests contain no athlete-specific values.

If you run this yourself, everything you generate stays local unless you choose to commit it.

Bring your own model

The LLM is optional. With no configuration the system runs with `NullLLM`: every deterministic feature works, and the only thing missing is the narrative paragraph — no prescription, score or rule verdict depends on a model being present.

To enable narration:

```
cp config/llm.example.toml config/llm.local.toml   # git-ignored
$EDITOR config/llm.local.toml                      # base_url, model, api_key
.venv/bin/python tools/llm_check.py                # show the effective settings
.venv/bin/python tools/llm_check.py --call         # send one test prompt
```

The default endpoint is **DeepSeek** (`https://api.deepseek.com/v1` , `deepseek-chat`), so a provider plus an API key is enough. Any other **OpenAI-compatible** endpoint works too (OpenAI, OpenRouter, vLLM, llama.cpp, Ollama) by setting `base_url` and `model` . The API key is optional because a local server may not need one.

Prefer not to put the key on disk? The environment overrides the file:

variable	meaning
`STRENGTH_COACH_LLM_PROVIDER`	`null` or `openai_compatible`
`STRENGTH_COACH_LLM_MODEL`	model name
`STRENGTH_COACH_LLM_BASE_URL`	endpoint root, e.g. `http://127.0.0.1:11434/v1`
`STRENGTH_COACH_LLM_API_KEY`	bearer token
`STRENGTH_COACH_LLM_TEMPERATURE` / `_MAX_TOKENS` / `_TIMEOUT_S`	sampling and limits
`STRENGTH_COACH_LLM_CONFIG`	path to a different settings file

The model **never computes**: every number is produced by Python and handed to it as a fact. Its output is recorded as a `run_event` , never written into a prescription or a score, so a bad model can produce bad prose but cannot change what you lift.

Repository map

```
config/
  exercises/taxonomy.v1.yaml   exercise identity + properties (single source of truth)
  rules/programme.v1.yaml      the constitution as citable, machine-checkable rules
  templates/session-types.v1.yaml  canonical session templates
  templates/inference.v1.yaml      imported-session type inference
  thresholds.v1.yaml           increments, rounding, e1RM window, fatigue multiplier
  llm.example.toml             LLM configuration template (the real file is git-ignored)
docs/                          the design record
data/                          your exports (git-ignored)
strength_coach/
  utils/                       permanent shared helpers (pure, tested)
  domain/                      pure domain arithmetic (e1RM, load)
  ingest/                      cleaning, flags, quarantine, sessions, readiness merge
  store/                       SQLite migrations, append-only repositories, read queries
  features/                    derived features, baselines, VBT calibration
  rules/                       constitution loader, predicates, policy verdicts
  prescribe/                   deterministic session-type engine and load solver
  critique/                    rubric over a stored session
  agents/                      LLM interface, config, narrative (NullLLM by default)
  trace/                       run lifecycle
  eval/                        backtest harness and labelled-set comparison
tools/profile_data.py          reproducible data profile
tools/baseline.py              reproducible strength baseline
tools/llm_check.py             shows the effective LLM settings; --call tests them
tests/                         unit + integration tests
```

The data model in one paragraph

The logs are messy in ways that are not obvious: `0` means five different things depending on the column, one weight column mixes unweighted / total-mass / added-load semantics, durations are stored in a reps column for isometric holds, and there are no timestamps so sessions must be reconstructed from dates alone. Repeated identical rows look like duplicates but are real sets. Every one of those decisions is written down with its evidence in `docs/DATA-QUALITY.md`, because a silent interpretation of any of them would corrupt every number downstream.

Reading order

1. `USER-GUIDE.md` — how to install it, run it and read the output
2. `ROADMAP.md` — where the project is right now, and what is next
3. `docs/DECISIONS.md` — decisions that are expensive to reverse
4. `docs/ARCHITECTURE.md` — layers, data flow, where the LLM sits
5. `docs/DATA-MODEL.md` — schemas, append-only records
6. `docs/DATA-QUALITY.md` — the data-quality ledger (`DQ-*`)
7. `docs/RULES-FORMAT.md` — citable rule IDs
8. `docs/VBT-CALIBRATION.md` — velocity targets and sparsity
9. `docs/V0-SCOPE.md` — the first end-to-end slice
10. `docs/OPEN-QUESTIONS.md` — open ambiguities, with defaults
11. `docs/SANITY-CHECKS.md` — external verification of the model
12. `docs/TRAINING-GUIDELINES.md` — evidence vs the programme

Contributor and agent instructions: `AGENTS.md`.

USER GUIDE

How to actually use this tool. For the design record see docs/, and for where the project is going see ROADMAP.md.

---

1. What it is, in one paragraph

It reads your training export, cleans it under a documented set of rules, reconstructs sessions, judges the most recent one against a fixed constitution, and prescribes the next session — squat, hinge, bench_squat or bench_hinge. **Python computes every number; the LLM only explains them.** If no model is configured, everything still works — you just lose the prose.

It is a coaching tool with a fixed methodology, not a chatbot that invents programmes. Where the data cannot support a judgement it says `unknown` or `n/a`, never a guess.

---

2. Requirements and install

- Python 3.11+
- Your exports as `data/raw/training_log.csv` and `data/raw/readiness.csv` (schema in data/README.md)

```
python3 -m venv .venv
.venv/bin/pip install -e ".[dev]"

.venv/bin/python -m pytest          # 610 public tests
```

Activate the environment before using the command:

```
source .venv/bin/activate          # bash / zsh
# or: . .venv/bin/activate.fish   # fish
strength-coach --version
```

`strength-coach` lives in `.venv/bin`, so it is only on `PATH` while the venv is active. If you would rather not activate it, call `.venv/bin/strength-coach ...` or `.venv/bin/python -m strength_coach.cli ...` directly — all three are identical.

2.1 Web UI

The same core is available in a local browser UI; the CLI remains the automation interface.

```
./run-web.sh                       # http://127.0.0.1:8000, opens a browser
./run-web.sh --no-open             # ...without opening one
./run-web.sh --port 8001           # another port
./run-web.sh --db other.db         # another athlete database
```

`run-web.sh` is a convenience launcher: it finds `.venv/bin/python` (falling back to `python3`), checks the dependencies, refuses a non-loopback `--host`, and then runs the same `strength-coach web` command. Everything else works unchanged:

```
./run-web.sh                       # http://127.0.0.1:8000, opens a browser
./run-web.sh --no-open             # no browser
./run-web.sh --port 8001           # another port

# or directly:
strength-coach web                 # http://127.0.0.1:8000
strength-coach web --open
strength-coach web --port 8001
```

It binds to **127.0.0.1 only** and has **no authentication**; do not expose it on a network. The pages are Today, Prescribe, Programme, Review, Readiness, Cardio, Import, Athlete, Terminal and Docs. Every load, RIR, warning and calibration number still comes from Python — the browser only displays and submits forms. Docs are rendered locally, so the UI works offline. **Import** accepts the training and readiness CSVs in the browser; the files are parsed in memory and stored in SQLite, not written to the repository.

```
.venv/bin/python -m strength_coach.cli web   # if the venv is not active
```

---

3. First run

```
strength-coach import
```

This reads the two CSVs into `data/coach.db` (git-ignored). It prints how many rows were inserted, how many were already present as overlap, how many sessions were reconstructed, and how many rows were quarantined. A new cumulative export is overlap-matched against the stored rows, so old history is not duplicated; identical rows inside one file are still kept as separate sets. Importing the same file twice writes nothing the second time — identity is the source line, not the content.

A quarantined row is **not** deleted: it is a row with no date or an exercise the taxonomy cannot resolve. It stays visible and counted, because silently dropping rows is how real violations disappear.

3.1 The workflow — how to actually use it

Think in four layers, not one big loop:

layer	how often	where
state — goals, constraints, suitability, e1RM, block	once, then when life changes	Athlete page / `state`
readiness — sleep, energy, soreness, HRV/RHR, optional wearable	before each session	Readiness page / `readiness set`
training — prescribe, perform, import	each session	Prescribe / Import
review + calibration — marks, reports, adjustments	after sessions, weekly	Review / Calibrate

The important rule: **readiness comes first, then the prescription.** A current or future workout is refused until readiness is ready enough; that is deliberate, because the readiness score changes the load and may trigger a deload.

First-time setup

1. **Import** the training and readiness exports (Import page or `strength-coach import`). Check the quarantine count; a row with no date or an unknown exercise is reported, not silently dropped. The full procedure, including when calibration happens, is §4.0.

2. **Record athlete state** (Athlete page or `strength-coach state`): primary goal, constraints, exercise suitability, and any e1RM overrides. Set the block start when the erector ramp should begin (`block --date YYYY-MM-DD`).
3. **Review a few recent sessions** so the rubric has marks. Do this while the sessions are fresh; reconstructed marks are kept but excluded from agreement. Calibration is **not** an import step — run it after this review (or weekly), not immediately after uploading.
4. **Optional device setup**: HRV/RHR are scored only after seven readings in the previous 28 days. Set a per-athlete baseline window, minimum and slopes on the Readiness page if the defaults do not fit. A wearable readiness score (0-100) is optional and only a 10% blend.

Before each workout

1. **Log readiness** for today: `bash strength-coach readiness set --date 2026-10-20 \ --sleep 4 --energy 4 --mood 4 --soreness 4 --stress 4` Device fields and the optional wearable score can be added in the same command or on the Readiness page. Any subset is valid, but a score needs at least three core fields plus sleep or energy.
2. **Check the readiness result.** The page/CLI shows the score, components, overrides, injury ceiling and whether a baseline was available. Do not use the old squat/bench/deadlift readiness fields; they are labels, not inputs.
3. **Prescribe the session**: `bash strength-coach run --date 2026-10-20 --session-type auto` Or use the Prescribe page. Check the warnings, rule citations, load sources and any constraints; a hard constraint or injury ceiling wins over the score. Poor readiness also removes one accessory set (main-lift volume is unchanged) in addition to the load/RIR cut.
4. **Optional:** print the rolling horizon with `strength-coach programme --weeks 4 --remaining` or the Programme page. Treat it as a forecast; the day's prescription is rebuilt when the day arrives.

After the workout

1. **Update the export and import it** (or use the Import page). A new cumulative export is overlap-matched against stored rows, so old history is not duplicated. A genuinely changed row is new, not silently merged.
2. **Review the session while it is fresh**: `bash strength-coach review --session 2026-10-20` Or use the Review page. Marks are append-only; leave an item blank if you do not remember.
3. **Catch up on missed reviews** with `strength-coach review --pending`; this is the routine that keeps calibration current.

Weekly or when something changes

- **Review** the new sessions and run the rubric calibration report (Review page or `strength-coach calibrate`). It is a report, not an automatic weight change.
- **Readiness calibration** (`strength-coach readiness calibrate`) compares the score with the historical labels. It is mimicry of those labels, not validation, and it does not auto-apply weights.
- **Baseline shadow** (`strength-coach readiness shadow`) compares the rolling-median HRV/RHR baseline with a trend projection. It is diagnostic only.
- **Symptoms/injury**: log a pre-session symptom when relevant; the injury signal can cap the readiness score and veto risky work. It never clears a constraint.
- **State changes** (new injury, goal change, exercise no longer suitable, new e1RM/test, block reset) go through the Athlete page or `state`. Never edit the database by hand.
- **Programme view** is a forecast, not a commitment; use it for planning, then prescribe the actual day.

The short loop

```
import  →  state (once)  →  readiness  →  prescribe  →  perform
                                              ↓
                                  import updated export
                                              ↓
                                        review marks
                                              ↓
                                  weekly calibration reports
```

CLI and UI are the same workflow

step	CLI	UI
import	`import`	Import
athlete state	`state`, `block`, `e1rm`, `test`	Athlete
readiness	`readiness set`	Readiness
prescription	`run --date ... --session-type auto`	Prescribe
programme	`programme --weeks N [--start-session-type T]`	Programme
review	`review --session ...`	Review
rubric report	`calibrate`	Review → Run rubric calibration
readiness report	`readiness calibrate`	Readiness → Run calibration
baseline shadow	`readiness shadow`	Readiness → Run baseline shadow
baseline settings	`readiness settings`	Readiness → Baseline settings

The Terminal page can run any of the CLI commands against the same database, but the dedicated pages are safer because they validate and show the result in context.

4. The routines

4.0 `import` — load a history file, and when to calibrate

```
strength-coach import
```

Reads `data/raw/training_log.csv` and `data/raw/readiness.csv`; the **Import** page uploads the same two files. A cumulative export is overlap-matched against stored rows, so old history is not duplicated. Quarantined rows (no date, or an exercise the taxonomy cannot resolve) are kept and counted, never silently dropped.

"Should I run a calibration after importing?" — not because of the import. Import adds sessions. Calibration compares the engine's rubric scores with **your marks**, so with no marks there is nothing to compare and `calibrate` prints `no marks yet`. The procedure is:

1. **Import and read the report.**

field	what to look at
new raw rows / overlapping	overlap means the export was already loaded; no duplicate sets
clean sets inserted / existing	how much training history is new
sessions written	reconstructed workouts

field	what to look at
quarantined	rows that could not be used; investigate a high count
readiness days	days merged from the readiness export
window	the date range actually covered
issues	flags such as unknown exercise names or sentinel values

1. **Review recent sessions** — `strength-coach review --pending` (up to 10 after the last reviewed session) or the **Review** page. The marks you enter are the calibration labels. Only `high`-confidence marks count toward the headline agreement. Do not reconstruct the distant past; `--pending` deliberately stops at the last reviewed session.

2. **Calibrate later, after the review.** `review` prints the report automatically; `strength-coach calibrate` reprints it, and the Review page has **Run rubric calibration**. Cadence: after a batch of reviews, weekly, or at the end of a block. Importing history does **not** require it, and re-importing does not invalidate existing marks.

So for a history file now: **import → check the report → review the recent sessions you actually remember → calibrate after that**, then keep reviewing new sessions and re-run the report on the same cadence.

Two different calibrations, both report-only:

report	compares	where
rubric calibration	your marks vs the engine's rubric scores	`strength-coach calibrate` / Review page
readiness calibration	the readiness score vs the old readiness labels	`strength-coach readiness calibrate` / Readiness page

Neither retunes anything automatically. A rubric change is a separate decision (`DEC-EVAL-03`); the report flags an item for review after five marks and is used to decide, not to auto-adjust.

First import also needs the one-time setup in §3.1: athlete state, block start, and a readiness entry before a current or future prescription.

4.1 `run` — critique the last session, prescribe a date

```
strength-coach run --date 2026-10-20
```

It critiques the most recent stored session on or before that date, prescribes the rolling session type for it, stores both inside a run trace, and prints the prescription with the rule or stored set that justifies every number.

Sample output (numbers are illustrative):

```
prescription 2026-10-20  (day B, full)
  exercise                 sets    reps     load  rir  refs
  ---------------------------------------------------------
  trap_bar_deadlift           4     3-5      158.5    2  PRG-RIR-001, set:2026-10-05/7821
  bench_press_narrow          4     3-5      106.5    2  PRG-RIR-001, set:2026-10-05/7835
  pull_up                     3     6-10         0    2  PRG-DP-001, set:2026-09-20/7012
  lateral_raise               3   10-12       25.0    2  PRG-DP-001, set:2025-11-10/6565
  split_squat                 3    8-10       67.5    2  PRG-DP-001, set:2026-06-05/7359
  note: bench_main: chose bench_press_narrow from [...]
  note: PRG-EREC-001 not assessable: missing symptom.flare, block.week_index

critique 2026-10-15: composite 0.88
  RIR_ADHERENCE             0.75   3/4 rated sets inside the effort window
  WEEKLY_SPINAL_COST        1.00   weighted spinal cost 0 against cap 3
  ESSENTIAL_WORK             n/a   no prescription template for day 'unknown'
  ...
```

Two programme details worth knowing: **The squat session opens with a power slot** (`box_jump` 3×3-5, maximal intent, full recovery, stop when height or quality drops — the "RIR 4" means far from failure), and **The hinge session's pull slot is interchangeable** between `pull_up` and `lat_pulldown`; the engine picks whichever you trained most recently and never compares their loads. Both are explained, with the evidence, in LONGEVITY-REVIEW.md.

Useful flags:

flag	effect
`--session-type T`	squat, hinge, bench_squat or bench_hinge (default: auto rolling)
`--llm on`	ask the configured model for a narrative (see §7)
`--force`	regenerate even though that date was already prescribed
`--db PATH`	use a different database

Re-running the same date writes nothing unless you pass `--force`. Two runs produce two append-only critique rows, by design: history is not rewritten.

If you have declared a block start (`block --date`), the week index drives a 4-week undulation cycle: week 3 puts main lifts on the `maximal` scheme (1-3 reps; hinges stay at 3-5), and weeks 2 and 4 put accessories on `volume` (6-8). The prescription notes which DUP week it used. This is a variety and fatigue-management tool — the evidence says undulating and linear periodization are comparable — and it can be switched off in `config/templates/undulation.v1.yaml`.

4.2 `review` — mark a session, sharpen the calibration

```
strength-coach review --session 2026-10-15
```

It prints the session (working sets, reps, loads, RIR), then the engine's scores, then asks you for a score per item:

```
  RIR_ADHERENCE           engine 0.75  score 0-1:
```

- a number `0`–`1` (0, 0.25, 0.5, 0.75, 1) — higher is always better
- **Enter** — skip this item (writes nothing; an earlier mark is not overwritten)
- `q` — stop and save what you have

Then it asks how well you remember the session: `high`, `low` or `unknown`. **Only `high` counts toward the headline agreement.** A mark from a session you cannot really remember is kept for the record but excluded — it is evidence of a guess, not of the rubric.

Finally it asks for two **optional** numbers: overall session RPE (1–10) and session length (minutes). Press Enter to skip either; they are stored append-only and can be corrected later. Why ask: sets, reps and load give volume, but not how hard the session felt or how long it took (no timestamps). Session RPE is a validated internal-load measure for resistance training; no model reads it yet, so it is capture for later review and a possible future fatigue model. Non-interactive form:

```
strength-coach review --session 2026-10-15 --session-rpe 7 --session-minutes 75
```

The Review page has the same two optional fields.

Catch up on a backlog in one go:

```
strength-coach review --pending            # unreviewed AFTER the last reviewed session, up to 10
strength-coach review --pending --limit 3
```

`run` prints how many sessions are unreviewed and points at this command, so a new workout does not silently leave the backlog behind. It deliberately stops at the last reviewed workout: the distant past does not need reconstructing. When manual workout logging lands (v0.2 §4.2), the same catch-up runs as part of concluding and logging the session.

You do not need ten sessions before using the app. Ten remembered sessions per item is the threshold for retuning a definition or weight (`DEC-EVAL-03`), not for using the tool. Calibration runs from the first review and updates after every session; the rubric simply is not re-fitted until there is enough evidence.

Marks are append-only: marking a session again adds a correction; the latest mark per item wins, older ones stay in the table.

You can also supply marks from a file instead of typing:

```
strength-coach review --session 2026-10-15 --scores private/eval/labelled.yaml
```

4.3 `calibrate` — the running picture

Run this **after you have reviewed sessions**, not as part of importing: with no marks it exits with `no marks yet`. It is a report; it never retunes the rubric.

```
strength-coach calibrate
```

```
session       confidence    compared  agreement     mae  disagreements
2026-10-05    high                 5        60%    0.23  ESSENTIAL_WORK, ROTATION
2023-12-13    low                  4        25%    0.41  ...

headline (high confidence only): 5 comparable items, agreement 60%, mae 0.23

per item (high confidence only)
  RIR_ADHERENCE              1       100%    0.06
  ROTATION                   1         0%    0.40   revisit this item's definition or weight
```

How to read it:

- **agreement** — share of items where your mark and the engine's score are within ±0.25 (one marking step).
- **mae** — mean absolute difference.
- **compared** — how many items were comparable. An item the engine could not measure (`n/a`) is never counted, so it can neither flatter nor punish the result.
- **"revisit this item"** appears only once an item has **five** marks and still disagrees by more than 0.25.

Nothing is retuned automatically. A rubric change is a decision, made after reading the report, and it bumps `RUBRIC_VERSION` so older marks stay interpretable.

4.4 `state` — record your goals, limits and suitability

```
strength-coach state --import private/athlete-state.yaml
strength-coach state
```

The web **Athlete** page edits all of this append-only: profile, constraints, suitability, e1RM overrides, strength tests, block start and symptoms. The CLI below remains the scripted route. This is the §11 athlete state: goals, constraints and per-exercise suitability. It exists so a hard constraint can veto a prescription deterministically, rather than living in a prompt.

- **constraint** — `hard_exclude` vetoes the exercise; `caution` and `monitor` are recorded but do not block. A resolved injury is `monitor`, not `hard_exclude`.
- **suitability** — `can_do: false` is a hard block; `will_do: false` is a preference the engine reports as a warning ("a substitution is needed"), never treats as a safety veto.
- The tables are append-only: re-running the import with `--force` adds a new version, and `state` always shows the latest.

If no state is recorded, a prescription is still produced but says so: "athlete state not recorded: constraints were not evaluated". Silence is not permission.

To rate one exercise without editing the YAML:

```
strength-coach state --rate EXERCISE_ID --can-do yes --will-do yes --feels-bad no
```

An unrated exercise is "unknown", which is why a main-lift substitution (see §4.8) is refused until its replacement has a rating.

4.4b `programme` — a rolling forecast

```
strength-coach programme --weeks 4
strength-coach programme --weeks 4 --remaining
```

It prints a dated rolling session-type forecast for the next N weeks, three sessions per week on Monday/Wednesday/Friday. Use `--start-session-type squat|hinge|bench_squat|bench_hinge|auto` to force the first session; the default is the rolling rule. This is a **forecast**: the day's prescription is rebuilt when the day arrives, using current e1RM, readiness and symptoms. It is not a fixed block and persists nothing. The first session is marked `fatigue_reduced` if the last two weeks exceeded the spinal cap or the last two sessions both flared.

4.5 `block` — declare or reset week 1

```
strength-coach block --date 2026-11-02          # declare week 1
strength-coach block --date 2026-11-02 --force  # reset: the ramp restarts at week 1
```

The block anchor drives the erector ramp. It is **declared, never inferred** (`DEC-RAMP-01`): a gap in the log cannot start a block, because gaps are non-adherence. Resetting is a genuine restart — the ramp goes back to week 1 rather than continuing at week 8 (`DEC-RAMP-02`).

4.6 `e1rm` — override the estimated max

```
strength-coach e1rm set --exercise ssb_squat --kg 160 --from 2026-11-01 \
    --reason "tested single"
strength-coach e1rm set --exercise ssb_squat --kg 150 --until 2026-12-01   # temporary
strength-coach e1rm list
```

Use an override when you know a value the formula cannot see: a tested single, a post-layoff starting point, or an estimate that disagrees with reality. From `--from` (or today) until `--until` (or forever), the override outranks the computed e1RM.

In the prescription it shows up as a normal load, with:

- a citation `e1rm_override:<id>` in its `refs` column, and
- a note: "using a manual e1RM override (160 kg)".

Worth knowing:

- **Append-only.** A correction is another `e1rm set` ; the latest applicable row wins and the older one stays in history.
- **Temporary is first-class.** `--until` expires it; after that the computed estimate returns on its own.
- **It only changes the derived load.** History, scores and verdicts are untouched.
- **Bulk entry** works through the state file: add an `e1rm_overrides:` list and run `strength-coach state --import private/athlete-state.yaml` .

Calibration report. `strength-coach e1rm calibrate` compares each measured `test` with the formula estimate the app would have used before it: per-lift median ratio, mean bias and MAE. The table includes every lift with a recent e1RM, even before any measured test: it shows both the **formula** and the **active measured override** as raw → effective (decayed), which source is used, the last set, and "no measured test; formula only". It also applies a **bounded correction** when at least 3 tests exist and the 95% bootstrap CI of the median ratio excludes 1.0: the formula e1RM is multiplied by the median ratio, clamped to ±10%, and cited as `MOD-E1RM-001` . Ambiguous evidence or a larger bias applies nothing. Measured overrides are never adjusted. The same table is on the Athlete page.

4.7 `symptom` — the pre-session back reading

```
strength-coach symptom --level 0                          # today, no symptoms
strength-coach symptom --date 2026-11-03 --level 2 --settled no --note "tight after sitting"
strength-coach symptom --show
```

One number for the whole back, recorded **pre-session**: `0` none, `1` awareness, `2` annoying, `3` limiting. `--settled` records whether it settled within 48 hours.

A reading counts as a **flare** when the level is 2 or more, or when a level of 1 did not settle within 48 hours. A flare makes `MOD-BACK-001` apply to the next prescription: erector work returns to the weeks 1-3 dose and main lifts are held at RIR 3.

With a reading **and** a declared block start (`block --date`), `PRG-EREC-001` stops returning `unknown` and grades the ramp, citing `symptom:<date>`. Without either input it still says `unknown` — no reading is never read as "no flare".

Correct a mistake by logging again; the latest entry for the day wins and the earlier one stays in history.

4.8 `run --propose` — the model may suggest, the rules decide

```
strength-coach run --date 2026-11-03 --llm on --propose
strength-coach run --date 2026-11-03 --llm on --propose --yes   # no prompt
```

The model sees the deterministic session and may propose a **structural** change:

- swap a non-essential item for a same-pattern alternative;
- change an item's set count within configured bounds;
- drop a non-essential item;
- reorder the session;
- pick a menu option Python computed: `keep` , `easier` , `harder` , `heavier` or `lighter` (RIR and load are re-solved together);
- **substitute an essential lift**, but only for a primary compound you have rated can-do/will-do with no "feels bad", at most once per proposal and no more often than every 42 days; a main-lift swap always asks for confirmation, even with `--yes` ;
- change the **rep scheme**: `maximal` (1-3, main lifts only and never the hinge), `strength` (3-5), `volume` (6-8), `accessory` (8-12). The load is solved for the scheme.

It may **not** supply a load, RIR, percentage or velocity. Any such field is rejected before the proposal is even applied, because the model never does arithmetic here.

Set counts are the one number it may change, within configured bounds. Menu options are bounded by the RIR rule itself: with a 1-4 window, `harder` at RIR 2 is RIR 1, and at RIR 1 it is not offered at all. If the window changes on evidence, the menu changes with it.

You see the proposal and its one-line reason, then accept or decline (`--yes` skips the prompt). If you accept, the prescription is stored with `approved_by = llm+policy` ; if you decline or the proposal is invalid, the deterministic session is kept, the run is marked `llm_downgraded` when the model was at fault, and the proposal plus reasons stay in the run trace.

A proposal is **not attempted** when a safety modifier already changed the session (a fatigue cut or a back flare). The safety decision outranks the suggestion.

4.9 `test` — a strength test that updates the e1RM

```
strength-coach test --exercise ssb_squat --kg 140 --reps 3 --yes
strength-coach test --exercise ssb_squat --kg 160 --reps 1 --max
strength-coach test --exercise ssb_squat --kg 140 --reps 3 --e1rm 165 --yes
strength-coach test --status
```

`test --status` shows the 3RM/5RM cadence: the next scheme and due date. The reminder interval is 10 weeks by default (`config/test-cadence.v1.yaml`); it never records a test automatically.

A **3RM or 5RM** is the default and the recommended test: it gives the e1RM the solver needs without loading the spine like a single. The command reports the estimate first, then records it as a `measured` e1RM override, so the next prescription uses it and cites `e1rm_override:<id>` .

- `--e1rm` lets you **override the computed value** with your own, after the test.
- `--date` sets when the result takes effect (default: today).

- A test is an **assessment, not a training set**: it is not counted as volume and does not enter the RIR rule, because a maximal test is RIR 0 by definition.

True 1RM is allowed, sparingly:

- requires `--max` (a single rep without it is rejected);
- asks for confirmation unless you pass `--yes`;
- is limited to one every **56 days** by default (`--force` overrides, and the config can change the interval);
- 3RM/5RM tests more often than every 28 days print a warning — testing becomes a stimulus.

The evidence behind this is in TRAINING-DYNAMICS.md.

4.10 Gaps, returns, and the injury signal

Gaps are detected, not declared. Every prescription compares its date with the last stored session. If the gap is long enough, the session is amended and says so:

gap	what changes
under 14 days	nothing
14-27 days	keep load, one extra RIR
28-55 days	load −5%, one extra RIR
56+ days	load −10%, one extra RIR, erector ramp steps back one stage

The amended items cite `MOD-GAP-001` and the note reads e.g. "MOD-GAP-001: 31-day gap; prescription amended". Bodyweight work loses a set instead of load. The DUP cycle counts only weeks with a session or a measured test, so a missed week does not advance it. The thresholds are config (`config/thresholds.v1.yaml`, `continuity`) and remain assumptions (§4.10 is documented in CONTINUITY.md).

Injury signal. Log a symptom number per region — low back by default:

```
strength-coach symptom --level 1                    # low back, today
strength-coach symptom --region knee --level 2      # a knee flare
strength-coach symptom --show                       # recent readings, all regions
```

`state` and every prescription then show a general signal plus one per region (`low_back`, `biceps`, `knee`). A region is `unknown` with no readings, `insufficient` when the last reading is more than 14 days old, `active` while it is flaring, `monitoring` in between, or `review` when there is enough evidence to look at a suitability rating.

`review` needs a symptom-free streak **and** enough **tolerated** sessions since the last flare. Tolerated means the session loaded the region and a reading in the following 7 days showed no flare; a session with no reading is `unverified` and does not count. The streak and the number of sessions both scale with how often you have flared before (28 days + 14 per prior flare, 4 sessions + 2 per prior flare, capped).

A `review` may recommend looking at a suitability rating; it never sets one. Every display says: "Not medical clearance. A symptom-free streak is weak evidence; the athlete decides." Readiness is not part of it. The regions and the lifts that load them are config (`config/injuries.v1.yaml`); the full definitions and evidence are in INJURY-SIGNAL.md.

4.10i Workout mode — start, log, end

The local UI has a **Workout** tab. Build a prescription on **Prescribe**, then choose **Start workout**.

- The seven self-report readiness fields are required; HRV, resting HR, bodyweight, body-fat and wearable score are optional.

- If readiness changed after the prescription was built, the UI asks before starting with the old plan; cancel and rebuild if you want a new prescription.
- The prescription snapshot is frozen. Manual sets are added against it; remove a set with the **Remove** button (a removal event, not a delete).
- **End workout** uses the measured timer unless you enter a duration override. Session RPE is required unless you tick **sRPE not recorded**.
- **Abandon** keeps the record but does not mark it completed.

During the workout, the plan is editable: adjust planned sets, swap an exercise, add or remove an exercise. The original prescription is frozen; every override is an event and the modified plan is shown separately. Actual sets can be edited inline after import; imported values show field-level conflicts and the last import can be undone.

CSV import: in Workout mode, paste or choose a file with columns `Exercise,Set,RPE,Value,Weight,Reps`. Preview first. RPE is converted with `RIR = 10 − RPE`, fractions round down, and RIR above 6 is capped at 6 and treated as a warm-up. `???` is missing. Import replaces the same exercise/set and leaves other manual sets alone. The file has no date, so it only imports into the active workout. On **End workout**, sets are materialised, actual vs modified plan and actual vs original prescription are shown, the deterministic rubric is persisted, and **Review workout** opens the rubric sheet. Saving marks runs calibration and closes the workout. The standard dated training export remains the Import-page backfill path. Workout CSV import is in-workout only. This plan is **closed**; CLI workout commands cover status/add/update/remove/end/import, while starting a workout remains UI-only.

4.10b `cardio` — capture the other half

Cardio is a separate exposure, not a strength session. It is captured first; it does **not** change a load yet and no fatigue model reads it.

```
strength-coach cardio --modality run --minutes 40 --avg-hr 150 --max-hr 170 --srpe 6
strength-coach cardio --modality walk --minutes 30
strength-coach cardio --show --weeks 12
```

`--date` defaults to today, `--distance-km` is optional, and sRPE is 1-10. `--show` prints the last N weeks: sessions, total and average minutes, weekly breakdown, modalities, and the WHO reference range (150-300 min/week) as **context**, not a target. The Cardio page in the web UI does the same. The evidence and the proposed prescription are in LONGEVITY-REVIEW.md.

4.10f `exercise` — catalogue listing/editing

```
strength-coach exercise list
strength-coach exercise set --id front_squat --json '{"pattern":"squat"}'
```

Edits and additions are written to `config/exercises/overrides.v1.yaml`; the checked-in taxonomy is not rewritten. The web **Exercises** tab is the form-based version.

4.10d Session types and rolling selection

`run` and the Prescribe page default to a rolling session type rather than a fixed A/B/C calendar. The types are `squat`, `hinge`, `bench_squat` and `bench_hinge`; the rule balances squat/hinge/bench emphasis over recent sessions. The four types are the only override labels:

```
strength-coach run --date 2026-10-20 --session-type auto
strength-coach run --date 2026-10-20 --session-type hinge
```

4.10e `select` — exercise selection

Apply it to a prescription with:

```
strength-coach run --date 2026-10-20 --session-type squat --apply-selection
strength-coach run --date 2026-10-20 --session-type squat --apply-selection --selection-confirm
strength-coach run --date 2026-10-20 --session-type squat --apply-selection --selection-llm
```

```
strength-coach select --date 2026-10-20 --session-type hinge
strength-coach select --date 2026-10-20 --session-type hinge --include-different
strength-coach select --date 2026-10-20 --session-type hinge --llm
```

Ranks the configured variants for the chosen session type from readiness, recent stress, rotation and preference. Constraints and `can_do: false` exclude a candidate; an active injury region penalises one. `--llm` lets the configured model choose **only from that filtered list**; Python validates and falls back deterministically on any error. Candidates now carry a comparability class:

- **A** — load-comparable; can apply automatically.
- **B** — near-comparable (same pattern, different grip/bar/ROM/stability); requires explicit confirmation with `--selection-confirm` or the Prescribe checkbox.
- **C** — different lift/load basis; hidden by default and **never applied automatically**.

The selector still re-solves the load from the selected lift's own history. Class C is shown only with `--include-different` (or "show different lifts" in the UI) for information. A display-only "own-history" ratio (latest e1RM versus that lift's own recent median) appears beside each candidate; it does not affect the score, ranking, class or prescription. The Prescribe **Apply selection** toggle applies A, and B only when confirmed.

4.10h `volume` — weekly hard-set ledger

The Today page also has a small **Descriptive progress** panel: sRPE coverage, performance ratio, weekly volume status and latest back symptom. It is monitoring only and never changes a prescription.

```
strength-coach volume --weeks 4 --date 2026-10-20
```

Working sets by movement pattern, average per week, total and target status. It reports drift. When a pattern is below MEV for three weeks, the engine adds **one accessory set** to that pattern (`MOD-VOL-002`); main lifts are never increased by this rule. The MRV cap still applies. The Readiness page's **Volume valve** sets whether poor readiness removes sets and how many.

4.10g `continuity` — return-to-training calibration

```
strength-coach continuity --date 2026-10-20
```

Shows per-band return count, e1RM retention median, bootstrap interval, fixed factor and applied factor. Fixed continuity remains the fallback when a band is under-powered. `--allow-relax-fixed on|off` changes whether a well-observed band may remove part of the fixed reduction; the same checkbox is in the Prescribe continuity panel.

4.10d `fatigue` — the strength model in shadow mode

```
strength-coach fatigue --date 2026-10-20
```

Per-lift fitness/fatigue state fitted from hard sets and your own e1RM observations, plus an **overall** relative-e1RM row. It shows R^2, span, fitness/fatigue/state with approximate 95% intervals, and a status (`ok`, `weak fit`, `insufficient`, `unstable`); it refuses parameters when the data are sparse or ill-conditioned. **No prescription reads it.** The Prescribe page shows the same table under **Strength fatigue model**. See FATIGUE-FITNESS.md §4a.

4.10f `validate` — retrospective walk-forward

```
strength-coach validate --max-dates 12
```

Replays past cutoffs using only earlier data. It checks:

- **continuity** against the fixed rule;
- the **strength fatigue** model against persistence;
- volume only descriptively, because no predictive volume model exists;
- selection is excluded, because there is no counterfactual.

The 12-date result was: fatigue relative MAE 0.158 vs 0.074 persistence; continuity 0.150 vs 0.128 fixed. Both lost to their baselines, so neither is adopted. The Prescribe page has the same report under **Retrospective validation**. This is not prospective validation; see RETROSPECTIVE-VALIDATION.md.

4.10e Deload policy

A deload is a **low-volume week**, not a new training block:

- a **scheduled suggestion** every 6 weeks from your declared `block_start_date`;
- you are asked at each workout in that week; default is **no**;
- declining skips only that workout; the next workout in the week asks again;
- after the week ends, the suggestion waits for the next cycle;
- also triggered reactively by two consecutive weeks over the spinal cap or two consecutive back flares — these reactive triggers are **automatic**;
- one set removed per item (never below two), load reduced 10%, RIR increased by one;
- cited as `MOD-DELOAD-001` in the prescription and shown on Prescribe/Programme.

Without a declared block anchor there is no scheduled calendar position, so only reactive deloads run. For scripts, `run --deload no|yes` controls the scheduled suggestion; the default non-interactive behaviour is **no deload**. The six-week interval is a configurable policy (`config/deload.v1.yaml`, `ASSUMPTION-32`), not a claim of a proven optimal schedule.

4.10c `measure` — periodic measurements

Readiness is daily; these are **outcomes** you test every few weeks. They never change a load.

```
strength-coach measure --metric vo2max --value 45.2 --method "step test"
strength-coach measure --metric grip_strength_kg --value 52 --method "dynamometer"
strength-coach measure --metric sit_to_stand_5_s --value 9.4
strength-coach measure --show
strength-coach measure --show --metric vo2max
```

The catalogue is `config/measurements.v1.yaml`: `vo2max`, `grip_strength_kg`, `sit_to_stand_5_s` (lower is better), `steps_per_day`, `sitting_hours_per_day` (lower is better). Each metric is declared once with its direction and aggregation; a later direction change is refused rather than silently reinterpreting old values. `--method` keeps the protocol next to the number, which matters for VO2max estimates.

Grip strength is **only** a measurement now; readiness no longer accepts it (`DEC-GRIP-01`). Log it here or on the Measures page.

The **Measures** page in the web UI does the same and shows the latest value and change per metric.

---

5. Reading the output

The prescription

Main lifts carry three **warm-up ramp sets** at roughly 50/70/85% of the working load; they are not counted as working volume. The Prescribe page shows them under the exercise.

column	meaning
exercise	canonical id from the taxonomy, not the raw log name
sets	prescribed working sets
reps	the rep range; add load only at the top of it
load	solved from your recent e1RM, **rounded down** to your increment
rir	target reps in reserve
recent stress	advisory decayed hard sets: `high`, `normal` or `low`; **never changes the load**
readiness	per-lift 0–1 score from global readiness + e1RM trend, RIR and VBT; can only **lower** a load
refs	the rule id **and** the stored set that justify the load

Loads come from the best usable e1RM in the last 84 days, inverted through the same Epley+RIR formula that produced it, then rounded down to 0.5 kg (your microplates). Rounding up would push the set above the target RIR, so it never happens. An e1RM older than 21 days is decayed by 1% per week (capped at 10%); this applies to measured overrides too, and cites `MOD-E1RM-DECAY-001`. If the continuity rule is already handling a 28+ day gap, the age decay is skipped to avoid counting the same absence twice.

`readiness` is the per-lift score: it starts from the global/category readiness and nudges it by recent e1RM, mean RIR and velocity-vs-calibration. A lower score can reduce this lift's load (and add RIR); a higher one never increases it. No per-lift readiness instrument is validated, so this is deliberately asymmetric (`docs/LIFT-READINESS.md`).

`recent stress` is a fixed exponential decay of your working sets (48 h half-life, high at 6 decayed sets). A lift prescribed within 48 h of high stress gets a warning saying so, but the load and RIR are unchanged — it is context for your judgement, not a fatigue model. The constants are in `config/stress.v1.yaml` and are assumptions (`ASSUMPTION-24`).

That e1RM is an **estimate, not a calibrated measurement**: Epley with RIR folded in is an approximation, and no bias correction is applied. A measured `test` writes an override and the next prescription cites it. A calibration report (measured vs predicted, bias/MAE per lift) is proposed but not implemented; see FATIGUE-FITNESS.md §6.

If a load is `-` and the output says `NO_LOAD`, there was no usable e1RM in the window. The prescription is then **not executable**, and it says so instead of inventing a number.

`CONSTRAINT_VETO:<exercise>` means a stored hard constraint blocks that exercise; the prescription is not executable until the constraint or the template changes. A warning that an exercise "you will not do" appears is a **preference**, not a veto — it asks for a substitution rather than blocking the session.

The critique

Six items are measured, two are not:

item	measures
`RIR_ADHERENCE`	share of rated primary-compound sets inside the allowed window (RIR 1–4)
`PROGRESSION_ADHERENCE`	accessories added load only after topping the rep range
`WEEKLY_SPINAL_COST`	the week's weighted spinal load against a cap of 3
`ESSENTIAL_WORK`	the day's essential lifts were present
`REDUNDANCY`	no hinge + erector stacking, no two hinge lifts
`ROTATION`	selection stable for self-directed sessions; expected rotation for app-assigned
`SESSION_DURATION`	**not measurable** — the export has no timestamps
`VELOCITY_LOSS`	**not measurable** — one mean velocity per set

`composite` is the weighted mean over the items that were measurable, re-normalised — a missing metric never drags the score to zero.

5.1 Readiness — how it is derived

Readiness can only reduce a prescribed load, so its derivation is written down in full in READINESS.md. In short:

- the export is read as text and then as numbers; `0` becomes missing in the fields where a physical zero is impossible (bodyweight, HRV, RHR, fat %, the three readiness scores);
- duplicate rows on one date are merged: **most complete wins, ties go to the latest**, and the merge note records which candidate won and why;
- a global 0.00–1.00 score is computed from sleep, energy, mood, stress and soreness; it needs at least three of those and either sleep or energy;
- HRV and RHR enter at half weight only with a 28-day baseline (7 readings); an optional **wearable readiness score (0–100)** is a 10% blend, never a replacement;
- the old sleep/energy rule remains a floor (at or below 2 still cuts loads by 5%);
- missing or insufficient readiness blocks a current/future workout rather than silently omitting the modifier;
- the three old per-lift readiness values are calibration labels, not inputs;
- readiness is **not per lift**: there is one global score plus manual **category** overrides (squat, hinge, horizontal push, …). An exercise inherits its category's factor. A validated per-lift readiness score does not exist.

Manual entry (any subset):

```
strength-coach readiness set --date 2026-10-20 --sleep 2 --energy 3 --soreness 1
strength-coach readiness set --date 2026-10-20 --energy 4 --note "slept badly"
```

The 1-5 fields and the device fields (HRV, resting HR, bodyweight, body fat, grip strength in kg, wearable readiness 0–100) are accepted in any combination. The old squat/bench/deadlift readiness fields are not. A manual value wins **per field**; an imported row fills the fields you did not provide. Re-logging a date appends a new version and the latest value wins — nothing is edited or overwritten. The command prints each field with its source (`manual` or `import`).

Bulk entry uses the same store through `state --import`:

```
readiness:
  - date: 2026-10-20
    sleep: 2
    energy: 3
```

Override the computed score at global or category scope (0.00–1.00, reason required):

```
strength-coach readiness override --date 2026-10-20 --scope global \
  --value 0.55 --reason "felt worse than the form"
strength-coach readiness calibrate     # report only; never auto-applies weights
```

A **current or future workout requires a ready assessment**; otherwise `run`/the web page refuses and tells you what is missing. Historic prescriptions remain reproducible. The web Readiness page also shows the computed score and has a **Run calibration** button.

To change which fields matter, the threshold, or the merge tie-break, edit `config/readiness.v1.yaml`. Only implemented values are accepted — anything else raises rather than silently doing nothing.

Per-athlete baseline settings override the config from a date:

```
strength-coach readiness settings --date 2026-10-20 \
  --baseline-days 14 --baseline-min-readings 5 \
  --hrv-slope 0.4 --rhr-slope 0.4 \
  --hrv-weight 0.5 --rhr-weight 0.5 --wearable-weight 0.1
strength-coach readiness shadow --date 2026-10-20   # diagnostic only
```

The same controls are on the Readiness page. Settings are append-only; the latest effective row wins. `readiness shadow` compares the rolling-median baseline with a trend projection and never changes a prescription.

5.2 Velocity targets — what they mean

When the export contains enough velocity data, a prescription item gains a `vel` column:

```
exercise                sets    reps     load  rir    vel  refs
bench_press                4     3-5     84.5    2   0.31  PRG-RIR-001, set:...
```

That number is the **calibrated median mean velocity** for that lift, rep count and effort band, taken from your own clean sets. It is not a population table. A target appears only when the calibration has **12 clean sets across 3 sessions** for the lift (or 24 in its movement family), and the basis next to it records the scope and sample size.

How to use it: the first working set should move at roughly the target. Noticeably faster means the load is light for the day; noticeably slower means accumulated fatigue or a heavy day. It is a reference, not a stop rule — the app does not yet measure within-set velocity loss.

What is deliberately not shown:

- a target when the data is thin — the item simply has no velocity, and the prescription says **unavailable** rather than guessing;
- a target for the **trap-bar deadlift**, because the velocity-to-RIR relationship for the hex bar failed accuracy criteria in the literature and in practice;
- a target for a session already modified by a fatigue or back-flare rule, because the load no longer matches the calibration.

The evidence and best practices are in VBT-EVIDENCE.md; the upgrade path (per-rep velocity, velocity loss, individual load-velocity profiles) is in VBT-CALIBRATION.md §9.

---

6. Configuration

Everything that changes a number lives in config, not code.

file	what it controls
`config/thresholds.v1.yaml`	load increment (0.5 kg), rounding, e1RM window (84 days), fatigue multiplier (0.95)
`config/templates/session-types.v1.yaml`	session slots, sets, rep ranges, rest
`config/rules/programme.v1.yaml`	the constitution: rule ids, severities, thresholds
`config/exercises/taxonomy.v1.yaml`	exercise identity and properties

Change an increment or a rep range and the golden test will fail on purpose — update the fixture only if the change is intended.

---

7. The LLM (optional)

With nothing configured, the system uses `NullLLM`: no network call, no narrative, everything else identical. To enable narration:

```
cp config/llm.example.toml config/llm.local.toml   # git-ignored
$EDITOR config/llm.local.toml                      # base_url, model, api_key
strength-coach run --date 2026-10-20 --llm on
```

The default endpoint is DeepSeek (`https://api.deepseek.com/v1` , `deepseek-chat`), so setting a provider and a key is enough. Any OpenAI-compatible endpoint works.

Check it before relying on it:

```
python tools/llm_check.py           # effective settings, key masked
python tools/llm_check.py --call    # one test prompt
```

Environment variables override the file: `STRENGTH_COACH_LLM_PROVIDER` , `_MODEL` , `_BASE_URL` , `_API_KEY` , `_TEMPERATURE` , `_MAX_TOKENS` , `_TIMEOUT_S` , and `_CONFIG` for a different file.

What the model may and may not do:

- it receives the facts as JSON and is told to use only those numbers;
- its output is stored as a `run_event` , **never** written into a prescription, a score or a verdict;
- a failed call is recorded and the deterministic run continues.

The model may **narrate or propose** within the rules (`run --propose`). It proposes a structured intent — a substitute, a set change, a scheme — and the policy engine accepts, adjusts or vetoes it; the numbers still come from Python, and the trace records `approved_by = llm+policy` when a proposal survives. The CLI remains the proposal surface; the web UI does not yet expose it.

---

8. The other tools

command	what it does
`python tools/profile_data.py`	regenerates the data profile (reproducibility check)

command	what it does
`python tools/baseline.py`	regenerates the strength baseline
`python tools/import_export.py`	same as `strength-coach import`, usable without install
`python tools/eval_prepare.py --session DATE`	prints a labelling sheet and writes a skeleton
`python tools/backtest.py`	agreement report from the database (`--labels` for a YAML file)

Tests: `.venv/bin/python -m pytest`, and the markers `-m golden` (locked numbers) and `-m integration` (touches the real CSVs).

---

9. Troubleshooting

- **`strength-coach: command not found`** — the venv is not active. Run `source .venv/bin/activate` first, or use `.venv/bin/strength-coach` directly. | symptom | cause and fix | |---|---| | `no stored session on or before <date>` | the date predates the log, or nothing is imported. Run `strength-coach import`. | | `already prescribed for <date>` | `run` is idempotent per date. Add `--force` to regenerate. | | `NO_LOAD: ... no usable e1RM` | no rated set for that lift in the last 84 days. Train it, or widen `e1rm_window_days`. | | `SPINAL_CAP ... exceeds cap` | the week already contains a hinge exposure; two hinge sessions cost 2+2 against a cap of 3. Move the day or accept the warning. | | `PRG-EREC-001 ... missing symptom.flare` | expected: there is no symptom input yet. It becomes assessable in v0.1. | | `PRG-DP-001 ... missing set.reps.<exercise>` | the scoped accessory has no comparable working set yet. Train it once; the range itself now comes from all day templates. | | `no marks yet` | run `strength-coach review --session DATE` first. | | `llm: off (NullLLM)` | no model configured; see §7. | | `call failed: HTTP 401` | wrong key, or the key is in the environment but not the file. Check `tools/llm_check.py`. |

---

10. Limits you should know

- **Days A, B and C** are all prescribable. The erector ramp still needs a declared block start (`block --date`); until then `PRG-EREC-001` is `unknown` on every day.
- **No symptom input.** Back-symptom rules return `unknown` rather than assuming no flare.
- **No timestamps.** Session duration is unmeasurable and is stored as such.
- **VBT targets are not prescribed** (`DEC-VBT-01`): the calibration exists, the step from a velocity band to "do this set at X" is not specified, so prescriptions carry no velocity target and no calibration id.
- **The spinal cost is load-blind.** A light good-morning at RIR 3 counts the same as a heavy one. Known gap, recorded in the roadmap.
- **Loads are estimates.** e1RM uses Epley with the RIR folded in (`ASSUMPTION-03`); treat the number as a target, not a measurement.
- **Calibration is thin.** One remembered session so far; the rubric is not tuned to it.
- **One day at a time, and one input path.** The CLI has a rolling forecast (`programme`); the web programme view is still planned. There is no manual logging or alternative file formats. Those are scoped in ROADMAP.md §4 (v0.2), including how missing readiness will be reported rather than assumed.

---

11. Privacy

This repository is public and contains no personal data.

- `data/` (exports, `coach.db`) is git-ignored.
- `private/` (evidence profile, baseline, your marks) is git-ignored.

- `config/llm.local.toml` and `.env` are git-ignored; keys are masked in every diagnostic.
- The code, schemas, rules and tests contain no athlete-specific values.

12. Using this for someone else

The app is local-first and one database per person. Nothing personal is committed, so a second athlete runs exactly the same code against their own database:

1. **Choose their database path.** Every command takes `--db /path/to/their.db`; the default `data/coach.db` belongs to the first athlete. Never point two people at one file.
2. **Create their state file.** Copy `config/athlete-state.example.yaml` to `private/athlete-state.yaml` (git-ignored) and fill it in. Leave `block_start_date: null` until they declare week 1.
3. **Import their state:** `bash strength-coach --db /path/to/their.db state --import private/athlete-state.yaml`
4. **Import their training export** into `data/raw/`, or point the importer at it, then: `bash strength-coach --db /path/to/their.db import strength-coach --db /path/to/their.db block --date YYYY-MM-DD strength-coach --db /path/to/their.db run --date YYYY-MM-DD --session-type squat strength-coach --db /path/to/their.db review --session YYYY-MM-DD strength-coach --db /path/to/their.db calibrate`
5. **Add their exercise vocabulary in YAML**, not Python: `config/exercises/taxonomy.v1.yaml`. Unknown names are reported, never silently dropped.

Privacy rules that make this safe:

- `data/` and `private/` are git-ignored; a tracked-file test fails if anything personal is committed.
- LLM keys live in `config/llm.local.toml` (git-ignored) or the environment.
- Their export, state, marks and database never touch the first athlete's files.

The portability test suite runs the full loop on two independent databases with different lifts, dates and days, and asserts neither database sees the other's artifacts.

ROADMAP

The resume point for this project. If you are picking this up cold — human or agent — read this file first, then `AGENTS.md` for the rules of the codebase.

Last updated: 2026-10-23 · **v0 complete, v0.1 done, v0.2 done, v0.3 done, v0.4 planned** · version `0.3.1-dev` · **611 public tests** (626 with the 15 private integration tests)

---

1. How to resume

```
cd strength-training-app
.venv/bin/python -m pytest            # 625 locally, including private integration
.venv/bin/python tools/profile_data.py   # data profile (needs data/raw/)
.venv/bin/python tools/baseline.py       # strength baseline
```

Then read, in order:

1. `USER-GUIDE.md` — how to use what exists
2. **this file** — where we are and what is next
3. `docs/DECISIONS.md` — decisions that are expensive to reverse
4. `docs/OPEN-QUESTIONS.md` — what is still ambiguous, with defaults
5. `docs/V0-SCOPE.md` — the agreed scope and its exclusions
6. `AGENTS.md` — invariants, layering, definition of done

Resume rule: update the status table below in the same commit as any phase change. If the table disagrees with the code, the table is wrong — fix it before adding features.

2. Status

#	Phase	Status	Artifacts
0	Design + approval	**done**	`docs/ARCHITECTURE.md`, `DATA-MODEL.md`, `DATA-QUALITY.md`, `RULES-FORMAT.md`, `VBT-CALIBRATION.md`, `MODULE-LAYOUT.md`, `V0-SCOPE.md`, `OPEN-QUESTIONS.md`, `SANITY-CHECKS.md`, `TRAINING-GUIDELINES.md`
1	Ingest (clean + sessions)	**done**	`ingest/clean.py`, `ingest/sessions.py`, `config/templates/inference.v1.yaml`
2	Derived features + VBT calibration	**done**	`features/derived.py`, `features/baseline.py`, `features/vbt_calibration.py`, `domain/e1rm.py`, `domain/load.py`, `domain/rir.py`
3	Store (SQLite, append-only)	**done**	`store/migrations/0001_initial.sql`, `store/db.py`, `store/repositories.py`, `store/importer.py`
4	Rules loader + policy engine	**done**	`rules/loader.py`, `registry.py`, `context.py`, `policy.py`; predicates for all five rules
5	Prescription engine (four session types) + deterministic fallback	**done**	`prescribe/templates.py`, `config.py`, `loads.py`, `engine.py`, `fallback.py`; `config/thresholds.v1.yaml`
5.5	Athlete state — DATA-MODEL §11 tables	**done**	`store/migrations/0003_athlete_state.sql`, `prescribe/constraints.py`, `cli state` / `cli block`; hard constraints veto

#	Phase	Status	Artifacts
6	Critique rubric + trace	**done**	`critique/facts.py`, `critique/rubric.py`, `trace/run.py`; 6 measurable items + 2 unmeasured
7	LLM layer (narrative only)	**done**	`agents/client.py`, `config.py`, `narrative.py`; `config/llm.example.toml`, `tools/llm_check.py`
8	Backtest harness + labelled eval set	**done**	`eval/harness.py`, `eval/labels.py`, `tools/eval_prepare.py`, `tools/backtest.py`; 5 high-confidence sessions, 75% agreement, MAE 0.20
9	CLI (`run` / `review` / `calibrate` / `import`)	**done**	`cli.py`; prescriptions persisted; `review --pending` catches up; calibration loop closed
v0.2	Input flexibility + programme view	**done**	readiness entry/score/devices/settings + gate (§4.1/4.9), continuity (§4.6), injury signal (§4.7), local web UI with import/state editing/Programme/Cardio/Terminal/calibration (§4.10/4.11), cardio capture + weekly summary (§4.13), labelled set (§4.14), session RPE/length, recent stress, per-lift readiness, e1RM calibration + bounded correction + age decay
v0.3	Strength/data depth	**done**	manual logging via live workout mode (§4.2/§4.21, closed), import adapters closed (§4.3), learned readiness weights evaluated and closed (§4.9), LLM proposal view closed as CLI-only (§4.11), continuity calibration (§4.12; calibrated factor rejected, fixed fallback), exercise selection apply + comparability + display-only history (§4.15; weakness/ratios in §4.20), strength fitness-fatigue shadow model (§4.16; retrospective rejected), rolling session types / upper-body balance / set reduction / volume management v1 / under-MEV accessory progression / warm-up ramp / 3RM cadence / descriptive progress dashboard (§4.17), live workout mode (§4.21, closed)
v0.4	Cardio + calibration depth	**planned**	cardio prescription and cardio impulse/fatigue model (§4.13 remaining); deeper labelled calibration set (§4.14)
v0.5	Video analysis	**planned**	video capture/analysis and its UI parity (§4.8, §4.11 video part)
v0.1	Complete	**done**	3.1-3.4, 3.6-3.7 done; 3.5 video moved to v0.2; 3.8 labelled set moved to §4.14
—	UI	**local web done; hosted UI deferred**	`strength_coach/web/`, `docs/WEB-UI.md`, `http://127.0.0.1:8000`

Version plan (2026-10-23)

version	boundary	status
v0.2	readiness, continuity, injury, web UI, cardio capture, measurements, session RPE/length, recent stress, per-lift readiness, e1RM calibration/decay	**done**
v0.3	everything strength/data-depth that is not cardio or video: manual logging, adapters closed, LLM proposal view CLI-only, continuity calibration, exercise selection, strength fitness-fatigue model	**done**
v0.4	cardio prescription/model plus the deeper labelled calibration set moved from v0.3	**planned**
v0.5	everything video: capture, analysis, UI parity	**planned**
future	LLM-guided suggested programme (§4.18)	**idea only; not scheduled**

Rule: cardio work does not enter v0.3; video work does not enter v0.3 or v0.4. The strength model (§4.16) is strength-only and closed with v0.3. The labelled calibration depth moves to v0.4.

What works end to end today

A real export can be cleaned, flagged, quarantined, grouped into sessions, given a programme day guess, calibrated for velocity targets, and **persisted**: one import writes `raw_row`, `clean_set`, `session` and `quarantine` inside a replayable `run` / `run_event` trace, and re-importing the same file writes no duplicate rows. The constitution can now be evaluated against that history: `rules` parses the YAML into hash-bearing `Rule` objects, the registry supplies one predicate per `check.fn`, and `policy` returns a verdict per rule with citable evidence. A session for **any programme day (A, B or C) is now generated**: loads solved from recent e1RM (or double progression for accessories), rounded down to the athlete's 0.5 kg microplates, with MOD-FAT-001 reducing loads when sleep or energy is <= 2. Every item cites a rule id and a stored set, and the prescription refuses to be executable if a load, the RIR window, the erector-day rule or the weekly spinal cap would be broken. A performed session can now be **critiqued**: six rubric items score it (RIR adherence, progression adherence, weekly spinal cost, essential work, redundancy, rotation) with evidence references, and two are stored as `measurable = 0` because the export cannot support them (session duration, within-set velocity loss). The critique is persisted append-only inside a `run` trace. An **optional narrative** can explain either artifact: the model receives the facts as JSON, may not compute, and its output is recorded as a `run_event` rather than written into a prescription or a score. With no model configured (`NullLLM`) the run is structurally identical and loses only the prose. There is **the full v0 slice runs**: `strength-coach run --date` imports if needed, critiques the last session, prescribes the date, persists both inside a run trace, and prints a rationale that cites a rule or a stored fact. `review` is the calibration loop: it shows the session beside the rubric's scores, takes marks, stores them append-only, and reports running agreement immediately. Two of the three marks are reconstructed from memory and excluded; on the one remembered session, 60% of items agree within ±0.25 (`DEC-EVAL-03/04`).

3. v0.1 — complete

v0 is complete (see §2 and the user guide). This is the agenda for the next version, in dependency order. Nothing here blocks using v0 as it stands.

3.1 Athlete state — the §11 tables (done)

Scope: `athlete_profile`, `athlete_constraint`, `exercise_suitability`, `metric_definition`, `symptom_log`, as numbered migration `0003_athlete_state.sql`, plus writers and a CLI to enter them.

Why it is first: it is the only way goals, injuries and preferences become state rather than something an agent remembers. Until it exists, no hard constraint can veto anything and an empty table is indistinguishable from "no constraint".

Acceptance: the five tables exist; a `hard_exclude` constraint vetoes a proposed exercise deterministically (tested); `exercise_suitability.will_do = 0` is surfaced as a warning, never a silent drop and never a safety veto; unrecorded state reports `none_recorded` and a warning, never "nothing is wrong".

Delivered: all five tables append-only; `strength-coach state [--import FILE]` and `strength-coach block --date`; `prescribe/constraints.py` applies constraints and suitability inside the engine's validation. `constraint` became `athlete_constraint` because the former is a reserved SQL keyword (`DEC-STATE-01`). The athlete's answers are imported from `private/athlete-state.yaml`: longevity goal, three resolved monitor-severity back tweaks, and 14 exercises all rated can-do/will-do. Also delivered: **manual e1RM overrides** (`e1rm set/list`, append-only, optional expiry) so a tested value outranks the computed estimate and is cited in the prescription.

Input: `private/docs/INFO-REQUESTS.md` §1-3, read back for confirmation before being written. **Status: the core answers are collected in `private/docs/ATHLETE-STATE.md`** (goals, back history, suitability, equipment, schedule). Two consequences are already decided: `longevity` joins the `primary_goal` enum, and training-log gaps are recorded as **non-adherence**, never as symptoms.

3.2 Days A and C (done)

Depends on 3.1. The SSB question (H1) is answered: the bar is available. The erector-ramp anchor is also settled in principle — **the athlete declares the official start, and can reset it at any time** (`DEC-RAMP-01`), so the anchor is stored, never inferred.

Scope: Day A and Day C templates; `block start [--date]` command that sets or resets week 1 as a new profile row (never an edit); **a reset restarts the ramp at week 1** (`DEC-RAMP-02`), not just the calendar; the ramp reads that row.

Acceptance: a template per day with its own essential lifts; `PRG-SCHED-001` enforced on both (no erector work on the hinge day, erector work allowed on A/C); a golden prescription per day, in the same style as `tests/fixtures/golden_day_b.yaml` .

Delivered: Day A (squat-led) and Day C (bench-led) templates with erector + accessory work; `run --session-type T` ; golden fixtures for the canonical types. Core work was added on 2026-10-20 (`ab_wheel` on A, `hanging_leg_raise` on C, `DEC-CORE-01` ; `docs/CORE-TRAINING.md`). Day A opens with a power slot (`box_jump` , `DEC-POWER-01`) and Day B with a vertical-pull variant slot (`pull_up` / `lat_pulldown` , `DEC-PULL-01/02` ; `docs/LONGEVITY-REVIEW.md`). Bodyweight erector work is prescribed at load 0 with an explicit warning, never as a 0.5 kg "progression". The athlete declares the block start whenever they choose, so the ramp stays `unknown` until then.

Consequence: `PRG-DP-001` should stop being `unknown` . It was unknown because its scope includes accessories that no single day prescribes, so the per-day rep-range map always lacked one exercise. Fixed in Q16 by pooling rep ranges across all three templates; see `docs/CORE-TRAINING.md` §5.4.

3.3 Symptom input (done)

Depends on 3.1 (`symptom_log`). This is the missing safety input (Q4): a manual `--back 0-3` per session, with `settled_within_48h` .

Timing is **pre-session only** and it is **one number for the whole back** for now; per-region detail is deferred (`DEC-SYMPTOM-01`).

Acceptance: `PRG-EREC-001` and `MOD-BACK-001` evaluate instead of returning `unknown` ; the stability gate uses real entries; every verdict cites the symptom rows it used.

Delivered: `symptom --level 0-3 [--settled yes|no] [--date]` and `symptom --show` ; the rules context derives the flare and the block week; `PRG-EREC-001` returns a real verdict once a reading and a declared block exist, citing `symptom:<date>` ; a flare makes `MOD-BACK-001` transform the prescription (erector back to the weeks 1-3 dose, main lifts at RIR 3). Multiple entries per day are allowed, latest wins (migration 0005).

3.4 The LLM may tweak the prescription — within the rules (done)

Today the model only narrates. In the next version, with `--llm on` , it may also propose a modified Day B session, subject to the same policy engine that judges a performed one. The standing rule (AGENTS.md §3.3) is what makes this safe: **the policy engine may veto or downgrade the model; the model may never veto the rules.**

What it may propose:

- substitute an accessory for one with the same movement pattern and compatible properties;
- choose between the day's allowed bench variants;
- adjust set counts within configured bounds;
- drop a non-essential item, or reorder items.

What it may **not** do:

- supply a raw load — every load is re-derived by the Python solver from the proposal's structured intent (AGENTS.md §3.2). The proposal schema has no field for a number on the bar;
- push RIR outside the rule window;
- add direct erector work on the hinge day (`PRG-SCHED-001`) or push the week over the spinal cap;
- invent an exercise outside the taxonomy, or one the athlete's constraints exclude.

How it is checked:

- the proposal is structured (exercise ids plus bounded deltas), never prose;
- `rules.policy` evaluates the proposed session against the constitution;
- `veto` → the deterministic fallback is used; `downgrade` → the reduction is applied; `warn` / `advisory` → recorded;
- `prescription.approved_by = "llm+policy"` , and the proposal plus its verdicts are stored in the `run` trace so the decision is replayable;
- with the model off or failing, the deterministic prescription is byte-identical.

Why it waits for 3.1: a substitution is only safe once the athlete-state tables say which exercises are excluded or merely unwelcome. Without them the model would be choosing from a pool that includes movements the athlete cannot or will not do.

Consent: the athlete wants **both substitutions and adjustments, but shown before applying**. No proposal is applied silently — `run --llm on` prints the difference and asks; a `--yes` flag covers scripting.

Decision needed at implementation time: this narrows `DEC-LLM-01` — prose stays event-only, but a structured proposal that passes policy may become the prescription. That needs a new `DEC-LLM-04` recording exactly which fields the model may influence and which it may not.

Acceptance:

- an illegal swap (erector work on the hinge day) is vetoed, the deterministic prescription is used, and the veto is visible in the trace;
- a legal substitution is accepted, persisted with `approved_by = llm+policy` , and its loads still come from the solver;
- a proposal carrying a raw load or an out-of-window RIR is rejected by validation before it reaches the engine;
- nothing is applied without an explicit confirmation, and declining leaves the deterministic prescription untouched;
- the existing LLM-off equivalence test passes unchanged;
- the golden fixture gains an accepted-proposal case and a vetoed one.

Delivered: `run --propose [--yes]` (requires `--llm on`). Parsing rejects any computed-number key — load, weight, RIR, percentage or velocity — anywhere in the payload, so a confused model cannot inject a weight. Substitutes are narrowed to the same movement pattern and filtered by stored constraints; the replacement is rebuilt by the engine's own `build_item` , so the load comes from the same solver as the deterministic path. The result is judged by `validate_prescription` and the constraint check before anything is stored. Acceptance is explicit (`--yes` for scripting), `approved_by = llm+policy` marks the artifact, and the proposal, verdict and reasons are `run_event` s. Proposals are not attempted on a session a safety modifier already changed. A proposal may also name one of a small menu of Python-computed options (keep/easier/harder/heavier/lighter); the menu's RIR width is the rule's window, so a documented window change widens it without a code change. `DEC-LLM-04/05` record the contract.

3.5 Video analysis — moved to §4.8

Video is a v0.2 feature; the full scope and its privacy constraints live in §4.8.

3.6 Another athlete, or a fresh start (done)

The tool should be usable by someone else with their own export, and by the athlete after a year away, without editing code or mixing data. This is **not** multi-tenancy: one database per person.

Scope: document the first-run path for a new athlete (export → state file → block start); verify nothing personal is baked into config or defaults; keep `--db` as the only switch needed; keep the exercise taxonomy editable so a different exercise vocabulary is a YAML edit.

Acceptance: a second database with a different athlete's synthetic history runs `import`, `run`, `review` and `calibrate` with no code change and no access to the first database; no personal value appears in any committed file; the user guide has a "new athlete" checklist.

Delivered: `tests/test_portability.py` runs the whole loop on two databases with different lifts, dates and days, and asserts neither sees the other's artifacts; a tracked-file test fails if anything under `private/` or `data/` (except the folder README) is committed; the guide has section 12 with the checklist, and `config/athlete-state.example.yaml` is a placeholder state file with no personal values. `DEC-PORT-01` records the one-database-per-person rule.

3.7 VBT prescription wiring (done)

Deferred by `DEC-VBT-01`. The calibration (band, sparsity ladder, versioning) already exists and is tested; what is missing is the rule that turns a band into a target:

- which rep in the set is targeted — the first, the last, or a mean;
- what tolerance counts as "on target" before a set is flagged;
- what to prescribe when the band is wide, or when the lift is `uncalibrated`.

Acceptance: `PrescriptionItem` carries `velocity_target` and `vbt_basis`; `prescription.calibration_ids` cites the rows used; an uncalibrated lift says so explicitly; the golden fixture gains velocity columns.

Do not start this before those three questions have answers — inventing them is what the deferral exists to avoid.

3.8 Labelled set — moved to §4.14

This is now a v0.2 item; the content and acceptance are in §4.14. v0.1 is complete.

3.9 Deliberately not planned

- A UI (deferred by request).
- Several athletes in one database, or a hosted service: local-first, one database per person (3.6 covers portability; §4.8 covers opt-in video).
- RAG over the programme: rules are cited by id, and exact lookup beats semantic search here.
- Automatic rubric retuning: the report proposes, a person decides.

4. v0.2 — input flexibility and the programme view

v0.1 assumes three things that the athlete has now explicitly challenged: that input arrives as one CSV in one known shape, that only one day is prescribed at a time, and that readiness is only available on days the export happened to record. v0.2 loosens all three.

Nothing here changes the constitution or the append-only rule. Missing data still yields `unknown`, never a default.

4.1 Readiness on demand — daily, occasional, partial (done)

The fatigue modifier is only as good as the readiness it sees. Today those rows come from a single exported CSV, so a missing day silently means no modifier, which is easy to mistake for fresh. That is the failure this item removes.

Scope: `readiness set --date ... --sleep --energy --soreness ...` accepting **any subset** of fields; bulk entry through `state --import`.

Precedence (decided): a manual value wins **per field**, the imported row fills the fields it does not provide, and the merge note records both values and which source won. Nothing is silently overridden.

Which fields drive the modifier is config (decided): MOD-FAT-001 reads sleep and energy today, and the list is a config value so it can grow. Other fields are recorded and reported, but do not move a load.

Robustness rule: an absent input never becomes a value. A prescription for a date without readiness says "readiness unknown for ; fatigue modifier not applied". A partial row applies only the fields it has.

Acceptance: a day with 2 of 7 metrics applies the modifier on those two only; a day with none leaves it unapplied **and says so**; a week-long gap changes nothing else; re-logging a date appends a version rather than editing it.

Data: `readiness_day` is currently one upserted row per date. This needs an append-only `readiness_entry` with latest-wins reads, so history is preserved.

Delivered: `readiness set` with any subset of fields, bulk entry through `state --import`, append-only `readiness_entry`, per-field manual-over-import precedence with a `sources` map, and the explicit readiness unknown warning. The derivation is in `docs/READINESS.md`; the changeable parts are config in `config/readiness.v1.yaml` (`DEC-READINESS-04`).

4.2 Logging workouts three ways (v0.3; manual path closed)

Three input paths, one downstream shape:

1. **the same export file** — already supported and idempotent by content hash;
2. **manual post-workout entry** — **per set** (decided): `log set --date --exercise --weight --reps --rir`, repeated per set. No exercise-level shorthand: the set is the unit of record everywhere else, and expanding a summary into sets would invent detail. Must produce exactly the same `clean_set` / `session` rows as an import;
3. **another file format** — see 4.3.

Manual entry is also the only path that can capture **session duration**, because the export has no timestamps. `session.duration_min` becomes measurable (`duration_basis = measured`), and the rubric's `SESSION_DURATION` flips from `measurable = 0` to a real score — the first time that item can move.

Manual entry must pass the same gates as cleaning: unknown exercise names are quarantined or reported (never dropped), `RIR 6` is a warm-up, and seconds-vs-reps follows the taxonomy.

Capture post-session fatigue (1-5) with the session; it feeds §4.12 and is not a rubric item.

Status 2026-10-23: manual logging is closed through live workout mode — Start from a prescription with the seven required readiness fields, manual/imported sets, inline correction, measured/editable duration, sRPE or not-recorded, end/abandon, plan overrides, materialisation into `clean_set` / `session`, and rubric/review. The standard dated training export remains the backfill path; the separate adapter work is §4.3.

Acceptance: a manually logged session is indistinguishable downstream from an imported one (same tables, same critique); concluding a workout also catches up unreviewed sessions via `review --pending`; duration is stored when given and stays `unavailable` when not — never estimated; the rubric scores duration only when it exists.

4.3 Column-mapping adapters for other exports (CLOSED 2026-10-23 — not needed)

Status: closed. The standard dated training export is the backfill path, and the workout CSV is imported in-workout. No additional adapter registry is needed. Unknown exercise names are still reported/quarantined, and extra raw columns remain in `raw_row`; neither justifies a second mapping layer.

4.4 Programme output — a rolling horizon, not a block (done: CLI + web)

Today `run` emits one day. The athlete wants either **the whole programme** or **what remains** of it.

Decided: there is **no fixed training block**. Training is continuous. A deload can be **scheduled** every six weeks from the declared block anchor or **reactive** when the data asks for it; the schedule is configurable and the reactive triggers still override it. That makes "remaining programme" simply "from today onward".

Scope: `programme --weeks N [--remaining] [--start-session-type T]` prints a rolling horizon. The horizon is a forecast, not a commitment, and `--remaining` starts from today. The declared `block_start_date` anchors the erector ramp and the scheduled deload; it does not bound the programme.

Scheduled deload: every six weeks from `block_start_date` by default, presented as a **suggestion** at workout time (default no; non-interactive default no). One set per item (minimum two), 10% load reduction and one RIR under `MOD-DELOAD-001`. Reactive deload triggers remain **automatic**: two consecutive weeks over the spinal cap, **or** two consecutive flaring sessions. A reactive deload emits `fallback_level = fatigue_reduced` and cites the trigger. The six-week interval is an assumption, not a scientific claim.

Planned loads (decided): **recomputed on the day**. A plan is a forecast; the day's prescription uses the current e1RM, readiness and symptoms. Planned rows are superseded when the day arrives, never edited.

Rules: progression between weeks uses the existing double-progression decisions for accessories and e1RM-derived loads for main lifts; the weekly spinal cap and the erector ramp apply across the horizon.

Acceptance: `programme --weeks N` prints a dated session-type forecast with canonical session types; `--start-session-type T` sets the first session; `--remaining` starts today; every planned session carries its rationale refs; the day's prescription recomputes and supersedes the plan; scheduled and reactive deloads fire on the configured triggers and say why; printing a programme needs no LLM.

4.5 Decisions taken (2026-10-17)

question	decision
Block length and deload	**No fixed block. Deload is scheduled every six weeks or reactive**, configurable; the reactive triggers remain
Planned loads	**Recomputed on the day**; a plan is a forecast and is superseded, not edited
Manual logging granularity	**Per set**; no exercise-level shorthand that would invent detail
Readiness fields for the modifier	**Sleep and energy, as a config list**; other fields are recorded and reported only
Manual vs imported readiness	**Manual wins per field**, import fills gaps, the merge note records both

These are recorded in the decision log as `DEC-PROGRAMME-01/02/05`, `DEC-LOG-01` and `DEC-READINESS-02/03`. The deload interval and reactive thresholds are explicit policy and remain config, not a claimed scientific finding.

4.6 Training continuity — missed and planned time off (done)

The programme must survive real life: a week lost to illness or work, a holiday, a planned deload, and a deliberate test week. Today nothing distinguishes "trained on Tuesday" from "did not train at all", so the DUP cycle and the erector ramp advance by calendar only.

Scope:

- **Detect gaps** from stored sessions. A session that was never logged must not be treated as a session at the old load; the next prescription should know how long the gap was.
- **Declared breaks.** A `break --from --to [--reason]` record (holiday, deload, test week) that freezes the DUP week and the ramp anchor for the duration. An undeclared break is inferred from the gap and reported.
- **Re-entry rules**, config-driven and separate for strength and erector work:
- 7-10 days off: keep the load, add one RIR for the first week back;
- 14+ days: reduce load ~10% for a week, then resume progression;
- 28+ days: re-enter the erector ramp at week 1 (or a declared re-anchor) rather than pretending the ramp continued.
- **Cycle semantics.** A missed week should not advance the DUP cycle; a declared break should pause it. This is the difference between "week 3 of the cycle" and "the third week of training".

Acceptance: a missed session does not advance the DUP week; a declared break freezes it; after a 3-week gap the prescription is a re-entry session with the reason cited; a declared test week does not count as a training week; no artifact is deleted or rewritten.

Decided: gaps are detected from history (a declared break is optional); thresholds 14/28/56 days; a missed week does not advance the DUP cycle; a test week counts as a trained week; the prescription states that it is amended because of the gap; the erector ramp pauses during a gap and steps back one stage after the long break; bodyweight work keeps its load but reduces volume and adds one RIR. The exact thresholds remain assumptions (`DEC-CONTINUITY-01`).

Delivered: gap detection from stored sessions, `MOD-GAP-001`, trained-week DUP cycle, ramp step-back after a long gap, bodyweight set reduction, and the message in the prescription. Open items recorded in Q12.

4.7 Injury signal — per region, informs suitability, never clears it (done)

The athlete asked whether a back that is "strong and recovered enough" could unlock exercises like the sumo deadlift. The app has no medical model and must not pretend to; what it can do is make the relevant history visible and require a human decision.

Scope: derive a **general injury signal plus one per region** from data already stored — symptom-free streak, flare count and settlement, whether recent sessions loaded the region and whether a later reading showed no flare, the erector-ramp stage, and the lifts that load the region. Surface it next to an unrated exercise and next to a proposal to change a main lift.

Hard rule: the signal **suggests a review, never a rating**. It cannot set `exercise_suitability`, cannot clear a constraint, and cannot substitute a lift. The athlete rates; the model may explain; the engine obeys.

Acceptance: the signal cites the symptom rows, window and tolerated count it used; an unrated lift shows "needs review" with that evidence; no automatic state change occurs; the athlete can rate in one command (`state --rate`).

Decided (`DEC-BACKSIGNAL-01`, amended in scope by `DEC-INJURY-01`): symptom-free **with tolerated loaded sessions** may recommend a rating; readiness is excluded; rest alone earns no credit; no data is **unknown**; the signal is always disclaimed and may recommend, never set, a rating. `DEC-INJURY-02` scales the window with prior flares; `DEC-INJURY-03` defines a tolerated session as loaded **and** followed by a reading with no flare.

Delivered (2026-10-19): `features/injury_signal.py` + `config/injuries.v1.yaml`, low back, biceps and knee, per-region status and scaled window, `unverified` sessions counted separately, `state`/prescription/web display, region-scoped symptom logging (`symptom --region`), and an advisory warning when a prescription loads a flaring non-back region. Evidence and definitions: `docs/INJURY-SIGNAL.md`. Q13 is answered; the remaining uncertainty is calibration (`ASSUMPTION-08/09/10`).

4.8 Video analysis (LLM) — moved to v0.5

Allow a video to be attached to a session or a single set for a vision-capable model to comment on: depth, bar path, tempo, rep count. It is a **commentary** feature, and the constraints matter more than the feature.

- **Privacy first.** Video is personal data. Files live under `private/` (git-ignored), are never committed, and never enter `data/coach.db` as bytes. Sending a video to a hosted model is an explicit per-video opt-in, and the output must say that the file left the machine. A local model keeps it local.
- **No numbers, no vetoes.** The analysis is stored as a `run_event` (new kind, `video_analysis`) with the file hash and model id. It cannot change a load, a rubric score or a policy verdict (AGENTS.md §3.2-3.3).
- **The log stays authoritative.** If the model says "8 reps", that is a comment; the logged set remains the record.
- **No default dependency.** Frame extraction needs ffmpeg or provider-side video support; decide per provider rather than adding a runtime dependency for everyone.

Acceptance: attaching a video produces exactly one event and changes no artifact; a run with no video, or with `--llm off`, stays byte-identical; video bytes never appear in the repository; every claim is labelled commentary.

Open questions: which sets get video, retention period, and per-provider capability.

Why v0.2 rather than v0.1: it needs a privacy decision (sending personal video to a hosted model), a provider-capability check, and a storage/retention policy. None of those belong in a version whose remaining work is calibration and documentation.

4.9 Readiness score and follow-ups (core + devices done; learned weights closed)

Implemented (2026-10-19/20):

- global 0.00-1.00 score from sleep, energy, mood, stress and soreness, with a three-field minimum and a sleep/energy requirement;
- category scores; the injury signal is a ceiling, never an average;
- global/category overrides at the score scale, append-only, reason required; override wins over computed readiness and changes the load, but never bypasses constraints or injury ceilings;
- 20% cap on combined readiness + continuity reduction (guardrail, no direct evidence);
- readiness-required gate for current/future CLI and web workouts;
- calibration report against the 306 black-box legacy labels, with main-lift load features, date split and baseline; report only, no auto-apply;
- baseline-relative HRV/RHR at 0.5 weight, only with 7 readings in the previous 28 days;
- optional device-computed wearable readiness (0-100) at a 10% blend.

Full design and evidence: `docs/READINESS-SCORE.md` and `docs/READINESS-WEIGHT-EVIDENCE.md`.

4.9.1 HRV and RHR in the score — done

Baseline-relative HRV/RHR are in the global score: 28-day median, minimum 7 readings, gap or direction handled explicitly, weight 0.5 each. Missing baseline is shown, not guessed.

4.9.4 Optional wearable score — done

A 0-100 score from a device such as Bevel is an optional 10% blend. It cannot override the subjective score, constraints or the injury ceiling.

4.9.2 Per-athlete score settings — done

Baseline window, minimum readings, HRV/RHR slopes and weights are append-only athlete rows, with athlete > config > code precedence. The effective source is shown in the score.

4.9.5 Shadow baseline comparison — done

`readiness shadow` compares the rolling-median baseline with a log-linear trend projection. It reports the normalised difference; it never changes a score or load.

4.9.3 Learned weights — closed

The constrained comparison (current equal weights, sleep-heavy, HRV-low) found no improvement in the held-out late period. The performance association was essentially zero in that window and sRPE associations were weak. Keep fixed transparent weights; do not fit individual weights from this dataset. Full method/results: `docs/READINESS-WEIGHT-EVIDENCE.md` and `DEC-READINESS-10`.

4.10 Local web UI (done)

A browser front end over the same core, so routine interactions do not require the CLI. The CLI remains the automation surface.

- `strength-coach web` serves on **127.0.0.1:8000** by default, no authentication, no external assets; docs are rendered locally.
- Pages: Today, Prescribe, Programme, Review, Readiness, Import, Athlete, Terminal, Docs.
- The API only serialises and routes; every load/score/warning still comes from the core.
- Tests cover the API round trips, docs rendering and the readiness/review writes.

Done: import, state editing, manual logging, Programme, Terminal. LLM proposal view closed as CLI-only; video remains v0.5.

4.11 Web UI parity (CLOSED for v0.3; video part v0.5)

Status: closed 2026-10-23. Manual logging, state editing, import, Programme, Terminal, exercise editing and calibration are in the UI. The LLM proposal view is deliberately **not** built: `run --propose` stays CLI-only, and the deterministic prescription, warnings and review remain the UI path. `workout start` is UI-only because it needs the full readiness form. Both are documented exceptions, not open items.

The local UI covers Today, Prescribe (rolling session types), Programme, Review, Readiness, Import, an editable Athlete page, an editable **Exercises** catalogue, Terminal and Docs. This item brings the remaining interactive CLI actions into the browser. The CLI stays as the automation interface; the UI must call the same core functions, never re-implement them.

Planned surfaces, in suggested order:

1. **Import** — **done**. Standard dated training/readiness CSVs and the in-workout workout CSV.
2. **State editing** — **done**. Profile, constraints, suitability ratings, e1RM overrides, strength tests, block start and symptoms; append-only, with an exercise picker and the current tables shown beside the forms.
3. **LLM proposals** — **closed as CLI-only.** `run --propose` keeps the proposal view, citations, warnings and policy veto in the terminal; the web UI does not duplicate it.

4. **Manual workout logging** (4.2) — **done** via Workout mode: per-set entry, plan overrides, import, end, review and calibration.
5. **Programme view** (4.4) — **done**. Rolling session-type horizon with recomputation on the day; canonical session types are the only label.
6. **Video** (4.8) — opt-in attachment, privacy warning, commentary only, never a score.
7. **Readiness settings** (4.9) — **done**. Baseline window, minimum readings, slopes and weights; shadow baseline report.
8. **Terminal** — **done**. One `strength-coach` command, no shell; bound to the UI database.
9. **Exercise catalogue** — **done**. Full taxonomy table, search/filter, all fields per row, edit/add form writing `config/exercises/overrides.v1.yaml`; ids immutable, log-name collisions refused; API tests and CLI `exercise list` / `exercise set`.

Constraints:

- all writes go through the same functions as the CLI; no business logic in JavaScript;
- destructive or diagnostic actions (import, overwrite, conclude) require an explicit confirmation step and show the resulting rows;
- uploads stay under private/ and are never committed; no arbitrary filesystem paths;
- the server stays local-only; adding authentication is out of scope for a single-user tool;
- every action reports warnings and refusals as part of the result, not as a toast that hides the reason;
- the terminal executes only `strength-coach`, never a shell, and refuses `web`, `--db` and interactive `review`.

Acceptance: every interactive CLI action has a browser path or a documented exception; the same inputs produce the same database rows and warnings through either surface; a failed action leaves no partial write; API tests cover each new write; no load, score or threshold is computed in the browser.

4.12 Continuity calibration from real returns (v0.3, implemented; calibrated factor rejected)

Estimate per-athlete return retention from actual returns. The fixed 14/28/56 bands remain per-band fallbacks until that band has enough returns and a tight bootstrap interval; when it does, the calibrated factor is used and cited in the prescription warning/notes.

Capture for every return:

- gap length and the last pre-gap session;
- first session back: load, reps, RIR, whether the planned work was completed;
- symptoms and readiness around the return;
- **post-session fatigue** on a 1-5 scale (1 fresh, 5 spent) as a separate session observation, not a rubric item — it is context, not execution quality.

Estimate two curves separately:

- **strength** — load at a fixed RIR versus days away;
- **capacity** — sets/reps tolerated without a flare or excessive fatigue versus days away.

Start descriptive (per gap band), then fit a simple retention curve with shrinkage toward the current defaults. Require a minimum number of returns before showing a suggested multiplier.

Output: per-band retention median, bootstrap 95% interval, return count, fixed factor, applied factor and status. The prescription keeps `MOD-GAP-001` and adds the calibrated factor in its warning/notes. A band with too few returns, a wide interval, or calibration disabled uses the fixed factor.

Retrospective walk-forward (2026-10-20): 12 returns; calibrated-factor ratio MAE 0.150 vs fixed-rule 0.128, so the calibrated factor **lost** and the fixed fallback stays in use. This is retrospective, not prospective. See RETROSPECTIVE-VALIDATION.md.

Acceptance: `strength-coach continuity` and `GET /api/continuity` show observed returns versus the fixed rule; the minimum-data rule is explicit; a sparse band keeps the fixed factor; a well-observed band applies the calibrated factor; the prescription cites the applied rule.

4.13 Cardio workouts and an impulse fatigue model (capture v0.2 done; cardio model v0.4; strength model is §4.16)

The app currently measures resistance training only. Cardio is a separate exposure with a different fatigue signature, and a single "fatigue score" would hide that. This item adds the data and a model, initially advisory and in shadow mode.

Delivered 2026-10-20 (`DEC-CARDIO-01`): append-only `cardio_log` with modality, duration, average/max HR, distance and sRPE; `strength-coach cardio` (log with `--date`, `--minutes`, `--avg-hr`, `--max-hr`, `--distance-km`, `--srpe`; `--show --weeks`); a Cardio page and `GET/POST /api/cardio`; a weekly summary with the WHO reference as context. No prescription reads it and no fatigue model exists yet.

Priority: the longevity review ranks cardio capture as the largest gap between the programme and the primary goal — cardiorespiratory fitness is the strongest modifiable mortality signal, and the app currently records none of it (`docs/LONGEVITY-REVIEW.md`). Capture is useful before the fatigue model exists; the model stays shadow-mode until there is data.

Scope

1. **Cardio capture** — modality, duration, average/max HR, HR zones, distance or power where available, session RPE, notes. Append-only; no free-text parsing in the core. **Done (migration 0011):** modality, duration, average/max HR, distance and sRPE; HR zones and power are not captured yet.
2. **Training impulse** — compute a documented impulse per session: HR-based TRIMP where the data supports it, otherwise session-RPE × duration. Say which one was used and why.
3. **Fatigue model** — start with an impulse-response / fitness-fatigue model: - a fitness (positive, slow-decaying) and fatigue (negative, faster-decaying) term; - at least two systems: cardiovascular and neuromuscular/spinal, rather than one global impulse; - time constants and weights are config, per athlete when enough data exists; - the model reports a **fatigue state**, not a diagnosis.
4. **Interaction with the existing layers** — cardio fatigue may inform readiness and load selection, but it must not override constraints, the injury signal or readiness overrides. Cardio that loads the knees/low back must be linked to the injury regions so it is not invisible exposure. Avoid double-counting the same session as both resistance and cardio.
5. **Shadow first** — compute and display the impulse and fatigue state for weeks without changing a prescription. Only then consider a load/RIR effect.

Evidence and limits

- The Banister impulse-response model relates training impulses to performance, but its parameters are ill-conditioned and unstable; a good fit does not make the parameters identifiable (Hellard 2006).
- A single impulse reduces different training types to one number; the proposed three-dimensional model separates energy-system-specific adaptation, which is the direction this item should take rather than one global fatigue scalar (Kontro 2026).
- HRV, sRPE and subjective wellness can be inputs to an impulse-response model, but the relationships are not a validated individual prescription (Piatrikova 2021).
- Fitness-fatigue models need longitudinal individual data; a generic parameter set is a prior, not a measurement (Wang 2025).

Open questions

1. Which cardio modalities are in scope (walking, cycling, rowing, running, intervals)?
2. Is TRIMP available from the athlete's devices, or is sRPE the only usable impulse?
3. Should the model have two systems (cardio/neuromuscular) or three (add metabolic/energy system), and how do they interact?

4. Do cardio sessions link to injury regions by pattern (e.g. running → knee) or only by explicit tags?
5. Does fatigue modify load, RIR, volume or only the displayed readiness context?
6. How many weeks of data before an individual time constant is shown?
7. Can a cardio session count as readiness/load evidence without a symptom follow-up?

Acceptance

A cardio session can be recorded and cited; its impulse is reproducible and labelled by method; the fatigue state is displayed with inputs and time constants; a shadow period changes no prescription; constraints and injury ceilings remain vetoes; no cardio session silently counts as resistance volume or vice versa.

4.14 Labelled calibration set (v0.4; moved from v0.3)

No code. Continue reviewing sessions while they are fresh; the rubric report needs roughly ten remembered sessions per item before changing a definition or weight (DEC-EVAL-03, DEC-EVAL-04). Reconstructed marks are stored but excluded from agreement.

Current state: 5 high-confidence sessions, 75% agreement, MAE 0.20; rotation is the weak item. The web Review page has the **Run rubric calibration** button; the CLI remains `strength-coach calibrate`.

Acceptance: ten high-confidence marks per item before a rubric change; every review appends, never edits; the report states the sample size and excludes low-confidence marks.

4.15 Exercise selection adapted to readiness and fatigue (v0.3; apply + comparability done)

Status: apply + comparability implemented 2026-10-20; tax metadata, accessory candidates and symptom/readiness suggestions implemented. Plan closed at suggestions + explicit apply; automatic withholding deferred to the 8-week/30-session review. The selector now carries tax metadata and `symptom` / `stress_readiness` triggers; substitutions remain explicit. See `READINESS-EXERCISE-SELECTION-PLAN.md` and `DEC-SELECT-06`.

Current implementation: `prescribe/selection.py` ranks the configured and catalogue variants per slot from readiness, recent stress, rotation, preference and the cached fitness-fatigue state; constraints and `can_do: false` exclude a candidate; injury is a penalty, not a veto. Candidates carry an A/B/C comparability class: A applies automatically, B requires confirmation, C is hidden by default and never automatic; the selected lift's own history is always used. A display-only own-history e1RM ratio is shown, but it does not enter the score. See `DEC-SELECT-03` and SELECTION-COMPARABILITY.md. It is exposed as `strength-coach select` and a Prescribe panel. With the explicit **Apply selection** toggle, the chosen candidate replaces the slot exercise and the load is re-solved from that lift's own history (`MOD-SELECT-001`). Without the toggle nothing changes. All synthetic raw variant names were canonicalised and their history rewritten (migration 0014, 114 entities, 3,607 audited remaps; migration 0015 split close-grip pulldown); trap-bar date-scoping and back-extension equivalence are preserved.

Weakness/injury tailoring — deferred. No positive weakness/injury targeting. Injury remains an exclusion/penalty/ceiling. Earliest review is after v0.5 and ~April 2027, and only with 12 sessions per target lift over 12 weeks, a pre-registered own-trend metric, a retrospective baseline win, and an opt-in N-of-1/alternating test; injury targeting remains unscheduled. See WEAKNESS-INJURY-SELECTION.md, `DEC-SELECT-04/05`, Q22.

Later phase — variant-group e1RM ratios. Variants stay distinct entities for now. A future model may express a variant's e1RM as a ratio of a parent lift (illustrative: front squat ≈60% of squat — a fictitious number, not a proposal). Any ratio must be a configured, per-athlete/phase prior with a CI, learned from paired sessions where possible, used only when the variant's own history is too sparse, and never presented as a measured e1RM. This needs a formal `variant_of` relation and a validation rule; it is not implemented.

Load handling — bands/chains. The log's weight figure already includes band/chain load, so the current total-load treatment is accepted for now and no prescription change is made. A future item should define whether an e1RM uses the total logged load, how band tension varies through range, and whether these variants need a separate comparison rule.

Goal. Choose among evidence-permitted exercises for a slot using the per-lift readiness score (`docs/LIFT-READINESS.md`), the recent-stress decay (option B), the strength fitness-fatigue state (§4.16), injury links, equipment, rep range and rotation — without letting an unvalidated model override safety.

Default: Python, deterministic. Score each allowed candidate:

- hard filters — constraint veto, `can_do: false`, injury region active, equipment missing;
- per-lift readiness and recent stress (lower readiness → prefer a lower-skill or less fatiguing variant, not a harder one);
- recent stress and the cached fitness-fatigue state;
- rotation/history (avoid repeating the same variant indefinitely);
- a preference/suitability tie-break (adherence is a real outcome).

Only one substitution per slot, cited as `MOD-SELECT-001`; the deterministic candidate list and scores are shown. Applying a substitution is explicit (UI/CLI toggle) and remains reviewable; it is never automatic.

Optional LLM (`--select=llm`). The model receives the **already-filtered candidate list** and returns one candidate plus a reason; Python validates and logs the choice. The LLM may not invent an exercise, a load, a rep count or a veto, and a missing/invalid answer falls back to the deterministic choice. Off by default; provenance recorded.

Evidence and limits. Specificity and adherence support stable exercise choices; there is no trial showing that algorithmic selection outperforms a fixed programme. Autoregulation evidence supports adjusting load/effort, not swapping exercises. Injury/suitability links are exposure maps, not causal (ASSUMPTION-10). Treat selection as a conservative, explainable preference, not an optimiser.

Dependencies: per-lift readiness (done), recent stress (done), session RPE/length (done), §4.16 model, exercise suitability state, taxonomy metadata.

Acceptance: constraints always win (done); selection is reproducible (done); LLM proposals are validated and cannot override policy (done); shadow mode changes no prescription (done); UI shows the chosen exercise and why (done); tests cover vetoes, ties, missing metadata and LLM fallback (done); explicit apply toggle and re-solve (done).

4.16 Strength fitness-fatigue model (v0.3; shadow only; retrospective rejected)

Status: shadow prototype implemented 2026-10-20; retrospective check failed 2026-10-20. `features/fatigue_fitness.py` fits a per-lift linear performance model from fixed fitness/fatigue half-lives and the athlete's own e1RM observations; it reports fitness, fatigue, state, R^2, condition and refuses sparse or ill-conditioned fits. It is exposed as `strength-coach fatigue` and on the Prescribe page. **No load reads it.** Option B (recent stress) and session RPE/length remain inputs.

Scope. Strength only. Cardio keeps its own impulse item (§4.13); do not collapse both into one fatigue scalar. Model per **lift**, with partial pooling across a movement family / muscle group, rather than a separate free model for every exercise with sparse data.

Candidate state model. Per lift (or family), an observed e1RM performance is a slow fitness term plus a fast fatigue term; training load is built from hard sets × proximity to failure, with session RPE × duration as a cross-check. Parameters are hierarchical and partially pooled; intervals are mandatory. The model must state when it is not identifiable.

Phased plan.

1. **Inputs**: session RPE/length (done), per-lift readiness (done), recent stress (done), e1RM calibration report (done), RIR/VBT. Add a per-lift soreness/readiness rating only if it earns its place.
2. **Shadow fit — done for this phase**: fixed half-lives + linear fit on stored e1RM series, approximate 95% intervals, family-slope shrinkage, and a chronological 80/20 backtest against a naive baseline; an **overall** relative-e1RM row pools lifts. No load change. **Still open:** a frozen-parameter prospective test on future sessions. **Retrospective walk-forward:** 25 lift-session predictions, relative MAE 0.158 vs persistence 0.074 (skill −112%); the model lost, so no load or adoption follows. Volume has no predictive model; selection has no counterfactual.
3. **Prospective validation**: freeze parameters, predict held-out sessions, compare with a fitness-only and a readiness-only baseline.
4. **Optional use**: only after validation, and only downward (extra caution), never to raise a load from an unvalidated model.

Evidence and limits. Direct strength evidence is one small weight-lifter study (Busso 1990); parameters are ill-conditioned and the fatigue term did not improve out-of-sample prediction in a 2025 re-analysis (Marchal 2025); load metric choice changes the fit (Vermeire 2021/2022); RIR and e1RM are noisy. Recovery after failure/volume is 24–72 h (Morán-Navarro 2017; Sousa 2024), which the current RIR window and 48–72 h spacing already manage. See `docs/FATIGUE-FITNESS.md` refs 93–100.

Open questions. Which load metric (hard sets, volume load, sRPE)? Per lift vs family? How many measured tests make a parameter identifiable? Does the model beat readiness-only prospectively? Does it change any decision in shadow mode?

Acceptance: reproducible fit with approximate intervals (done); refuses to show parameters when unidentified (done); shadow mode changes no prescription (done); per-lift, family-shrunk and overall views done; chronological backtest done; UI shows inputs, state, intervals, backtest and “not identifiable” honestly (done); no cardio in the strength model (done); frozen-parameter prospective test remains open.

4.17 Rolling session types and strength-programme balance (v0.3; volume management v1 done)

Status: implemented 2026-10-20. The rigid A/B/C calendar is replaced for new prescriptions by four session types:

type	main work	upper body	direct erector
`squat`	SSB squat + power	bench support + row	yes
`hinge`	trap-bar deadlift	narrow/pause bench + row + pull	no
`bench_squat`	bench + squat support	overhead press + row	no
`bench_hinge`	bench + single-leg RDL	overhead press + row + pull	no

The rolling rule targets two squat, two hinge and two bench-emphasis sessions per six, uses the recent `session_type` history plus forecast choices, and falls back to the preferred emphasis cycle when the counts are tied. A missed session shifts the deficit; it does not advance a calendar. The legacy A/B/C fields and overrides were removed by migration 0023 (`DEC-SESSION-01`); imported sessions without enough evidence stay `unknown`.

Balance changes: every session includes bench or another upper-body press/pull, plus a horizontal row. After the 2026-10-20 review, squat-emphasis SSB is 5×3–5, the bench+squat squat support is 4×3–5, and the trap-bar hinge is 5×3–5. This restores ~7 squat sets/week and raises heavy hinge to ~5 sets/week without a second heavy exposure; back extension stays at 3 sets/week. The hinge gets a second exposure through the single-leg RDL support on the bench+hinge session, without stacking another high-fatigue spinal lift. Only the squat-emphasis session carries direct erector work, keeping a normal week at squat 1 + hinge 2 = 3 against the spinal cap. Poor readiness can remove one accessory or main-lift set

(`include_essential: true` by default), and the Readiness page exposes that **volume valve**. A weekly **hard-set ledger** reports sets per movement pattern against broad MEV/MAV/MRV landmarks; non-essential sets are trimmed only when the recent week plus the proposed session exceeds MRV (`MOD-VOL-001`). Under-MEV adds one accessory set after three weeks below MEV (`MOD-VOL-002`), while main lifts are unchanged by that rule; poor readiness can remove one main-lift or accessory set. Main lifts carry a 50/70/85 warm-up ramp, and `test --status` reports the 10-week 3RM/5RM cadence. Power work is bodyweight/implement-only.

Acceptance (done): session templates parse; every session has a row and an upper-body push/pull; hinge sessions contain no direct erector work; the two-week sequence balances squat/hinge/bench; canonical types resolve for history; the rolling forecast and UI expose the selected type; set reduction is cited in the `MOD-FAT-001` warning; power carries no external load; full tests pass.

Volume management v1 boundary (done 2026-10-20). The evidence check, hard-set ledger, configured MEV/MAV/MRV landmarks, conservative MRV cap, readiness volume valve and UI/CLI/API are complete. The following are explicitly **not** claimed as done:

- individual MRV calibration — closed as not estimable (`DEC-VOLUME-04`);
- main-lift volume titration below MEV;
- a fully autoregulated deload (a scheduled/reactive policy now exists);
- per-muscle rather than movement-pattern volume;
- set quality/load/tonnage rather than raw hard sets;
- prospective validation against e1RM or performance;
- a forecast-aware cap that accounts for planned-but-not-performed future sessions.

These are documented boundaries, not implied claims; individual MRV is closed and the rest stay open or deferred.

4.18 LLM-guided suggested programme (future; not scheduled)

Idea: the model could eventually propose a suggested programme — session emphasis, exercise choices, weekly volume allocation and progression structure — rather than only narrating a Python-built prescription.

This is explicitly future work, not a v0.3 deliverable. The existing invariants still apply:

- the model receives a structured state/evidence summary and a bounded menu of allowed exercises, sessions and volume bands;
- it may propose **choices**, never loads, reps, RIR, percentages or safety vetoes;
- Python validates every proposal against the constitution, constraints, injury signal, volume landmarks and comparability classes;
- a suggested programme is shown as a diffable proposal with evidence and assumptions, and is never applied without explicit approval;
- every proposal and approval is audited; the deterministic programme remains the fallback;
- the proposal must not invent evidence or hide uncertainty.

Prerequisite: a stable programme schema, a validated volume/selection policy and enough reviewed sessions to evaluate whether the suggestions are better than the fixed programme. **Status:** idea only; no implementation, no phase assigned.

Next-phase design: the final lean non-cardio fatigue-management plan is **closed** (`docs/NEXT-PHASE-FATIGUE-PLAN.md`). It used the existing data plus an sRPE/duration backfill for descriptive calibration and implemented the optional deload plus four deterministic programme changes, and closed individual MRV as not estimable. No programme experiments, no individualised MRV and no randomised deload trial. The remaining deferred items stay listed in §4.19.

4.19 Consolidated next actions

priority	action	status
P1	Scheduled deload / broader fatigue-management policy	done — `DEC-PROGRAMME-05`
P1	3RM/5RM calibration cadence and explicit warm-up prescription	done — 10-week reminder + 50/70/85 ramp
P2	Parent-variant e1RM ratios	moved to §4.20, unscheduled
P2	Weakness/injury-targeted selection	display-only context done; targeting deferred — earliest after v0.5 / ~April 2027; gates in `DEC-SELECT-05`
P2	Strength-review implementation closeout	**review closed**; future work in §4.20
P2	Individualized MRV / main-lift volume experimentation	**closed** — not estimable; no experiment (`DEC-VOLUME-04`)
P2	Prospective validation of continuity, volume, selection and fatigue models	moved to §4.20 — no candidate; current models failed the retrospective baseline
P1 (v0.3)	Live workout mode	**closed** — full UI flow; CLI start remains a documented UI-only exception
P1 (v0.3)	Manual workout logging	**closed** — live Workout mode
P1 (v0.3)	Import adapters	**closed** — standard export + workout CSV cover the paths
P1 (v0.3)	LLM proposal view	**closed** — CLI-only by decision
P2	Learned readiness weights	**closed** — fixed weights retained (`DEC-READINESS-10`)
v0.4	Deeper labelled calibration set	moved to v0.4
v0.4	Cardio prescription and cardio fatigue interaction	planned
v0.5	Video capture/analysis and UI parity	planned
future	LLM-guided suggested programme (§4.18)	idea only

4.20 Post-review follow-ups (unscheduled)

Moved out of the strength-programme review so it can close. These are **not** v0.3 work.

1. **Prospective validation gate.** No current candidate. A model may enter a frozen prospective period only after it beats the fixed/persistence baseline in a retrospective walk-forward. The current continuity and fatigue models failed; they remain fixed/shadow-only. A prospective run is 8–12 weeks / 24–36 sessions with no retuning and no load effect.
2. **Parent-variant e1RM ratios.** Needs a formal parent relation, enough paired sessions and a validated ratio with uncertainty. It is a prior/display aid, never a silent load conversion. No schedule.
3. **Weakness/injury targeting.** No positive targeting. A display-only own-history context is implemented. Weakness targeting is gated by DEC-SELECT-05 (earliest after v0.5 / ~April 2027); injury targeting is unscheduled and remains constraint/penalty/ceiling only.

Open decisions register

decision	state	where
Prospective validation of a future continuity/fatigue candidate	gate only; current candidates rejected	§4.20.1, DEC-VALIDATION-01
Parent-variant e1RM ratios	unscheduled; needs paired data and validation	§4.20.2

decision	state	where
Weakness targeting	deferred to after v0.5 / ~April 2027	§4.20.3, Q22, DEC-SELECT-05
Selection A/B boundary (pause/narrow, SSB/high-bar)	dormant policy decision; current default is conservative (A only same exercise)	Q21, docs/SELECTION-COMPARABILITY.md
Cardio model	v0.4	§4.13, Q14

4.21 Live workout mode (v0.3, CLOSED 2026-10-23)

Status: closed. The approved workout flow is implemented; the only remaining parity exception is `workout start` in the CLI, because starting requires the full pre-workout readiness form and the displayed prescription. That is not part of this plan.

slice	scope	status
1	Schema/state machine, Start form with required readiness, manual sets, timer, duration/sRPE, End/Abandon, API/UI	**done**
2	Materialise workout sets into `clean_set` / `session`; actual-vs-prescribed comparison	**done**
3	CSV import inside workout: paste/file, preview, conflicts, RPE→RIR, replace/undo	**done**
4	Plan overrides: swap/add/remove exercise and sets	**done**
5	Deterministic rubric review + human marks; post-processing pipeline	**done**

Workout sets materialise into `session` / `clean_set` on completion. The end screen compares actual vs the modified plan and the original prescription, runs the deterministic rubric, and closes on review marks. Actual-set edit, plan overrides, in-workout CSV import with conflict preview/undo, and the exercise-details popup are done. The standard dated training export remains the backfill path. CLI workout commands are partial (no start).

5. Known gaps and risks

gap	impact	mitigation / status
No persistence	cannot replay a past day; work is lost between runs	**resolved** — `run` / `run_event` trace + append-only tables (phase 3)
Erector ramp has no anchor	`PRG-EREC-001` returns `unknown`; Day A/C cannot be prescribed safely	rule implemented; `unknown` names both missing inputs (`OPEN-QUESTIONS` Q3, Q4)
No athlete state	no injury, goal or preference can veto a prescription	**resolved** — §11 tables and `state` command
No symptom input	back-flare modifier and the ramp's stability gate cannot run	**resolved** — pre-session 0-3 input, `symptom_log`, `MOD-BACK-001`
Imported sessions have no timestamps	imported duration unmeasurable; two imported sessions on one date indistinguishable	workout mode measures duration; imports remain date-only (`OPEN-QUESTIONS` Q5)
Only one session type can be prescribed	The other templates are blocked by the erector-ramp anchor (Q3), the SSB question (H1) and the missing §11 tables	**resolved** — all four session types prescribable; the ramp anchor is optional and `unknown` is explicit
Only ~45% of dates map to a programme day	day-specific rules cannot be scored on the rest	day inference reports explicit reasons; do not force a bucket
Velocity is per-set only	within-set velocity loss not computable	labelled decay proxy; `DEC-VELOCITY-01`; post-hoc rep-level decision in `docs/VBT-REP-LEVEL.md` (`DEC-VBT-03`)

gap	impact	mitigation / status
Rubric calibrated on **five high-confidence sessions**	75% agreement, MAE 0.20; rotation is the weak item (40%, n=5); two old marks are reconstructed and excluded	`DEC-EVAL-04`; needs ~10 remembered marks per item before changing weights
Spinal cost ignores load and effort	a 50 kg good-morning at RIR 3 counts the same as a maximal one	ASSUMPTION-02 / Q8; a weighted-by-e1RM or RIR version needs the athlete's decision
VBT targets not prescribed	a prescription carries no `velocity_target`	**resolved** — 22 targets where trustworthy; trap bar withheld; `DEC-VBT-02`
ACWR contested	cannot be a hard rule	advisory only; `DEC-ACWR-01`

6. Idea backlog (not approved, not scheduled)

Recorded so good ideas are not lost, explicitly **not** commitments:

- `sets_per_muscle_group_per_week` feature, so volume is audited in sets rather than sessions.
- Readiness composite with a `metric_definition` table so new metrics need no code change.
- Rotation scoring split by session provenance (app-assigned vs self-directed).
- Grip-strength left/right asymmetry as a readiness input and a unilateral-work trigger.
- Per-lift manual velocity overrides with expiry.
- Regime-change detection (a lift's volume collapsing while a substitute rises).
- LLM-guided suggested programme under the §4.18 guardrails (idea only).
- Wider proposal licence (more than one increment on the menu) once the menu has proved itself in real sessions.
- ~~A declared 3RM/5RM test session.~~ **Delivered in v0.1** (`DEC-TEST-01`): `strength-coach test`, 3RM/5RM by default, `--max` for a confirmed 1RM. Evidence in `docs/TRAINING-DYNAMICS.md`.
- ~~Rep-scheme menu.~~ **Delivered** (`DEC-REPS-01`): named schemes including `maximal` 1-3 (main lifts only, not the hinge).
- ~~Weekly undulation (DUP).~~ **Delivered** (`DEC-UNDUL-01`): 4-week cycle keyed to the declared anchor, config-disabled by default when no anchor exists.

7. How to change this roadmap

- Phase status changes go in the table, in the same commit as the work.
- New phases get a number and an acceptance criterion before any code is written.
- Anything that changes a rule, a threshold that affects a load, or a recorded decision must also append to `docs/DECISIONS.md`.
- Personal goals, injuries and preferences are **not** recorded here — this repository is public. They live in the git-ignored `private/`.

AGENTS.md — how to work in this repository

Instructions for any AI agent (or human) contributing to this project. Read this before editing anything. It encodes decisions that are easy to "helpfully" undo.

---

1. What this project is

A local-first strength-coaching tool with a **fixed methodology**. It is not a chatbot that invents programmes. Python computes; the LLM explains. If the LLM is unavailable, the system must still emit a safe, valid prescription.

2. Current status — read this first

v0, v0.1, v0.2 and v0.3 are complete; v0.4 is next. Cardio is v0.4, video v0.5 (see ROADMAP). The deeper labelled calibration set moves to v0.4 (§4.14).

Read `ROADMAP.md` first — it holds the live phase status, the next actions and the resume checklist. `docs/DECISIONS.md` holds every decision that is expensive to reverse.

What exists today is:

- `docs/` — the proposal and design record (public).
- `private/docs/` — documents derived from a real training log (git-ignored, never pushed). `EVIDENCE.md`, `BASELINE.md`, `INFO-REQUESTS.md` live there.
- `data/raw/` — raw exports (git-ignored).
- `strength_coach/utils/` — permanent shared helpers (built early on purpose; see §4).
- `config/exercises/taxonomy.v1.yaml` — exercise identity + properties.
- `tests/` — tests for the above.
- `tools/profile_data.py` — regenerates `private/docs/EVIDENCE.md`.
- `tools/baseline.py` — regenerates the numbers in `private/docs/BASELINE.md`.
- `strength_coach/domain/` — e1RM, load arithmetic, RIR classes (tested).
- `strength_coach/ingest/` — cleaning, flags, quarantine, session reconstruction.
- `strength_coach/features/` — derived metrics, per-lift baseline, VBT calibration, per-region injury signal, cardio capture summary, periodic measurement log, optional session RPE/length, e1RM calibration/decay, per-lift readiness, strength fitness-fatigue shadow model, scheduled/reactive deload policy, standardised warm-up ramps, 3RM/5RM cadence, under-MEV accessory progression, main-lift readiness set reduction and a descriptive progress dashboard.
- `strength_coach/store/` — SQLite migrations, append-only repositories, import orchestration.
- `strength_coach/workout.py` — live workout session, linked prescription snapshot, manual/imported sets, plan overrides, timed/manual end and rubric link.
- `data/coach.db` is created on demand and is git-ignored.
- `strength_coach/rules/` — constitution loader, one predicate per `check.fn`, policy verdicts.
- `strength_coach/prescribe/` — deterministic four-session-type engine, load solver, fallback, and the constraint check that applies stored athlete state. `config/thresholds.v1.yaml` holds the increments and e1RM window.
- Athlete state lives in the §11 tables (`athlete_profile`, `athlete_constraint`, `exercise_suitability`, `metric_definition`, `symptom_log`), entered with `strength-coach state`.
- `strength_coach/critique/` — deterministic rubric over a stored session; unmeasurable metrics are stored `measurable = 0`.
- `strength_coach/trace/` — run lifecycle around `run` / `run_event`.

- `strength_coach/eval/` — backtest harness, labelled-set comparison, and the database-backed marks that make calibration continuous (`rubric_mark`, append-only).
- `strength_coach/cli.py` — `import`, `run`, `review`, `calibrate`, `state`, `block`, `e1rm`, `test`, `symptom`, `cardio`, `measure`, `fatigue`, `select`, `exercise`, `readiness`, `programme`, `validate`, `workout`, `web`. `review` (with `--pending`) is the routine that keeps calibration current.
- `strength_coach/web/` — local-only HTTP UI and JSON API over the same core; no authentication, bind 127.0.0.1.
- `strength_coach/agents/` — LLM interface, config, narrative and structured proposals. `NullLLM` is the default; the model may explain or propose, but never compute. Keys live in the git-ignored `config/llm.local.toml` or the environment (`config/llm.example.toml` is the committed template).
- `ROADMAP.md` — phase status and next actions (update it in the same commit as the work).

Build order, current status and the next actions live in `ROADMAP.md` — do not duplicate them here, because two copies drift. Do not add a hosted UI. The local web UI exists because the user asked for it and must stay 127.0.0.1-only.

3. Non-negotiable invariants

Break any of these and the result is wrong even if the tests pass.

1. **The constitution (`docs/RULES-FORMAT.md`, `config/rules/`) is the source of truth.** Never silently reinterpret an ambiguous rule. Raise it as an open question.
2. **Never let the LLM do arithmetic.** Every number in a prescription comes from a Python function via a tool call.
3. **The LLM may never veto the rules.** The policy engine may veto or downgrade an LLM suggestion; the reverse is not allowed.
4. **Readiness scales are 1–5, higher is better.** `Stress = 5` means no stress. Any code assuming "stress 5 = bad" is a bug. There is a test for this.
5. **`0` is not one thing.** See §5.
6. **No fabricated science.** If a threshold is unknown, put it in config as an assumption and mark it `ASSUMPTION-n`. Do not invent a citation.
7. **Append-only records.** Prescriptions, critiques, calibrations and rule versions are never mutated; corrections create a new row with a pointer back.
8. **Never deduplicate training rows.** Identical repeated sets are real sets. See `docs/DATA-QUALITY.md` DQ-DUP-01.
9. **Unknown exercise names must not be silently dropped.** Resolve or report.
10. **Every non-obvious decision gets a one-line comment stating the domain reason.**
11. **User goals, injuries, tightness and preferences are STATE, not prompt text.** They live in the `athlete_profile` / `constraint` / `exercise_suitability` tables (`docs/DATA-MODEL.md` §11) so a hard constraint can veto a prescription deterministically, whatever an agent says.
12. **An empty/incomplete input produces an `unknown` verdict, never `pass`.**
13. **Never commit personal data.** `data/` and `private/` are git-ignored and this repository is public. If a test needs a realistic value, synthesise it. Tests that read `data/raw` and assert values derived from it are private (`tests/*_integration.py`, ignored).

4. Code layout rules

- Import direction points **down**: `utils` ← `domain` ← `ingest` / `store` ← `features` ← `rules` / `prescribe` / `critique` ← `agents` / `eval`. Nothing lower may import anything higher.
- **If a computation is needed in two places, it belongs in `utils` (pure) or `domain` (domain arithmetic) and is imported.** Never re-implement. The user asked for this explicitly.
- Config, not code: exercise taxonomy, day templates, rule documents, thresholds, rep ranges, load increments, calibration overrides.
- Prefer small pure functions with explicit inputs. No global mutable state.

- Type-hint public functions. Docstrings explain why, not just what.

5. The sentinel rules (memorise these)

`0` means different things per column. Never apply a global rule.

Column	`0` means	Handling
`Avg_velocity`, `Avg_power`, `Avg_rom`, `TUT`	device not used	sentinel → NaN
`Weight_Used`	unweighted bodyweight movement	**valid**, preserve
`RIR`	true failure	**valid**, preserve. `RIR 6` = warm-up (`DEC-RIR-01`)
readiness `Weight`, `HRV`, `RHR`, `Fat_Percent`	missing	sentinel → NaN
readiness `*_Readiness`	missing	sentinel → NaN
readiness `Sleep` / `Energy` / `Mood` / `Soreness` / `Stress` / `Nutrition` / `Hydration`	impossible (1–5 scale)	no sentinel exists

Do **not** reuse `dates.NULL_DATE_TOKENS` for numeric columns: it contains `"0"` and doing so silently erased genuine `RIR = 0` observations during development.

6. Exercise identity

- Resolve every raw name through `utils.exercises`; never branch on the raw log string.
- Recorded decisions are encoded and **must not be reverted**:
- `DEC-TRAPBAR-02` — "Competition Deadlift" is **date-scoped**: sumo deadlift before ~2026-08-17, trap-bar deadlift from then on and forward. Never relabel the whole history. The switch date is config (ASSUMPTION-05) and may change, especially with manual logging.
- `DEC-BACKEXT-02` (amends `DEC-BACKEXT-01`) — back extension ≡ hyperextension **including historical rows**. `Hyperextensions` rows keep their own identity, but they **do** count as back-extension history for load and dose lookups.
- `DEC-RIR-01` — `RIR 6` marks a **warm-up**. Not effort. Excluded from compliance, e1RM adjustment and working volume, but preserved and reported.
- `DEC-LOAD-02` — bodyweight progression acts on **added load**; total mass is used only for an e1RM.
- `DEC-WINDOW-01` — the trap bar's RIR window is the default (it used to be 3-4). The per-lift override map is kept, empty, so changing it back is config, not code.
- `DEC-RIR-02` — the effort window is **RIR 1-4 with a 2-3 target**; RIR 0 is forbidden. The evidence is in `docs/RIR-EVIDENCE.md`.
- There is a standing directive: **back extension is the same as hyperextension, including its history.** When prescribing, prefer `back_extension`.
- Adding an exercise = editing YAML, not Python. If you need a new property, add it to the entity schema and document it in `docs/DATA-MODEL.md`.

7. Definition of done for any change

1. `python -m pytest` passes.
2. New behaviour has a test that fails without it.
3. If the change affects measured facts, `python tools/profile_data.py` still matches `private/docs/EVIDENCE.md` — or you have updated both.
4. New/changed config has a comment explaining the domain reason.
5. If you made a judgement call, it is written down as an assumption, not just in code.

7.1 How to run tests without wasting ten minutes

The full suite includes costly integration and golden tests. While iterating, run only what the change can affect:

```
.venv/bin/python -m pytest tests/test_prescribe.py tests/test_rules_policy.py
.venv/bin/python -m pytest --lf                      # only the last failures
.venv/bin/python -m pytest -m "not integration and not golden"   # broad, ~2 min
.venv/bin/python -m pytest -n 0                      # serial, for debugging a failure
```

`pytest` runs in parallel by default (`pytest-xdist` , one worker per core). A full run is about 4-5 minutes; the fast lane is under 2. Run the **full** suite once, before handing off or committing — not after every edit. If a test fails, fix it and re-run that file/marker first; a full re-run is the last step, not the first reflex.

Tests assert **contracts, not incidental order**: e.g. "power work precedes the main lift" rather than "item 0 is the box jump". Only the golden fixtures pin full item order, on purpose, because the prescription is a user-facing document. When a template changes, update the fixture deliberately; if a test fails only because an unrelated item moved position, the test is too literal.

A test that imports the real export is `integration`-marked and must not run in the default fast lane. If a public test needs data, synthesise it with `make_training` / `make_readiness` — never read `data/raw` .

8. Style

- Python 3.11+. `from __future__ import annotations` .
- Small modules, explicit dataclasses, no clever metaprogramming.
- Exceptions over silent fallbacks; if a fallback is correct, name it and log it.
- Do not add a dependency without a reason in the PR text. Current runtime deps: `pandas` , `numpy` , `PyYAML` . Test dep: `pytest` .

9. Documentation map

File	Contents
`USER-GUIDE.md`	how to install, run and read the output (the human-facing entry point)
`ROADMAP.md`	live phase status, next actions, resume checklist
`docs/DECISIONS.md`	append-only decision log (`DEC-*`) with where each is enforced
`docs/REFERENCES.md`	consolidated scientific bibliography with DOIs and where each is used
`docs/ARCHITECTURE.md`	layers, data flow, where the LLM sits
`docs/DATA-MODEL.md`	canonical schemas, tables, append-only records
`docs/DATA-QUALITY.md`	the decision ledger (DQ-*) and open sentinel questions
`docs/READINESS.md`	how readiness is derived, merged and used; what is config vs rule
`docs/READINESS-SCORE.md`	proposed global and per-lift readiness scoring, evidence and phases
`docs/READINESS-WEIGHT-EVIDENCE.md`	readiness-weight evidence and held-out comparison; learned weights closed
`docs/CONTINUITY.md`	gap detection and re-entry rules; the injury signal is separate
`docs/INJURY-SIGNAL.md`	per-region injury signals, what "tolerated" loading means, injury-to-exercise links
`docs/CORE-TRAINING.md`	core/trunk training evidence and the proposed programme amendment
`docs/LONGEVITY-REVIEW.md`	modality review against the longevity-first goal: gaps, doses, priorities

File	Contents
`docs/FATIGUE-FITNESS.md`	fatigue-fitness evidence, strength recovery, e1RM calibration and why no model drives loads
`docs/LIFT-READINESS.md`	per-lift readiness evidence, components, weights and downward-only rule
`docs/EXERCISE-VARIANTS.md`	how to add a canonical variant, slot wiring, history caveat, selection tuning
`docs/SELECTION-COMPARABILITY.md`	evidence and proposed three-class guard for variant substitution
`docs/READINESS-EXERCISE-SELECTION-PLAN.md`	locked symptom/readiness substitution plan plus app-wide readiness policy and periodic-review panel design
`docs/STRENGTH-PROGRAM-REVIEW.md`	evidence check and critique of the strength programme (review only)
`docs/VOLUME-MANAGEMENT.md`	MEV/MAV/MRV evidence, configured landmarks, ledger and MRV cap
`docs/NEXT-PHASE-FATIGUE-PLAN.md`	final lean non-cardio fatigue plan: sRPE/duration backfill, descriptive calibration, optional deload; no experiments
`docs/RETROSPECTIVE-VALIDATION.md`	retrospective walk-forward method/result; not prospective; no model adoption
`docs/WEAKNESS-INJURY-SELECTION.md`	evidence and deferral of weakness/injury-targeted selection
`docs/WEB-UI.md`	local web UI surface, API and its limitations
`docs/RIR-EVIDENCE.md`	why the effort window is RIR 1-4 with a 2-3 target
`docs/SPINAL-COST.md`	evidence behind the ordinal spinal-cost ratings
`docs/TRAINING-DYNAMICS.md`	evidence on 1RM testing, rep ranges and DUP; proposal, not decision
`docs/WORKOUT-TYPE-READINESS-EVIDENCE.md`	evidence/decision: readiness and injury adjust load/sets, not the planned rep scheme
`docs/RULES-FORMAT.md`	citable rule IDs + 5 encoded rules
`docs/VBT-CALIBRATION.md`	clean set, sparsity, overrides, versioning, current wiring
`docs/VBT-EVIDENCE.md`	VBT evidence, best practices, what our data can and cannot support
`docs/VBT-REP-LEVEL.md`	post-hoc rep-level VBT: benefits, limitations and decision not to build manual entry
`docs/MODULE-LAYOUT.md`	file layout and test strategy
`docs/V0-SCOPE.md`	smallest end-to-end slice and its exclusions
`docs/OPEN-QUESTIONS.md`	ranked design questions blocking the build
`docs/SANITY-CHECKS.md`	external verification of classification/formulas
`docs/TRAINING-GUIDELINES.md`	programme vs published evidence, with conflicts
`private/docs/EVIDENCE.md`	measured profile of a real export (personal)
`private/docs/BASELINE.md`	current strength baseline (personal)
`private/docs/INFO-REQUESTS.md`	goals/injury/suitability backlog (personal)

10. CLI/UI parity — standing instruction

Everything built in the CLI must also be available in the local web UI **where practicable**, in the same change. The UI must call the same core function as the CLI, never re-implement it. If a capability cannot reasonably live in the UI (interactive prompt, destructive migration, host-level operation), record the exception in `docs/WEB-UI.md` and `ROADMAP.md` §4.11 rather than leaving it silently missing.

The in-browser terminal is the fallback for uncovered commands: it runs **only** `strength-coach`, never a shell, and is bound to the UI database.

11. Things an agent should ask instead of doing

Standing rule: when in doubt, ask the user. Do not silently choose an interpretation, a variant, a threshold, a data migration or a programme change because it seems convenient. State the ambiguity, the options and your recommendation, then wait.

- Adding a new hard rule → ask; the constitution is versioned and owned by the user.
- Changing a threshold that affects a prescription → ask.
- Interpreting an ambiguous rule → add to `OPEN-QUESTIONS.md`, do not pick silently.
- Phase status changes → update `ROADMAP.md` in the same commit, never in a separate one.
- "Fixing" data that looks wrong (e.g. `Reps_Done = 1015`) → flag it, quarantine it, do not silently delete it.

Changelog

[0.3.1] — unreleased

Changed

- Replaced the legacy A/B/C session fields and overrides with four canonical session types (`squat`, `hinge`, `bench_squat`, `bench_hinge`). Migration 0023 reparses imported sessions from their preserved raw exercise composition, leaves unresolved sessions `unknown`, and drops `session_day` / `day_guess`. `--day` and `--start-day` are removed; use `--session-type`.

Added

- Workout mode foundation: start a workout from a prescription, required pre-workout readiness, manual set entry, removable sets, timer, editable measured duration, session RPE, end and abandon.
- Workout session/API/UI; a prescription snapshot is frozen at start so later regeneration cannot rewrite what was shown.
- Completion materialises workout sets into `session` / `clean_set`, shows actual vs modified plan and original prescription, persists the deterministic rubric, and closes the workout when review marks are saved.
- In-workout plan overrides: adjust planned sets, swap/add/remove exercises, append-only events.
- In-workout CSV import (paste or file): `Exercise,Set,RPE,Value,Weight,Reps`; RPE→RIR mapping, preview/conflicts/apply/undo, replace matching sets.
- Inline actual-set edit, review-page planned-vs-actual summary, exercise details popup, and partial workout CLI (status/add/update/remove/end/import).

[0.3.0] — 2026-10-20

Added

- Retrospective walk-forward validation report (`strength-coach validate`, API and Prescribe panel). It is explicitly retrospective, not prospective.
- Display-only own-history e1RM context on exercise-selection candidates. It never affects the score, ranking, comparability class or prescription.
- Selection A/B/C comparability guard; A applies, B requires confirmation, C is never automatic.
- Readiness volume valve, weekly volume ledger, MRV cap, under-MEV accessory progression.
- Scheduled deload suggestion + automatic reactive deload.
- 50/70/85 warm-up ramp and 10-week 3RM/5RM cadence.
- Readiness score, per-lift readiness, HRV/RHR baselines and e1RM calibration/decay.
- Descriptive progress dashboard and local web coverage for these surfaces.

Changed / decided

- Strength fatigue model remains **shadow-only**; it lost the retrospective persistence baseline.
- Calibrated continuity remains **rejected**; fixed 14/28/56 bands stay in use.
- No weakness/injury-targeted selection. Injury remains exclusion/penalty/ceiling.
- Strength-programme review closed; future items moved to ROADMAP §4.20.
- Session duration remains manual/optional; no automatic timestamp feature.

Validation

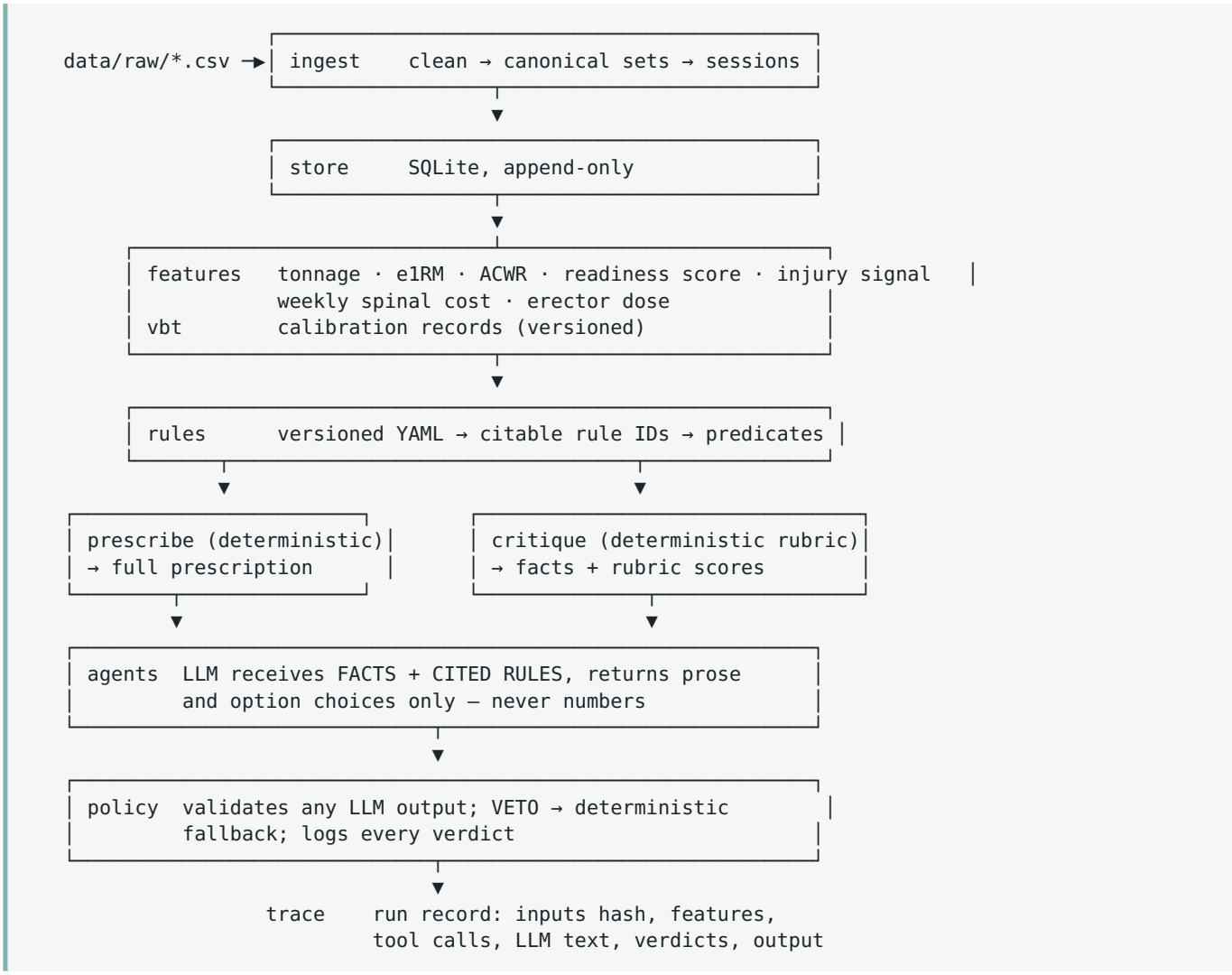
- 604 tests pass locally, including 15 private integration tests (589 public).

ARCHITECTURE

One page. The long version is the rest of `docs/`.

1. Shape

Hybrid by construction: a deterministic Python core, with an LLM layer bolted on the outside that may only explain and select among pre-approved options.



2. Layers and what each may not do

Layer	Owns	Must never
`utils`	pure helpers: dates, numeric sentinels, stats, exercise resolution	import domain code
`domain`	e1RM, load rounding, RIR logic, velocity logic, progression, fatigue mods	do I/O
`ingest`	CSV → raw → cleaned → sets → sessions	invent values; drop rows silently
`store`	SQLite schema, append-only writes	mutate a prescription/critique

Layer	Owns	Must never
`features`	feature store, VBT calibration	contain rule thresholds (those are config)
`rules`	load/hash/version rule docs; evaluate predicates	be edited at runtime
`prescribe`	deterministic prescription from template + state + rules	call an LLM
`critique`	deterministic rubric scores from rules + facts	call an LLM
`policy`	veto/downgrade checks	let the LLM override a veto
`agents`	prompt assembly, tool calls, prose	compute a load, invent a rule, cite an uncited source
`eval`	backtest harness, labelled rubric set	write to production tables

3. Where the LLM sits (and where it must not)

Sits: the why. Narrative critique, rubric rationale, rationale sentences that cite stable rule IDs, and choosing among a small set of pre-approved session options.

Does not sit: any number. Loads, reps, RIR targets, velocity targets, rest, time allocation, tonnage, e1RM, progression decisions, session-type selection logic, or the rubric arithmetic. Those are Python, always.

Three hard boundaries:

1. **Numbers arrive by tool call.** If the prescription agent wants a load, it calls `compute_load_from_velocity(...)` or `next_load(...)`; the tool returns the value and the agent may only echo it.
2. **Every LLM output is validated.** The policy engine re-checks the composed prescription against the hard rules. Failure ⇒ deterministic fallback prescription is used, and the run is marked `llm_downgraded`.
3. **The LLM is optional at runtime.** `NullLLM` returns the deterministic prescription with no narrative. This path is exercised by a test, so "LLM unavailable" is a tested state, not a hope.

4. Run lifecycle and trace

A run is identified by `run_id` and records, in order: the input window, the hash of the two CSVs' relevant rows, the code version, the config/rule versions, the computed features, the candidate prescriptions, every tool call with arguments and result, the raw LLM output, each policy verdict, and the final artifacts. Replaying a day means re-running with the same `run_id` inputs and diffing.

5. Why SQLite and not "just CSVs"

The append-only requirement (`prescription` → `critique` compares prescribed vs performed) and the replay requirement both want referential integrity and cheap atomic writes. SQLite is a single local file, needs no server, and keeps the whole audit trail in one place. DataFrames are for computation; SQLite is for memory.

6. Deliberate omissions

- No UI (explicitly deferred).
- No vector database for v0: the rules document is small and cited by ID, so exact retrieval by ID beats semantic search. RAG is planned only for the programme prose, where chunk IDs must be stable.
- No multi-user, no cloud, no auth.

Local web UI

`strength_coach/web/` is a thin HTTP layer above the same core. It serialises prescriptions, critiques, readiness score/overrides and calibration; it never computes a load or a score. The server binds to 127.0.0.1 and has no authentication. The CLI remains the automation surface.

CONTINUITY AND BACK SIGNAL

Two related rules that keep the programme honest about time: what to do after a gap, and what the app may say about back recovery. Both were decided with the athlete and are implemented in the engine; the open questions are at the end.

---

1. Continuity after a gap

What the evidence says

- **Two weeks of cessation** already reduced fast-concentric and eccentric knee strength in trained sprinters (Yamashita et al. 2023).
- **Eight weeks of cessation** reduced strength and power adaptations in cyclists (Rønnestad et al. 2016).
- **Muscle size is retained longer** than strength in older adults: no significant loss at 12-24 weeks, significant loss at 31-52 weeks (Grgic 2022).
- There is **no clean meta-analysis of cut-points**. The thresholds below are adapted from these findings, not measured (`ASSUMPTION-07`).

The rules (implemented)

gap	adjustment
< 14 days	none
14-27 days	keep load; +1 RIR for the first week back
28-55 days	load −5%; +1 RIR
≥ 56 days	load −10%; +1 RIR; erector ramp steps back one stage

- The gap is measured from the **last stored session** to the prescribed date; no declaration is required.
- Every amended item cites `MOD-GAP-001`, and the prescription warns e.g. "MOD-GAP-001: 31-day gap; prescription amended".
- **Bodyweight work** cannot shed load, so it loses one set instead.
- **DUP**: only weeks with a session (or a measured test) count. A missed week does not advance the cycle.
- **Erector ramp**: pauses during the gap; after the long break it steps back one stage. Rest settles symptoms but does not build load tolerance.

Retrospective walk-forward (2026-10-20)

On 12 observed returns, the calibrated factor had ratio MAE 0.150 versus 0.128 for the fixed rule; the calibrated version lost. The fixed fallback therefore remains in use. This is retrospective, not prospective; see RETROSPECTIVE-VALIDATION.md.

Open questions — parked for calibration

The arithmetic is frozen (`DEC-CONTINUITY-02`) and these are inputs to the calibration in ROADMAP §4.12, not things to change by opinion:

1. Is −5%/−10% right, and should the reduction apply to sets as well as load?
2. Does bodyweight work need a different rule from barbell work beyond losing a set?
3. Should a measured test count for the **erector ramp**? It already counts in the trained-week index, so the ramp advances today; a test is high load but low volume, so this is unresolved.

2. Back-progression signal (now one region of the injury signal)

The single back summary was replaced on 2026-10-19 by a general signal plus one per region (low back, biceps, knee, extensible in config). The back rules are unchanged and still read **low back only**. The full design, evidence and definitions are in INJURY-SIGNAL.md; this section keeps only what continuity needs.

What the evidence says

- Relapse after an episode of low back pain is **common**: 69% within 12 months in a prospective inception cohort, median 139 days (da Silva et al. 2019); 38% in another cohort (Huang et al. 2025).
- **More than two previous episodes predict recurrence** (da Silva 2019), which is why the window scales with prior flares rather than being fixed (`DEC-INJURY-02`).
- There is **no evidence that four symptom-free weeks means clearance**. Four weeks is the base of a configurable window (`ASSUMPTION-06` , `ASSUMPTION-08`).
- Rest helps symptoms but does not preserve strength; the signal therefore counts **loaded sessions**, not calendar.

The rules (implemented)

- Status is one of `unknown` (no entries), `insufficient` (last entry older than 14 days), `active` (latest reading is a flare), `monitoring` , or `review` .
- `review` requires the **scaled** symptom-free window and the **scaled** number of **tolerated** loaded sessions since the last flare, with a current level of 0. Base: 28 days and 4 tolerated sessions; +14 days and +2 sessions per prior flare, capped at 84 days and 12 sessions.
- A loaded session is **tolerated** only when a reading exists in the 7 days after it and no reading in that window is a flare. An unlogged session is `unverified` and never counts.
- Readiness is **excluded**: it already drives `MOD-FAT-001` and is the wrong signal for a structural judgement.
- The signal may **recommend** reviewing a suitability rating; it never sets one, never clears a constraint, and never substitutes a lift.
- Every display carries: "Not medical clearance. A symptom-free streak is weak evidence; the athlete decides..."
- It is shown by `state` and in the prescription notes.

Open questions

The former three (window length, tolerated-loading definition, per-lift detail) were answered on 2026-10-19: the window scales, tolerance means "loaded and followed by a clean reading", and the signal names the lifts that load the region. What remains is calibration, not design: the step sizes and the 7-day verification window are assumptions, and INJURY-SIGNAL.md §7 lists what would replace them.

Full bibliography: `docs/REFERENCES.md` §6 and §7. Decisions: `DEC-CONTINUITY-01` , `DEC-CONTINUITY-02` , `DEC-BACKSIGNAL-01` , `DEC-INJURY-01/02/03` . Implementations: `prescribe/engine.py` (`_apply_continuity` , `_injury_warnings`), `features/injury_signal.py` .

---

3. Calibration from real returns (implemented 2026-10-20)

`features/continuity_calibration.py` estimates, per fixed band (short/medium/long), the ratio of the first return session's best working e1RM to the last pre-gap session's best working e1RM for the same lift. A return pair is excluded if the gap is outside the configured range or the ratio is implausible.

Each band reports:

- number of returns;
- median retention and a deterministic bootstrap 95% interval;
- the fixed factor and the applied factor;
- `ok`, `insufficient`, `uncertain`, `disabled` or `report_only`.

A calibrated factor is applied only when the band has at least `min_returns_per_band` and the whole dataset has `min_total_returns`, the interval is no wider than `max_ci_width`, and calibration is enabled. Otherwise the fixed 14/28/56 rule is used. The factor is bounded to `[factor_floor, 1.0]`; `allow_relax_fixed` decides whether the model may remove part of a fixed reduction. The config is `config/continuity-calibration.v1.yaml`; `MOD-GAP-001` remains the cited rule and the warning names the calibrated factor and its return count. `allow_relax_fixed` can be changed from the Prescribe continuity panel or with `continuity --allow-relax-fixed on|off`.

This is **strength retention**, not capacity retention, and not the e1RM age decay. A future capacity curve needs post-session fatigue/completion data; the current model only uses e1RM observations already in `clean_set`. `strength-coach continuity` and the Prescribe continuity panel show the report; `GET /api/continuity` is the same data.

CORE TRAINING — is the programme doing enough?

Short answer: **before 2026-10-20 the prescribed programme contained no dedicated core work at all.** The performed log did (planks, carries, ab wheel, hanging leg raise), so the omission was in the templates, not in the athlete's practice or equipment. The evidence supported adding a small, progressive block — but it does **not** support claiming that core work prevents back pain, and it does not justify a large isolation circuit.

The amendment was **approved and applied on 2026-10-20** (`DEC-CORE-01`): Day A now carries `ab_wheel`, Day C carries `hanging_leg_raise`, both progress through `PRG-DP-001`, and the constitution is at **1.2.0**. This note keeps the evidence and the reasoning; the sections that describe the gap are written in the past tense where the change has landed.

Decisions/assumptions: `DEC-CORE-01`, `ASSUMPTION-23`. Config: `config/templates/days.v1.yaml`, `config/rules/programme.v1.yaml`. Private detail: `private/docs/CORE-AUDIT.md` (git-ignored).

---

1. What the programme prescribes

session type	main work	trunk-adjacent work	dedicated core (since 2026-10-20)
squat	SSB squat 5×3-5, bench 4×3-5	back extension 3×10-12	**ab wheel 3×6-10**
hinge	trap-bar deadlift 5×3-5, narrow/pause bench 4×3-5	none direct	—
bench+squat	bench 4×3-5, SSB squat support 4×3-5	none direct	**ab wheel 3×6-10**
bench+hinge	bench 4×3-5, single-leg RDL 3×8-10	none direct	**hanging leg raise 3×8-12**

Before the change, the prescribed week contained **6 sets of back extension** (spinal extension) and heavy squat/hinge bracing, and **zero anti-extension, anti-rotation, lateral or carry work**. The taxonomy already carried the movements — `plank` (seconds), `ab_wheel` (reps), `hanging_leg_raise` (reps), `carry` (seconds), plus synthetic `pattern:core_*` buckets — but no day template named one and the constitution's accessory collection excluded them. It was an omission, not a recorded decision: there was no `DEC-*` or open question that said "no core work".

2. What "core work" means here

Five distinct demands, not one:

1. **Anti-extension** — resisting lumbar extension under load (plank, ab wheel, rollout).
2. **Anti-rotation / anti-lateral flexion** — resisting twist and side bend (Pallof, suitcase carry, side plank).
3. **Flexion** — actively shortening the trunk (hanging leg raise, sit-up).
4. **Loaded carry** — walking under load with a braced trunk (farmer's, suitcase).
5. **Bracing during compounds** — the squats, hinges and back extensions already train this.

The programme covers (5) and some spinal extension. It does not cover (1)-(4). "Enough" has to be judged against that coverage, not against a set count alone.

3. What the evidence says

3.1 Core exercise is not superior to other exercise for low back pain

- A meta-analysis of five trials (414 participants) found core stability exercise better than general exercise for pain and disability in the **short term**, but **no significant difference at six months** (Wang 2012).
- The Cochrane review of motor control exercise concluded it is better than minimal intervention, but **not superior to other forms of exercise**, and that the choice should depend on preference, cost and safety (Saragiotto 2016).

Implication: there is no evidence-based reason to prescribe this particular core programme to treat or prevent back pain. Any block is justified by capacity and performance, not by a medical claim.

3.2 Exercise does not clearly prevent recurrence of low back pain

A randomised trial in people recently recovered from an episode found a 12-week exercise and education programme did **not** reduce recurrence compared with an education booklet (63% vs 57%; HR 1.09, 95% CI 0.7-1.8) (Ferreira 2021).

Implication: the injury signal's "tolerated loading" framing stays; core work must not be sold as back insurance.

3.3 Core training does improve athletic performance — modestly

A 2026 meta-analysis of 49 studies reported positive effects of core training on speed (effect size 0.60) and jumping (0.49) across sports (Kayantaş 2026). The populations are mostly athletes and the outcomes are speed/jump, not barbell strength, so this supports some core work for a lifter, not a large one and not a claim about the squat or deadlift.

3.4 Trunk endurance is associated with lower overuse-injury risk (observational)

In 545 naval cadets, failing a 60-second prone plank was associated with greater odds of an overuse injury (OR 3.3, 95% CI 1.2-8.8) (Lopes 2021). This is a prospective association in a young military cohort, not proof that plank training prevents injury, and it does not set a target.

3.5 Loaded carries load the trunk hard (mechanistic)

Surface EMG during farmer's and suitcase carries showed high bilateral activation of longissimus, multifidus, rectus abdominis and external oblique, with suitcase carries adding ipsilateral activation (Ellestad 2024). Mechanistic only: it shows the exercise does what it looks like it does.

3.6 Longer planks are more fatiguing, not more productive

In a randomised acute study, only a 3-minute plank produced increases in muscle thickness, echo intensity and a drop in peak force; 1- and 2-minute holds did not, and perceived effort rose with duration (Aleais 2022). Mechanistic and acute, but it argues for **30-60 s holds with added load or leverage**, not for chasing 2-3 minute planks.

3.7 What the evidence does not say

- There is no trial of core training in **this** population (a strength trainee with a history of resolved low-back tweaks) measuring lift performance or flare rate.
- There is no validated minimum dose. The dose below is a synthesis and is marked `ASSUMPTION-23`.
- No study shows that a specific core exercise is protective, or that omitting core work is harmful.

4. A defensible dose

For a longevity-first athlete training three days a week, with no current symptoms:

element	recommendation	basis
frequency	2 of 3 sessions (the hinge day stays as it is)	cheap, recoverable, avoids adding fatigue before the heaviest pull
volume	2-3 sets per slot, 4-6 sets/week total	enough for a stimulus; the evidence gives no validated dose
coverage	one anti-extension + one anti-rotation/lateral per week	the two demands the compounds do not train
holds	30-60 s; progress by load or leverage	longer holds only add fatigue (§3.6)
progression	reps to top of range, then load (ab wheel, leg raise)	existing `PRG-DP-001` logic
timing	end of session	never before the main lift; it is accessory work
claim	capacity and performance, **not** injury prevention	§3.1-3.2

A reasonable instance: **Day A: ab wheel 3×6-10; Day C: hanging leg raise 3×8-12.** Both are reps-based, both already exist in the taxonomy, and both can progress by load or lever. Planks and carries stay optional/self-directed until a time-based progression rule exists (see §5.3).

5. What changed

5.1 Day templates

One core slot was added to Day A and one to Day C (applied 2026-10-20):

```
# Day A, after the erector slot
- id: core_anti_extension
  exercise: ab_wheel
  sets: 3
  reps: {low: 6, high: 10}
  rir: {low: 2, high: 3}
  rest_s: 90
  essential: false
  load_source: double_progression

# Day C, after the erector slot
- id: core_flexion
  exercise: hanging_leg_raise
  sets: 3
  reps: {low: 8, high: 12}
  rir: {low: 2, high: 3}
  rest_s: 90
  essential: false
  load_source: double_progression
```

5.2 Constitution collection

`PRG-DP-001` scopes `$accessories_and_back_extension` and already carried `increment_kg.per_exercise.ab_wheel: 2.5` — but `ab_wheel` was not in the collection and not in any template, so that config was dead. Both exercises are now in the collection and the constitution is at **1.2.0**. Time-based core (plank, carry) is deliberately still excluded.

5.3 No new hard rule

A rule like "every session must contain core work" has no evidence base and would be a veto with nothing to veto — the template either includes the slot or it does not. The guarantee is a test over the templates, not a constitutional rule. Time-based core (plank, carry) stays out of `PRG-DP-001` until a progression rule for seconds exists; otherwise "hold until top of range" would mean "hold longer", which §3.6 argues against.

5.4 Adjacent defect: `PRG-DP-001` was never assessable

Fixed 2026-10-20. The rule asked for a rep range for every scoped exercise, but `rep_ranges` was built from the **current day's** template. Each day was missing at least one scoped accessory (A: `split_squat`, B: `back_extension`, C: `lateral_raise`), so the rule returned `unknown` on every day and double-progression decisions fell back to "hold load".

- **Cause:** `DayTemplate.rep_ranges()` (`prescribe/templates.py`) was per-day, while the rule scope is the whole collection.
- **Fix:** `templates.all_rep_ranges()` pools the map across A/B/C; the engine passes it to the rule context. On the current store the decision is still "hold load" for all three original accessories, so no prescribed load changed on the day of the fix.

6. Recommendation (applied)

1. **Added core work** — two reps-based slots (Day A anti-extension, Day C flexion), 6 sets/week, framed as capacity/performance.
2. **Amended the constitution collection** (`accessories_and_back_extension` gains `ab_wheel` and `hanging_leg_raise`; version 1.2.0) so progression is governed by the existing rule.
3. **Fixed the per-day rep-range defect** so `PRG-DP-001` can evaluate.
4. **No new hard rule**, no rubric item, no claim about back-pain prevention.

What would change this: a trial in strength trainees linking core work to lift performance or flare rate; a validated minimum dose; or a time-based progression rule that makes planks and carries first-class prescribed work.

7. Open questions (resolved)

- Q15: should the programme prescribe core work, and at what dose? **Yes — two slots, as above.**
- Q16: should `PRG-DP-001` build rep ranges from all day templates? **Yes — fixed.**

Both are marked answered in `OPEN-QUESTIONS.md`.

DATA MODEL

Storage: one SQLite file (`data/coach.db`), written through numbered migrations in `strength_coach/store/migrations/`. Computation is done in pandas; SQLite holds state.

Rules that apply to every table: `created_at` is set by the store, never by caller; artifacts are **append-only** (a correction inserts a new row and sets `supersedes_id`); every row that came from a CSV keeps `source_row` so an error is traceable to a line.

---

1. `raw_row` — untouched intake

column	type	notes
raw_id	INTEGER PK	
table_name	TEXT	`training` | `readiness`
source_row	INTEGER	1-based line in the CSV
file_sha256	TEXT	which import produced it
payload	TEXT	the row as JSON, verbatim

Nothing is ever deleted. `(null)` dates live here permanently. A cumulative re-export reuses the existing raw row when its canonical payload is identical; duplicates within one file still get separate rows (`DEC-IMPORT-01`, DQ-DUP-01).

2. `clean_set` — canonical training record

One row per set, not per exercise or session.

column	type	notes
set_id	INTEGER PK	
raw_id	INTEGER FK	traceability
session_id	TEXT FK	e.g. `<YYYY-MM-DD>`
occurred_on	DATE	parsed; null dates are quarantined (see §4)
exercise_id	TEXT	canonical id from taxonomy
raw_exercise	TEXT	as logged
load_kg	REAL	normalised to kg
load_basis	TEXT	`external` | `bodyweight_total` | `added` | `unknown`
external_load_kg	REAL	what was actually added; 0 for pure bodyweight
reps	INTEGER	
reps_unit	TEXT	`reps` | `seconds`
rir	INTEGER	null when missing
rir_source	TEXT	`logged` | `logged_warmup` (DEC-RIR-01) | `assumed_default` | `missing`

column	type	notes
velocity_mps	REAL	null when device unused
velocity_present	INTEGER	0/1 — one flag for the whole telemetry block; a `0.000000` reading is absent, not zero (DQ-SENT-01)
power_w	REAL	null
rom_mm	REAL	null
tut_s	REAL	null
form_score	INTEGER	null; direction unconfirmed (Q5)
tonnage_kg	REAL	null when `load_basis != external` or `reps_unit = seconds`
is_working_set	INTEGER	excludes warm-up ramps
flags	TEXT	JSON list: `REPS_TYPO`, `LOAD_AMBIGUOUS`, `SECONDS_NOT_REPS`…

`tonnage_kg` is deliberately nullable: `weight × reps` on a plank or a pull-up is meaningless and must not be summed.

3. `session` — reconstructed

column	type	notes
session_id	TEXT PK	date-based
occurred_on	DATE	
workout_label	TEXT	`Sheiko` | `Freestyle` | `Template` — provenance for rotation
set_count	INTEGER	
exercise_count	INTEGER	
main_lifts	TEXT	JSON list of canonical ids
working_set_count	INTEGER	working sets only (DEC-RIR-01 excludes warm-ups)
session_type	TEXT	`squat` | `hinge` | `bench_squat` | `bench_hinge` | `unknown`
tonnage_kg	REAL	sum of non-null `tonnage_kg`
duration_min	REAL	**null by default** — no timestamps exist
duration_basis	TEXT	`measured` | `unavailable`
flags	TEXT	`POSSIBLE_DOUBLE_SESSION`, `NO_MAIN_LIFT`, …

4. Quarantine

`raw_row` rows that cannot be placed (rows with a null date) stay in `raw_row` and get a `quarantine` row with a reason. They are counted in the run report. They are never silently dropped, because doing so hid exactly 8 false `RIR 0` bench violations and 6 for trap bar.

column	type	notes
quarantine_id	INTEGER PK	
raw_id	INTEGER FK	the line that could not be placed
reason	TEXT	`null_date` | `unknown_exercise`
created_at	TEXT	set by the store

5. `readiness_day`

column	notes
on_date PK	
sleep, energy, mood, soreness, stress, nutrition, hydration	1–5, higher better, null when absent
squat_readiness, bench_readiness, deadlift_readiness	1–5 or null
hrv_ms, rhr_bpm, bodyweight_kg, fat_pct, grip_strength_kg	null when 0 sentinel
wearable_score	0–100 device-computed readiness; null when absent; 0 is a valid worst score
composite	derived, 0–1, with `composite_basis` listing which inputs existed
merge_note	JSON: which duplicate rows were considered and why one won

Duplicate-readiness merge rule is DQ-RD-01; implemented as most-complete-row, ties to the latest (`DEC-READINESS-01` , `ingest/readiness.py`). `composite` stays null until the metric directions in Q6 part 2 are declared.

5.1 `readiness_entry` (append-only)

column	notes
entry_id PK	opaque id
on_date	the day the reading describes
field	one of the readiness fields
value	numeric; 1–5 for score fields, positive for device fields (including grip strength in kg), 0–100 for wearable_score
source	`manual` (currently the only writer)
recorded_at	when it was entered
note	optional

Manual readings are one row per field, never an update. `queries.readiness_day` overlays the latest entry per (date, field) over the imported `readiness_day` row, so a manual value wins **per field** and the imported fields around it survive (`DEC-READINESS-03`). The returned dict includes `sources` and `manual_merge_note` (imported value, manual value, winner) so the override is auditable. `readiness_day` remains the imported, derived row.

5.2 `readiness_override` (append-only)

column	notes
override_id PK	opaque id
effective_on	the day the override applies to
scope	`global` or `category:<name>`
value	0.00–1.00, same scale as the computed score
reason	required
source	user
created_at	when entered

The latest row per (date, scope) wins. A category override wins over global, which wins over the computed score; the winning value goes through the load-factor table, but never bypasses a constraint or injury ceiling. The legacy `squat_readiness` / `bench_readiness` / `deadlift_readiness` columns are calibration labels, not manual inputs.

The full derivation — sentinels, merge, storage, and what actually moves a load — is `docs/READINESS.md`. Which fields drive MOD-FAT-001, the threshold and the merge tie-break are config (`config/readiness.v1.yaml`, `DEC-READINESS-04`).

5.3 `readiness_setting` (append-only, per athlete)

column	notes
setting_id PK	opaque id
effective_from	date the setting takes effect
baseline_days / baseline_min_readings	HRV/RHR baseline window and minimum
hrv_slope / rhr_slope	ratio half-width for normalisation
hrv_weight / rhr_weight / wearable_weight	weights/blend
note	optional
source	user
created_at	when entered

The latest row with `effective_from <= date` wins. It overrides `config/readiness.v1.yaml`; no row means config/defaults. Append-only, because changing a baseline changes a prescribed load.

6. `vbt_calibration` — append-only, versioned

column	notes
calibration_id PK	
exercise_id / vbt_family	which target this calibrates
scope	`lift` | `family` | `global_prior`
rep_count	target applies at this rep count
rir_low / rir_high	the window bounds, e.g. 2 and 3 (stored as two columns so SQL can compare)
velocity_median, velocity_mad	m/s
velocity_low, velocity_high	robust band with relative floor
n_sets, n_sessions, window_start, window_end	provenance
method_version	calibration algorithm version
params_json	k, floor, exclusions — hashed
source	`computed` | `override`
override_reason	required when `source = override`
valid_from, valid_until	overrides may expire
supersedes_id	

A prescription stores the `calibration_id` it used, so a past call is reproducible even after recalibration.

7. `rule_document` / `rule`

column	notes
document_id PK-part	`programme.constitution`; the key is (document_id, version), because a new version is a new row
version	semver
effective_from	date
source_hash	hash of the YAML
body_yaml	the full document

`rule` holds one row per rule ID (`PRG-RIR-001` ...) with `severity`, `scope`, `check_fn`, `citations`, and `depends_on_data_decisions`. Rules are immutable; a change is a new document version.

8. `prescription` / `prescription_item` — append-only

prescription column	notes
prescription_id PK	
run_id	trace
for_date	
session_type	squat/hinge/bench_squat/bench_hinge
fallback_level	`full` | `fatigue_reduced` | `llm_downgraded`
rule_document_version	
calibration_ids	JSON
approved_by	`deterministic` | `llm+policy`

item column	notes
item_id PK	
prescription_id FK	
order_index	
exercise_id	
sets, reps_low, reps_high	
load_kg, load_basis, rounding_increment	
rir_target	
velocity_target	nullable, with `vbt_basis`
rest_s, time_cap_s	
essential	0/1
rationale_refs	JSON list of rule IDs + data-point ids
substitution_for	nullable

9. `critique` / `rubric_score`

`critique` stores the narrative plus `facts_hash`; `rubric_score` stores one row per rubric item (`RIR_ADHERENCE` …) with `score`, `weight`, `evidence_refs`, `rule_ids`, and `measurable` (0/1). Metrics that the data cannot support — session duration, within-set velocity loss — are stored with `measurable = 0` rather than a guessed score.

10. `run` — the trace

column	notes
run_id PK	
started_at, finished_at	
input_hash	hash of the source rows used
code_version, config_hash	
window_start, window_end	
llm_model, llm_tokens	null when `NullLLM`
status	`ok` | `llm_failed` | `llm_downgraded` | `error`

`run_event` is the append-only log: ordered `(run_id, seq, kind, payload)` where `kind` is `input` | `feature` | `tool_call` | `llm_output` | `policy_verdict` | `artifact`.

---

11. Athlete state — goals, constraints, suitability, metric definitions

These four tables exist so that you (goals, injuries, tightness, preferences) are **state**, not prompt text. A hard constraint stored here can veto a prescription deterministically, no matter what the LLM says. They are empty until the user answers `docs/INFO-REQUESTS.md`.

11.1 `athlete_profile` — singleton, versioned

column	notes
profile_id PK	
effective_from / effective_to	goals change; old prescriptions must still explain themselves
primary_goal	`strength` | `hypertrophy` | `body_comp` | `pain_free` | `longevity` | `competition`
goal_rank	JSON ordered list; a conflict between erector ramp and squat volume is resolved by this
focus_lifts	JSON list of exercise ids that matter most
competition_date	nullable
sessions_per_week_target	the intended frequency, so the rubric can separate intent from execution
max_session_minutes	the real ceiling, used once duration becomes measurable
block_start_date	week 1 of the **erector ramp**, as the athlete declared it; it does not bound the programme (DEC-PROGRAMME-01). Resetting means a new profile row, never an edit (DEC-RAMP-01/02)

11.2 `athlete_constraint` — injuries, contraindications, mobility limits

Stored as `athlete_constraint`: `constraint` is a reserved SQL keyword (`DEC-STATE-01`).

column	notes
constraint_id PK	
kind	`injury` | `contraindication` | `mobility` | `asymmetry`
region	`low_back` | `mid_back` | `shoulder` | `knee` | `hip` | `ankle` | …
severity	`hard_exclude` | `caution` | `monitor`
forbidden_patterns	JSON, e.g. `["loaded_spinal_flexion"]`
forbidden_exercises	JSON list of exercise ids
rom_limit	nullable, e.g. depth constraint
since / settled_on	dates; drives the "has it settled in 24-48h" logic
evidence	free text from the user, kept verbatim
source	`user` | `clinician`

`severity: hard_exclude` is evaluated by the policy engine with **veto** severity and is never overridable by an agent.

11.3 `exercise_suitability` — living per-exercise rating

column	notes
exercise_id PK-part	
rated_on	last time the user revised it
can_do	0/1 — physically possible
will_do	0/1 — preference; the system may propose around a 0 but must not silently drop it
feels_bad	0/1 — subjective, distinct from contraindicated
note	free text
source	`user` | `observed`

Deliberate distinction: `feels_bad` is a preference the prescription may work around; `constraint.severity = hard_exclude` is a safety rule that vetoes. Conflating them would let a preference masquerade as a medical restriction, or the reverse.

11.4 `metric_definition` — the normalisation layer for new readiness inputs

Every readiness metric, existing or future (Bevel score, HRV, RHR, grip strength L/R), is declared here before use:

column	notes
metric_key PK	e.g. `bevel_readiness`, `grip_left_kg`
label	human name
scale_min / scale_max	bounds
higher_is_better	**0/1 — mandatory.** The documented trap (stress 5 = good) is exactly this field
aggregation	`mean` | `max` | `best_of_3` | `last`
unit	`ms`, `bpm`, `kg`, `score`, `1-5`

column	notes
source	device / app
active_from	

Adding a new readiness metric therefore requires **no code change and no prompt change**: insert a row, and the composite picks it up. A metric with no row is rejected rather than assumed to be higher-is-better.

11.6 `e1rm_override` — a manual estimate that outranks the computed one

column	notes
override_id PK	
exercise_id	canonical id
e1rm_kg	must be positive
effective_from / expires_on	an override may be temporary; NULL expiry means "until replaced"
reason	why it exists — a tested single, a post-layoff start
source	`user` | `measured` | `clinician`
created_at	set by the store

Append-only: a correction is a new row, and the latest row whose window contains the prescription date wins. A prescription cites the row as `e1rm_override:<id>`, so the number is auditable even after the override is replaced.

11.5 `symptom_log` — the manual input the constitution needs

column	notes
logged_on	
timing	`pre_session` | `post_session` | `next_morning`
region	`low_back` | `biceps` | `knee` | … — one reading per region; the configured regions live in `config/injuries.v1.yaml`
level	0–3 agreed scale (see INFO-REQUESTS §6)
settled_within_48h	nullable boolean
note	

v0.1 recorded **one pre-session number for the whole back** (`DEC-SYMPTOM-01`). Since 2026-10-19 the same 0-3 scale is recorded **per region** (`DEC-SYMPTOM-03`): `low_back` , `biceps` , `knee` , and any region added to the injuries config. The column already existed, so no migration is needed — but the region is now load-bearing, not decorative. Every back rule reads `low_back` only, so a knee reading cannot trigger `MOD-BACK-001` or reset the back signal. Multiple entries per day are allowed and the latest wins (migration 0005), so a correction is an insert. A reading is a **flare** at level >= 2, or at level 1 that did not settle within 48 hours (`DEC-SYMPTOM-02`); no reading is `unknown` , never "no flare".

The per-region signal built from this table is documented in `INJURY-SIGNAL.md` ; which exercises load a region is config, not schema.

Until this table has rows, MOD-BACK-001 and the PRG-EREC-001 stability gate return `unknown` — visibly, in the run report.

---

15. `exercise_remap_log` — one-off variant identity rewrite

Migration `0014` canonicalised synthetic `pattern:*` identities in `clean_set` to named variant entities. `clean_set` is normally append-only; the migration temporarily drops its update trigger, records every change here, then recreates the trigger. Raw rows are untouched.

column	notes
raw_id	source raw row
old_exercise_id	previous synthetic or canonical id
new_exercise_id	canonical variant id
reason	migration reason
created_at	

Exceptions: `Competition Deadlift` keeps date-scoped sumo/trap-bar resolution; `Hyperextensions` keeps its own id and remains `history_equivalent_of: back_extension`. Migration `0015` later split `close_grip_pulldown` from `lat_pulldown` using the same audit table.

---

16. `workout_session`, `workout_set`, `workout_event` — live workout mode

Migration `0017_workout` adds the live workout lifecycle. A workout is operational state: the prescription and the sets are immutable, while the workout status and end fields are updated until completion.

`workout_session`:

column	notes
workout_id	UUID
prescription_id	nullable link; the snapshot is `plan_json` because the displayed prescription is frozen at start
for_date / session_type	copied from the plan
status	`in_progress` | `completed` | `abandoned`
plan_json	frozen prescription payload; later regeneration cannot rewrite it
plan_modified_json	latest in-workout plan after swaps/adds/removes/set edits; null until changed
readiness_json	validated seven required fields plus optional device fields
started_at / ended_at	ISO timestamps
duration_measured_s / duration_min / duration_source	measured timer and optional manual override
session_rpe / session_rpe_recorded	required unless explicitly not recorded
notes	
critique_id	deterministic rubric persisted at End workout
reviewed_at	set when human review marks are saved

`workout_set` is append-only. A removed set is an event, not a DELETE:

column	notes
set_id	autoincrement

column	notes
workout_id	parent
exercise_id / set_index	one set per exercise per index
load_kg / reps / rir / velocity	actual values
source	`manual` | `import` | `device`
prescription_item_index	index into the frozen plan, when matched

`workout_event` records `start`, `add_set`, `update_set`, `remove_set`, `restore_set`, `end`, `abandon`, `reviewed`, `import_applied` and `import_undone` with an optional `target_set_id`; it is append-only. Actual-set corrections and imports replay these events rather than mutating `workout_set`.

On completion, `workout.py` materialises the workout into the normal pipeline once:

- a `session` row with `session_id = workout_id`, `source = 'workout'`, `prescription_id`, `workout_id`, measured duration and sRPE;
- one `raw_row` + `clean_set` row per actual set, preserving `prescription_item_index`;
- optional `session_feedback`.

This keeps e1RM, volume and review on one actual-session source of truth. In-workout CSV import writes `workout_set` rows (or append-only `update_set` events when replacing a set); RPE is converted to RIR at import, with RIR 6 marking a warm-up. Date-less backfill is not supported.

DATA-QUALITY DECISIONS

The decision ledger. Every `DQ-*` id is referenced from code and rules, so a decision can be changed in one place.

Status values: `CONFIRMED` (settled), `NEEDS CONFIRMATION` (a rule is proposed and must be approved by the athlete), `ASSUMPTION` (no source can settle it; recorded so it is visible).

> **Note on evidence.** This repository is public, so the measured profile of a real training export is not included. Every claim below was derived from a multi-year Sheiko Gold export and is stated in structural terms — the shape of the problem, not anyone's numbers. Reproduce your own with `python tools/profile_data.py`.

---

1. Sentinels — the core trap

`0` means a different thing in almost every column. **There is no global rule.** Applying one is the single most likely way to corrupt every downstream metric.

DQ-SENT-01 — telemetry zeros `CONFIRMED`

`Avg_velocity`, `Avg_power`, `Avg_rom` and `TUT` are all zero on exactly the same rows, and `velocity == 0` never coexists with `TUT > 0`. So the whole telemetry block is present or absent together, and one flag `telemetry_present = Avg_velocity > 0` describes it.

Action: coerce to null. Exclude from VBT, ROM and TUT features. Never treat as a slow rep — that would make an unmeasured set look like a fatigued one.

DQ-SENT-02 — readiness zeros `CONFIRMED`

`HRV`, `RHR`, `Weight`, `Fat_Percent` and the per-lift `*_Readiness` fields use 0 for missing. This is provable rather than assumed: a bodyweight of 0 kg, an HRV of 0 ms, a resting heart rate of 0 bpm and 0% body fat are all physiologically impossible, and the recorded non-zero bodyweight range is a narrow band consistent with one person.

Action: coerce to null. The 1–5 wellness fields have no zero bucket, so a 0 there is a hard data error, not a sentinel.

DQ-RIR-01 — `RIR 6` marks a warm-up `CONFIRMED`

The athlete confirmed that `RIR = 6` labels a **warm-up set**. That is a much stronger statement than "unknown", and it changes what the value may influence:

- a warm-up is not evidence of effort, so it must not enter compliance scoring;
- it must not contribute an RIR adjustment to an e1RM (extrapolating a ramp-up set to a maximum is meaningless);
- it must not count toward working-set volume, spinal dose or progression decisions;
- but it **must remain visible**: the warm-up share of a session is itself a coaching signal (how much of the logged work was ramp-up).

In the reference export this relabels roughly a quarter of all rows, so the practical working volume is materially smaller than the raw row count suggests. That is exactly the kind of error that is invisible in aggregate and biases every prescription, which is why it was the highest-ranked open question.

Action: `domain.rir.classify` maps 6 to class `warmup`, sets `is_working_set = False`, and records `rir_source = logged_warmup`. The value itself is preserved.

DQ-RIR-02 — `RIR 0–1` remain suspect but in scope `NEEDS CONFIRMATION`

These are genuine effort ratings (unlike the warm-up marker), so they are **not** excluded — the hard "no RIR 0–1 on main lifts" rule must be able to see them. They are flagged `SUSPECT_RIR` and reported separately from clean effort.

What remains open is whether an RIR 0 on an isolation accessory is a real failure set (which the constitution does not forbid) or an unset field. The default treats it as real.

DQ-RIR-03 — `RIR 5` is treated as a genuine easy working set `ASSUMPTION`

Nothing indicates it is a sentinel, and it is used far less than the warm-up marker. This is the least consequential assumption in this file.

2. Bodyweight load convention `NEEDS CONFIRMATION`

One weight column carries **three different meanings**, sometimes for the same exercise:

meaning	what it looks like
unweighted bodyweight movement	`0`
bodyweight captured by the device	a value within a few percent of the athlete's mass
added external load	a small number, well below body mass

Disambiguation key: the readiness file records bodyweight, so a logged value near body mass is total mass, while a value far below it is a plate.

DQ-BW-01 — proposed rule

Resolve `load_basis` per row, in this order:

1. `exercise.time_based` (plank, carry) → tonnage undefined; score on seconds.
2. `Weight_Used == 0` → `bodyweight_total`, `external_load_kg = 0`.
3. `exercise.bodyweight_type` **and** `0.5 × BW < weight ≤ 1.15 × BW` → `bodyweight_total`.
4. `exercise.bodyweight_type` **and** `0 < weight ≤ 0.5 × BW` → `added`.
5. otherwise → `external`.

`BW` is bodyweight on the nearest date within a configurable tolerance (±7 days by default), because weighing is sporadic but body mass is stable across a week. If no bodyweight is on record, `load_basis = unknown` and the row is excluded from e1RM but still counted for sets, reps and rotation.

Why those thresholds: real values cluster either far below body mass (added plates) or within a few percent of it (captured total), so the cut sits in empty space and is not sensitive. They are config, not code.

Open sub-question: for a bodyweight movement logged at total mass, is progression tracked on **total mass** or on **added load**? This changes the double-progression increment for every pull-up prescription. Proposed default: total mass for e1RM, added load for increments.

DQ-LOAD-02 — progression acts on ADDED load `CONFIRMED`

The athlete chose **added load** as the progression axis for bodyweight movements. So a pull-up is programmed as "bodyweight + 2.5 kg", never as a total mass, and the double-progression increment applies to the plate.

This is why `ResolvedLoad` carries both `total_load_kg` and `external_load_kg`: `progression_load()` returns the added load for a bodyweight movement — with a device-captured total-mass reading resolving to **0 kg added**, because the bar is the athlete — while total mass remains the correct input for an e1RM.

DQ-BW-02 — `0` vs "not recorded" `ASSUMPTION`

`Weight_Used = 0` on a bodyweight exercise is read as "unweighted", not "unknown". The alternative discards a large number of real repetitions.

3. Units and types

DQ-UNIT-01 — mixed-case units `CONFIRMED`

Only `kg` and `KG` appear. No unit-less rows were observed, contrary to the brief, but the reader normalises case-insensitively and defaults to kg with a `LOAD_AMBIGUOUS` flag rather than guessing if a blank ever appears.

DQ-REPS-01 — `Reps_Done` is not always repetitions `CONFIRMED`

Plank and carry families store **seconds**: values in the tens-to-hundreds range are isometric hold durations, not repetitions. There are also genuine typos (a four-digit rep count on a push-up variant).

Action: `reps_unit` follows the taxonomy (`rep_semantics`); tonnage is null for `seconds` ; a typo threshold flags implausible rep counts. Flagged rows are retained and excluded from features — never deleted.

4. Duplicates

DQ-DUP-01 — training rows are not duplicates `CONFIRMED`

This contradicts a natural assumption, so it matters.

If you group fully identical rows, the group sizes trace a textbook set structure: mostly pairs, then threes, then fours and fives, with a thin tail. That is what a session with straight sets looks like — e.g. a squat at the same load and RIR three times, then two bench doubles at one load and two singles at another.

Action: never deduplicate the cleaned set table. Blanket deduplication deletes real volume. A row is only collapsed when it is identical **and** carries identical non-zero telemetry, which is effectively impossible for genuinely distinct sets.

DQ-RD-01 — readiness conflicts `NEEDS CONFIRMATION`

Multiple readiness rows per date occur. Only a minority are exact duplicates; the rest are **mutually contradictory** measurements for the same day (different HRV, different hydration). This is not duplication, it is multiple or edited entries.

Proposed merge, in order: 1. drop rows where every field is a sentinel; 2. prefer the row with the most non-null fields; 3. tie-break on latest source row; 4. record the discarded rows in `merge_note` .

Step 2 assumes "more complete = more authoritative", which is why it needs confirmation.

5. Sessions

DQ-SESSION-01 — no timestamps exist `CONFIRMED`

The date column is calendar-date only, with a null placeholder. There is no time column, so "same date + **time gap** + exercise sequence" cannot be implemented. The logging app's own documentation says a generated workout is valid for a bounded window, i.e. one workout is one session, so date is the best available boundary.

Action: one session per calendar date; a `POSSIBLE_DOUBLE_SESSION` flag when the set count is implausible for one session or when two main-lift families appear that no day template contains; `duration_min` is null with `duration_basis = unavailable`.

DQ-SESSION-02 — day classification is retrofitted `NEEDS CONFIRMATION`

The canonical session type is inferred from which main lift and support work is present. Squat and bench_squat both contain squat + bench, so they are distinguished by emphasis and support; bench_squat and bench_hinge are distinguished by squat-support versus hinge-support exercises. Ambiguous sessions get `unknown`, and no type-specific rule is scored.

DQ-NULLDATE-01 — rows with a null date `CONFIRMED`

A contiguous block of rows with a null date forms a plausible session fragment, all carrying the same suspicious RIR value and no telemetry — most likely an orphaned early import.

Action: quarantine, exclude from sessions, report the count in every run. Do not delete: quarantined rows otherwise leak into violation counts and fabricate failures.

DQ-IMPORT-01 — overlap across exports is not deduplication

A cumulative re-export contains rows already stored. The importer matches each new row to a stored payload and reuses the existing `raw_id`; that is **overlap detection across files**, not content deduplication. Two identical rows inside the same new file still create two raw rows and two clean sets, preserving DQ-DUP-01. A correction with a changed load/reps/date is a new row, not a silent merge.

6. Flags that must not be resolved silently

issue	handling
implausible rep counts (e.g. four digits)	`REPS_TYPO` flag, excluded from features, reported
implausible load × rep combinations	`REPS_TYPO` flag
dates with an implausible number of sets	`POSSIBLE_DOUBLE_SESSION`
`Form` 1–4 with unknown direction	stored, **not** used in scoring until the scale is confirmed
`ExertionLoad` semantics unclear	unused; not trusted as a fatigue proxy

7. The general principle

Every row either becomes a feature, gets an explicit flag, or is quarantined with a reason. Nothing is dropped silently, and nothing is "cleaned up" by inference. A missing value stays missing and propagates as `unknown` — because a plausible-looking substitute is worse than a visible gap in a system whose whole purpose is to justify each number it prints.

DECISION LOG

Append-only. Every decision that is expensive to reverse, or that an agent might "helpfully" undo, gets an id, a reason and a pointer to where it is enforced.

Two id namespaces:

- `DEC-*` — a decision the athlete or the project made.
- `DQ-*` — a data-quality finding, documented in `docs/DATA-QUALITY.md`.

Numbers are never reused. Superseding a decision adds a new row; it never edits the old one.

| `DEC-PROGRAMME-05` | 2026-10-20 | Add a scheduled low-volume deload week every 6 weeks from `block_start_date`, alongside the two reactive triggers. A deload removes one set per item (minimum two), reduces load 10% and adds one RIR under `MOD-DELOAD-001` | The review identified missing planned fatigue management. The evidence for a fixed calendar is weak, so this is a configurable policy, not a claimed optimum: six weeks is an assumption, reactive triggers still override, and the scheduled trigger requires a declared block anchor. | `config/deload.v1.yaml`, `features/deload.py`, `prescribe/deload.py`, `MOD-DELOAD-001`, `ASSUMPTION-32`, `ROADMAP.md` §4.4/§4.17 | | `DEC-VOLUME-04` | 2026-10-20 | Close individual MRV calibration as not estimable from the available log. Keep the descriptive volume–performance report and monitoring dashboard; do not add experimental volume manipulation or automatic main-lift volume control | A single athlete on a mostly fixed programme lacks independent volume variation and a counterfactual. Associations can supply priors and flags, not an MRV threshold. The athlete explicitly rejected becoming an experiment. | `docs/VOLUME-MANAGEMENT.md`, `docs/NEXT-PHASE-FATIGUE-PLAN.md`, `private/docs/VOLUME-PERFORMANCE.md`, `features/progress.py`, `/api/progress` | | `DEC-READINESS-10` | 2026-10-20 | Keep fixed, transparent readiness weights; do not learn individual weights from the existing data | A constrained held-out comparison of current, sleep-heavy and HRV-low schemes found no improvement. Readiness is already used to change load/sets, so learned weights would partly fit the programme’s feedback loop; the data lack an independent outcome and sufficient poor-readiness variance. | `config/readiness.v1.yaml`, `docs/READINESS-WEIGHT-EVIDENCE.md`, `ROADMAP.md` §4.9.3 | | `DEC-VBT-03` | 2026-10-20 | Do not add rep-level VBT logging or manual per-rep entry. Keep set-level VBT; preserve automatic per-rep device exports raw for a possible retrospective analysis | Post-workout rep-level velocity cannot stop sets or adjust load in real time; manual entry is high-burden and noisy, and one athlete cannot validate an individual threshold. The evidence-supported use of velocity loss is live set termination. Automatic exports are worth archiving at low cost. | `docs/VBT-REP-LEVEL.md`, `docs/VBT-EVIDENCE.md`, `docs/VBT-CALIBRATION.md` | | `DEC-SELECT-06` | 2026-10-20 | Lock the exercise-substitution plan: symptom/active-region exclusions are automatic and replacement is explicit; Tier 2 requires high recent stress plus readiness below 0.70 and at least 0.10 below the prior 28-day median (min 10 readings, fallback 0.65); mandatory tax dimensions are axial+systemic+skill for main lifts and axial+skill for accessories; no new rule ID | Exercise substitution is a larger intervention than load/RIR reduction and the evidence for daily selection is weak. A rolling baseline avoids a fixed cut-point chasing population norms while the absolute guard prevents baseline drift. Explicit replacement preserves autonomy; no new hard rule until the behaviour is tested. | `docs/READINESS-EXERCISE-SELECTION-PLAN.md`, `ASSUMPTION-34/35`, future `prescribe/selection.py`, `MOD-SELECT-001` | | `DEC-REVIEW-01` | 2026-10-20 | Add a local Periodic reviews panel with an append-only `maintenance_review` record. Reviews are prompts; the only write is a human completed/snoozed decision, never an automatic threshold or prescription change | The athlete will not remember 12-week readiness reviews, rubric-label gaps or test due dates. A local prompt with an explicit decision keeps those reviews visible and auditable without turning them into automatic tuning. Threshold changes still require the documented held-out review. | `strength_coach/store/migrations/0020_maintenance_review.sql`, `features/reviews.py`, `/api/reviews`, Today panel, `config/reviews.v1.yaml`, `docs/READINESS-EXERCISE-SELECTION-PLAN.md` §12 | | `DEC-SELECT-07` | 2026-10-20 | Close the substitution plan at **suggestions + explicit apply**. Do not enable automatic symptom withholding yet. Revisit after 8–12 weeks or ~30

sessions via the Exercise substitution review task | The tax values are reviewed and the suggestion path is implemented, but automatic withholding is a training change that can leave a session without its main lift. The evidence supports injury-aware suggestions, not automatic exercise removal. A recorded review is required before changing the behaviour. | `docs/READINESS-EXERCISE-SELECTION-PLAN.md` §14, `config/reviews.v1.yaml` (`exercise_substitution`), `features/reviews.py`, `/api/selection` | | `DEC-REPS-02` | 2026-10-20 | Do not change the planned rep scheme from daily readiness. Readiness adjusts load/RIR/sets; injury may exclude/modify exercises; a planned maximal exposure can be explicitly skipped or downgraded, but not silently converted into another session type | DUP and linear periodisation are comparable; autoregulated load has not clearly beaten standardised loading; strength is load-specific and hypertrophy is rep-range-tolerant. Switching the goal on a bad day changes the stimulus without evidence of benefit. Injury is a movement-safety problem, not a reason to redefine the block. | `docs/WORKOUT-TYPE-READINESS-EVIDENCE.md`, `config/templates/schemes.v1.yaml`, `config/templates/undulation.v1.yaml`, `prescribe/engine.py` | | `DEC-SESSION-01` | 2026-10-20 | Drop the legacy A/B/C session fields and overrides. Use four canonical session types everywhere: squat, hinge, bench_squat, bench_hinge; imported sessions with insufficient evidence stay `unknown`. Reparse imported sessions from their preserved raw rows/exercise composition; infer old prescriptions from their items. `--day`, `--start-day`, `session_day` and `day_guess` are removed | The legacy labels could not distinguish bench+squat from bench+hinge, which are now distinct session types. Keeping two identities forced every forecast, API, workout and rule to translate between them. Raw rows are preserved, so the old labels are not the only evidence; historical C rows can be resolved from exercise composition and left unknown when they cannot. New prescriptions must have a real type, not a compatibility label. | migration 0023, `ingest/sessions.py`, `prescribe/schedule.py`, `prescribe/engine.py`, `features/programme.py`, `workout.py`, `/api/prescribe`, `/api/programme`, `/api/selection`, CLI `run` / `programme` / `select` | | id | date | decision | why | enforced at | |---|---|---|---|---| | `DEC-PROGRAMME-06` | 2026-10-20 | Add a standardised 3-set main-lift warm-up ramp at ~50/70/85% of the working load (5/3/2 reps), stored with each essential item. Warm-up sets are not working volume | Comparability and session preparation, not a claim of an optimal warm-up; the ramp is config-driven (`config/warmup.v1.yaml`) and persisted separately so it cannot be mistaken for volume. | `config/warmup.v1.yaml`, `prescribe/warmup.py`, `PrescriptionItem.warmup`, migration 0016, `docs/NEXT-PHASE-FATIGUE-PLAN.md` | | `DEC-PROGRAMME-07` | 2026-10-20 | Add a 3RM/5RM test reminder every 10 weeks, alternating schemes. It is a reminder only; no automatic test is written | Strength outcomes remain useful even when computed e1RM is the primary observational outcome; the cadence prevents testing from becoming an unplanned stimulus. | `config/test-cadence.v1.yaml`, `features/test_cadence.py`, `strength-coach test --status`, prescription notes, `docs/NEXT-PHASE-FATIGUE-PLAN.md` | | `DEC-VOLUME-03` | 2026-10-20 | Add one accessory set when a movement pattern has been below MEV for three weeks (`MOD-VOL-002`), up to the configured per-item maximum; main lifts are excluded. Poor readiness may reduce main-lift sets one at a time | This is a small deterministic progression, not an experiment. It addresses the observed under-MEV squat/hinge/pull patterns without inventing an individual MRV or changing main-lift volume upward. | `config/volume.v1.yaml`, `features/volume.py`, `prescribe/engine.py` (`_apply_under_mev`), `MOD-VOL-002`, `docs/VOLUME-MANAGEMENT.md`, `ASSUMPTION-33` | | `DEC-PROGRAMME-04` | 2026-10-20 | Restore squat volume and raise heavy-hinge volume without adding a second heavy hinge: squat session SSB 5×3–5, bench+squat SSB support 4×3–5, trap-bar hinge 5×3–5; keep back extension 3 and single-leg RDL 3 | The rolling programme had cut squat to ~5.5 sets/week and left heavy hinge at ~4. Evidence favors volume over extra frequency when volume is matched, and adding a second heavy hinge would exceed the spinal-exposure cap. Adding sets to the existing trap-bar exposure keeps the cap at squat 1 + hinge 2 = 3 while restoring ~7 squat sets/week and raising hinge to ~5. | `config/templates/session-types.v1.yaml`, `tests/test_session_types.py`, `ROADMAP.md` §4.17, `docs/VOLUME-MANAGEMENT.md` | | `DEC-VOLUME-02` | 2026-10-20 | Add configurable per-pattern MEV/MAV/MRV landmarks and a **conservative MRV cap** that trims non-essential sets; under-MEV is report-only. Do not auto-add volume or auto-deload | MEV/MRV are coaching constructs, not measured individual constants. A one-directional cap prevents accidental volume stacking while respecting the lack of validation for automatic increases. The readiness valve remains a separate recovery response. | `config/volume.v1.yaml`, `features/volume.py`, `prescribe/engine.py` (`_apply_volume_cap`), `MOD-VOL-001`, `docs/VOLUME-MANAGEMENT.md`, `ASSUMPTION-31` | | `DEC-VOLUME-01` | 2026-10-20 | Add a weekly hard-set **volume ledger** by movement pattern with advisory config targets, and expose the readiness volume valve in the Web UI. The ledger does not change loads | Volume drift was invisible while per-lift load, set reduction and volume targets were separate; a ledger makes over/under-volume obvious without inventing automatic

caps. The valve is the only applied volume change and remains explicit/config-gated. | `config/volume.v1.yaml`, `features/volume.py`, `cli.py` (`volume`), `GET /api/volume`, Prescribe volume panel, `POST /api/readiness/volume`, Readiness page | | `DEC-PROGRAMME-03` | 2026-10-20 | Replace the rigid A/B/C calendar for new prescriptions with four rolling session types (squat, hinge, bench+squat, bench+hinge); keep A/B/C as historical/legacy labels. Balance squat/hinge/bench emphasis over six sessions and add a horizontal row plus vertical press | The fixed week left the hinge at four sets and one exposure while the bench had three; adding another hinge to the same template conflicts with the spinal cap. A rolling type layer lets missed sessions shift the deficit, keeps every session with an upper-body push/pull, and confines direct erector work to the squat emphasis. | `config/templates/session-types.v1.yaml`, `prescribe/schedule.py`, `prescribe/engine.py`, `features/programme.py`, web/CLI session-type selectors, ROADMAP §4.17 | | `DEC-READINESS-08` | 2026-10-20 | Poor readiness may also remove one set from accessories **and main lifts**, gated by `readiness.set_reduction`; `include_essential` is now enabled by default after the 2026-10-20 programme change. Power work is bodyweight/implement-only | A load cut alone can leave a fatigued athlete with the same volume; a single accessory set is a conservative, configurable valve. It is not evidence of a strength benefit, so it is separate from the load/RIR modifier and can be disabled. | `config/readiness.v1.yaml`, `features/readiness.py`, `prescribe/engine.py` (`_apply_readiness_score`), `docs/READINESS.md`, tests | | `DEC-CONTINUITY-03` | 2026-10-20 | Implement per-band **return-to-training calibration** from observed e1RM retention. Use a calibrated factor only when the band has enough returns and a tight bootstrap interval; otherwise use the fixed 14/28/56 rule. `MOD-GAP-001` remains the cited rule | The fixed percentages were assumptions. The athlete's own returns can estimate retention without pretending to a fatigue model. Per-band minimums and a bootstrap interval make under-powered bands fall back explicitly; bounding to `[factor_floor, 1.0]` and a config switch keep it reversible. Capacity retention and post-session fatigue remain future work. | `config/continuity-calibration.v1.yaml`, `features/continuity_calibration.py`, `prescribe/engine.py` (`_apply_continuity`), `cli.py` (`continuity`), `GET /api/continuity`, Prescribe panel | | `DEC-VALIDATION-01` | 2026-10-20 | The retrospective walk-forward rejects the calibrated continuity factor and the current strength fatigue model for load use: both lost to fixed/persistence baselines. Keep continuity fixed and fatigue shadow-only; do not start a prospective period for these candidates | 12 returns and 25 lift-session predictions are small, but the candidates were worse than simple baselines. A prospective test is only justified for a candidate that first beats the retrospective baseline. This is retrospective, not prospective. | `features/retrospective_validation.py`, `cli validate`, Prescribe panel, `docs/RETROSPECTIVE-VALIDATION.md` | | `DEC-SELECT-05` | 2026-10-20 | Weakness/injury targeting is not scheduled before v0.5 and not before ~April 2027; revisit only with 12 sessions per target lift over 12 weeks, a pre-registered own-trend weakness metric, a retrospective baseline win, and an opt-in N-of-1/alternating test. Injury targeting remains unscheduled and requires a clinician-defined rehab goal | A date alone is not evidence. The gates prevent a low-data, post-hoc weakness label from becoming a programme rule, and keep injury in the constraint layer rather than turning it into a treatment selector. | `docs/WEAKNESS-INJURY-SELECTION.md`, Q22, ROADMAP §4.19 | | `DEC-SELECT-04` | 2026-10-20 | Selection will **not** target weakness or injury. Injury may exclude/penalise/cap, never positively select. A relative-weakness display, if added, must be display-only and must not change ranking. Any future targeting must clear a retrospective baseline and a prospective/N-of-1 test | Individual response is heterogeneous and weakness/asymmetry measures are inconsistent; injury links are exposure maps, not treatment rules. Incentivising a lift because it supposedly targets an injury is unsafe; current evidence does not show weakness-targeted selection improves outcomes. | `docs/WEAKNESS-INJURY-SELECTION.md`, `docs/SELECTION-COMPARABILITY.md`, `prescribe/selection.py`, tests | | `DEC-SELECT-03` | 2026-10-20 | Variant substitution uses three comparability classes: A load-comparable, B near-comparable (explicit confirmation), C different lift/load basis (never automatic). The selector hides C by default; loads always come from the selected lift's own history | Same-pattern variants differ in load–velocity relationships, mechanics and implement; transferring a load is invalid, and the selector had only a pattern check. A hard class is simpler and auditable than a similarity score. B is the conservative default for changed grip/bar/ROM; C covers different implements/load bases. | `config/selection-comparability.v1.yaml`, `prescribe/selection.py`, `prescribe/engine.py`, `cli.py` (`select`, `run --selection-confirm`), web Prescribe panel, tests | | `DEC-SELECT-02` | 2026-10-20 | Selection may **apply** at most one substitution per slot when the athlete ticks **Apply selection** (or `run --apply-selection`); an LLM may choose only from the Python-filtered candidates. The item is rebuilt with the selected lift's own e1RM/history and cites `MOD-SELECT-001`. No automatic selection | Applying a recommendation without a rebuild would copy a load or silently keep the old lift; rebuilding through `build_item` keeps the same solver,

constraints, injury ceiling and validators. Explicit opt-in preserves the fixed-programme default and makes the provenance visible. | `config/rules/programme.v1.yaml` (1.7.0), `prescribe/engine.py`, `prescribe/selection.py`, `agents/selection.py`, `/api/prescribe`, `cli.py` (`run --apply-selection`), `MOD-SELECT-001` | | `DEC-VARIANT-03` | 2026-10-20 | Apply the approved pattern/muscle audit: Reverse Nordic → `knee_extension`; Split/Bulgarian squat → `squat_unilateral`; Tib Bar Calves → new `dorsiflexion` pattern with **no readiness category**; correct face-pull/rear-delt/rotator-cuff/T-bar-row/row/speed-bench primary muscles; keep landmine press `vertical_push` and dumbbell pullover `lat_isolation` | The audit found a genuine quadriceps eccentric, a unilateral squat, and an ankle-dorsiflexion lift that the old patterns mislabelled. Changing identity would be unnecessary; pattern/muscle properties are the right layer. Tib bar gets no category rather than a false plantar-flexion category. | `config/exercises/taxonomy.v1.yaml`, `tests/test_exercises.py`, `docs/VARIANT-PATTERN-AUDIT.md` | | `DEC-VARIANT-02` | 2026-10-20 | Split **Close Grip Pulldown** from `lat_pulldown`; it is a grip variant, not the same lift. Keep `Bench Press (medium grip)` as its own entity. Future variant-group e1RM ratios are roadmap-only | Narrower grip changes leverage and usable load, so sharing one e1RM/history would blur a real difference. The same logic applies to medium-grip bench. Parent-e1RM ratios (e.g. front squat as a fraction of squat) are a future modelling prior, not a measured fact. | `config/exercises/taxonomy.v1.yaml`, migration `0015`, `prescribe/selection.py`, `config/variants.v1.yaml`, `ROADMAP.md` | | `DEC-VARIANT-01` | 2026-10-20 | Canonicalise **all 114 synthetic raw variant names** from the log, rewrite `clean_set` identities in a one-off audited migration (`0014`, `exercise_remap_log`), and keep the raw rows untouched. Trap-bar date-scoping and back-extension equivalence are explicit exceptions | The log already contains the variant names and muscle groups; leaving them as `pattern:*` made history, readiness and selection blind to the actual lift. A mapping generated from `Main_Muscle` / `Mechanics` / `Force` /name semantics is reproducible and auditable; the remap log makes the exception reversible. `clean_set` is normally append-only, so the migration is a deliberate, logged exception. | `config/exercises/taxonomy.v1.yaml`, `strength_coach/store/migrations/0014_variant_identity.sql`, `config/variants.mapping.yaml`, `docs/EXERCISE-VARIANTS.md` | | `DEC-SELECT-01` | 2026-10-20 | Add a **shadow exercise-selection** score for configured variant slots: readiness, recent stress, rotation and preference; constraints/`can_do:false` exclude, injury penalises; optional LLM may only choose from the filtered candidates. **No prescription change** | Fixed selection cannot use the readiness/fatigue signals already computed, but an unvalidated selector must not swap exercises yet. Ranking only configured variants keeps it safe and explainable; the LLM is a chooser over a Python-filtered list, never a source of exercises or loads. The full fatigue model is cached by database version, so the selector can use it without a per-request fit. | `config/selection.v1.yaml`, `prescribe/selection.py`, `agents/selection.py`, `cli.py` (`select`), `/api/selection`, Prescribe panel, `ASSUMPTION-29` | | `DEC-FFM-03` | 2026-10-20 | Add **family-slope shrinkage** and a chronological **80/20 backtest** to the shadow model; still no load effect | Per-lift fits are sparse; shrinking slopes toward a movement-family fit is a simple partial-pooling compromise. A chronological holdout and naive-baseline comparison make the model's predictive value visible instead of relying on in-sample R^2. | `features/fatigue_fitness.py`, `config/fatigue-fitness.v1.yaml`, `cli.py` (`fatigue`), Prescribe model panel | | `DEC-FFM-02` | 2026-10-20 | Extend the shadow model with an **overall** relative-e1RM row and approximate 95% intervals; still **no load effect** | Per-lift fits are sparse; a pooled relative-performance view uses all lifts without pretending the absolute kg scale is common. Linear-propagation intervals expose parameter uncertainty. Neither justifies changing a load before a prospective backtest. | `features/fatigue_fitness.py`, `cli.py` (`fatigue`), `web/server.py` (`/api/fatigue`), Programme page | | `DEC-E1RM-DECAY-01` | 2026-10-20 | Old e1RM estimates **and** measured overrides are decayed after a **21-day grace**, by **1% per week**, capped at **10%**; modifier `MOD-E1RM-DECAY-001`, constitution **1.6.0** | The 84-day window can select a stale best. Short-term detraining leaves maximal strength largely intact, then strength declines gradually; there is no single validated rate. A bounded linear decay is conservative and auditable. Continuity remains a separate gap modifier. | `config/e1rm-decay.v1.yaml`, `features/e1rm_decay.py`, `prescribe/engine.py` (`_latest_e1rm`), `ASSUMPTION-27` | | `DEC-FFM-01` | 2026-10-20 | Implement a **strength-only, per-lift fitness-fatigue shadow model**: fixed 42-day fitness / 4-day fatigue half-lives, hard-set load, linear fit to e1RM observations, status/refusal for sparse or ill-conditioned data. **No load effect** | The evidence supports a diagnostic but not a controller; a four-parameter fit is not identifiable from ~3 sessions/week. Fixing half-lives and refusing bad fits makes the uncertainty visible. Load changes require a separate decision after intervals and prospective validation. | `config/fatigue-fitness.v1.yaml`, `features/fatigue_fitness.py`, `cli.py` (`fatigue`), `web/server.py` (`/api/fatigue`), Programme page, `ASSUMPTION-28` | | `DEC-E1RM-CAL-02` | 2026-10-20 | **Apply** the e1RM calibration factor when a lift has ≥3

measured tests, the 95% bootstrap CI of measured/formula excludes 1.0, and the median bias is within ±10%; factor = median ratio clamped to ±10%. Measured overrides are never adjusted. Config flag `apply`; modifier `MOD-E1RM-001`; constitution **1.5.0** | Report alone leaves a consistent bias in the load. A bounded robust ratio is the smallest defensible correction; a curve fit on a handful of tests would overfit. CI and bound stop the correction when the evidence is ambiguous or the protocol mismatch is large. | `config/e1rm-calibration.v1.yaml`, `features/e1rm_calibration.py`, `prescribe/engine.py` (`_latest_e1rm`), `config/rules/programme.v1.yaml` (`MOD-E1RM-001`), `ASSUMPTION-26` | | `DEC-LIFT-READY-01` | 2026-10-20 | Add a **per-lift readiness score** (global base + e1RM trend, RIR and VBT deviations; weights 0.20/0.15/0.15) that can only **lower** a load, never raise one; exposed in the prescription notes/UI | There is no validated per-lift readiness instrument; meta-session autoregulation review found subjective readiness measures often lacked validity. RIR-based autoregulation and bar velocity are legitimate fatigue/performance inputs, but combining them is an assumption. Downward-only keeps an unvalidated composite from making training harder; the global score remains primary. | `features/lift_readiness.py`, `prescribe/engine.py` (`_apply_readiness_score`), `docs/LIFT-READINESS.md`, `ASSUMPTION-25` | | `DEC-E1RM-CAL-01` | 2026-10-20 | Add an **e1RM calibration report**: measured tests vs the formula estimate the app would have used, per-lift median ratio, bias and MAE; no auto-correction (superseded by `DEC-E1RM-CAL-02` for the bounded correction) | The Epley+RIR estimate is an approximation and RIR error is load-dependent; measured tests already override it. A correction fitted to a handful of tests would overfit, so the report quantifies the gap without changing loads. | `features/e1rm_calibration.py`, `cli.py` (`e1rm calibrate`), Athlete page | | `DEC-STRESS-01` | 2026-10-20 | Add an **advisory recent-stress** indicator: per-lift decayed hard sets (48 h half-life, 6/3 thresholds) shown on Prescribe/Programme, with a warning when a lift is scheduled within 48 h of high stress. No load or RIR change | A workout's fatigue is real but a fitted fitness-fatigue model is not identifiable from this data (docs/FATIGUE-FITNESS.md §2-4). A transparent fixed decay gives the athlete context without pretending to a validated controller. The constants are `ASSUMPTION-24`; every warning says the load is unchanged. | `config/stress.v1.yaml`, `features/recent_stress.py`, `prescribe/engine.py` (`notes.recent_stress`), `cli.py` (`run` output), Prescribe/Programme pages | | `DEC-DURATION-01` | 2026-10-20 | Session duration stays **manual/optional** (`session_feedback.duration_min`); imported sessions have no timestamp and duration is not inferred from set counts or proxies. No timestamp feature is planned | The export is date-only, so automatic duration would be fabricated. Optional feedback captures what the athlete remembers without changing prescriptions. If manual logging is built, it can capture start/end later; until then the gap remains explicit. | `store/migrations/0013_session_feedback.sql`, `cli review --session-minutes`, web Review, Q5 | | `DEC-SRPE-01` | 2026-10-20 | Session RPE (1–10) and session length (minutes) are **optional** review fields stored append-only in `session_feedback`; blank is valid and no model reads them yet | Volume load and hard-set counts can be derived from sets, but perceived effort and elapsed time cannot: the export has no timestamps, and sRPE is a validated internal-load measure for resistance training (Day 2004). Capturing it at the review stage is one skippable prompt, and it is the missing variable for any future fatigue model. Storing it does not change a prescription. | `store/migrations/0013_session_feedback.sql`, `features/session_feedback.py`, `cli.py` (`review --session-rpe/--session-minutes`), `web/server.py` (`/api/review`), `Review` page | | `DEC-GRIP-01` | 2026-10-20 | Grip strength is **removed from readiness** (CLI flag, web form, validation, ingest mapping, score inputs) and lives only as a periodic `measurement`; the `readiness_day` / `readiness_entry` columns remain for historical rows but are inert | Grip is a prognostic **capacity** marker associated with mortality (Celis-Morales 2018), not a day-to-day readiness signal, and the readiness review already excluded it from the score. Keeping a second entry point implied it was a daily input; dynamometer effort and time of day add noise to a number that can reduce a load. The measurement table gives it the right cadence and direction. | `utils/readiness.py` (`MEASURE_FIELDS`), `cli.py` (`readiness`), `web/static/app.js` (`DEVICE_FIELDS`), `ingest/readiness.py`, `features/readiness_score.py`, `docs/READINESS.md`, `config/measurements.v1.yaml` | | `DEC-MEASURE-01` | 2026-10-20 | Periodic outcomes (VO2max, grip, 5x sit-to-stand, steps, sitting) are a **separate append-only `measurement` table**, declared in `metric_definition` from `config/measurements.v1.yaml`; they never change a load | The app could measure strength but not the primary goal. Readiness is daily state that may reduce a load; an outcome taken every few weeks must not. Direction (`higher_is_better`) is declared once and a later change is refused, because silently reinterpreting stored values is the same class of bug as reading stress 5 as bad. Plausible ranges catch unit errors, and `method` keeps the protocol with the number. Grip stays in readiness for daily context and in measurements for deliberate testing; the two series are never mixed. | `store/migrations/0012_measurement.sql`, `features/measurements.py`, `cli.py` (`measure`), `web/`

`server.py` (`/api/measurements`), `config/measurements.v1.yaml` | | `DEC-POWER-01` | 2026-10-20 | Day A prescribes **`box_jump` 3×3-5 first**, before the main lift, RIR 4-5, full recovery; the slot is excluded from weekly undulation and from the double-progression collection, and `box_jump` becomes a real taxonomy entity | Power declines with age and power training modestly improves physical function versus strength training in older adults (SMD 0.30, low certainty); 3-5 reps with maximal intent and full recovery is the conventional power prescription. A power set cannot be scored from per-set average velocity, so it carries no rule citation and no load progression; quality/height is the progression. Excluding it from undulation preserves its intent when accessory reps undulate. | `config/templates/days.v1.yaml`, `config/exercises/taxonomy.v1.yaml`, `prescribe/templates.py` (`Slot.power`, `bodyweight` load source), `prescribe/engine.py` (`_apply_undulation`, `build_item`), `tests/fixtures/golden_day_a.yaml` | | `DEC-PULL-02` | 2026-10-20 | The vertical-pull slot is a **variant**: `pull_up` and `lat_pulldown` are interchangeable in Day B, and both join `accessories_and_back_extension` (constitution **1.4.0**); each progresses on its own load history | They train the same vertical-pull pattern with similar latissimus activation, and lat-pulldown training improves pull-up endurance. But they are not identical: chin-ups show higher biceps and erector activity while the pulldown shows higher rectus abdominis activity, and 1RM lat-pull and pull-up correlate strongly (~0.78) without being the same strength. Substitutable **stimulus**, not interchangeable loads; the engine picks the most recently trained variant deterministically and never compares their loads. | `config/exercises/taxonomy.v1.yaml`, `config/templates/days.v1.yaml`, `config/rules/programme.v1.yaml`, `prescribe/engine.py` (`_choose_exercise`), `docs/LONGEVITY-REVIEW.md` | | `DEC-PULL-01` | 2026-10-20 | Day B gains **`pull_up` 3×6-10**, and `pull_up` joins `accessories_and_back_extension` (constitution **1.3.0**) so `PRG-DP-001` governs progression | The prescribed upper body was entirely pressing (bench ×2-3, lateral raise) while the log already contained rows and pull-ups: no template pulled at all. One vertical pull closes a coverage/balance gap and gives the grip and upper back some work without inventing a new rule. It is bodyweight, so progression acts on reps first and added load later (`DEC-LOAD-02`). | `config/templates/days.v1.yaml`, `config/rules/programme.v1.yaml`, `tests/fixtures/golden_day_b.yaml`, `tests/test_prescribe.py`, `docs/LONGEVITY-REVIEW.md` | | `DEC-CARDIO-01` | 2026-10-20 | Cardio is captured as its **own append-only exposure** (`cardio_log`: modality, duration, average/max HR, distance, sRPE). No prescription and no fatigue model read it yet; the WHO reference is displayed as **context**, not a target | Cardiorespiratory fitness is the strongest modifiable mortality signal in the evidence, and the app recorded none of it. Capture is the prerequisite for any honest prescription or fatigue model; building the model before the data would be a fabricated parameter set. sRPE is included because HR is not always available. Shown alongside the WHO guidance so the athlete can see where they are, with the disclaimer that it is population guidance, not a personal prescription. | `store/migrations/0011_cardio_log.sql`, `store/repositories.py` (`append_cardio`), `features/cardio.py`, `cli.py` (`cardio`), `web/server.py` (`/api/cardio`), `ROADMAP.md` §4.13 | | `DEC-CORE-01` | 2026-10-20 | The programme adds **two reps-based core slots** — Day A `ab_wheel` 3×6-10, Day C `hanging_leg_raise` 3×8-12 — and both join `accessories_and_back_extension` so `PRG-DP-001` governs progression (constitution **1.2.0**). No hard core rule is added; planks and carries stay self-directed | The prescribed programme had no dedicated core work, while the athlete already trains it voluntarily and the rules carried dead `ab_wheel` config. The evidence supports a **small** block for capacity/performance (core training improves speed and jump; trunk endurance is associated with fewer overuse injuries) but **not** for back-pain prevention (exercise did not reduce recurrence; core exercise is not superior to other exercise). Two reps-based slots keep progression inside the existing double-progression rule and avoid "progress a hold by holding longer", which the evidence does not support. The template, not a new veto, is the guarantee. | `config/templates/days.v1.yaml`, `config/rules/programme.v1.yaml`, `tests/test_prescribe.py`, `docs/CORE-TRAINING.md` | | `DEC-SYMPTOM-03` | 2026-10-19 | Symptom readings are **region-scoped**: `symptom_log.region` may hold any configured injury region, and every back rule reads `low_back` only | Once a knee or biceps reading can exist, an unfiltered "latest symptom" would let a knee flare trigger `MOD-BACK-001` and reset the back signal. Scoping the query is the difference between a multi-injury record and a back record that is silently corrupted by unrelated input. Unknown regions are rejected at the CLI so a typo cannot create a region that never loads. | `store/queries.py` (`latest_symptom`, `symptom_for_date`, `symptom_entries`), `rules/context.py`, `cli.py` (`symptom --region`) | | `DEC-INJURY-03` | 2026-10-19 | A session is **tolerated** only when a symptom reading exists in the 7 days after it and **no reading in that window is a flare**; a loaded session with no reading is `unverified` and never satisfies the review requirement | "Trained a lot" is not evidence that the loading was tolerated. Counting loaded sessions alone rewards behaviour rather than response, and treating a missing reading as "no flare" is the

inference invariant 12 forbids. A flare in the window disqualifies the session even if a later reading is clean. The 7-day window and the 24-48 h settlement question are assumptions (`ASSUMPTION-09`). | `features/injury_signal.py` (`_verified_tolerated`), `config/injuries.v1.yaml` (`verify_within_days`) | | `DEC-INJURY-02` | 2026-10-19 | The review window and the tolerated-session requirement **scale with prior flares**: 28 days + 14 per prior flare (cap 84) and 4 + 2 tolerated sessions (cap 12) | Recurrence after low back pain is common (69% within 12 months, median 139 days), and **more than two previous episodes predict recurrence** (da Silva 2019). A fixed four-week certificate therefore over-states recovery for the athlete who flares repeatedly. The direction is evidence-based; the step sizes are not measured (`ASSUMPTION-08`). `prior_flares` excludes the flare that started the current streak, because that is the event being recovered from. | `config/injuries.v1.yaml` , `features/injury_signal.py` (`window_days` , `required_sessions`) | | `DEC-INJURY-01` | 2026-10-19 | The back-progression summary becomes a **general injury signal plus one per region** (low back, biceps, knee; more by config). Each region names the lifts that load it (direct weight 2, indirect 1) and a prescription that loads a flaring non-back region gets an advisory warning | The athlete asked for a general signal and a per-injury signal, starting with low back, biceps and knee, and for the injury to be linked to exercise selection. A back-only summary could not speak about a knee or biceps flare, and a prescription could load a flaring region silently. This amends the **scope** of `DEC-BACKSIGNAL-01` ; its authority is unchanged — the signal may recommend, never set, a rating, and it is always disclaimed. The injury-to-lift link is exposure, not causation (`ASSUMPTION-10`). | `config/injuries.v1.yaml` , `features/injury_signal.py` , `prescribe/engine.py` (`_injury_warnings`), `web/server.py` | | `DEC-BACKEXT-02` | 2026-10-17 | Historical `Hyperextensions` rows **do** count as back-extension history; this amends the "must not be projected backwards" half of `DEC-BACKEXT-01` | The athlete corrected the original reading: the movement is the same, so its history is usable, including for the erector dose ramp. Identity is still preserved — the rows keep their own id and no equivalence is inferred for any other purpose — so the change is scoped to history lookups and the ramp. | taxonomy `history_equivalent_of` ; `utils.exercises.history_equivalents` ; `prescribe/engine.py` history lookups | | `DEC-TRAPBAR-02` | 2026-10-17 | "Competition Deadlift" is **date-scoped**: it was a **sumo deadlift** until the switch around 2026-08-17 (ASSUMPTION-05) and is the **trap-bar deadlift** from then on and going forward; the switch date may change | The earlier blanket equivalence (`DEC-TRAPBAR-01`) would have counted ~485 sumo pulls as trap-bar history, inflating the trap-bar e1RM and corrupting its velocity calibration. A lift whose meaning changed is not a rename; it is two identities with a boundary. Resolving by row date keeps the sumo history usable under its own name and starts the trap-bar record at the switch. The date is the athlete's approximation and is config, especially because manual logging makes the implement explicit. | `config/exercises/taxonomy.v1.yaml` (`historical_canonical` / `switch_date`), `utils/exercises.py` (`resolve` with date), `ingest/clean.py` | | `DEC-TRAPBAR-01` | 2026-10-17 | "Competition Deadlift" in the log **is** the trap-bar deadlift | The logging app records it under the powerlifting name. Without this the hinge main lift appeared to have no history at all. | `config/exercises/taxonomy.v1.yaml` (`alias_overrides`) | | `DEC-BACKEXT-01` | 2026-10-17 | Back extension ≡ hyperextension **going forward only**; historical rows keep their own identity | Same movement from now on, but the historical rows may not have used the same bench, so projecting the equivalence backwards would silently fabricate erector-dose history. | taxonomy `alias_overrides` + `forward_alias` ; `utils.exercises.forward_name` | | `DEC-LLM-06` | 2026-10-17 | A model may substitute an **essential lift** only for a **rated primary compound** of the same pattern, at most once per proposal and no more often than every 42 days; a main-lift swap always requires explicit confirmation, even with `--yes` | The athlete asked for main-lift substitution with more caution. A main lift is the programme's identity, so the replacement must be something the athlete has explicitly rated can-do/will-do with no "feels bad"; an unrated lift is refused. One per proposal and a 42-day interval keep it a considered change rather than churn, and the trace records `main_substitution: true` so the interval is enforceable. | `agents/proposal.py` (`allowed_substitutes` , `assess`), `cli.py` (`--propose`), `config/thresholds.v1.yaml` | | `DEC-VBT-02` | 2026-10-17 | VBT targets are wired from the **mean-velocity** calibration when it is trustworthy (12 sets/3 sessions, or a family pool of 24); no within-set velocity loss or MVT is inferred; **hex-bar targets are withheld**; a missing target is stated, never guessed | `DEC-VBT-01` deferred wiring because the step from a band to a prescription was unspecified. The evidence now supports that step for mean velocity (reliable L-V relationships, 20-30% velocity loss for strength, no hypertrophy advantage above 25%), while explicitly not supporting RIR-from-velocity for the hex bar. Wiring only what the data supports keeps the no-fabrication rule: the prescription cites the calibration row and says "unavailable" when there is none. Per-rep loss and MVT wait for richer input. | `prescribe/engine.py` (`_attach_vbt_targets`), `features/vbt_calibration.py` , `docs/VBT-EVIDENCE.md` , `docs/VBT-CALIBRATION.md` | |

`DEC-PORT-01` | 2026-10-17 | One database per person, selected only by `--db`; a new athlete uses the same code, their own state file and their own export, with a new-athlete checklist in the guide | "Usable by someone else" is only true if nothing personal is baked into defaults, config or the database path. Keeping one database per person avoids cross-contamination and keeps the append-only history meaningful for each athlete. The tracked-file check enforces the privacy half of that promise mechanically. | `tests/test_portability.py`, `config/athlete-state.example.yaml`, `USER-GUIDE.md` §12 | | `DEC-UNDUL-01` | 2026-10-17 | Weekly undulation is an optional, config-driven 4-week cycle (week 3 main lifts at `maximal`, weeks 2/4 accessories at `volume`), keyed to the declared ramp anchor; with no anchor, day templates apply | The evidence (29-study meta-analysis and older-adult trial) shows undulating and linear periodization produce comparable outcomes, so this cannot be sold as an optimisation. It is included as a variety/fatigue tool because the athlete asked for more dynamic programming, and it is disabled automatically until week 1 is declared so it never guesses a cycle position. | `config/templates/undulation.v1.yaml`, `prescribe/undulation.py`, `prescribe/engine.py` (`_apply_undulation`) | | `DEC-REPS-01` | 2026-10-17 | Reps are chosen from **named schemes** — `maximal` (1-3), `strength` (3-5), `volume` (6-8), `accessory` (8-12) — never written as a number. `maximal` is main-lift only and excluded on hinge lifts | Strength is load-specific, so a 1-3 scheme is a legitimate maximal-strength exposure; keeping 3-5 as the default avoids making near-maximal work routine. The hinge exclusion is a judgement from the athlete's back history: a near-maximal trap-bar pull concentrates the most spinal load of anything in the programme, and the evidence base does not justify that risk for a longevity-first goal. The model names a scheme; Python re-solves the load, so the no-arithmetic rule still holds. | `config/templates/schemes.v1.yaml`, `prescribe/schemes.py`, `agents/proposal.py` (`set_scheme`) | | `DEC-TEST-01` | 2026-10-17 | A strength test is a declared assessment: 3RM/5RM by default, a true 1RM only with `--max` plus confirmation and a minimum interval, recorded as a `measured` e1RM override and never as a training set | A submaximal test gives the same load precision the solver uses (it rounds to 0.5 kg) at far lower spinal cost, and the evidence on true-1RM testing is thin on injury risk but non-zero (case reports in older/osteoporotic populations). The test set is deliberately not written into clean_sets: a maximal test is RIR 0, which PRG-RIR-001 forbids, and counting it as volume would corrupt dose and progression. Recording it as an override keeps provenance (`source = measured`) and lets the athlete supply their own value after the test. | `cli.py` (`test`), `store/repositories.py` (`append_e1rm_override`), `config/thresholds.v1.yaml` (`testing`), `docs/TRAINING-DYNAMICS.md` | | `DEC-CONTINUITY-02` | 2026-10-19 | Freeze the re-entry arithmetic (14/28/56 days, +1 RIR, −5%/−10%) and calibrate it from the athlete's own returns; capture post-session fatigue as a separate observation | The current numbers are adapted from detraining evidence, not measured cut-points. Changing them on opinion would swap one guess for another. Real returns can estimate strength and capacity retention per athlete; until there are enough, the defaults stay. Fatigue belongs with the session context, not the execution rubric, so it cannot distort agreement. | ROADMAP §4.12, `docs/CONTINUITY.md` | | `DEC-CONTINUITY-01` | 2026-10-17 | Gaps are **detected from stored history** before every prescription; a declared break is optional. Re-entry thresholds: **14 days** (keep load, +1 RIR), **28 days** (load −5%, +1 RIR), **56 days** (load −10%, re-enter the ramp one step back). A missed week does not advance the DUP cycle; a test week counts as a trained week. The prescription says it is amended because of the gap | The programme must not prescribe as if nothing happened, and the athlete will not always declare a break. The thresholds are adapted from detraining evidence: two weeks of cessation already reduced fast/eccentric strength in trained sprinters (Yamashita 2023), eight weeks reduced strength in cyclists (Rønnestad 2016), and muscle size is largely retained for 12-24 weeks in older adults (Grgic 2022). The exact cut-points are assumptions, not findings. Counting only trained weeks for DUP is a design choice to stop a missed week becoming "progress". | `prescribe/engine.py`, `config/thresholds.v1.yaml`, `docs/REFERENCES.md` | | `DEC-BACKSIGNAL-01` | 2026-10-17 | A **back-progression summary** may recommend a suitability rating after **4 weeks symptom-free with spinal loading and no flare**, but never sets one; readiness is excluded (it is a daily modifier); rest alone earns no credit; no data means **unknown**; every display carries a disclaimer | Four weeks is a pragmatic assumption: relapse after low-back pain is common (about two-thirds within a year), so a symptom-free streak is weak positive evidence, not clearance. Rest helps symptoms but does not build load tolerance, so the summary must count loaded sessions, not calendar. Readiness already drives MOD-FAT-001 and is the wrong signal for a structural judgement. The summary may recommend; the athlete rates; the engine obeys. | ROADMAP §4.7, `docs/OPEN-QUESTIONS.md` Q11 | | `DEC-TRAPBAR-03` | 2026-10-17 | Sumo is a **replacement** for the trap-bar hinge, not a rotation and not an accessory; no hinge-variant slot is planned | A second hinge in one session violates the no-stacking rule, and on A/C it would sit

next to direct erector work. A rotation would need a new variant-slot mechanism for a lift the athlete can already reach through main-lift substitution; the athlete chose replacement, which keeps the programme one hinge per session and one decision point. | roadmap §4.7 / OPEN-QUESTIONS Q10; main-lift substitution in `agents/proposal.py` | | `DEC-SPINE-01` | 2026-10-17 | Sumo deadlift carries **spinal_cost 2.5**, between the trap bar (2) and conventional (3), on the programme owner's instruction; the half-step is a design choice, not a measurement | The evidence supports sumo being lower in sagittal demand than a conventional pull but does not rank it against the trap bar. The athlete chose to place it between them rather than tie it with the trap bar. Fractional cost is a deliberate half-step on an ordinal scale; whether the cap arithmetic should accept half-steps is an open question (Q11). | `config/exercises/taxonomy.v1.yaml`, `docs/SPINAL-COST.md` | | `DEC-RIR-02` | 2026-10-17 | The effort window on primary compounds is **RIR 1-4 with a 2-3 target**; RIR 0, max singles and in-block 1RM testing stay forbidden | The 2-3 floor was a conservative choice, not an evidence claim. Strength gains are similar across a wide RIR range (meta-regression, null slopes), failure is not required and may be slightly worse for dynamic strength, and self-reported RIR carries roughly ±1-2 reps of error, so a tight veto punishes estimation error. RIR 0 stays banned: it is the grinding set the rule exists to prevent. 2-3 remains the target because it is a good buffer for accuracy and fatigue. | `config/rules/programme.v1.yaml` (PRG-RIR-001), `domain/rir.py` (`ALLOWED_RIR_WINDOW`), `prescribe/engine.py` (`RIR_OUTSIDE_WINDOW`), `docs/RIR-EVIDENCE.md` | | `DEC-RIR-01` | 2026-10-17 | `RIR 6` marks a **warm-up set**, not an easy set | The athlete confirmed the meaning. It carried a quarter of all rows, so treating it as effort inflated volume, fatigue and e1RM. | `domain/rir.py` (`classify`) | | `DEC-LOAD-02` | 2026-10-17 | Bodyweight progression acts on **added load**; total mass is used only for e1RM | The athlete's choice. Programming a pull-up as "bodyweight + 2.5 kg" is actionable; programming it as a three-digit total mass is not. | `domain/load.py` (`progression_load`) | | `DEC-WINDOW-01` | 2026-10-17 | The trap bar's RIR window is amended from 3-4 to the default **2-3** | The tighter window fired on ~43% of trap-bar sets because of window definition, not effort: the athlete trains the hinge at ~RIR 2. One window now covers every primary compound. | `domain/rir.py` (`PER_EXERCISE_RIR_TARGET`, now empty) + `config/rules/programme.v1.yaml` | | `DEC-DUP-01` | 2026-10-17 | Training rows are **never** deduplicated | Identical repeated rows trace a textbook set structure (mostly pairs, then threes, fours, fives). Blanket dedupe would delete real volume. | `ingest/clean.py` (no dedupe step, by design) | | `DEC-NULLDATE-01` | 2026-10-17 | Rows with a null date are **quarantined**, not deleted | They form a plausible session fragment and would otherwise leak into violation counts. Visible and counted beats silently removed. | `ingest/clean.py` (`quarantine`) | | `DEC-ACWR-01` | 2026-10-17 | ACWR is **advisory only** and may never veto a prescription | The "sweet spot" interpretation of acute:chronic ratio is actively contested in the literature. | planned `rules/policy.py` severity `advisory` | | `DEC-VELOCITY-01` | 2026-10-17 | Within-set velocity loss is reported as **not measurable**, with a labelled decay proxy | The export stores one average velocity per set; a mean has no within-set variance. | `docs/SANITY-CHECKS.md` §C | | `DEC-PRIVACY-01` | 2026-10-17 | No training data, bodyweight, injury history or derived metrics in this repository | The repository is public. Personal documents live in the git-ignored `private/`. | `.gitignore`, `AGENTS.md` §3.13 | | `DEC-STORE-01` | 2026-10-17 | Store identity is **(which file, which line)**, never content: `raw_row` is keyed by (table, source_row, file_sha256) and `clean_set` by `raw_id` | Content-keyed deduplication is exactly what `DEC-DUP-01` forbids, and it would also merge a re-export into the original instead of recording it as a separate provenance. Keying on the source line makes a re-import a no-op while two identical sets stay two rows. | `store/migrations/0001_initial.sql` (`UNIQUE` keys), `store/repositories.py` | | `DEC-STORE-02` | 2026-10-17 | Append-only is enforced by **SQLite triggers**, not by convention | A comment cannot stop a scripted `UPDATE`. The triggers make a mutation fail loudly, which is the only version of append-only that survives an agent "fixing" a row. `session` and `run` are deliberately exempt: the first is derived and recomputed, the second must record when it finished. | `store/migrations/0001_initial.sql` (`*_no_update` / `*_no_delete`) | | `DEC-RULES-01` | 2026-10-17 | A rule whose inputs are absent returns **`unknown`**, never `pass`, and every evidence reference must resolve or evaluation raises | An unassessable rule counted as compliance is the failure mode the whole policy layer exists to prevent. Unresolvable evidence is a hallucination even when Python produced it, so it fails loudly rather than being stored. | `rules/registry.py` (per-predicate missing inputs), `rules/policy.py` (`UnresolvedReference`) | | `DEC-RULES-02` | 2026-10-17 | Only an **approved** constitution version is published; a draft is refused rather than given a fabricated `effective_from` | "Which version was in force on this date" must stay answerable. `rule_document.effective_from` is NOT NULL for that reason, and inventing a date for an unapproved draft would make the answer wrong. | `rules/policy.py` (`publish`) | | `DEC-READINESS-01` | 2026-10-17 | Readiness is materialised as one row per date using the **most-complete row, ties to the latest**, and the composite

is left empty | The prescription engine needs a readiness signal for MOD-FAT-001, and leaving 1,013 dates in raw form would mean the modifier never fires. Completeness-equals-authority is an assumption (ASSUMPTION-04), so `merge_note` records every candidate and why one won. The composite needs declared metric directions (Q6 part 2) and is left null rather than guessed. | `ingest/readiness.py`, `store/repositories.py` (`upsert_readiness_days`) | | `DEC-CRITIQUE-01` | 2026-10-17 | A rubric item the data cannot support is stored `measurable = 0` with a NULL score, and the composite is a weighted mean **renormalised over the measured items only** | Treating a missing metric as 0.0 would punish every session for something the export never recorded; treating it as 1.0 would flatter it. Renormalising keeps the composite honest and the `measurable` flag keeps the gap visible. | `critique/rubric.py` (`_unmeasurable`, `Critique.composite`), `store/migrations/0001_initial.sql` (`CHECK measurable = 1 OR score IS NULL`) | | `DEC-CRITIQUE-02` | 2026-10-17 | Rotation is scored **split by workout label**: app-assigned rotation is neutral, self-directed churn is penalised | The logging app deliberately rotates variations between Sheiko sessions, so penalising rotation there would flag the programme's own design. In a self-directed session the opposite is true: changing lifts every week is the failure pattern. The label is the only provenance the export carries. | `critique/rubric.py` (`_rotation`, `APP_ASSIGNED_LABELS`) | | `DEC-E1RM-01` | 2026-10-17 | A manual `e1rm_override` **outranks the computed estimate** for the dates it covers, and the prescription must cite it | The athlete may know a tested value the formula cannot see, especially after a layoff. Overrides are append-only state with an optional expiry, so a temporary re-entry value cannot silently become permanent. Citing `e1rm_override:<id>` keeps the number auditable after it is replaced, and the override changes only the derived load — never history or a score. | `store/migrations/0004_e1rm_override.sql`, `prescribe/engine.py` (`_latest_e1rm`), `cli.py` (`e1rm`) | | `DEC-STATE-01` | 2026-10-17 | The constraint table is named **`athlete_constraint`**, and all five §11 tables are append-only with latest-row-wins reads | `constraint` is a reserved SQL keyword, so the spec's name would need quoting everywhere and invite bugs; the rename is documented rather than silently diverging. Append-only applies the same rule as every other artifact: a re-rating of an exercise or a goal change is a new row, and `block start --force` appends a new profile version instead of editing history. An unrecorded state reports `none_recorded` and a warning, never "nothing is wrong". | `store/migrations/0003_athlete_state.sql`, `prescribe/constraints.py`, `cli.py` (`state`, `block`) | | `DEC-RAMP-02` | 2026-10-17 | Resetting the block anchor **restarts the erector ramp at week 1**, not just the calendar | "Reset means reset." Quietly continuing at week 8 after a declared restart would defeat the ramp's purpose, which is to rebuild tolerance from a known low point. Implemented as a new `block_start_date` with a new ramp phase, so the old phase remains visible in history. | `docs/DATA-MODEL.md` §11.1, roadmap 3.2 | | `DEC-SYMPTOM-02` | 2026-10-17 | A symptom reading is a **flare** at level >= 2, or at level 1 that did not settle within 48 h; no reading is `unknown` | The constitution's wording is "symptoms rise and do not settle in 24-48 h", so severity alone is not enough — an unsettled level 1 counts. Treating a missing reading as "no flare" would let the safety rules pass by silence, which invariant 12 forbids. | `rules/context.py` (`flare_from`), `prescribe/engine.py` (`_apply_back_flare`), `rules/registry.py` (`erector_dose_ramp`) | | `DEC-SYMPTOM-01` | 2026-10-17 | v0.1 records **one pre-session back number**, not per-region | The athlete's three events were all described as "back", and a single number is what the ramp's stability gate and the back-flare modifier actually need. Per-region detail can be added without a migration by writing the existing `region` column with finer values later. | `docs/DATA-MODEL.md` §11.5, roadmap 3.3 | | `DEC-RAMP-01` | 2026-10-17 | Week 1 of the erector block is **set explicitly by the athlete**, stored as a resettable value, and never inferred from dates or the log | The athlete asked to control the official start and to be able to reset it. Inferring a boundary from session gaps would be wrong for a different reason too: the gaps are known to be non-adherence, not programme breaks, so they carry no block signal at all. A reset creates a new profile row rather than editing history. | `docs/DATA-MODEL.md` §11.1 (`block_start_date`), roadmap 3.2 command | | `DEC-VBT-01` | 2026-10-17 | Velocity targets are **not** prescribed in v0; `prescription.calibration_ids` stays empty and the wiring is deferred to v0.1 | The calibration code is sound (band, sparsity ladder, versioning, tests), but the step from "this lift at this rep count has a median 0.62 m/s band" to "do this set at this velocity" is not specified: which rep in the set, what tolerance counts as on-target, and what to prescribe when the band is wide or absent. Inventing those would put a fabricated number on the athlete's screen, which is worse than an explicitly missing one. `docs/V0-SCOPE.md` keeps the step visible. | `docs/V0-SCOPE.md` (steps 4 and §2), `cli.py` (`calibration_ids=[]`), `prescribe/engine.py` (notes) | | `DEC-EVAL-05` | 2026-10-17 | Calibration marks live in an append-only **`rubric_mark`** table; the YAML file is import/export only, and the latest row per (session, item) is the one compared | Calibration is a loop, not an audit: every review must add evidence without rewriting history.

Append-only means a correction is a new row, a NULL score writes nothing (so a skipped item cannot erase an earlier judgement), and `rubric_version` ties each mark to the rubric it judged. The YAML format remains, because it is how marks are shared and reviewed outside the tool. | `store/migrations/0002_rubric_mark.sql`, `eval/store.py`, `store/queries.py` (`latest_rubric_marks`) | | `DEC-EVAL-04` | 2026-10-17 | Every label carries **confidence**, and only `high` counts toward agreement; reconstructed marks are kept but excluded | A mark from a session the athlete cannot really remember is a guess, and tuning a rubric against it would encode the guess as ground truth. Keeping the low-confidence marks preserves the record and still shows what a reconstructed reading looks like, without letting it move the headline. | `eval/labels.py` (`CONFIDENCE_LEVELS`), `eval/harness.py` (`reliable_sessions`) | | `DEC-EVAL-03` | 2026-10-17 | The rubric is **not retuned** to a 3-session labelled set; disagreements are recorded and revisited at ~10 marks | Fitting weights and anchors to twelve item scores from one marker would reproduce that marker's scale use, not validate the rubric. The first pass already showed why: an ambiguous item direction produced 17% agreement, and fixing the protocol (not the rubric) raised it to 42%. Tuning now would confound the two. | `private/eval/` (report), ROADMAP "Known gaps" | | `DEC-PROGRAMME-01` | 2026-10-17 | There is **no fixed programme block**. Progression is continuous and a **deload is reactive** — triggered by data, never by a calendar. **Amended by `DEC-PROGRAMME-05`:** a configurable scheduled deload now exists alongside the reactive triggers. | The athlete rejected calendar deloads: a deload when the data says you are fine is wasted weeks, and a fixed block end would force a periodisation decision the evidence does not support. The original triggers were two consecutive weeks over the spinal cap, or two consecutive flaring sessions. `block_start_date` anchors the ramp and the scheduled deload. | roadmap §4.4, `config/deload.v1.yaml`, `features/deload.py` | | `DEC-PROGRAMME-02` | 2026-10-17 | Planned loads are **recomputed on the day**; a programme is a forecast whose rows are superseded, never edited | A load frozen at generation is stale by the time the day arrives, especially after a layoff or a flare. The append-only rule already forces supersession, so the plan can stay visible as the forecast it was while the day uses current e1RM, readiness and symptoms. | roadmap §4.4, prescription supersession | | `DEC-LOG-01` | 2026-10-17 | Manual workout logging is **per set**; there is no exercise-level shorthand | Every downstream number (tonnage, RIR compliance, set counts) is built on sets. Expanding "4x3 @ 150" into four sets would invent per-set RIR and velocity that the athlete never entered. Sets are the unit of record; typing is the accepted cost. | roadmap §4.2 | | `DEC-READINESS-05` | 2026-10-17 | The readiness modifier's fields and threshold live in `config/readiness.v1.yaml`; the "enforced at" pointer in `DEC-READINESS-02` predates that file | DEC-READINESS-02 decided that the modifiers' inputs are a config list; this row records where that list actually lives, so somebody following the earlier pointer is not sent to the rule document instead of the config. | `config/readiness.v1.yaml`, `features/readiness.py` | | `DEC-IMPORT-01` | 2026-10-19 | A new cumulative export is matched against stored raw payloads; an identical row reuses its existing raw row so clean sets/sessions are not duplicated. Duplicates **inside one file** still become separate rows | A fresh export has a different file hash, so file/line identity alone would double all history. Content matching across files is overlap detection, not deduplication: the first occurrence of each stub in a file still creates a row, preserving DQ-DUP-01. A changed row (different load/reps/date) is genuinely new and is not silently merged. | `store/repositories.py` (`store_raw_rows`), `store/importer.py` | | `DEC-READINESS-08` | 2026-10-20 | Per-athlete readiness score settings are append-only rows with athlete > config > code precedence. The baseline stays a rolling median; a trend-projected baseline is reported only in shadow mode | The athlete needs to tune window, minimum readings, slopes and weights without code changes. A learned baseline is not validated yet, so it must not silently move loads. | `readiness_setting` (migration 0010), `features/readiness_score.py`, `features/readiness_baseline.py` | | `DEC-READINESS-06` | 2026-10-19 | The old per-lift readiness fields (squat/bench/deadlift) are **black-box calibration labels**, not inputs; remove them from new input surfaces and keep the 306 historical days read-only. Computed readiness is overridable at global and category scope; an override wins over the computed score and changes the load, but cannot bypass constraints or injury ceilings | The athlete recognised the three values as outputs of an older external calculation and wants them used to calibrate the new model, not re-entered. Keeping them as labels preserves the only reference set. Overrides need to win over a model the athlete knows is wrong today, while safety constraints and injury ceilings remain vetoes. | ROADMAP §4.9.5, `docs/READINESS-SCORE.md` §2.8-2.9 | | `DEC-READINESS-07` | 2026-10-19 | Overrides use the same 0.00-1.00 scale as the computed score; combined readiness + continuity reduction is capped at 20%; main-lift load features enter calibration only | The athlete asked for the override to match the score scale. No study defines a safe stack of readiness and return-to-training reductions, so 20% is a guardrail, not a finding; constraints, flares and

injury ceilings stay outside it. The old labels were load-derived, so main-lift load history is a calibration feature, never a production input. | `docs/READINESS-SCORE.md` §2.3a, §2.8; `DEC-READINESS-06` | | `DEC-READINESS-04` | 2026-10-17 | The readiness derivation is documented in `docs/READINESS.md`, and its changeable parts are config: modifier fields, threshold, merge strategy and tie-break | Readiness can reduce a load, so it must be explainable end to end. The mechanical steps (sentinel mapping, coercion) stay in code because they are data-quality rules; the choices a user may reasonably want different live in `config/readiness.v1.yaml`. Unimplemented values raise instead of being ignored, so a config edit never silently does nothing. Changing a modifier field or threshold changes a prescribed load and must be recorded here. | `docs/READINESS.md`, `config/readiness.v1.yaml`, `features/readiness.py` | | `DEC-READINESS-02` | 2026-10-17 | The fatigue modifier reads a **config list** of readiness fields (sleep and energy today); other fields are recorded and reported but never move a load | Keeping the list in config lets it grow without a code change, and matches MOD-FAT-001's wording. Silently widening it (soreness, stress) would multiply false downgrades without evidence. | `config/rules/programme.v1.yaml` (MOD-FAT-001), roadmap §4.1 | | `DEC-READINESS-03` | 2026-10-17 | Manual readiness wins **per field**; the imported row fills the fields it does not provide, and the merge note records both values and the source | The athlete is present on the day and the export is a delayed summary. Per-field precedence avoids discarding a full imported row because one field was typed. Recording both keeps the decision auditable rather than silent. | roadmap §4.1, `readiness_day.merge_note` | | `DEC-EVAL-01` | 2026-10-17 | Human marks live in **git-ignored `private/eval/`**; the committed `eval/labelled.example.yaml` is synthetic | The marks are a judgement about a real athlete's sessions and the keys are real training dates. The format still needs to be public so anyone can reproduce the harness, so the format is committed and the data is not. | `eval/labelled.example.yaml`, `.gitignore` (`private/`) | | `DEC-EVAL-02` | 2026-10-17 | Agreement is `\|human - engine\| <= 0.25` per item, and an item the engine could not measure is **excluded** from the denominator | 0.25 is one marking step, so the tolerance matches the granularity a human was asked for. Counting an unmeasurable item as disagreement would punish the rubric for a gap in the export, which is exactly what `measurable = 0` exists to prevent. | `strength_coach/eval/harness.py` (`compare`), `docs/V0-SCOPE.md` §3 | | `DEC-LLM-05` | 2026-10-17 | A proposal may **name a menu option** that Python computed (keep / easier / harder / heavier / lighter), but it still may not write a load or RIR. The menu's width is the PRG-RIR-001 window, so relaxing that rule widens the model's choices without a code change | A menu keeps provenance: every number is computed by the solver, and the model's contribution is a choice among valid ones — the useful part of a model's judgement without the unverifiable part. Bounding RIR options by the rule means the constitution, not a prompt, decides what is choosable; if evidence later justifies an RIR 1-4 window, the menu follows automatically. Load options are always real increments; RIR options are re-solved from the same e1RM, so the pair stays coherent. | `agents/proposal.py` (`menu`), `prescribe/engine.py` (`RIR_OUTSIDE_WINDOW` validation) | | `DEC-LLM-04` | 2026-10-17 | A model proposal is a **structural** edit (substitute a non-essential same-pattern item, bounded set-count change, drop a non-essential item, reorder) that is rebuilt by the engine and judged by the same validators; any load/RIR/percentage/velocity field is rejected before parsing, and proposals are only attempted on an undowngraded session | "Never let the LLM do arithmetic" must be enforced structurally, not by trusting the prompt: rejecting computed-number keys means a compromised or confused model cannot inject a weight. A substitution reuses `build_item`, so the load comes from the same solver as the deterministic path. A session already modified by MOD-FAT-001/MOD-BACK-001 is closed to proposals because the safety decision outranks model convenience. Acceptance is explicit (prompt or `--yes`), and the stored prescription records `approved_by = llm+policy`, so the artifact says who chose it. | `agents/proposal.py`, `cli.py` (`--propose`), `prescribe/engine.py` (`build_item`, `validate_prescription`) | | `DEC-LLM-03` | 2026-10-17 | The default model endpoint is **DeepSeek** (`deepseek-chat`), while `provider` still defaults to `"null"` | Choosing a provider should not also mean choosing an endpoint: DeepSeek is the athlete's preferred provider, so `provider` + an API key is enough to enable narration. Keeping the provider default null preserves the zero-config guarantee — an unconfigured checkout never makes a network call. | `agents/config.py` (`DEFAULT_BASE_URL`, `DEFAULT_MODEL`), `config/llm.example.toml`, `tools/llm_check.py` | | `DEC-LLM-01` | 2026-10-17 | LLM output is recorded as a **`run_event`** and never written into a prescription, score or verdict; `NullLLM` is the default | The architecture claims a bad or missing model cannot change what the athlete lifts. Writing prose into `critique.narrative` would blur that line, so the deterministic narrative stays in the artifact and the model's version is an event with its own model id and token count. A run with no model differs only in that event. | `agents/narrative.py`, `store/migrations/0001_initial.sql` (`run_event`) | | `DEC-LLM-02` | 2026-10-17 | LLM

settings resolve **environment > git-ignored file > defaults**, and only an OpenAI-compatible adapter is implemented | The default must be a working system with no model at all, so `provider = "null"` is the default and an incomplete config degrades to `NullLLM` rather than erroring. The key is never printed (masked in diagnostics) and never committed: `config/llm.local.toml` is git-ignored, `llm.example.toml` is the template. One adapter covers hosted and local endpoints without adding a dependency. | `agents/config.py`, `.gitignore`, `config/llm.example.toml`, `tools/llm_check.py` | | `DEC-PRESCRIBE-01` | 2026-10-17 | Prescribed loads are solved from a recent e1RM and rounded **down** to 0.5 kg | The athlete owns 0.25 kg microplates, so 0.25 kg per side is the smallest real jump; 2.5 kg steps would leave most of that resolution unused. Rounding down because rounding up would push the set above the target RIR — the failure direction that costs a session. The 84-day e1RM window stops a two-year-old peak from prescribing today's load. | `config/thresholds.v1.yaml`, `prescribe/loads.py`, `domain/e1rm.py` (`load_for_reps`) | | `DEC-PRESCRIBE-02` | 2026-10-17 | MOD-FAT-001 applies the **smaller** reduction (5%, not 10%), and only reduces | The rule allows 5-10%. Taking 5% first keeps the session completable as written, and a reduce-only modifier cannot accidentally increase a load through rounding. Every application cites `readiness:<date>` so the downgrade is explainable. | `config/thresholds.v1.yaml`, `prescribe/engine.py` | | `DEC-TRAPBAR-01` (blanket identity) | `DEC-TRAPBAR-02` | 2026-10-17 | The name did not mean one lift for the whole history; the correction is date-scoped rather than a different alias. | | `DEC-STORE-03` | 2026-10-17 | Readiness is persisted as `raw_row` only until the duplicate-row merge rule (DQ-RD-01) is decided | Materialising `readiness_day` requires choosing which of several conflicting rows wins. That is open question Q6; writing a guess into storage would make it indistinguishable from a measured fact. | `store/importer.py` (readiness imported raw, not merged) |

Open questions that are not decided

See `docs/OPEN-QUESTIONS.md`. Each has a documented default so work can continue without an answer.

Superseded

id	superseded by	date	why
`DEC-BACKEXT-01` (historical-rows clause)	`DEC-BACKEXT-02`	2026-10-17	The original projection-before-history caution was overruled by the athlete: the rows are the same movement and their history is usable.
`DEC-STORE-03`	`DEC-READINESS-01`	2026-10-17	Readiness is now materialised because the prescription engine needs it for MOD-FAT-001. The merge rule it was waiting for is the documented Q6 default, and the `readiness_day` table already existed for exactly this.

Exercise variants — configuration

Variants are **configuration**, not code. A variant slot is a template slot with more than one allowed canonical exercise; the engine picks one, and the shadow selector can rank them.

0. Web Exercises tab

The web UI's **Exercises** tab shows every entity and writes edits/additions to `config/exercises/overrides.v1.yaml`. The checked-in taxonomy stays intact, so comments and the base audit trail survive. IDs are immutable.

1. Add the canonical exercise

`config/exercises/taxonomy.v1.yaml`:

```
  front_squat:
    display: "Front Squat"
    log_names: ["Front Squat"]
    pattern: squat
    main_lift: false
    primary_compound: true          # the RIR window applies
    programme_slot_of: ssb_squat    # informational: used where SSB was programmed
    vbt_family: squat
    loadable: true
    time_based: false
```

property	why it matters
`log_names`	exact names resolved at import; add every spelling
`pattern`	readiness category, selection family, injury links
`primary_compound`	applies `PRG-RIR-001`
`bodyweight_type`	`load 0` is a real bodyweight load
`loadable`	external load is meaningful
`vbt_family`	VBT calibration family
`history_equivalent_of`	**only** for the same movement (e.g. back extension/hyperextension); it merges history and load

Do not use `history_equivalent_of` for a different variant with different loads (pull-up vs lat pulldown).

2. Add it to a slot

`config/templates/session-types.v1.yaml`:

```
      - id: squat_main
        exercise_variants: [ssb_squat, front_squat]
        sets: 4
        reps: {low: 3, high: 5}
        rir: {low: 2, high: 3}
        rest_s: 180
        essential: true
        load_source: e1rm
```

- `exercise` **or** `exercise_variants`, not both.
- every id must exist in the taxonomy;
- all variants share the slot's sets/reps/RIR/rest;
- `essential: true` treats whichever is selected as essential work.

The engine still chooses the most recently trained variant. The shadow selection panel ranks all configured variants but does not change the prescription.

3. History caveat

Taxonomy changes apply to **future imports**. Existing `clean_set` rows keep the canonical id they had at import. If a raw name was previously stored as `pattern:squat`, adding `front_squat` will not relabel that history.

Options:

- use the existing canonical id instead of introducing a duplicate;
- `history_equivalent_of` only if it is genuinely the same movement;
- a deliberate re-resolution migration (not built);
- rebuild the database from exports in a fresh setup (loses marks/state).

4. Canonicalisation and history rewrite (2026-10-20)

All synthetic raw names in the current log were canonicalised: **114 new entities** were generated from the log's `Main_Muscle`, `Mechanics`, `Force` and name semantics, and `clean_set` identities were rewritten by migration `0014` (audit table `exercise_remap_log`, **3,607** rows). The raw rows are untouched.

Correction: `Close Grip Pulldown` was initially reused as `lat_pulldown`; it is a grip variant with different leverage/load, so migration `0015` split it into `close_grip_pulldown` (**15** rows) and added it to the pull-slot candidates. The same principle applies to `Bench Press (medium grip)`, which remains its own entity.

Exceptions preserved:

- `Competition Deadlift` keeps date-scoped sumo/trap-bar resolution; no blanket rewrite.
- `Hyperextensions` keeps its own id and remains `history_equivalent_of: back_extension`.

The generated mapping is kept for audit at `config/variants.mapping.yaml`. A pre-migration database backup was made at `data/coach.db.pre-variant-20261020` (git-ignored).

`config/variants.v1.yaml` lists the squat/bench/pull variants as selection candidates. The template still decides the default prescription; ticking **Apply selection** in Prescribe (or `run --apply-selection`) applies at most one substitution per slot and re-solves the load from the selected lift (`MOD-SELECT-001`).

5. Selection tuning

`config/selection.v1.yaml` holds the weights and rotation window. The selector uses readiness, recent stress, rotation, suitability preference and the fitness-fatigue state. The model is **cached per database version** (file mtime/size), so the selector does not pay the fit cost on every request. Selection comparability is implemented as A/B/C classes (`DEC-SELECT-03`); see SELECTION-COMPARABILITY.md. Symptom and stress/readiness suggestions add tax metadata and a trigger, but remain explicit; see READINESS-EXERCISE-SELECTION-PLAN.md. Per-exercise preference comes from Athlete suitability, not config.

6. Validation checklist

- [ ] canonical entity resolves: `strength-coach state` or a fresh import;
- [ ] new variant has history (or accept that it starts at no history);
- [ ] template parses: `strength-coach select --date YYYY-MM-DD --session-type T`;
- [ ] golden/rep-range tests updated if the template changed;
- [ ] server restarted.

Fatigue, fitness, and strength: evidence and options

Status: research note (2026-10-20). No model is implemented. The current programme changes loads from readiness and history; it does not carry a session-level fatigue state forward. This note asks whether it should, what the evidence supports, and what the alternatives are.

Short answer: **a workout does contribute to fatigue, but the evidence for a strength-specific fitness-fatigue model is not strong enough to let one change prescriptions.** A shadow model now fits per-lift fitness/fatigue state for display; it does not read into a load. The app also captures the practical consequence through self-reported readiness.

---

1. What the app does today

mechanism	what it does	what it does not do
readiness score	global 0–1 from sleep/energy/mood/stress/soreness (plus HRV/RHR baseline and optional wearable blend); category overrides	no per-lift readiness; no post-session fatigue
`MOD-FAT-001`	reduces load up to 5% and raises RIR when readiness is low	does not know what you lifted yesterday
e1RM	best usable Epley+RIR estimate in the **84-day** window, or a measured override	does not decay the estimate after a hard session
gap rules	14/28/56-day bands reduce load and step back the erector ramp	a 48-hour turnaround is "no gap"
deload	scheduled/reactive low-volume week can reduce sets, load and RIR	no accumulated-load or fitness-fatigue state drives the trigger

So a very hard Monday does not directly change Wednesday's prescription. Only the readiness you report on Wednesday does. That is conservative and defensible, but it means the app treats a workout as a **history record**, not as a stress that decays.

---

2. The classic model and the strength evidence

The fitness-fatigue (Banister) model represents performance as a slow positive adaptation minus a fast negative response:

```
P(t) = P0 + k1 * Σ w_i * exp(-(t - t_i)/τ1) - k2 * Σ w_i * exp(-(t - t_i)/τ2)
```

- `w_i` — training load for session `i` (TRIMP, volume load, sRPE, …)
- `τ1` — fitness decay (slow), `τ2` — fatigue decay (fast)
- `k1`, `k2` — fitness and fatigue multipliers
- `P0` — baseline performance

The strength-specific evidence is thin. The most direct study is small: six elite weight-lifters over one year, with individually fitted parameters; predicted and actual performance correlated in each subject, but the fatigue index "remains to be clarified" (Busso 1990, ref 94). Applications and validations are overwhelmingly endurance-based. We found no trial showing that changing strength loads because of a fitted fitness-fatigue model improves strength outcomes.

The model is also statistically fragile:

- Parameters are ill-conditioned and sensitive to starting values, fitting method and the load metric (Hellard 2006, ref 64; Vermeire 2022, ref 97).
- In a Bayesian re-analysis of two independent datasets, the model was non-identifiable and the fatigue component **did not improve out-of-sample prediction** over a fitness-only model (Marchal 2025, ref 98).
- The choice of load metric changes when training is estimated to have its peak positive effect by up to ~20 days (Vermeire 2021, ref 96).
- The app has an additional problem: its training log has **no session-RPE for strength** and no frequent maximal performance test, so the two inputs a fit would rely on are sparse or noisy.

This does not make the physiology wrong. It means the model is a **descriptive, data-hungry approximation**, not a validated individual controller.

---

3. What strength recovery actually looks like

The acute evidence is more useful than the model:

- Training to failure slowed recovery of jump height and bar velocity at **24–48 h** (measurements extended to 72 h), compared with the same volume split into sets of 5 (Morán-Navarro 2017, ref 99).
- A review of 24 recovery studies concludes that failure, higher volume, lower-body/multi-joint work, eccentric emphasis and long-muscle-length positions increase recovery demands, with wide individual variation (Sousa 2024, ref 100).
- Three full-body sessions per week already leave ~48–72 h between similar work. That spacing is a practical fatigue-management rule, not a modelled state.

The app's programme is deliberately nowhere near failure (RIR 1–4, target 2–3), which is the single most effective recovery lever in that literature. A fatigue model would mostly be re-deriving what the RIR window and the 48–72 h spacing already enforce.

---

4. Why a fitted strength model would be hard here

A per-lift model needs:

input	what the app has	problem
session load	sets × reps × load × RIR	no consensus metric; volume load ignores proximity to failure, RIR is ordinal
performance observation	e1RM from working sets; measured test every ~8–12 weeks	e1RM is itself an estimate with RIR error; tests are sparse
training dates	yes	3 sessions/week, little within-week variation
fatigue observation	self-reported readiness	global, not per lift; not a performance measure

RIR estimates are systematically imperfect: predicted RIR differed from actual by ~2 reps at 1 RIR, ~3.7 at 3 RIR and ~5 at 5 RIR in one squat study (Zourdos 2021, ref 101); errors were similar across single- and multi-joint exercises and training experience (Remmert 2023, ref 102). That error propagates directly into any e1RM series used to fit a curve.

With ~3 sessions/week and a performance observation every few weeks, a four-parameter model has more parameters than the data can identify. Fitting it would produce confident-looking curves whose fatigue half-life is an artefact of the priors.

4a. Shadow prototype (2026-10-20)

`features/fatigue_fitness.py` implements the first phase:

- per lift, session load = hard working sets (RIR ≤ 4);
- fixed half-lives: fitness 42 days, fatigue 4 days;
- a linear performance model `e1RM ≈ P0 + k1·fitness − k2·fatigue` is fitted to the athlete's own e1RM observations;
- it refuses (status `insufficient` or `unstable`) when there are fewer than 8 observations, less than 28 days of span, rank < 3, or a condition number above 1e8;
- only canonical lift ids are modelled (synthetic pattern ids would average unrelated movements); an accessory with too few observations is hidden, while a primary lift is shown as `insufficient`;
- an **overall** row pools lifts by relative e1RM (each lift divided by its median) against total hard-set load; its state is a ratio, not kg;
- a **family** slope is fitted on relative e1RM across lifts with the same movement pattern, and each lift's slopes are shrunk toward it (weight `n / (n + pooling_k)`);
- a chronological **80/20 holdout** reports backtest MAE against the training-mean baseline; a model that cannot beat the naive baseline is still displayed, but not trusted;
- `R²`, condition number, and approximate 95% intervals from the fit covariance are shown; intervals are linear-propagation approximations, not Bayesian posteriors;
- `strength-coach fatigue` and the Programme page display it;
- **no prescription reads it.**

Why fixed half-lives: fitting four parameters (two half-lives + two coefficients) at once is the identifiability trap described in §2–4. Intervals, family shrinkage and a chronological backtest are present. A retrospective walk-forward check (2026-10-20) was also run: 25 lift-session predictions had relative MAE 0.158 versus 0.074 for persistence (skill −112%), so the model was **not adopted** and remains shadow-only. A prospective test is not started because the candidate failed the retrospective screen. See RETROSPECTIVE-VALIDATION.md.

5. Options

A. Status quo — readiness only (recommended)

Keep the current design: readiness is the athlete's integrated fatigue signal, and the gap rules handle long breaks. No modelled fatigue, no new failure mode. The cost is that a hard session is not automatically reflected in the next prescription unless the athlete reports it.

B. Advisory recent-stress indicator (applied 2026-10-20)

A fixed, transparent decay of recent hard sets per lift (48 h half-life, high ≥ 6 decayed sets, normal ≥ 3), shown on Prescribe/Programme as "recent stress: high/normal/low", with a warning when a lift is scheduled within 48 h of high stress. No fitting, no load change. The constants are `ASSUMPTION-24` (`DEC-STRESS-01`). It is a prompt to check readiness, not a controller.

C. Fitted per-lift fitness-fatigue model

Fit on the athlete's own e1RM series with hierarchical priors, validate prospectively, freeze parameters, and only then consider load changes. This is a research project, not a config edit. The evidence above says it is premature now.

D. Collect the data first (partially applied)

Applied 2026-10-20 (`DEC-SRPE-01`): optional session RPE (1–10) and session length are captured at review in `session_feedback`, stored append-only, and never change a load. The remaining part — optional per-lift soreness/readiness after the session — is still open. After ~6 months and enough measured tests, revisit B/C. This is the lowest-risk way to make a future model identifiable.

---

6. e1RM calibration — a separate, smaller question

Applied 2026-10-20: a report comparing measured tests with the pre-test formula estimate, per lift (median ratio, mean bias, MAE), and a **bounded correction** when ≥3 tests and the 95% bootstrap CI excludes 1.0 (`DEC-E1RM-CAL-01/02`, `MOD-E1RM-001`). `e1rm calibrate` and the Athlete page show both. Measured overrides still win. The value remains an estimate:

- Epley with RIR folded in (`epley_plus_rir`), marked as an approximation (`ASSUMPTION-03`);
- the **best usable** estimate in the 84-day window is used for loads, not the latest;
- a measured `test` writes an `e1rm_override` and outranks the formula;
- `--e1rm` can override a test if the athlete knows better.

A calibration report would compare measured tests with the estimate the app would have used before the test: per-lift bias (measured − predicted), MAE, and the spread. That is useful and cheap; a **bounded median-ratio correction** is justified only when the confidence interval excludes 1.0 and the bias is within ±10% (implemented, `DEC-E1RM-CAL-02`). A curve fit to a handful of tests would overfit; the single robust ratio is the limit.

Recommended order: report first (now implemented), manual override remains the control. If the report shows a consistent per-lift bias over several tests, add a config-level correction with an assumption, not a learned model.

---

7. Exercise selection and per-lift readiness

Exercise selection is template-driven, not fatigue-driven. Each day has fixed slots and a main lift; variant slots (e.g. `bench_press_narrow` / `bench_press_pause`, `pull_up` / `lat_pulldown`) resolve to the variant trained most recently. Constraints and suitability can veto or warn. Readiness changes the **load and RIR** of the selected lift, not which lift is selected.

Readiness is global plus categories, with an advisory per-lift score. The global score and **category** overrides (`squat`, `hinge`, `horizontal_push`, …) remain the base. A per-lift score (`docs/LIFT-READINESS.md`, `DEC-LIFT-READY-01`) adds e1RM trend, RIR and VBT deviations; it can only lower a load. There is still no validated per-lift readiness questionnaire, so this is an assumption, not a measurement. The old `squat_readiness` / `bench_readiness` / `deadlift_readiness` fields are legacy labels and are excluded.

If per-lift readiness is wanted later, the honest version is a lift-specific **soreness/readiness rating** logged before the session, used as a bounded modifier and validated against the session outcome — not inferred from the global score.

---

8. Recommendation

1. **Do not let a fitted fatigue-fitness model change loads now.** A shadow, fixed-half-life prototype is implemented and displays per-lift state; the strength-specific evidence is small, the model is non-identifiable in this data regime, and the current RIR window + 48–72 h spacing already manages acute fatigue.
2. **Option B is applied** (2026-10-20): fixed decay, advisory only, no load change (`DEC-STRESS-01`, `ASSUMPTION-24`).
3. **Add session RPE for strength** as a low-burden input; it is the missing load variable for any future model and is useful on its own.
4. **e1RM calibration report** (predicted vs measured, bias/MAE), with the bounded correction applied only when the CI excludes 1.0 — implemented.
5. Revisit a fitted model only after ~6 months of the richer data and a prospective backtest.

See `OPEN-QUESTIONS.md` Q19 for the decision record and the default.

INJURY SIGNAL — from a symptom number to a decision the athlete can inspect

The app records a 0-3 symptom level per region. That number alone is not a recovery status: it says how today feels, not whether a tissue has tolerated training. This document explains what the injury signal does with the number, what evidence stands behind each choice, and where the model is coarse on purpose.

It replaces the single back-progression summary described in the earlier version of CONTINUITY.md §2. Decisions: `DEC-INJURY-01`, `DEC-INJURY-02`, `DEC-INJURY-03`, `DEC-SYMPTOM-03`. Config: `config/injuries.v1.yaml`. Code: `strength_coach/features/injury_signal.py`.

> **This is not medical clearance.** A symptom-free streak is weak positive evidence. The signal may suggest reviewing a suitability rating; it never sets one, never clears a constraint, and never substitutes a lift.

---

1. One general signal, one per region

The athlete asked for a general signal plus a per-injury signal, starting with **low back**, **biceps** and **knee**, with more injuries addable later. The shape is therefore:

- a **general** status, the worst of the regions (a live flare outranks a review suggestion, which outranks monitoring, which outranks "no data");
- one **region signal** each, with its own streak, its own window and its own tolerance count.

Regions are configuration, not code. Adding an injury means appending a block to `config/injuries.v1.yaml` and logging against it with `strength-coach symptom --region`. The region name is the same string written to `symptom_log.region`.

The per-region summary is what the athlete reads. It deliberately carries the numbers it used (window, required sessions, tolerated vs loaded) so it can be argued with.

The signal is **history**, not permission. The veto mechanism is still `athlete_constraint` (`severity = hard_exclude`) plus `exercise_suitability`; a `review` here never edits either. An injury can therefore exist in three places without contradiction: a constraint (what is forbidden), a suitability rating (what the athlete will do), and a signal (what the loading history shows).

---

2. What "tolerated loading" means

This was the open question. There are three candidate definitions, and only the third is honest.

(a) Any loading counts. This is the weakest option. It credits a month of training at any intensity and cannot distinguish a hard hinge session from a warm-up. It also rewards the behaviour the signal is meant to check, not the response to it.

(b) Loading above a minimum dose counts. Necessary but not sufficient. A region is "loaded" when the session's work sets on linked exercises reach a threshold: direct loading weighs 2, indirect 1, and the default threshold is 2. So one direct set, or two indirect sets, makes a session count. This stops a single unloaded set from being evidence, but it still says nothing about whether the loading was tolerated.

(c) Loading followed by evidence of no flare counts. This is the definition used. A loaded session is **tolerated** only when:

1. the session loaded the region at or above the minimum dose, and
2. a symptom reading for that region exists in the **7 days after** the session, and
3. **no reading in that window is a flare**.

A session with no follow-up reading is **unverified**. It is counted and displayed, but it never satisfies the review requirement. Missing input is `unknown`, not `pass` (AGENTS.md invariant 12) — the same rule that makes an empty symptom log `unknown` rather than "no flare".

Two consequences are deliberate:

- **A pre-session log on the next training day is enough.** It is the natural logging cadence, and it reflects the state after the previous session. The window is 7 days so a normal week of training self-verifies.
- **A flare in the window disqualifies the session even if a later reading is clean.** A flare that appears the next day is evidence the load was too much; waiting it out does not make the session tolerated.

What this definition cannot tell you

- **It is not adaptation.** Symptoms can stay quiet while tissue adapts, and can flare while tissue is adapting normally. The signal measures symptom response to load, not healing.
- **The readings are self-reported and unblinded.** There is no sham control and no clinician in the loop.
- **A late flare is invisible.** A reaction that appears after the 7-day window is attributed to something else.
- **It inherits the scale's coarseness.** A 0-3 scale cannot reproduce the pain-monitoring rules used in tendinopathy rehabilitation (which work on a 0-10 scale and a 24-hour response). The app therefore uses the direction of the response, not a pain ceiling.

The evidence for pain-guided rather than rest-based progression comes from tendinopathy, not from the low back: in a randomised trial of 38 patients with Achilles tendinopathy, continuing tendon-loading activity under a pain-monitoring model produced no worse outcomes than six weeks of active rest (Silbernagel 2007). For patellofemoral pain, a 12-week programme built on activity modification, load management and pain monitoring produced 86% treatment success at 12 weeks and 81% at 12 months in 151 adolescents (Rathleff 2019). Neither trial is about back pain, and neither validates a 7-day verification window; the window is a design choice (`ASSUMPTION-09`).

---

3. Why the window scales with prior flares

Four weeks was never a finding. It is now the **base**, and it grows with the number of previous flares for that region:

```
window_days      = min(28 + 14 × prior_flares, 84)
tolerated_needed = min( 4 +  2 × prior_flares, 12)
```

`prior_flares` counts flares **before** the one that started the current streak. The flare that ended the streak is the event being recovered from, not evidence of a recurring problem.

The evidence for scaling is reasonably direct. In a prospective inception cohort of 250 people who had recovered from an episode of low back pain, 69% (95% CI 62-74) had a recurrence within 12 months, median time to recurrence 139 days, and **more than two previous episodes predicted recurrence** ($p < 0.01$), along with awkward postures and more than five hours of daily sitting (da Silva 2019). A 2025 cohort found 38% recurrence (Huang 2025).

So a prior-episode count is a real risk marker, and a fixed certificate for everyone over-states recovery for the person who flares repeatedly. The **size** of the step (14 days, 2 sessions) is not measured (`ASSUMPTION-08`): the literature gives a direction and a predictor, not a dose-response curve for required symptom-free time.

There is also no evidence that any fixed symptom-free period means tissue clearance. The window is a prompt to look, not a threshold of healing.

---

4. Linking an injury to the lifts that load it

The athlete asked to cross-check the injury against exercise selection: link the injury to the lift. Every region therefore declares which exercises load it, by canonical taxonomy property rather than by log name:

```
low_back:
  links:
    direct:   {patterns: [erector_extension, hinge, hinge_unilateral], flags: [direct_erector]}
    indirect: {patterns: [squat, ...], muscles: [erectors], flags: [high_fatigue_spinal]}
```

Direct loading weighs 2, indirect 1. The mapping is **exposure, not causation** (`ASSUMPTION-10`): "this lift loads that region" is a mechanical statement, and it does not claim the lift caused the injury or that avoiding it prevents recurrence.

4.1 Low back

Linked: erector work, all hinges (trap bar, sumo, conventional, RDL, good morning, rack pull), squats and carries as indirect. This reuses the existing taxonomy properties, so the `SPINAL-COST` work and this signal cannot drift apart.

The evidence does not support banning loading. Exercise treatment is probably effective for chronic low back pain compared with no treatment (moderate certainty, mean difference −15.2 on a 0-100 pain scale) (Hayden 2021). What matters for this signal is the response to loading, which is exactly what the tolerated count measures. The one caveat is that the cohort evidence is about episodes of pain, not about heavily loaded barbell work; the app cannot say a heavy hinge is safe, only that it was loaded and followed by a quiet reading.

4.2 Biceps / elbow flexors

Linked: elbow flexion, vertical pulls (pull-ups, chin-ups) and horizontal pulls (rows).

This is the weakest evidence of the three. A scoping review of physical-therapy interventions for long-head biceps tendinopathy found 14 eligible articles, mostly of modalities, and concluded there is "a dearth of evidence detailing the conservative management" (McDevitt 2024). Exercise and eccentric training appear among the interventions used, but no loading protocol is established.

The model also cannot see grip: a supinated curl and a neutral-grip row are not distinguished, and grip and shoulder position change biceps demand. That makes the linkage coarse rather than precise, which is stated rather than hidden.

Distal biceps **repair** is a different case and is not modelled: return to sport after repair averages 6.3 months in a review of 42 studies and 1093 patients (Boufadel 2025). A repaired tendon is a surgical timeline, not a load-tolerance signal, and the app should not pretend otherwise.

4.3 Knee

Linked directly: squats, split squats, knee extension, knee flexion, Nordic hamstring work and jump/landing. Indirectly: hinges and carries.

Exercise is the standard treatment for patellar tendinopathy, but the evidence is weak: a Cochrane review of seven trials and 211 athletes found very-low-certainty evidence for pain versus no treatment, and **no trial measured adverse events** (Lopes 2025). That absence is a reason to keep the signal advisory and to keep recording symptoms, not a reason to assume exercise is risk-free.

For patellofemoral pain the load-management approach has better support (Rathleff 2019), and it is the model this signal follows: reduce or modify loading when symptoms rise, progress when they settle.

Knee loading is the least specific of the three: patellofemoral pain, patellar tendinopathy and meniscal symptoms are different tissues with different aggravating movements, and the app records one "knee" region. The signal is therefore a general knee-load history, not a diagnosis-specific one.

---

5. Statuses

status	meaning
`unknown`	no reading for the region; nothing can be said
`insufficient`	the last reading is older than 14 days; a stale reading cannot speak for today
`active`	the latest reading is a flare (level ≥ 2, or level 1 that did not settle in 48 h)
`monitoring`	readings exist, no active flare, and the review bar is not met
`review`	streak ≥ window **and** tolerated sessions ≥ requirement, with no active flare

`review` produces a recommendation to consider re-rating the lifts that load the region. It does not change any rating. A flaring region adds an advisory warning to a prescription that loads it; the low back is excluded from that warning because `MOD-BACK-001` already reports it and a duplicate would bury the modifier.

---

6. Adding an injury

1. Append a region to `config/injuries.v1.yaml` with its `links` and, if it needs different numbers, any of the window keys.
2. Log against it: `strength-coach symptom --region <name> --level <0-3>`.
3. If the region needs a movement the taxonomy cannot express, add the property to the taxonomy (and document it in DATA-MODEL.md), not a special case in code.

An unknown `--region` is rejected with the list of configured regions: a typo must not create a silent region that never loads.

---

7. What would change this design

- Enough returns to estimate **per-region tolerance curves** from the athlete's own data, instead of fixed steps (the same logic as ROADMAP §4.12 for gap arithmetic).
- A symptom input that records **which movement** provoked the symptoms, which would replace the exposure map with a direct attribution.
- A validated per-region scale (the current 0-3 is the athlete's, and the 24-48 h settlement question is the only timing information).
- Evidence that a specific loading pattern is protective or harmful for a region; today the link is mechanical, and the app should not be read as claiming more.

The proposed readiness score uses this signal as a **ceiling**, never an average: a high readiness score cannot raise a lift above its region's injury status. See `READINESS-SCORE.md` §2.2 and §2.6.

Per-lift readiness — evidence and implementation

Status: implemented 2026-10-20 (`DEC-LIFT-READY-01`), advisory and downward-only. The score is a bounded extension of the global readiness score, not a validated instrument.

---

1. Why a per-lift score at all

The global score answers "how ready is the athlete today". A session is not uniform: the squat may feel fine while the hinge is still fatigued from two days ago. The app already has **category** overrides (squat, hinge, horizontal push, …), but no per-exercise estimate.

This implementation exists to make the available signals explicit. It does **not** claim that the composite is valid; the evidence for per-lift readiness is weak.

2. What the evidence supports

input	evidence	what it can and cannot say
Subjective wellness (global score)	Subjective measures track training load better than most objective markers (Saw 2016, ref 47); Hooper 1995, Govus 2018	Best-supported general readiness signal; not lift-specific
RIR / RPE-based autoregulation	A systematic review of 14 studies found RIR/RPE-based autoregulation increased 1RM, but the methods and comparisons were heterogeneous (Larsen 2021, ref 106)	Effort is a legitimate adjustment input; RIR is not a validated per-lift readiness measure
Bar velocity	Mean propulsive velocity in the squat is sensitive to acute fatigue, especially at 40% and 80% 1RM (Moura 2024, ref 107)	A performance/fatigue marker at a standardised load; the app's VBT data are sparse and not always at a fixed load
e1RM trend	The load itself is derived from e1RM; a falling e1RM at similar RIR indicates impaired performance	Noisy because e1RM is an Epley+RIR estimate
Meta-session autoregulation	A scoping review found subjective readiness measures often lacked validity evidence and fidelity was rarely reported (Ibrahim 2024, ref 108)	Strong caution: do not treat a per-lift readiness score as established

Conclusion: there is no validated per-lift readiness questionnaire or composite. Combining global readiness with recent RIR, velocity and e1RM is a **transparent assumption**, not a finding. That is why the score can only lower a load.

3. The score

```
base   = global readiness score (or category override), capped by the injury ceiling
score  = base
       + 0.20 * (e1rm_score   - 0.5)
       + 0.15 * (rir_score    - 0.5)
       + 0.15 * (vbt_score    - 0.5)
```

- Neutral components are **0.5**, so a normal session leaves the base unchanged.
- Missing components are omitted (no renormalisation needed: they contribute zero).
- `score` is clamped to 0–1.

component	source	neutral	low / high
`e1rm`	latest session's best e1RM vs the median of the previous 28 days	ratio 1.0	±10% maps to 0 / 1
`rir`	mean RIR of the last session's working sets	RIR 2 (target 2–3)	RIR 0 → 0.1; RIR 5 → 1.0
`vbt`	mean actual velocity vs the VBT calibration median at the same rep count	ratio 1.0	±10% maps to 0 / 1

Status bands: `ready ≥ 0.75`, `monitor ≥ 0.50`, `low < 0.50`.

4. How the engine uses it

- The per-lift score is computed for the exercises actually prescribed, from history **up to** the prescription date.
- If the per-lift score is **lower** than the global/category score, the readiness factor is recomputed from the lower score (load −5%, RIR +1 at the same bands). The injury ceiling still applies.
- If it is **higher**, it is ignored: the engine never uses an unvalidated composite to make a session harder.
- The scores and components are attached to `notes.per_lift_readiness` and shown on the Prescribe page and in `run` output. They are **not** a new rule and do not appear in the constitution.

5. Limitations

- **It is retrospective.** Today's RIR/velocity do not exist before the session; the score uses the last session. It is a prior, not a measurement.
- **RIR is imprecise.** Predictions can be off by ~2–5 reps depending on proximity to failure (Zourdos 2021, ref 101; Remmert 2023, ref 102), which feeds directly into the `rir` component.
- **VBT is sparse and load-dependent.** A velocity target only exists for lifts with enough clean sets; the app does not prescribe velocity targets for the trap bar (`DEC-VBT-01`), and mixed loads make the comparison approximate.
- **No per-lift soreness/pain input.** The global soreness field and the injury signal stand in for it. A true per-lift readiness score would need a lift-specific pre-session rating.
- **Weights are arbitrary.** 0.20/0.15/0.15 were chosen to let the subjective base dominate; they are `ASSUMPTION-25`, not fitted values.
- **No prospective validation.** There is no trial showing that reducing a lift's load on this score improves outcomes. Downward-only is the safety compromise.

6. What would make it stronger

1. A lift-specific pre-session soreness/readiness rating (1–5), logged before the session.
2. A standardised velocity check at a fixed load, so VBT is comparable across sessions.
3. More measured e1RM tests, enabling an e1RM calibration report and a better trend baseline.
4. Prospective validation: does a low per-lift score predict a missed rep or a worse session?

Until then, the global readiness score remains the primary signal and this score is a transparent second opinion.

References

Refs 47, 101–103, 106–108 in `REFERENCES.md`.

LONGEVITY REVIEW — do the training modalities match the goal?

The athlete's primary goal is **longevity**, with **strength** as the tie-breaker, three sessions a week under a 90-minute ceiling, and a history of resolved low-back tweaks. This review asks a different question from the programme audit: not "are the lifts correct?" but "are the modalities that matter for a long, capable life present at all?"

Short answer: **the resistance-training half is strong and already above the dose associated with lower mortality; the aerobic half was absent, and power and assessment are still missing.** Adding more lifting is not the answer — cardio and a small amount of power/upper-back work are.

Status (2026-10-20): **cardio capture**, the **pulling slot** and the **power slot** have been applied (`DEC-CARDIO-01`, `DEC-PULL-01/02`, `DEC-POWER-01`); the cardio prescription and the assessment metrics remain proposals. This document keeps the evidence and the reasoning; the sections below say what has landed.

---

1. What the evidence says matters

modality	key evidence	size of effect
Resistance training	Muscle-strengthening activity is associated with **10-17% lower all-cause mortality**, with the largest reduction around **30-60 min/week** (J-shaped) (Momma 2022)	moderate; plateau above ~60 min/week
Cardiorespiratory fitness	Each **1-MET** higher fitness = **13% lower** all-cause mortality (Kodama 2009); elite vs low fitness HR **0.20**, no upper limit of benefit (Mandsager 2018)	**largest single modifiable signal**
Aerobic + strength combined	Combined activity is associated with lower mortality than either alone (Momma 2022); guidelines: 150-300 min moderate or 75-150 min vigorous + 2+ strengthening days (WHO 2020)	large, dose-dependent
Power / explosive work	Power training modestly improves physical function vs traditional strength training in older adults (SMD 0.30, low certainty) (Balachandran 2022); larger gains in muscle power and activity tests (El Hadouchi 2022)	small-moderate; cheap to add
Balance / falls	Balance and functional exercise reduces the number of people who fall by about **22%** (RR 0.78) (Sherrington 2019)	large, but evidence is in **60+**
Grip strength	Higher grip strength is associated with lower all-cause and cause-specific mortality (Celis-Morales 2018)	prognostic **marker**, not a proven training target
Intervals for fitness	4×4 min at 90-95% HRmax improved VO2max more than the same work at moderate intensity (Helgerud 2007); sprint intervals added to lifting did **not** blunt strength and raised VO2max (SMD 0.78) (Ferraro-Farro 2026)	large for fitness; interference minimal at this dose
Protein	~**1.6 g/kg/day** maximises resistance-training gains; more adds nothing (Morton 2018)	nutrition, outside the app

Two conclusions follow immediately:

1. **The programme's lifting dose is not the problem.** Three sessions of 60-80 minutes is far above the 30-60 min/week associated with the mortality benefit. The J-shape means adding lifting volume for longevity is not supported; the existing volume is for strength and is reasonable for that.

2. **Cardiorespiratory fitness is the biggest missing lever.** It is the strongest modifiable predictor in the evidence, and before 2026-10-20 the app captured **no cardio at all**. Capture now exists (`cardio_log`, CLI, Cardio page); a prescription and fitness estimate do not.

2. What the programme actually contains

quality	prescribed?	evidence that it matters	verdict
Maximal/strength	yes: squat, bench, hinge 1-2×/week, RIR 1-4	strong for strength; mortality benefit already exceeded	**covered**
Hypertrophy/accessory	modest: 6-9 sets/week	supports strength and lean mass	**adequate, not a priority**
Aerobic base / VO2max	**captured, not prescribed**	strongest longevity signal	**capture done; prescription missing — priority 1**
High-intensity intervals	**no**	efficient VO2max gain, minimal interference	**missing — priority 1**
Power / explosiveness	**yes: `box_jump` on Day A**	modest function benefit; declines with age	**covered (2026-10-20)**
Pulling (vertical/horizontal)	**yes: `pull_up` / `lat_pulldown` on Day B**	structural balance, shoulder/ upper-back; no mortality claim	**covered (2026-10-20)**
Grip / loaded carry	**no**	grip is prognostic; carries load the trunk	**optional — priority 3**
Balance	**no**	falls prevention in 60+	**age-dependent — priority 3**
Mobility / stretching	no	no injury-prevention evidence as a standalone	**not needed**
Assessment (VO2max, grip, sit-to-stand, steps)	**no**	you cannot track what you do not measure	**missing — priority 2**

The programme is also **narrower than the athlete's actual practice**. The log shows rows, pull-ups, vertical pressing, jumps, calf work and carries that no template prescribes; the engine therefore cannot progress, critique or protect what is already being done. That is a product gap as much as a training one.

3. Ranked recommendations

3.1 Cardio — the highest-value addition

Target: accumulate **150-300 min/week of moderate activity or 75-150 min of vigorous**, plus the existing two-to-three lifting sessions (WHO 2020). A practical structure for three lifting days:

- **2 × 30-40 min zone 2** (conversational pace; bike, rower, incline walk, easy run) on non-lifting days or after lifting;
- **1 × 4 × 4 min at 90-95% HRmax** (3 min easy between), once a week, **not** immediately before a heavy lower-body session;
- start at the current fitness level and add 10% per week; walking counts.

Why not more: the mortality benefit is dose-dependent but flattens; the goal is a high fitness floor, not endurance sport. The interference literature supports this dose alongside lifting (Ferraro-Farro 2026).

What the app needs first: cardio capture (modality, duration, HR, sRPE), then a prescription and a fatigue model. That is ROADMAP §4.13. **Capture is done** (2026-10-20): `cardio_log`, `strength-coach cardio`, the Cardio page and `GET/POST /api/cardio`. The prescription and fatigue model are still open, and the injury signal still cannot see running/cycling load.

3.2 Power — one small slot (applied)

Applied 2026-10-20: Day A opens with `box_jump` 3×3-5, RIR 4-5, full recovery (`DEC-POWER-01`). It is excluded from undulation and carries no double-progression citation.

Target: 1-2×/week, **2-3 sets of 3-5 reps**, maximal intent, full recovery, stop when velocity drops. Put it **before** the main lift on a non-hinge day (e.g. Day A or C), or as its own short block.

Options with the available equipment: trap-bar jump (light), box jump, broad jump, medicine-ball throw if available. The log already contains jump work, so the movement is familiar.

Caveat: the strongest evidence is in adults 60+; for a younger lifter the benefit is plausible but not established. It is cheap, low-fatigue, and does not replace strength work.

Why it needed a taxonomy entity: a synthetic `pattern:power_jump` cannot be prescribed safely because synthetic ids carry no taxonomy entity, so the engine cannot tell it is bodyweight. `box_jump` is now a real entity; it is the only canonicalised jump, while other jump variants stay synthetic.

3.3 Pulling — balance the pressing (applied)

Applied 2026-10-20: Day B prescribes a vertical-pull slot, `pull_up` **or** `lat_pulldown` 3×6-10 (`DEC-PULL-01`, `DEC-PULL-02`); the engine picks whichever was trained most recently. A horizontal row remains self-directed; the athlete already trains both.

Target: 2-3 sets of 8-12, twice a week: pull-up / chin-up (bodyweight or added load) and a row (barbell, dumbbell, cable or inverted).

What it fixes: the programme currently has **no pull at all** — three pressing slots (bench ×2-3, lateral raise) against zero pulling, which is an unbalanced upper-body prescription regardless of mortality evidence. It also gives the grip and upper back the longevity argument cannot supply on its own.

3.4 Assessment — make the goal measurable (applied)

Applied 2026-10-20 (`DEC-MEASURE-01`): the `measure` command and Measures page record periodic outcomes in an append-only table, with direction declared once. The default catalogue is:

Add optional, low-burden measures so the programme can be judged against its own objective:

- **VO2max estimate** (submaximal step test or 1.5-mile/2-km test) every 8-12 weeks;
- **grip strength** (dynamometer) — prognostic marker;
- **5× sit-to-stand** and **30-s chair stand** — function/power;
- **steps/day** and **sitting time** — context, not a prescription.

None of these changes a load. They answer "is the longevity programme working?" in a way an e1RM cannot.

3.5 Balance and loaded carries — optional, age-dependent

Balance training has good falls evidence in adults 60+, and almost none in younger lifters. Include it if the athlete is over ~60 or reports unsteadiness; otherwise treat it as optional. Carries are already performed and load the trunk; one carry slot (3×30-40 s) is a reasonable optional addition, but grip strength is a **marker**, not a proven target — do not sell it as mortality reduction.

3.6 What not to add

- **More lifting volume** for longevity: the mortality association plateaus around 30-60 min/week; the current programme is far above it.

- **More core work**: just added, and the evidence does not support a larger block.
- **Stretching as injury prevention**: no standalone evidence; keep mobility specific to the lifts if needed.
- **A single "fatigue score"** for cardio plus lifting: the two exposures have different fatigue signatures (ROADMAP §4.13).

4. What is outside the app's scope

Nutrition (protein ~1.6 g/kg/day, Morton 2018), sleep, alcohol and smoking are large longevity levers but are **not training prescriptions**. The app should say so once and not pretend to coach them; a food or sleep log would be a different product with its own data-quality problems.

5. Sequence (1 and 3 applied)

1. **Cardio capture** (ROADMAP §4.13): table, CLI/web entry, display; no prescription change. **Done 2026-10-20.**
2. **Cardio prescription** after capture exists: a weekly target and one interval session, config-driven, with injury links for running/cycling.
3. **Pull slot**: `pull_up` (or a row entity) on Day B or C — small, no new taxonomy needed. **Done 2026-10-20:** vertical-pull variant slot (`pull_up` / `lat_pulldown`) on Day B.
4. **Power slot**: needs a real taxonomy entity (e.g. `box_jump`) and a placement decision. **Done 2026-10-20:** `box_jump` 3×3-5 first on Day A.
5. **Assessment metrics**: optional, no load effect. **Done 2026-10-20:** `measure` + Measures page.
6. **Balance/carry**: only with an age/needs trigger.

6. Open questions

- Q17: should the programme add a pulling slot, a power slot, or both, and where? **Applied:** pulling variants and power.
- Q18: which longevity assessment metrics should the app record, and how often?

Cardio scope is Q14 and ROADMAP §4.13. This review only ranks the priorities.

7. Limitations

- Most mortality evidence is observational; effect sizes are associations, not proof that adding the modality causes the benefit.
- The power and balance trials are in older adults; the athlete's age is not recorded, so the recommendation is age-conditional.
- The cardio dose is a guideline synthesis, not an individual prescription; it needs the athlete's current fitness and preferences, neither of which is in the app yet.
- Nothing here changes the constitution. The strength methodology remains the source of truth.

MODULE LAYOUT AND TEST STRATEGY

1. File layout

`[x]` exists today, `[ ]` planned or superseded. This file is the original layout plan; the current tree in `AGENTS.md` and `README.md` is authoritative.

```
pyproject.toml                        [x] deps, pytest config, ruff
AGENTS.md                             [x] instructions for contributors/agents
README.md                             [x] overview + quickstart
config/
  exercises/taxonomy.v1.yaml          [x] exercise identity + properties
  rules/programme.v1.yaml             [x] the constitution, 5 encoded rules + modifiers
  calibration/overrides.v1.yaml       [ ] per-lift manual velocity overrides
  templates/session-types.v1.yaml     [x] canonical session templates
templates/inference.v1.yaml         [x] imported-session type inference
  thresholds.v1.yaml                  [x] increment 0.5 kg, rounding, e1RM window, fatigue
multiplier
  readiness.v1.yaml                   [x] modifier fields/threshold, merge strategy (DEC-
READINESS-04)
  athlete-state.example.yaml          [x] placeholder state file for a new athlete (DEC-PORT-01)
  templates/schemes.v1.yaml           [x] named rep schemes (DEC-REPS-01)
  templates/undulation.v1.yaml        [x] 4-week DUP cycle (DEC-UNDUL-01)
  llm.example.toml                    [x] LLM template; llm.local.toml is git-ignored (may hold a
key)
data/raw/training_log.csv             [x] immutable source
data/raw/readiness.csv                [x] immutable source
data/coach.db                         [x] SQLite store (created on demand)
docs/*.md                             [x] the proposal + evidence
strength_coach/
  utils/                              [x] pure shared helpers (permanent)
    paths.py io_utils.py dates.py numeric.py stats.py exercises.py reporting.py
  domain/                             [x] pure domain arithmetic
    e1rm.py                           [x] Epley + RIR-adjusted, rep caps
    load.py                           [x] load-basis resolution, added-load progression
    rir.py                            [x] RIR classes incl. warm-up (DEC-RIR-01)
    velocity.py                       [ ] presence, decay proxy
    progression.py                    [ ] double-progression decision
    fatigue.py                        [x] readiness score/overrides + modifiers
    spinal.py                         [ ] exposure counting, erector dose
  ingest/                             [x] partially
    clean.py                          [x] sentinels, load basis, units, flags, quarantine
    sessions.py                       [x] reconstruction + day inference
    readers.py readiness.py           [ ] readiness merge rule (DQ-RD-01)
  store/                              [x] SQLite, append-only
    migrations/0001_initial.sql       [x] schema + append-only triggers (single source of truth)
    migrations/0002_rubric_mark.sql   [x] human rubric marks, append-only (continuous calibration)
    migrations/0003_athlete_state.sql [x] §11: profile, constraint, suitability, metrics, symptoms
    migrations/0004_e1rm_override.sql [x] manual e1RM that outranks the computed estimate
    migrations/0005_symptom_log_append.sql [x] allow multiple symptom entries per day (latest wins)
    db.py                             [x] connection, numbered migrations, checksums
    repositories.py                   [x] append-only writers, explicit identity keys
    importer.py                       [x] clean export -> raw_row/clean_set/session/run
  features/                           [x] partially
    baseline.py                       [x] per-lift/session baseline
    derived.py                        [x] e1RM, velocity flags, effort windows
    vbt_calibration.py                [x] clean-set filters + sparsity ladder
    readiness.py                      [x] readiness config: fields, threshold, merge strategy
  prescribe/schemes.py                [x] named rep schemes and their applicability
  prescribe/undulation.py             [x] weekly undulation cycle loader
    session.py weekly.py              [ ] weekly aggregates
  rules/                              [x] versioned constitution + policy
    loader.py                         [x] YAML -> Rule objects, content hash, strict validation
    registry.py                       [x] one predicate per check.fn (5 rules)
    context.py                        [x] immutable view of stored sets/sessions
    policy.py                         [x] verdicts, severity effects, evidence validation, publish
    rag.py                            [ ] deferred (rules are cited by id, not retrieved)
  prescribe/                          [x] deterministic session engine
    templates.py                      [x] session slots from config/templates/session-types.v1.yaml
    config.py                         [x] thresholds: increment, rounding, e1RM window
    loads.py                          [x] e1RM -> rounded load, never up
    engine.py                         [x] build + validate an executable session
    fallback.py                       [x] deterministic fallback, load scaling
```

```
    constraints.py                  [x] stored constraints veto, suitability warns
    templates.py engine.py fallback.py
  critique/                         [x] deterministic rubric
    facts.py                        [x] one session -> immutable, citable facts
    rubric.py                       [x] 6 measured items + 2 unmeasured, persistence
  trace/                            [x] run lifecycle
    run.py                          [x] start/event/finish; closes failed runs as error
  eval/                             [x] backtest harness + calibration loop
    labels.py                       [x] labelled-set format, strict validation, confidence
    harness.py                      [x] per-item comparison, agreement and MAE
    store.py                        [x] marks in the database: append-only, latest wins
    report.py                       [x] shared sheet and report renderers
  cli.py                            [x] import / run / review / calibrate / state / block / e1rm /
test / symptom
  agents/                           [x] LLM interface + narrative
    client.py                       [x] protocol, NullLLM, OpenAI-compatible adapter
    config.py                       [x] file + environment settings, masked diagnostics
    narrative.py                    [x] facts -> prompt -> prose, recorded as an event
    proposal.py                     [x] structured proposal + menu schema, policy check (DEC-
LLM-04/05)
    facts.py rubric.py
  (proposal/narrative implemented; no further agents planned)
  eval/                             [x]
    backtest.py labelled_rubric.yaml scenarios/
  cli.py                            [x]
tools/
  profile_data.py                   [x] regenerates docs/EVIDENCE.md
  baseline.py                       [x] regenerates docs/BASELINE.md numbers
tests/
  conftest.py                       [x] fixtures + raw-CSV availability guard
  test_dates.py test_numeric.py     [x] sentinels
  test_stats.py                     [x] robust stats
  test_exercises.py                 [x] taxonomy + recorded decisions
  test_io_utils.py                  [x] raw reading fidelity
  test_e1rm.py test_load.py         [x] domain arithmetic
  test_clean.py test_sessions.py    [x] cleaning + reconstruction
  test_rir.py test_vbt_calibration.py [x] RIR classes + calibration ladder
  test_rules.py test_policy.py      [x] rule loader and policy
  test_prescribe.py                 [x] prescription engine
  test_critique.py                  [x] rubric
  golden/                           [x] snapshot prescriptions
```

2. Where code is reused

The rule from the brief: **if a calculation is needed in more than one place, it lives in one tested module and is imported.** Concretely:

Calculation	Single home	Used by
date parsing + null tokens	`utils.dates`	ingest, baseline, every window function
numeric coercion + zero sentinels	`utils.numeric`	ingest, baseline, readiness normalisation
robust median/MAD/band	`utils.stats`	VBT calibration, e1RM series, readiness composite
exercise identity + properties	`utils.exercises`	ingest, features, rules, prescription, critique, tools
e1RM (Epley/RIR/rep caps)	`domain.e1rm`	features.baseline, critique, prescription rationale
load-basis + rounding	`domain.load`	ingest.clean, prescription engine, critique
RIR window & suspect values	`domain.rir`	rules registry, critique rubric, features
velocity presence/decay	`domain.velocity`	features.vbt_calibration, critique
double-progression decision	`domain.progression`	prescription engine, critique

Calculation	Single home	Used by
weekly spinal cost	`domain.spinal`	features.weekly, rules.PRG-SPINE-001, critique

Note the deliberate consequence: **the critique and the prescription call the same progression and spinal functions.** They cannot disagree about what the rule says.

3. Test strategy

Tests that read `data/raw` are git-ignored (`tests/*_integration.py`): they assert counts and values derived from the real export, so committing them would leak personal data. They run locally when the export is present; the public suite is synthetic-only.

Five layers, cheapest first.

1. **Unit tests on pure functions** — every function in `utils` and `domain`. Table-driven, no I/O. Assert the domain reason, not just the output: e.g. "RIR 0 is preserved" rather than "returns 0".
2. **Property tests** (planned, `hypothesis`) — rounding is idempotent and never rounds up; `robust_band(low <= median <= high)` ; e1RM is monotonic in load and in reps.
3. **Integration tests against the real CSVs** — marked `@pytest.mark.integration` . These assert structural invariants: every exercise name resolves, null-date rows are quarantined, `velocity == 0` never coexists with measured `TUT` , and clean VBT counts per lift stay stable. A silent change in cleaning becomes a test failure.
4. **Golden/snapshot tests** — full prescriptions for a handful of historical dates, stored as YAML. Any change to the engine that alters a prescribed load shows up as a diff to be reviewed, not a surprise later.
5. **Backtest harness** — replays a historical week using only data available up to that day, then checks executability and rubric agreement against a hand-labelled set (see `docs/V0-SCOPE.md` and the eval design in `docs/ARCHITECTURE.md`).

Non-obvious tests that already exist because they caught real bugs

- `test_rir_zero_is_preserved` — reusing the date null tokens for numerics silently erased genuine `RIR = 0` observations.
- `test_leg_curls_are_knee_flexion_not_biceps` — regex ordering sent `Lying Leg Curls` to the biceps family.
- `test_historical_hyperextension_keeps_identity_but_counts_as_history` — pins DEC-BACKEXT-02: the entity id is preserved, but its rows count as back-extension history.
- `test_zero_dispersion_collapses_without_floor` — documents the MAD-collapse hazard that the VBT module must guard against.

What is deliberately not tested

- The LLM's prose. It is validated structurally (every citation must resolve, every number must match a tool result), never by string comparison.
- Anything requiring a live network call. The LLM client is injected; tests use a fake.

Next-phase fatigue-management plan — final lean version

Status: closed 2026-10-20. Phase 0 backfill done; descriptive sRPE and readiness–performance reports done; optional-deload suggestion implemented; four programme changes implemented. Deferred items are explicit. **Date:** 2026-10-20 **Scope:** non-cardio fatigue-management work. This supersedes the earlier experimental draft; the feedback was that the programme must not become an experiment.

---

1. Feedback incorporated

The plan now follows these constraints:

- use the data already available plus a one-time sRPE/duration backfill;
- use the computed e1RM as the primary observational outcome; 3RM/5RM tests are a reminder cadence, not a requirement;
- treat the three known low-back tweaks as the current back-history evidence; revisit only if a future episode does not fit the pattern;
- do not invent a “bad session” label;
- do not experiment with or manipulate the training programme;
- do not run a scheduled-deload trial;
- do not build individualised MRV or automatic volume control;
- keep the scheduled deload as an optional suggestion;
- use evidence-informed defaults where data cannot calibrate a decision;
- if a small programme-specific measurement check is needed, limit it to one month and do not change training.

The repeated principle: **measure and document; do not experiment on the athlete.**

---

2. Direct answers

1. What extra information will be available later?

Assuming sRPE and duration are backfilled, the usable dataset will contain:

- session RPE, 1–10;
- session duration;
- actual reps and load;
- computed e1RM from the existing sets;
- daily readiness;
- the three known low-back tweaks;
- future sessions logged normally, if the athlete chooses.

It will **not** contain:

- measured 3RM/5RM tests unless performed for another reason;
- a manual “bad session” label;
- a randomized deload comparison;
- enough volume variation to identify an individual MRV;
- a counterfactual for causal claims.

That is enough for **descriptive calibration**, not for automatic decision rules.

2. The three low-back tweaks

Use them as a **qualitative case series**:

- all three occurred in the historical record;
- three events are too few for statistical calibration;
- they justify keeping the existing reactive back rules;
- if a future episode does not fit the expected pattern, revisit the model then.

No new symptom schema or daily symptom logging is required for this plan. Future episodes can be recorded through the existing mechanism when they happen.

3. Will there be a “bad session” label?

No. A label is not necessary and would be another subjective input.

Instead, derive an observational measure:

```
performance_ratio = actual reps at the prescribed load / expected reps from the current e1RM
```

Use it descriptively. Do not convert it into an automatic trigger in this phase.

sRPE and back episodes provide independent context, but they do not create a validated “bad session” definition.

4. Scheduled-deload calibration

Ignore it for now, as requested.

The scheduled deload remains an **optional suggestion**, not a trial and not an automatic action. Its effects will not be estimated experimentally.

5. Experimenting with the programme

No. The plan does **not**:

- randomise training;
- alter volume to estimate MRV;
- run deload versus no-deload periods;
- use the athlete as a guinea pig.

The deterministic programme changes — warm-up ramp, 3RM/5RM reminder, under-MEV accessory progression and main-lift readiness set reduction — are implemented rules, not experiments. The programme otherwise stays as written unless the athlete independently chooses the optional deload. Observations are recorded around the existing programme.

---

3. Evidence used

topic	evidence	conclusion for this plan
Session RPE	Valid global session-load measure (Foster 2001), not a validated fatigue/deload trigger	Calibrate descriptively; do not automate

topic	evidence	conclusion for this plan
Autoregulation	Mixed evidence, no consistent advantage over standardised training (meta-analysis; review)	Keep decisions with the athlete; no automatic triggers
Volume	Dose-response with diminishing returns (2025 meta-regression; Schoenfeld 2017)	No individual MRV from fixed-programme data
MRV/MEV	Coaching constructs, not measured thresholds (RP)	Document the heuristic; no individualisation
Deload	Evidence for a fixed schedule is weak; the 2026 trial is narrow (trial)	Optional suggestion only; no trial
Computed e1RM	Epley + RIR estimate, not a measured test	Accept as an observational outcome with caveats

Additional evidence note: a single athlete's observational data cannot identify a causal deload or MRV effect without deliberate variation or randomisation. Since neither is acceptable here, the correct output is a descriptive report, not a calibrated controller.

---

4. What can and cannot be calibrated

Can be calibrated descriptively

1. **sRPE scale use** - distribution and central tendency; - relationship with hard sets, tonnage and duration; - expected sRPE with uncertainty.
2. **Readiness–performance association** - readiness versus performance ratio; - exercise- and load-adjusted if the data allow; - clearly labelled associational, not causal.
3. **Back-episode case review** - what preceded each episode; - what the programme was doing; - whether the existing reactive rule would have fired.
4. **Data quality** - coverage, missingness, recall bias, scale outliers.

Cannot be calibrated

- individual MRV;
- optimal weekly volume;
- causal effect of a deload;
- a validated "bad session" threshold;
- a readiness threshold for main-lift volume reduction;
- a performance-decline trigger that is proven to improve outcomes.

These are documented as **not estimable** from the available data, rather than approximated with false precision.

---

5. Data requirements

Backfill

A single editable table, planned for the Review/Import UI:

field	rule
session date / id	existing session
sRPE	1–10, optional
duration	minutes, optional
note	optional
missing	leave blank; never invent

- all sessions with a plausible memory;
- append-only corrections;
- no provenance flag;
- phase-start cutoff recorded in the analysis plan, not the schema;
- report coverage.

Existing data

- actual reps and loads;
- computed e1RM;
- readiness;
- three known low-back tweaks.

Not required

- measured 3RM/5RM tests;
- daily symptom logging;
- a bad-session label;
- new readiness metrics;
- a block anchor for this descriptive analysis.

---

6. Final lean plan

Phase 0 — Backfill and audit (done)

Goal: make the existing record usable.

1. Build the editable sRPE/duration backfill table (later implementation).
2. Backfill all sessions with plausible recall.
3. Audit: - number of sessions with sRPE; - number with duration; - distribution; - missingness; - recall-bias caveat.
4. Freeze a phase-start date for retrospective/prospective separation.

No programme change in this phase. The programme changes are listed in §6a.

Phase 6a — Programme changes implemented

These are deterministic programme changes, not experiments:

change	implementation
Standardised warm-up	three ramp sets at ~50/70/85% of the working load (5/3/2 reps) on main lifts; not counted as volume (`DEC-PROGRAMME-06`)
3RM/5RM cadence	a 10-week reminder, alternating schemes; `strength-coach test --status` (`DEC-PROGRAMME-07`)

change	implementation
Under-MEV accessory progression	one accessory set when a pattern is below MEV for three weeks (`MOD-VOL-002` , `DEC-VOLUME-03`)
Main-lift readiness volume	poor readiness can remove one set from main lifts as well as accessories (`include_essential: true`)
Volume/MRV	no individual MRV and no experimental volume manipulation

Phase 1 — Optional one-month measurement check (not triggered; closed)

Only if the backfill leaves a specific, concrete gap.

Examples:

- sRPE is present but duration is too sparse to compute session load;
- the scale is being used inconsistently;
- one lift's performance ratio cannot be computed.

For at most **one month**:

- log sRPE/duration normally;
- do not change sets, load, exercises or deload policy;
- no experimental conditions.

If no specific gap exists, **skip this phase**.

Phase 2 — Descriptive calibration report (sRPE + readiness descriptive done)

Produce, with uncertainty and limitations:

1. sRPE coverage and distribution.
2. sRPE versus hard sets, tonnage and duration.
3. Readiness versus performance ratio.
4. Three low-back case reviews.
5. A statement of what cannot be estimated.

Retrospective result: continuity MAE 0.150 vs 0.128 fixed; fatigue MAE 0.158 vs 0.074 persistence. Both candidates lost, so neither is adopted. See RETROSPECTIVE-VALIDATION.md.

Do not produce:

- an automatic fatigue score;
- an MRV estimate;
- a deload effect estimate;
- a new trigger.

Phase 3 — Optional deload suggestion (implemented)

The scheduled deload remains an optional suggestion, not a calibration:

- due every six weeks from a declared anchor;
- ask at workout time;
- default no;
- skip only that workout;
- reactive triggers remain automatic;
- no trial, no randomisation.

The optional-deload suggestion is implemented; the reactive triggers remain automatic.

Phase 4 — Stop and document (closed)

After the descriptive report:

- do not adopt a model that lost its retrospective baseline; no prospective period is started for continuity or fatigue in this phase;
- keep the evidence-based defaults;
- keep the reactive rules;
- keep the optional deload;
- do not add individualised MRV or main-lift volume control;
- revisit only if a specific observed problem appears.

---

7. What the report will say if the data are insufficient

It will say so directly:

- “sRPE coverage is too low to estimate a scale relationship.”
- “Readiness–performance association is descriptive only.”
- “Two back episodes are a case series, not a model.”
- “Deload effect and individual MRV are not estimable.”
- “No automatic trigger or individualised volume change is justified.”
- “The deterministic programme changes in §6a are separate from this calibration.”

That is a valid outcome. It is better than overfitting a model to a single athlete’s fixed programme.

---

8. Decisions

decision	choice
Backfill fields	sRPE + duration
Backfill scope	all sessions with plausible memory
Missing values	blank
Provenance flag	none; use phase-start cutoff
Post-session signal	sRPE only
Bad-session label	not needed; derive performance ratio
Measured strength tests	computed e1RM is the primary outcome; 3RM/5RM reminder every 10 weeks
Back episodes	three known low-back tweaks as a case series; revisit only if pattern changes
Scheduled deload	optional suggestion; no trial
Reactive deload	automatic
Volume/MRV work	under-MEV accessory progression implemented; individual MRV deferred
Programme experiments	none
One-month check	only if a specific data gap exists; no programme change

---

9. Closure decisions

1. **Backfill tool:** not built. The processed export/YAML were imported directly, then removed; no bulk table is needed now.
2. **Phase-start cutoff:** not needed; there is no prospective experimental phase.
3. **Report format:** the private descriptive report is sufficient.
4. **One-month check:** not triggered; no specific data gap justified it.

Plan closed. Phase 0 backfill, Phase 2 descriptive reports, the volume–performance baseline, the small progress dashboard, the optional-deload policy and the four programme changes in §6a are complete. No programme experiments were made.

Deliberately deferred, not blockers: individualized MRV (closed as not estimable), main-lift volume experimentation and prospective validation of the fatigue/volume models. These require data or designs the athlete explicitly did not want.

OPEN QUESTIONS — ranked by how much they change the design

These are genuine ambiguities in the programme brief, the data, or both. Each states the default this project will implement if nobody answers, so a missing answer is survivable and produces an `unknown` verdict rather than a silent guess.

> Quantitative evidence for each question is kept out of this public repository. The questions themselves are methodology questions and apply to any athlete using the system.

Q1. Sessions have no timestamps — accepted: date-only, manual duration only

The date column is calendar-date only. So "same date + time gap" is impossible, two sessions on one day are indistinguishable, and a squat-plus-bench day is ambiguous between the squat day and the bench-plus-squat day unless erector or power work is present.

Blocks: session/day identity, scheduling rules, duration scoring. **Default:** one session per date; day inferred from the main lift present, with the ambiguous pair separated by erector/power content; ambiguous -> `unknown`, and no day-specific rule is scored when the day is unknown.

Q2. What is a "high-fatigue spinal exposure"? Default implemented: weighted cost

The brief caps "more than ~3 per week" without defining an exposure. Measured on **distinct exercise types** the cap almost never fires, which makes it decorative; measured **per set** it fires almost every week, which makes it meaningless.

The likely intent is distinct high-fatigue exercise types per week — but then three heavy rack-pull sets count the same as three light back-extension sets, which is probably not the intent either.

Blocks: the spinal-cap rule and the redundancy rubric line. **Default:** distinct high-fatigue exercise types per ISO week, cap 3, with an ordinal weighted cost reported alongside as context (ASSUMPTION-02).

Q3. The erector ramp needs an anchor — default implemented: optional anchor

The 8-12 week dose ramp (bodyweight -> add load -> add a set -> consolidate) requires knowing which week of the block today is. Nothing in the source export marks a block boundary, and the historical erector rows deliberately do **not** count toward the dose.

Blocks: the erector ramp rule, squat/bench session prescription, the "weekly erector dose vs plan" rubric line. **Default:** block starts on the first Monday after approval; until then, bodyweight only, phase labelled `unanchored`.

Q4. ~~There is no symptom field.~~ Answered: symptom_log exists

The most safety-relevant rules — "if back symptoms rise and do not settle in 24-48 h", "progress only if no symptom flare" — depend on data that does not exist in either export. The form-quality column is the only candidate and its direction is unknown, so it is not used as a proxy.

Blocks: the back-flare modifier, the erector stability gate, the "symptoms reported" rubric line. **Applied:** manual pre-session 0-3 entries in `symptom_log`; `MOD-BACK-001` and the stability gate read it. Missing entries still return `unknown`.

Q5. ~~Session duration is unmeasurable.~~ Answered: manual/optional only

The rubric wants "session duration vs 60-80 min cap". There are no timestamps. The options are a set-count proxy or dropping the metric — and a proxy presented as minutes would be a fabricated score.

Applied partially: `session_feedback.duration_min` records optional session length at review; there are still no timestamps, so imported sessions cannot be timed automatically. The rubric line remains unmeasured.

Q6. Readiness conflicts and new metrics — merge + score implemented

Two sub-questions: 1. **Merge** — multiple readiness rows per date exist, and most are conflicting rather than identical. Proposed rule: most-complete row, tie-break on latest. That assumes completeness equals authority. 2. **New metrics** — a composite readiness score, HRV and RHR are planned. Each needs a declared scale, direction (does higher mean better?) and aggregation before it can enter the composite. Grip strength is a periodic measurement, not a readiness input (`DEC-GRIP-01`).

Blocks: the readiness composite. The merge half no longer blocks anything. **Default:** most-complete-row merge with conflicts logged; new metrics enter as `metric_definition` rows with explicit `scale_min`, `scale_max`, `higher_is_better` and `aggregation`, and stay dormant until populated. **Status:** the merge default is implemented (`ingest/readiness.py`, `DEC-READINESS-01`); every date carries its `merge_note`. The global/category score, HRV/RHR baseline and wearable blend are implemented separately (`features/readiness_score.py`); `readiness_day.composite` still stays null by design. `ASSUMPTION-04`: completeness is treated as authority.

Q7. Does "main lifts at RIR 2-3" forbid sets easier? Default implemented

The constitution states a window, but `PRG-RIR-001`'s `on_fail` code is `RIR_FLOOR`, and the danger it names is "no RIR 0-1 sets, no max singles". A set at RIR 4-5 is easier than prescribed, not more dangerous, so treating it as a veto would make the rule fire on almost every session and bury the real violations.

Blocks: nothing — the rule runs today with the default below. **Default:** the floor is a veto (RIR below the window, plus max singles at >= 90% of a multi-rep e1RM); a set above the window is counted and reported in the verdict details but does not fail the rule. Making easy sets violations is then a config change (`on_fail.action`) plus a predicate branch, not a rewrite.

Q8. What counts as one "high-fatigue spinal exposure"? Default implemented

ASSUMPTION-02 fixes the weighting (ordinal `spinal_cost`) but not the unit:

- per **set** — one 12-set trap-bar session is twelve exposures, so the cap fires every week;
- per **exercise type per week** — the cap almost never fires, which makes it decorative;
- per **exercise per session** — two hinge sessions in a week cost 2+2 = 4 against a cap of 3.

Blocks: how strict `PRG-SPINE-001` is. It does not block the build. **Default:** one exposure = one exercise in one session, weighted by `spinal_cost` (`rules/registry.py` `weekly_spinal_cap`). Sets do not multiply an exposure.

Q9. ~~Can the RIR window itself be widened?~~

Closed by `DEC-RIR-02`: the window is now RIR 1-4 with a 2-3 target, based on the evidence collected in `docs/RIR-EVIDENCE.md`. Changing it again is still a constitution change.

Q10. ~~Can the sumo deadlift be an accessory?~~ Closed

Replacement only (`DEC-TRAPBAR-03`). Sumo may replace the trap bar as the Day B hinge, but it is not added alongside it or as an accessory: a second hinge violates the no-stacking rule (`PRG-RED-001`), and on A/C it would sit next to direct erector work. No hinge-variant slot and no rotation are planned. The path is main-lift substitution (`DEC-LLM-06`), once sumo is rated.

Q11. Sumo replacement and the 2.5 cost — Closed

Closed 2026-10-20. Sumo is a replacement for the trap-bar hinge (`DEC-TRAPBAR-03`), not a rotation or accessory. The 2.5 spinal cost is accepted as the owner's ordinal design choice (`DEC-SPINE-01`); it is not a measured value and will not be recalibrated.

There is **no formal trial and no open question**. If you want to use sumo, run it as the hinge for a normal 4–6 week block, using ordinary training judgement:

- normal working-set/RIR rules, conservative progression;
- log a symptom reading only when the back actually feels different;
- continue at 0–1, or level 2 that settles within about 48 hours;
- reduce or stop at level 3, persistent level 2, or new neurological symptoms;
- one video or coach check in the first 1–2 sessions; not every session.

Evidence: no RCT ranks sumo against trap bar for injury; they differ in muscle activity and joint moments, and trap-bar/squat selection matters less than heavy loading and target muscles. See references 132–135 and `docs/SPINAL-COST.md`.

Q12. ~~How should re-entry be calibrated from real returns?~~ Implemented; calibrated factor rejected

Now rejected for use; the fixed arithmetic stays. 14/28/56 days, +1 RIR and −5%/−10% remain the defaults. They are evidence-informed assumptions, not measured cut-points; the calibrated factor was tested and lost.

The plan is to calibrate them on the athlete's own returns (ROADMAP §4.12):

- record each gap and the first sessions back, with load, reps, RIR, completion, symptoms, readiness and **post-session fatigue**;
- estimate how strength (load at a fixed RIR) and capacity (work tolerated without a flare) taper with time away;
- replace a fixed band only when there are enough returns to support it, and only as a configurable, cited change.

Applied then rejected: the calibration exists, but the 2026-10-20 retrospective walk-forward had ratio MAE 0.150 vs 0.128 for the fixed rule. The calibrated factor is not adopted; fixed bands remain. See RETROSPECTIVE-VALIDATION.md.

Q13. ~~Is four weeks the right back-signal window?~~ Answered (2026-10-19)

The owner answered all three parts, and the signal is now per region (`DEC-INJURY-01/02/03`; `docs/INJURY-SIGNAL.md`):

1. **The window scales** with prior flares: 28 days + 14 per prior flare (cap 84), with 4 + 2 tolerated sessions per flare (cap 12). More than two previous episodes predict recurrence (da Silva 2019), so a fixed certificate over-states recovery for someone who flares repeatedly.
2. **"Tolerated" means loaded and followed by a clean reading**, not merely loaded. A session with no post-session reading is counted but `unverified`, and never satisfies the review requirement — missing input is `unknown`, not `pass`.
3. **The signal names lifts.** A general signal plus one per region (low back, biceps, knee), with the exercise links in `config/injuries.v1.yaml`; adding an injury is a config change.

Residual uncertainty is calibration, not design: the step sizes and the 7-day verification window are assumptions (`ASSUMPTION-08`, `ASSUMPTION-09`), and the injury-to-lift link is exposure, not causation (`ASSUMPTION-10`).

Q14. What cardio load and fatigue model should the app use?

Planned in ROADMAP §4.13. Open: which modalities, whether TRIMP or sRPE is the impulse, two vs three fatigue systems, whether cardio modifies load or only readiness context, and how many weeks of individual data are needed before showing a time constant. No implementation.

Q15. ~~Should the programme prescribe dedicated core work, and at what dose?~~ Answered (2026-10-20)

Before the answer, the prescribed days contained **no dedicated core work**: four slots per day (main lifts, back extension, lateral raise, split squat), with six sets of spinal-extension work per week. The taxonomy and rules already knew the movements — `PRG-DP-001` even carried an `ab_wheel` increment — but no template or collection referenced them, so the config was dead.

The evidence supports a **small** block, not a large one: core exercise is not superior to other exercise for low back pain at six months (Wang 2012; Saragiotto 2016), exercise did not reduce recurrence in a randomised trial (Ferreira 2021), core training modestly improves speed/jump in athletes (Kayantaş 2026), and trunk endurance is associated with lower overuse injury (Lopes 2021, observational).

Applied (`DEC-CORE-01`, constitution 1.2.0): two reps-based slots — Day A `ab_wheel` 3×6-10, Day C `hanging_leg_raise` 3×8-12 — added to `accessories_and_back_extension` so `PRG-DP-001` progresses them. No new hard rule; planks and carries stay self-directed because they are time-based. Full reasoning: `docs/CORE-TRAINING.md`.

Q16. ~~Why is `PRG-DP-001` never assessable?~~ Answered (2026-10-20)

The rule scoped the whole `$accessories_and_back_extension` collection, but `rep_ranges` was built from the current day's template (`DayTemplate.rep_ranges`). Every day lacked at least one scoped exercise (A: `split_squat`, B: `back_extension`, C: `lateral_raise`), so the rule returned `unknown` every day and "add increment" could never fire.

Applied: `templates.all_rep_ranges()` pools the map across all three day templates, and the engine passes it to the rule context. On the current store the decision is still "hold load", so no prescribed load changed on the day of the fix. See `docs/CORE-TRAINING.md` §5.4.

Q17. ~~Should the programme add pulling and/or power slots?~~ Answered (2026-10-20)

- **Pulling — applied (`DEC-PULL-01/02`, constitution 1.4.0).** The prescribed upper body was entirely pressing (bench ×2-3, lateral raise) with no pull in any template, while the log contained rows, pull-ups and lat pulldowns. Day B now has a **vertical-pull variant slot**: `pull_up` **or** `lat_pulldown` 3×6-10, whichever was trained most recently; both are in `accessories_and_back_extension` so `PRG-DP-001` progresses each on its own history. They are substitutable for the stimulus but not identical (chin-ups load the biceps/erectors more, pulldowns the rectus abdominis more; 1RM correlation ~0.78), and their loads are never compared.
- **Power — applied (`DEC-POWER-01`).** Day A opens with `box_jump` 3×3-5, RIR 4-5, full recovery, before the main lift. `box_jump` is now a real taxonomy entity; the slot is excluded from weekly undulation and carries no double-progression citation because power cannot be scored from per-set average velocity.

Full reasoning and evidence: `docs/LONGEVITY-REVIEW.md`.

Q18. ~~Which longevity assessment metrics should the app record?~~ Answered (2026-10-20)

Applied (`DEC-MEASURE-01`). A generic append-only `measurement` table with a config catalogue (`config/measurements.v1.yaml`): `vo2max`, `grip_strength_kg`, `sit_to_stand_5_s`, `steps_per_day`, `sitting_hours_per_day`. Each metric is declared in `metric_definition` with a direction and aggregation; a later direction change is refused. The CLI is `strength-coach measure` and the web UI has a Measures page.

Grip strength was previously a readiness field; it has since been removed from readiness (`DEC-GRIP-01`) and is measurement-only. None of these metrics changes a load.

Q19. ~~Should a fatigue-fitness model drive strength loads?~~ Answered: no load effect

Answered 2026-10-20. The model is shadow-only; the retrospective check lost to persistence (0.158 vs 0.074), so it does not drive loads. A future candidate must pass §4.20's gate.

Research note: `docs/FATIGUE-FITNESS.md`.

At the time of the question, the app had no post-session fatigue state: readiness was the athlete's integrated signal, and gap rules handled long breaks. The model now exists in shadow but does not change a prescription.

Options:

- **A (default, recommended):** readiness only. No modelled fatigue.
- **B:** a fixed-decay "recent stress" indicator, advisory only, no load effect. **Applied 2026-10-20 (`DEC-STRESS-01`):** shown on Prescribe/Programme; still no load change.
- **C:** a per-lift fitness-fatigue model fitted to the athlete's e1RM series, then validated prospectively before it may change a load.
- **D:** collect session RPE for strength (and optional per-lift soreness) first; revisit C. **Partially applied 2026-10-20 (`DEC-SRPE-01`):** optional session RPE and length are captured at review; no model reads them yet.

Why not C now: the strength-specific evidence is one small weight-lifter study; parameter identifiability is poor; the fatigue component did not improve out-of-sample prediction in a 2025 re-analysis; and the log still lacks frequent performance tests. The current RIR window and 48–72 h spacing already manage acute fatigue.

Default if unanswered: no modelled fatigue (A). Recommendation: B or D, then revisit.

Q20. ~~Should e1RM estimates be calibrated against test results?~~ Answered: bounded correction applied

Answered 2026-10-20. Epley+RIR estimates were used as `best_usable` in an 84-day window, and a measured `test` became an override. The report and bounded correction now exist; a correction is applied only when the CI excludes 1.0 and the bias is within ±10%.

Applied 2026-10-20 (`DEC-E1RM-CAL-01/02`): report plus a bounded correction when the CI excludes 1.0; `e1rm calibrate` and the Athlete page. Measured overrides remain the control.

Q21. How should exercise selection decide what is comparable?

The selector can swap configured variants but has no explicit comparability class. Open: where the A/B boundary sits (pause/narrow bench, SSB/high-bar), whether the class is per exercise or per pair, what confirmation means, and whether parent-variant e1RM ratios can ever be used.

Proposed in SELECTION-COMPARABILITY.md: class A load-comparable, class B near-comparable (confirmation required), class C different lift (never automatic). **Implemented 2026-10-20**; the exact A/B boundary remains open.

Q22. Should selection target weakness or injury?

Deferred 2026-10-20; earliest review after v0.5 (~April 2027). No positive weakness/injury targeting. Injury remains a constraint/ceiling/penalty; it must never boost a candidate. A relative-weakness display, if added, is display-only. Revisit only with 12 sessions per target lift over 12 weeks, a pre-registered weakness definition, a retrospective baseline win, and an opt-in N-of-1/alternating test. See WEAKNESS-INJURY-SELECTION.md.

Closed

- ~~What do the extreme `RIR` values mean?~~ **Closed** — `RIR 6` marks a **warm-up set** (`DEC-RIR-01`), so it is excluded from effort, e1RM and working volume while staying visible. `RIR 0-1` remain in scope but flagged (`DQ-RIR-02`).
- ~~Bodyweight movements: total mass or added load?~~ **Closed** — progression acts on **added load** (`DEC-LOAD-02`); total mass is used only for an e1RM.
- ~~Is there a trap-bar deadlift in the data?~~ **Closed, then corrected** — the log name "Competition Deadlift" meant a **sumo** pull until ~2026-08-17 and the trap bar after that (`DEC-TRAPBAR-02`); the history is resolved by date, not relabelled.
- ~~Which exercise is "back extension"?~~ **Closed** — synonymous with the older hyperextension name, **including historical rows**: they keep their own identity but count as back-extension history (`DEC-BACKEXT-02`, amending `DEC-BACKEXT-01`).

Decisions taken as defaults unless told otherwise

- Training rows are **never** deduplicated — repeated identical rows are real sets.
- ACWR is advisory only and may never veto a prescription (the "sweet spot" reading is contested).
- Within-set velocity loss is reported as not measurable, with a labelled set-to-set decay proxy instead.
- Exercise-rotation scoring is split by session provenance, because in a real export the logging app deliberately rotates variations while self-directed sessions are stable.

Readiness- and symptom-driven exercise selection — implementation plan

Status: closed 2026-10-20. Tax metadata, candidate catalogue, review panel and explicit suggestions are implemented. Automatic symptom withholding is deliberately deferred to the follow-up review. **Scope:** exercise substitution inside an already-prescribed session. Load/RIR/set adjustment stays the first response to fatigue.

---

1. Question

Should readiness, recent stress and symptoms change the **exercise**, not only the load?

Example: the back is not well, or the same lift is heavily loaded — should the prescription offer a less taxing variant?

2. Position

Yes, but conservatively:

1. **Symptom/injury** may trigger a protective substitution or removal.
2. **Recent stress + low readiness** may trigger a suggestion of a lower-tax, same-pattern variant.
3. **Readiness alone** does not change the exercise; it changes load/RIR/sets.
4. Every substitution is **explicit** and comparability-guarded.
5. No class C substitution is ever automatic.
6. No LLM chooses an exercise outside the Python-filtered candidates.

The hierarchy is:

```
constraints / injury veto
  > symptom-specific protective substitution
  > recent-stress + readiness suggestion
  > load/RIR/set reduction
  > no change
```

---

3. Why not simply swap exercises on readiness

- Per-lift readiness is not a validated instrument.
- The evidence for autoregulated **load/volume** is mixed; evidence for daily **exercise selection** is much weaker.
- One small study found a small autoregulated-selection benefit in lean mass and bench, but no squat benefit; it was not a weakness/injury algorithm (Rauch 2020).
- Exercise specificity matters: frequent arbitrary variation can dilute practice.
- A change of exercise changes the load basis, skill demand and comparison; it is a larger intervention than a load cut.
- Pain-guided modification is defensible, but it is **injury management**, not generic readiness selection.

This is consistent with `WEAKNESS-INJURY-SELECTION.md` and `SELECTION-COMPARABILITY.md`.

---

4. Current building blocks

Already present:

- `prescribe/selection.py` ranks configured variants by readiness, recent stress, rotation, preference and the shadow fatigue model.
- A/B/C comparability classes:
- A: load-comparable;
- B: near-comparable, confirmation required;
- C: different lift/load basis, never automatic.
- Explicit apply only (`Apply selection` , `run --apply-selection`); load is re-solved from the selected lift's own history (`MOD-SELECT-001`).
- Injury signal, constraints and suitability already exclude/penalise candidates.
- Back-flare rule currently reduces erector work and main-lift RIR; it does **not** select a substitute.

Missing:

- an explicit **tax/risk metadata** for candidates;
- a clear symptom-specific substitution rule;
- a readiness/stress trigger that is separate from general readiness;
- reasons and audit fields for “why this variant was suggested”;
- tests for “should not substitute” cases.

---

5. Proposed design

5.1 Candidate tax metadata

Add a config map, not taxonomy code:

`config/selection-tax.v1.yaml`

```
schema_version: "1.0.0"
# Ordinal, per exercise: 1 = low, 3 = high.
# Missing means unknown, not low.
exercises:
  ssb_squat: {axial: 2, systemic: 2, skill: 2, rom: 2}
  front_squat: {axial: 2, systemic: 3, skill: 3, rom: 2}
  hack_squat: {axial: 1, systemic: 2, skill: 1, rom: 2}
  trap_bar_deadlift: {axial: 2, systemic: 3, skill: 2, rom: 2}
  back_extension: {axial: 1, systemic: 1, skill: 1, rom: 1}
```

Rules:

- values are ordinal, not measured;
- unknown metadata blocks a “less taxing” claim;
- `spinal_cost` remains the safety input, not replaced by this map;
- the map is reviewed when exercises are added.

Mandatory dimensions for a lower-tax claim:

- **main lifts:** axial, systemic, skill;
- **accessories:** axial, skill;
- **ROM:** record and flag material changes, but do not require it. It is a stimulus dimension, not a simple fatigue score.

A candidate is lower-tax only if it is **not higher on any mandatory dimension** and is lower on at least one.

5.2 Trigger tiers

Tier 1 — symptom/injury (strongest)

Trigger when:

- pre-session symptom flare is true, or the injury signal marks the region active; and
- the current exercise is linked to that region or has high axial/erector cost.

Behaviour:

- a hard constraint veto is **automatic** — the exercise is not prescribed;
- a symptom-linked exercise that is not a hard constraint is **safety-blocked** in the suggestion and flagged in the UI; automatic withholding is **not enabled** in this slice;
- a same-pattern candidate is suggested when it has:
- no link to the active region;
- lower axial/erector cost;
- class A or B only;
- an executable load from its own history;
- if no safe candidate exists, **do not invent one**; show “no safe substitution” and leave the decision to the athlete/clinician.

The hard-constraint exclusion is automatic; the symptom **suggestion and replacement remain explicit**. Automatic symptom withholding is the next gate, not a current behaviour.

Tier 2 — recent stress + a meaningful readiness drop

Trigger when **all**:

- the current lift has high recent stress (`recent_stress.status == "high"`); and
- readiness is below **0.70** in absolute terms; and
- readiness is at least **0.10 below the rolling 28-day median**; and
- the current lift is high tax (axial/systemic/skill); and
- a same-pattern candidate is lower-tax on at least one mandatory dimension and not higher on the others.

Baseline rules:

- use the previous 28 days, excluding today;
- require at least **10 prior readings**;
- if the baseline is insufficient, fall back to a fixed **0.65** and mark the fallback in the notes;
- update slowly; never re-centre on the last few days.

The absolute guard prevents a chronically low baseline from normalising poor readiness; the relative drop prevents reacting to a normal day that is merely below 0.70.

Behaviour:

- show a suggestion, not an automatic change;
- require confirmation (class B always; class A also explicit in this phase);
- keep the main lift slot; do not change session type.

Tier 3 — readiness alone

No exercise substitution.

Use existing:

- load/RIR reduction;
- readiness set reduction;
- optional deload.

This avoids turning a noisy readiness score into a different programme.

5.3 Candidate constraints

A candidate is excluded if:

- class C (unless explicitly shown, never applied);
- hard constraint veto;
- not `can_do` / suitability fails;
- linked to an active injury region;
- no executable load from its own history;
- not in the configured slot catalogue;
- tax metadata unknown for a “lower-tax” claim.

5.4 Reasons and audit

The selection result should include:

- `trigger`: `symptom` | `stress_readiness` | `none`;
- `current_tax`, `candidate_tax`;
- `reason` in plain language;
- `comparison_class` and `requires_confirmation`;
- `applied` and `confirmed_by`;
- `no_safe_substitution` when applicable.

The prescription cites `MOD-SELECT-001`; a future rule ID for the protective trigger can be added only if the user approves a new hard rule.

---

6. UX and API plan

Prescribe panel

Add a **Substitution** section:

```
Back symptoms active.
Current: SSB squat (axial 2, skill 2)
Suggested: hack squat (axial 1, skill 1)
[Apply substitution] [Keep current]
```

Rules:

- explicit button;
- B always requires confirmation;
- C never shown as applicable;
- if no safe candidate: “no safe same-pattern substitution”;
- show that load will be re-solved from the candidate’s history.

CLI

- `select` already previews; add the tax/trigger columns.
- `run --apply-selection --selection-confirm` remains the explicit path.
- Do not add an automatic flag for symptom substitutions.
- Add a `--no-selection` explicit opt-out if needed (already implicit default).

API

- `GET /api/selection` returns trigger/tax/reason fields.
- `POST /api/prescribe` applies selection only with existing explicit flags.
- No new automatic mutation path.

---

7. Tests

1. **Symptom trigger** - active back flare + SSB: suggests a lower-axial same-pattern candidate or reports none; - no class C applied; - no automatic application.
2. **Stress + readiness trigger** - high recent stress + low readiness: suggestion appears; - normal stress + low readiness: no exercise suggestion; load/set reduction only; - high stress + normal readiness: no suggestion.
3. **Comparability** - A applies only when explicit; - B requires confirmation; - C never applies automatically and is hidden by default.
4. **Load** - suggested lift is rebuilt through `build_item`, not copied; - unknown load is reported, not guessed.
5. **No candidate** - no same-pattern lower-tax candidate: no invented substitution; - message says so.
6. **Audit** - trigger, tax, class, reason and confirmation are visible.
7. **Safety** - constraints/injury veto wins over the selector; - no change to main-lift volume beyond existing readiness rules.

---

8. Rollout

Phase 0 — Config and shadow

- add tax config;
- compute trigger/tax/reason in selection;
- no UI apply;
- tests prove “no suggestion” cases.

Phase 1 — Symptom-specific suggestions

- show protective suggestions in Prescribe/select;
- explicit apply only;
- log trigger and outcome.

Phase 2 — Stress/readiness suggestions

- add the dual-trigger suggestion;
- keep readiness-alone as no substitution;

- no automatic application.

Phase 3 — Review

After 12–16 weeks, review:

- substitution uptake;
- back symptoms/flares;
- session completion;
- main-lift e1RM;
- whether suggestions were accepted or ignored.

No automatic promotion to a rule.

---

9. Non-goals

- No free exercise generation.
- No LLM-only selection.
- No class C automatic substitution.
- No injury prediction.
- No “target the weak muscle” selection.
- No readiness-only exercise swap.
- No change to the programme template or session type.
- No substitution without re-solving the load.

---

10. Locked decisions

decision	locked choice
Tier 2 readiness	`< 0.70` absolute **and** at least `0.10` below the 28-day median
Baseline	previous 28 days, excluding today, minimum 10 readings; fallback to `0.65`
High stress	existing `recent_stress.status == "high"` for the current lift
Symptom	hard veto automatic; symptom-linked exercise safety-blocked, suggested replacement explicit; automatic withholding deferred
Mandatory tax dimensions	main: axial + systemic + skill; accessories: axial + skill; ROM optional
Rule ID	no new rule; applied substitutions cite `MOD-SELECT-001`; exclusions cite existing injury/constraint rules; trigger recorded in notes

These values are policy assumptions, not validated cut-points. Changes require updating this document, the config, tests and the decision log.

The review panel (§12), tax metadata, candidate catalogue and explicit substitution suggestions are implemented. Automatic symptom withholding remains deferred to the follow-up review; see §14.

---

11. App-wide readiness threshold policy

The exercise-substitution plan is one consumer of a broader rule:

> **Personal baseline for context and high-cost, reversible choices; fixed absolute thresholds for safety and load.**

11.1 What stays absolute

threshold	treatment
sleep/energy ≤ 2	fixed safety floor
load factor bands (`0.80 / 0.60 / 0.40`)	fixed intervention dose
set-reduction score `< 0.50`	fixed volume valve
injury/symptom rules	fixed safety rules
deload triggers	separate policy; never readiness-relative

A rolling baseline must never normalise a poor absolute state.

11.2 What may use a rolling baseline

decision	use
HRV/RHR components	existing 28-day baseline, minimum 7 readings
readiness warnings	absolute score **and** deviation from 28-day median
exercise substitution Tier 2	absolute `< 0.70` **and** at least `0.10` below the median
“falling readiness” display	relative trend only
per-lift readiness	show own-history context; keep action thresholds fixed

Per-lift readiness is already relative (global base, e1RM trend, RIR, VBT). Do not add a second rolling baseline to its action thresholds; that would make it difficult to interpret.

11.3 Periodic review, not daily tuning

Thresholds are reviewed at a pre-specified cadence:

- every **12–16 weeks**, or
- after roughly **40–50 new sessions**, whichever comes first;
- plus after a material programme change or a new back pattern.

The review may:

1. re-run readiness–performance and readiness–sRPE associations;
2. compare the current absolute threshold with the absolute + relative gate;
3. use held-out sessions;
4. change a threshold only if it beats the current one out of sample and does not increase symptoms or missed sessions;
5. otherwise keep the fixed value and record why.

The review must not:

- refit weekly;
- change thresholds automatically;
- use the same data to choose and validate a cutoff;
- turn a weak association into a rule.

11.4 Baseline rules

- window: previous **28 days**, excluding today;
- minimum: **10** readiness readings;
- insufficient baseline → use the fixed fallback and mark it;
- slow update; no re-centring on the last few days.

---

12. Website: periodic review reminders

The app should not rely on the athlete remembering reviews. Add a small **Periodic reviews** section to Today (or a “Maintenance” panel) that is local-only and non-intrusive.

12.1 Purpose

Tell the athlete:

- what is due;
- why it is due;
- what to do;
- what **not** to do;
- how to mark it reviewed.

No automatic threshold changes. The panel is a prompt, not a controller.

12.2 Suggested task cards

task	due calculation	suggested action
Readiness threshold review	12 weeks or 40 new sessions since last completion	Re-run readiness associations, compare absolute vs relative gates, keep unless held-out improvement
Rubric labels	show count; target 10 high-confidence marks per measurable item	Review recent sessions; current coverage 5–6 per item
3RM/5RM test	10 weeks from last measured test	Run `strength-coach test --status`; schedule a test if due
sRPE/duration coverage	monthly check; target 80% of recent sessions	Review recent sessions with missing sRPE/duration
Back-pattern review	after a new low-back episode or a pattern that does not fit the existing case series	Review the episode against the three known tweaks; do not auto-change rules
Readiness baseline sufficiency	fewer than 10 readings in the 28-day window	Keep the fixed fallback; record the gap

12.3 Card content

Each card should show:

- **status:** due / due soon / not yet / insufficient data;
- **last completed** (or “never”);
- **why:** one sentence;
- **what to do:** 2–4 concrete steps;
- **what not to do:** e.g. “do not change the threshold without held-out evidence”;
- **links:** the relevant doc/section;
- **actions:** `Mark reviewed` and optionally `Snooze 2 weeks`.

No card should apply a programme change itself.

12.4 Data model

No implementation yet. The minimal design is an append-only table:

```
maintenance_review
  review_id
  task_key          -- readiness_thresholds | rubric_labels | strength_test | srpe_coverage |
back_case
  completed_on
  decision          -- kept | changed | insufficient | not_applicable
  note
  created_at
```

Rules:

- latest row per `task_key` is the completion record;
- corrections append;
- due dates are computed from the latest completion plus the interval;
- task definitions and intervals live in config, not code;
- no external notifications; the local UI is the reminder.

12.5 Behaviour

- The panel is visible on Today, collapsed when nothing is due.
- Due items are not modal; they do not block a workout.
- “Mark reviewed” records a decision and note; it does not silently change a threshold.
- “Snooze 2 weeks” records a snooze, not a completion.
- If data are insufficient, the card says so and does not manufacture a due date.
- The panel must not turn into a dashboard of unstable rolling correlations.

12.6 Review checklist content

When a readiness threshold review is opened, show:

1. current absolute threshold and fallback;
2. current 28-day baseline and reading count;
3. readiness–performance association on held-out sessions;
4. readiness–sRPE association;
5. how many sessions would have triggered the current vs proposed rule;
6. decision: - keep; - change with a pre-specified rationale; - insufficient data.
7. record the decision in `maintenance_review`.

This makes the reminder useful and auditable, rather than a generic “review due” badge.

---

13. Implementation gates

Before the review panel is built:

1. decide whether `maintenance_review` is the right persistence;
2. define intervals and snooze behaviour in config;
3. decide where the panel lives (Today vs a separate Maintenance page);
4. keep all actions non-prescriptive.

No code has been written for this section.

---

14. Closure and follow-up

14.1 Implemented

- ordinal tax metadata for main lifts and accessories (`config/selection-tax.v1.yaml`);
- accessory candidate lists in `config/variants.v1.yaml`;
- symptom and stress/readiness **suggestions** in `/api/selection` and the Prescribe panel;
- current vs candidate tax, trigger, safety-blocked flag and reasoning;
- explicit apply only; load re-solved from the selected lift's history;
- no automatic substitution;
- no automatic symptom withholding.

14.2 Evidence boundary

The evidence supports:

- load/volume autoregulation better than daily exercise selection;
- pain-guided modification as injury management, not a general readiness rule;
- same-pattern variants over unrelated exercises;
- comparability classes because load bases differ.

The evidence does **not** establish:

- a validated per-lift readiness threshold;
- that daily exercise substitution improves outcomes;
- that an ordinal tax map is a measured fatigue model;
- that automatic withholding is safer than a visible suggestion.

The one small autoregulated-selection study found a small lean-mass/bench advantage and no squat advantage (Rauch 2020); that is not enough for automatic removal.

14.3 What is needed before automatic withholding

1. **8–12 weeks or ~30 sessions** of suggestion use.
2. Record: - trigger frequency; - accepted vs ignored suggestions; - back symptoms/flares; - session completion; - main-lift performance.
3. No increase in symptoms or abandoned sessions.
4. The tax table remains accepted after real use.
5. A decision that automatic withholding is preferable to the current safety-blocked suggestion.

The **Exercise substitution review** task is now in the periodic reviews dashboard with an 8-week/30-session interval. It records a decision and note; it does not change behaviour.

14.4 Plan status

Closed for the suggestion slice. Automatic withholding is a separate, evidence-gated follow-up, not unfinished implementation.

READINESS SCORE — global, category and calibration

Status: implemented (2026-10-19; HRV/RHR + wearable 2026-10-20) for the global/category score, overrides, the readiness-required gate, the 20% combined cap, the calibration report, baseline-relative HRV/RHR, the optional device-computed wearable score, per-athlete baseline settings and the shadow baseline report. Learned weight adoption was evaluated and closed on 2026-10-20; see `READINESS-WEIGHT-EVIDENCE.md`.

The score replaces the old sleep/energy-only trigger, but the old trigger remains as a safety floor. The three legacy per-lift values are calibration labels, never inputs.

This document records the evidence, the model, what is assumed rather than measured, and the remaining path. Decisions live in `docs/DECISIONS.md`.

1. What the evidence supports

claim	evidence	strength / limit
Subjective self-reports are more sensitive to acute and chronic training load than most objective measures	Saw et al. 2016 systematic review (56 studies)	Moderate; heterogeneous measures, no universal questionnaire
A multi-item wellness index can track recovery and staleness	Hooper et al. 1995 (daily fatigue/stress/sleep/muscle soreness in swimmers); Bell et al. 2025 (Hooper index during a squat overreaching protocol)	Moderate for trend; no validated weighting of items
Pretraining wellness is associated with subsequent training load	Govus et al. 2018 (American football)	Observational; not a prescription rule
Sleep restriction impairs maximal strength in compound movements	Knowles et al. 2018 systematic review (17 studies)	Moderate/weak quality; single-joint movements less affected
HRV-guided training improves vagal HRV indices, but performance benefits are small and not clearly significant; resting HR was not improved	Manresa-Rocamora et al. 2021 meta-analysis	Supports HRV as context/trend, not a hard daily gate
HRV interpretation depends on the decision framework; no standard multimodal rule	Schaffarczyk & Sperlich 2026 SWOT review	Weak for a specific formula; supports transparency
Perceived readiness can relate to resistance-training performance	Munteanu et al. 2026 (28 resistance-trained females, two mesocycles)	Emerging, small; not a validated per-lift scale
Grip strength is a strength/health marker related to upper-body power	Silder et al. 2024 (military); LaMonte et al. 2026 (mortality)	Chronic capacity, **not** acute daily readiness
Injury history and symptom response matter more than a calendar	injury-signal evidence, refs 40-46	Supports a ceiling, not an average

DOIs: Saw 2016 `10.1136/bjsports-2015-094758`; Hooper 1995 `10.1249/00005768-199501000-00019`; Govus 2018 `10.1123/ijspp.2016-0714`; Knowles 2018 `10.1016/j.jsams.2018.01.012`; Manresa-Rocamora 2021 `10.3390/ijerph181910299`; Schaffarczyk 2026 `10.3389/fspor.2026.1858271`; Munteanu 2026 `10.1007/s40279-026-02459-8`; Bell 2025 `10.3390/jfmk10020227`; Silder 2024 `10.1093/milmed/usae227`; Long 2026 `10.1371/journal.pone.0358036`; Toohey 2017 `10.1136/bjsports-2017-097500`; Lou 2026 `10.1038/`

s41598-026-59571-y ; Yuan 2026 10.52082/jssm.2026.172 ; Impellizzeri 2020 10.1123/ijspp.2019-0864 ; Herr 2026 10.4085/1062-6050-0619.25 . Grip and mortality context: LaMonte 2026 10.1001/jamanetworkopen.2025.59367 . Injury evidence: refs 40-46 in REFERENCES.

What is absent

- **No validated per-lift readiness questionnaire.** `squat_readiness` , `bench_readiness` and `deadlift_readiness` are self-ratings; treating them as measured lift readiness is a design assumption, not a finding.
- **No evidence for specific composite weights or cut-points.** Equal weights and the thresholds below are transparent starting points, not science.
- **No evidence that grip strength tracks day-to-day readiness.** It is slow capacity.
- **No validated per-lift readiness score.** A per-lift advisory score now combines the global base with e1RM trend, RIR and VBT and can only lower a load (`docs/LIFT-READINESS.md`); it is an assumption, not validation.
- **No trial of a readiness score overriding an injury signal.** The safe rule is that it cannot: injury is a ceiling/veto, readiness is a down-adjustment.

---

2. Proposed model

2.1 Global readiness, 0-1

Inputs and directions (higher score = more ready):

input	scale	direction	role
sleep, energy, mood, stress	1-5	higher better	core subjective
soreness	1-5	higher = less sore	core subjective
HRV	ms	higher better, only relative to baseline	context, low weight
RHR	bpm	**lower** better, only relative to baseline	context, low weight
wearable score	0-100	higher better, device-defined	optional 10% blend; never a replacement
grip strength	kg	higher better	**moved out of readiness**: measured with `measure` , never a daily score input (`DEC-GRIP-01`)

For a field present on the date:

- 1-5: `normalised = value / 5` (so a neutral 4/5 is 0.80 and a 5/5 is 1.00);
- HRV: `median(previous 28 days)` baseline; ratio clipped to 0.5-1.5, mapped to 0-1;
- RHR: same baseline, ratio **inverted** before mapping.

```
global = sum(weight_i * normalised_i) / sum(weight_i)   over present, eligible fields
```

Defaults (config, not code): weight 1 for the five subjective fields; weight 0.5 for HRV and RHR when a baseline exists; no score with fewer than **three** subjective fields (report "insufficient", not a number). Missing fields are excluded, never zero. If a wearable score is present, the final score is `0.90 * subjective/device score + 0.10 * wearable/100` .

Athlete settings. A `readiness_setting` row effective on the date overrides the config window, minimum readings, slopes and weights. Precedence is **athlete > config > code default**; the score report says `athlete score setting` when one is active. Settings are append-only.

Every display shows the contributors and their weights, a confidence label based on how many fields were present, and the baseline dates used for HRV/RHR.

2.1a Equal weights vs explicit weights

Equal weights (recommended first). Every present subjective field counts once; HRV/RHR count half when a baseline exists. The score is the plain mean of the normalised values.

- why: no field's weight is validated, so equal weights are the least arbitrary choice;
- it is transparent and cannot overfit a short history;
- it is robust when a field is missing or noisy;
- weakness: it assumes a bad night's sleep and a slightly low mood matter equally, and it can dilute one severe field.

Example with sleep 2 (0.25), energy 4 (0.75), mood 4 (0.75), stress 4 (0.75), soreness 4 (0.75): equal score = **0.65**. With explicit weights sleep ×2, energy ×1.5 and the rest ×1, the score falls to about **0.60**. Neither number is evidence; the first is the default.

Explicit weights (later, per athlete). Give each field a weight or a floor, e.g. sleep 2, energy 1.5, HRV 0.5, because the evidence is strongest for sleep and weakest for HRV. This can be more faithful to the athlete, but:

- the weights are not validated;
- with a handful of sessions they fit noise, not readiness;
- they make the score harder to explain and easier to game.

Recommendation: equal weights in shadow mode; explicit weights only as per-athlete config after enough scored sessions, with the contributors still shown. Keep the **individual safety floor** regardless of weights: sleep or energy at or below 2 still applies the old 5% cut even if the composite is high. That prevents a single catastrophic field being averaged away.

2.1b HRV and RHR: implementation

HRV and RHR enter the score only when the athlete has a personal baseline:

- baseline = median of the previous **28 days**, at least **7** readings (imported or manual);
- HRV is higher-is-better, RHR lower-is-better; both are normalised around 0.5 at baseline;
- a 50% deviation from baseline maps to 0 or 1; the slope is a heuristic, not a clinical cutoff;
- each device field has weight **0.5**, against 1.0 for each subjective field;
- if the baseline is missing, the field is shown as present but **not scored**; it is never replaced by an arbitrary population value;
- a day with only device data is still insufficient: three subjective fields plus sleep or energy are required.

This is deliberately low-weight. HRV-guided training improves vagal indices, but performance benefits are small and not clearly significant, and there is no standard decision framework (Manresa-Rocamora 2021; Schaffarczyk 2026).

2.1b1 Per-athlete baseline settings

The rolling median is the default. An athlete can set the baseline window, minimum readings, slopes and weights in `readiness_setting`; the latest row effective on the score date wins. Nothing is auto-fitted. The CLI/UI show the effective values and their source.

The **shadow report** compares the shipped rolling median with a trend-projected baseline (log-linear fit over the same prior readings), reporting the normalised difference. It is diagnostic only: it does not change a score, a factor, or a load. The projection is capped to 0.5-1.5× the rolling median so an extrapolated trend cannot produce an absurd target.

2.1c Optional wearable readiness score

A device such as Bevel can supply its own readiness score (0-100). The app treats it as an **optional context blend**, not as a second truth:

- absent: no effect;
- present: `score = 0.90 * computed_score + 0.10 * wearable/100`;
- 0 is valid (worst), blank is missing; the device threshold is not reinterpreted;
- global and category overrides still win; the injury ceiling still caps the result;
- the UI and score report label it `wearable_score` and show the 10% weight.

Why only 10%: the algorithm is proprietary and cannot be audited; it may double-count sleep, HRV and soreness that are already in the subjective score; and different devices disagree. Wearable data are useful context and can predict performance in groups, but they do not replace the athlete's own report for an individual session (Spetz 2025; Khodr 2024; Zhang 2026).

2.2 Per-lift readiness

```
lift_readiness = min(global_readiness,
                     category_override_if_present,
                     injury_ceiling_if_linked)
```

- The old `squat_readiness` / `bench_readiness` / `deadlift_readiness` fields are **labels, not inputs** (section 2.8). They are not read here.
- A **category override** is the manual per-lift input: the athlete may set `category:hinge`, `category:squat`, etc. for the day. This is how the score stays overridable without keeping the old fields in the form.
- `injury_ceiling` comes from the injury signal for the region linked to that exercise; direct loading uses the table below, indirect loading adds one step less severe.

injury status	direct ceiling	indirect ceiling
active flare	0.50	0.60
monitoring	0.80	0.85
review	1.00 (no cap)	1.00
insufficient / unknown	1.00, with a warning	1.00

These numbers are **assumptions** (`ASSUMPTION-13`). A high readiness score can never lift an item above its injury ceiling, and an injury signal can never increase readiness.

2.2a Minimum data before a score is shown

Decision: require **at least three of the five core subjective fields** for a numeric global score. With one or two, show the individual fields and `insufficient`, not a composite. For a readiness-driven load change, require at least one of **sleep or energy** to be present; otherwise the score can be displayed but does not move a load. The existing sleep/energy floor still applies independently.

For a lift with no global score: use its **category override** if present, capped by the injury ceiling; otherwise no readiness factor. The injury ceiling still applies either way. This is `ASSUMPTION-15`; it prevents a two-field day from masquerading as a full score.

2.3 From score to load

readiness	load factor	RIR
≥ 0.80	1.00	unchanged
0.60-0.80	0.95	+1
0.40-0.60	0.90	+1
< 0.40	0.85	+2

Cut-points and factors are `ASSUMPTION-14`. The current sleep/energy rule stays as a **floor**: if either is at or below 2, the old 5% reduction still applies even if the composite is high. Final factor is the **strongest single reduction**, not a product of every rule; total readiness + continuity reduction is capped (proposal: 20%) unless a flare or constraint says otherwise. This prevents three rules silently compounding into a session that is too easy.

2.3a Combined reduction cap

Several rules can fire on the same day: the readiness score, the old sleep/energy floor, a return-from-gap adjustment, a back flare and a per-lift injury ceiling. Without a cap, three small reductions can silently become a deload that nobody intended.

Proposal:

- **Readiness vs the old sleep/energy floor:** take the **stronger single reduction**, not the product. They are two views of the same signal.
- **Readiness + continuity:** these are independent risks, so combine them, but cap the total load reduction at **20%**. The cap applies to the global load factor only.
- **Not capped:** hard constraints, back-flare RIR rules and per-lift injury ceilings. Those are safety layers; they are allowed to be more restrictive and must never be diluted by a cap.
- **Trace:** if the cap bites, say so and show the uncapped factors.

Why 20%, and what the evidence says: **there is no direct study of a numerical cap for stacked readiness + return-to-training reductions.** Autoregulation research supports adjusting load to readiness but shows only small advantages over standard prescription, and the reviews do not define a safe stacking ceiling (Hickmott 2022; Hickmott 2025). The cap is therefore a guardrail, not a finding. Twenty percent is deliberately modest: it honours two bad signals without turning the session into a planned deload. Fifteen or twenty-five percent would be equally defensible; the number is `ASSUMPTION-21`.

2.4 Interaction order

1. Hard constraints and suitability vetoes (unchanged; readiness cannot override).
2. Back-flare `MOD-BACK-001` (unchanged; wins over readiness).
3. Per-lift readiness: `min(global, self, injury ceiling)`.
4. Apply the factor to that item; items without a lift-specific value use the global factor.
5. Continuity (`MOD-GAP-001`) applies as today, but the combined reduction is capped.
6. Cite `readiness:<date>`, `readiness_score:<value>`, the contributors, and the injury region that set a ceiling.

A `review` injury status does **not** raise a score or clear a constraint. An `active` region produces a warning and a ceiling; the hard-exclude constraint remains the only veto.

2.5 Grip strength

Stored as a capacity measurement, shown as a 28-day trend. It does not move a daily load in this proposal. If later evidence supports it, it can enter the global score as a slow baseline term; that would be a separate decision.

2.6 Injury attention by movement/muscle category

Decision: the attention indicator is per movement/muscle category, not only per injury region. Regions remain the evidence unit (a flare is a low-back flare); the category is the training unit the athlete actually chooses between.

A category is configuration, for example:

category	matches	typically linked regions
hinge	patterns `hinge`, `hinge_unilateral`, `hinge_ballistic`; glutes/hamstrings/erectors	low back (direct), knee (indirect)
squat	`squat`, `squat_unilateral`; quads/glutes	knee (direct), low back (indirect)
horizontal push	`horizontal_push`, `horizontal_push_bw`; chest/triceps/front delts	none configured by default
vertical push	`vertical_push`; delts/triceps	none configured by default
horizontal pull	`horizontal_pull`; lats/rhomboids/biceps	biceps (direct)
vertical pull	`vertical_pull`; lats/biceps	biceps (direct)
elbow flexion	`elbow_flexion`; biceps/brachialis	biceps (direct)
knee extension	`knee_extension`; quads	knee (direct)
dorsiflexion	`dorsiflexion`; tibialis anterior	ankle/shin (if configured)
knee flexion	`knee_flexion`, `hamstring_eccentric`; hamstrings	knee (direct)
spinal extension	`erector_extension`; erectors	low back (direct)
core / carry	core patterns or `carry`	low back (indirect), knee (indirect)

The exact list is config; patterns and muscles must exist in the taxonomy or the config is rejected. An exercise can match a category by pattern **or** primary muscle; a category can also list explicit exercise ids. A category may be **muscle-only** (for example elbow flexors: biceps/brachialis) when no single movement pattern captures it.

For each category:

1. collect the configured regions linked to its patterns/muscles;
2. take the **worst** region status (active > review > monitoring > insufficient/unknown);
3. show the drivers: region, direct/indirect, latest level, prior flares, tolerated/unverified sessions;
4. if no region is linked, say "no injury configured for this category" — unknown, not fine.

No probability, no load effect, no second penalty. A category's attention is display and education only; the injury signal and constraints remain the decision layer.

2.7 Injury risk indicator (why it is not a probability)

This is **not** a probability of injury and must not be called one. The evidence supports a transparent **attention indicator**, not individual prediction.

Evidence

- Previous injury is a risk factor for subsequent/recurrent injury; ACL history was associated with later hamstring injury (RR 2.25, 95% CI 1.34-3.76) (Toohey 2017); prior low-back episodes predict recurrence (da Silva 2019).
- Machine-learning models report pooled sensitivity 0.79 / specificity 0.71 across 10 small models, with substantial methodological heterogeneity and risk of bias (Lou 2026). A 15-study review found AUCs up to 0.91 but called for large prospective validation (Yuan 2026).
- The acute:chronic workload ratio has **not** been shown to be causal for injury; treating it as one is an overinterpretation (Impellizzeri 2020).
- Wellness/readiness as an injury predictor is case-level and observational, not validated (Herr 2026).

Proposed indicator

For each **movement/muscle category**, produce an attention tier with drivers, not a percentage:

tier	drivers (examples)
high attention	active flare, or ≥3 prior flares with recent unverified/un tolerated loading
medium attention	monitoring, 1-2 prior flares, or a sudden load increase without follow-up readings
low attention	review status, no recent flare, tolerated loading

It is **display and education only** in the first phase. It does not change a load, substitute a lift, or clear a constraint. It must not feed back into the readiness score as a second penalty. A future probability model is out of scope until prospective, calibrated validation exists.

Current calibration report (2026-10-19)

`readiness calibrate` on the 306 labelled days, date split 70/30:

label	n	MAE readiness-only	MAE + load	baseline	rank r
squat	306	0.102	0.101	0.119	0.11
bench	306	0.069	0.068	0.071	0.00
deadlift	306	0.106	0.111	0.118	0.04

Read: the score is close in absolute error but barely ranks the old labels; main-lift load features add almost nothing. That is evidence about **mimicry**, not readiness. The fitted weights are not applied. If this remains weak, the old label may be mostly a different construct, not a target worth chasing.

2.8 Legacy per-lift readiness labels (black box)

The three imported fields `squat_readiness`, `bench_readiness` and `deadlift_readiness` are **not inputs** in the new model. They are outputs of an older external calculation — training load plus other readiness data — whose formula is a black box. The athlete could override the old value, which is why it is a useful reference but not ground truth.

What we know from the data:

- 306 days carry all three values;
- all are 1-5, higher better;
- means: squat 3.53, bench 3.79, deadlift 3.56.

Plan:

1. **Remove them from new input surfaces**: readiness CLI, readiness web form and bulk state import. Stop asking the athlete for them.
2. **Keep the historical values read-only** in `readiness_day` /raw rows as calibration labels. They are not deleted and never become model inputs at prediction time.
3. **Calibrate against them in shadow mode.** They can tell us whether the new score mimics the old app, not whether it is physiologically correct. That distinction is stated wherever the calibration is reported.
4. If the old labels depend on training-load features, include those features in calibration (recent acute/chronic load, category-specific sets/tonnage, days since the category was trained). We cannot recover the black-box formula; we can only fit an approximation.

Main-lift load features

Calibration must include the main-lift load history, because the old labels were derived from training load as well as readiness. Features are built **only from data available before the session** (no leakage from the session being predicted):

- readiness inputs for the day (sleep, energy, mood, stress, soreness; HRV/RHR relative to baseline);
- main lift: last load and e1RM estimate at a comparable RIR, and the change over the last two sessions;
- main-lift acute load (7 days) and chronic load (28 days): sets, tonnage and best set, relative to the athlete's 28-day baseline;
- days since that main lift was trained;
- last session's RIR and mean velocity for that lift, when present;
- missingness indicators, so "no data" is not silently zero.

Keep the feature set small and pre-specified; 306 labelled days will overfit quickly. Use a date split and ridge/linear regression, not a black box. Report the global-only baseline next to the load-feature model, and per-label MAE/rank correlation.

Calibration shape

One shared global component plus three category adjustments, fitted against the three labels:

```
label_category ~ f(global readiness inputs,
                   recent training load,
                   category-specific load,
                   days since category trained)
                   + category adjustment
```

- fit on a **date split** (earlier dates train, later dates test), not a random split;
- report MAE and rank correlation for each label and for the pooled labels;
- compare against a baseline (global score alone) before adding load features;
- only 306 labelled days exist, so start simple (linear/ridge); no deep model, no black box;
- the fitted weights are proposals, never auto-applied; the athlete adopts them.

Only squat, bench and deadlift have labels. Other movement categories have no calibration target until the athlete provides one.

2.9 Readiness overrides

Computed readiness is a model output, so the athlete must be able to correct it. Overrides are append-only, dated and reasoned.

- **Scopes:** `global` and `category:<name>` (for example `category:hinge`).
- **Value:** 0.00-1.00, the same scale as the computed score (higher = more ready). No separate 1-5 override scale.
- **Precedence:** a category override wins over the global override, which wins over the computed score.
- **Effect:** the winning score goes through the same load-factor table, so an override **does change the prescribed load**.
- **Safety ceiling:** an override may raise or lower readiness, but it **cannot** raise a lift above an injury ceiling or bypass a hard constraint/back-flare rule. Those remain the veto layer. This is the one place where "override wins" does not mean "override everything".
- **Trace:** the prescription cites `readiness_override:<id>` and the reason. A later override appends; the earlier one stays in history.

A manual override is a statement about today's readiness, not a new programme rule. It expires with the date unless re-entered.

3. Why this shape

- It separates **subjective readiness** (what the athlete feels today) from **injury history** (what the tissue has tolerated) instead of averaging them into one number.
- It preserves the existing safety floors and the injury signal's "may recommend, never clear" rule.
- It is transparent: the score, its inputs, weights and ceilings are all visible.
- It starts in **shadow mode**, so the first phase changes no prescription.

---

4. What is implemented

Delivered on 2026-10-19:

- global score from sleep, energy, mood, stress, soreness; 3-field minimum and sleep/energy requirement;
- category scores and overrides at global/category scope, 0.00–1.00, reason required;
- injury ceiling per item, applied before the factor;
- load factor table and the old sleep/energy floor;
- 20% combined readiness + continuity cap, with a warning when it bites;
- readiness-required gate for current/future CLI and web workouts;
- calibration report (`readiness calibrate`) with main-lift load features, date split, ridge/linear fit and baseline comparison;
- HRV/RHR baseline-relative components (0.5 weight, 7 readings in 28 days);
- optional wearable score at a 10% blend.

Closed:

- learned weight adoption — the held-out comparison found no improvement; fixed transparent weights are retained (`DEC-READINESS-10`, `READINESS-WEIGHT-EVIDENCE.md`).

5. Decisions and open questions

Periodic review and the app-wide threshold policy are specified in `READINESS-EXERCISE-SELECTION-PLAN.md` §11–13.

Decided:

- **Equal weights retained**; explicit per-athlete weights and learned weights are not adopted; the sleep/energy floor stays.
- **Cut-points and injury caps accepted.**
- **Per-lift mapping accepted**: squat → `squat` / `squat_unilateral`, bench → `horizontal_push` / `horizontal_push_bw`, deadlift → `hinge`; unmapped lifts inherit global.
- **Minimum data**: three core subjective fields, with sleep or energy for a load effect.
- **Grip is capacity only**, never a daily score input.
- **Injury attention is per movement/muscle category**, not only per region; display only.
- **The three old per-lift fields are black-box labels**, kept read-only for calibration and removed from new input surfaces.
- **Overrides cover global and category scores**, win over computed readiness, change the load, but never bypass constraints or injury ceilings.

Still open:

1. Is a 1–5 soreness scale direction "higher = less sore" intuitive enough, or should the UI invert the label?
2. The exact category list and exercise-to-category mapping; the table above is a starting proposal, not final.
3. The exact regularisation/feature weighting in calibration; the feature set is specified, the fit is not.
4. Whether a category with no linked injury should say `unknown` or `not configured`; both avoid "fine", but they mean different things.

Decided: overrides use 0.00–1.00 like the score and require a reason; the combined readiness + continuity cap is 20%, applying to loads only.

6. Assumptions

- `ASSUMPTION-11` — equal weights over present subjective fields are a transparent starting point, not a validated composite.
- `ASSUMPTION-12` — per-lift readiness is constructed from global + self-rating + injury ceiling; no validated per-lift instrument exists.
- `ASSUMPTION-13` — the injury ceilings are design choices, not measurements.
- `ASSUMPTION-14` — readiness cut-points/factors and the combined-reduction cap are design choices; the cut-points and injury caps are approved for the plan.
- `ASSUMPTION-15` — three core fields are the minimum for a numeric score; this is a design choice, not a finding.
- `ASSUMPTION-16` — the injury attention tiers are descriptive, not probabilistic; they do not predict a new injury.
- `ASSUMPTION-17` — the attention indicator is grouped by movement/muscle category, with the region as the underlying evidence unit; the category list and links are config, not science.
- `ASSUMPTION-18` — the old per-lift readiness values are a black-box reference, not ground truth; calibration measures mimicry, not validity.
- `ASSUMPTION-19` — readiness overrides are bounded by constraints and injury ceilings; "override wins" applies to the computed score only.
- `ASSUMPTION-20` — the calibration features and date split are design choices until tested.
- `ASSUMPTION-21` — a 20% cap on combined readiness + continuity reduction is a guardrail, not an evidence-based threshold.
- `ASSUMPTION-22` — main-lift load features are used only in calibration and only from data available before the session.

Evidence references: `docs/REFERENCES.md` §8. Injury evidence: `docs/INJURY-SIGNAL.md`.

Readiness weights — evidence and sensitivity check

Status: closed. Fixed, transparent weights are retained; learned individual weights are not adopted. **Date:** 2026-10-20

This note records the evidence for readiness weighting, the constrained comparison requested before any learned model, and the decision not to learn weights from the existing data.

1. Evidence

claim	evidence	strength / limit
Subjective self-reports are more sensitive to acute/chronic training load than most objective measures	Saw et al. 2016 systematic review, 56 studies	Moderate; heterogeneous measures, no universal questionnaire
A multi-item wellness index can track recovery/staleness	Hooper et al. 1995; Bell et al. 2025 (squat overreaching)	Moderate for trend; **no validated weighting of items**
Pretraining wellness is associated with subsequent training load	Govus et al. 2018	Observational; not a prescription rule
Sleep restriction impairs maximal strength in compound movements	Knowles et al. 2018 systematic review	Moderate/weak quality; single-joint less affected
HRV-guided training improves vagal indices; performance benefit is small/not clearly significant	Manresa-Rocamora et al. 2021 meta-analysis	HRV is context/trend, not a hard daily gate
Perceived readiness can relate to performance	Munteanu et al. 2026, 28 trained females, two mesocycles	Emerging, small; not a validated per-lift scale
No validated composite weights or cut-points	Readiness-score review	The current equal weights and thresholds are policy, not measurement
No validated learned individual readiness weights	No trial or validation study identified	A learned model would require prospective, held-out validation

Sources and DOIs are recorded in `READINESS-SCORE.md` §1.

What the evidence supports

- tracking wellness as a **trend**;
- using sleep/energy as a safety floor;
- using HRV/RHR as low-weight context;
- keeping the score transparent.

What it does not support

- specific weights for sleep, energy, mood, soreness or HRV;
- a learned individual weight vector from one athlete’s fixed programme;
- using readiness to justify automatic main-lift volume changes.

2. Constrained sensitivity check

Rather than fit free weights, three transparent schemes were compared:

scheme	subjective core	HRV/RHR
current	each weight 1.0	0.5
sleep-heavy	sleep/energy 2.0, rest 1.0	0.5
HRV-low	each weight 1.0	0.25

Outcomes:

- performance ratio: session best e1RM / prior 84-day best for the same lift;
- sRPE where available.

The late period (from 2025-07-01) is the held-out evaluation window; it was not used to choose the schemes.

Results

All sessions with readiness + performance (n = 349):

scheme	performance ρ	sRPE ρ
current	0.118	-0.054
sleep-heavy	0.107	-0.040
HRV-low	0.118	-0.054

Held-out late period (n = 100; all with sRPE):

scheme	performance ρ	sRPE ρ
current	-0.004	-0.075
sleep-heavy	-0.057	-0.052
HRV-low	-0.017	-0.061

Readiness below vs at/above 0.75 in the late period:

scheme	low median ratio	high median ratio
current	0.968 (n=37)	0.981 (n=63)
sleep-heavy	0.980 (n=54)	0.961 (n=46)
HRV-low	0.955 (n=28)	0.980 (n=72)

Interpretation

- No alternative scheme improves the association out of sample.
- The current equal-weight score performs as well as the alternatives in the held-out period.
- All associations are weak; this is consistent with readiness being context, not a precise predictor.
- The late-period performance association is essentially zero, so there is no justification for a more complex weighting.

3. Decision

Keep the fixed score: equal subjective weights, HRV/RHR at 0.5, sleep/energy safety floor.

Do not learn individual weights from this dataset. The dataset is large in sessions but small in independent variation, and readiness already influences the prescription.

Revisit learned weights only if:

- prospective data include an independent outcome;
- enough poor-readiness days exist;
- a held-out comparison beats the fixed score;
- the result does not increase symptoms or missed sessions.

---

4. Limitations

- Retrospective and observational.
- Performance ratio is model-derived from e1RM, not a measured test.
- Readiness is used to change load/sets, creating circularity.
- HRV/RHR coverage and wearable blending vary by date.
- sRPE is available for 173 sessions, not all.
- The late window has only 100 sessions.

This is enough to reject a more complex weighting scheme; it is not enough to validate a different one.

---

5. What stays

- Fixed transparent readiness score.
- Sleep/energy floor.
- Per-lift readiness as a downward-only advisory.
- Descriptive readiness–performance monitoring on the progress dashboard.
- No learned weights, no automatic main-lift volume change from readiness.

READINESS — how the daily signal is derived

Readiness is the one input that can reduce a prescribed load, so its derivation is written down here in full. Two parts are **rules** (fixed in code, because getting them wrong corrupts history) and two are **preferences** (config, because a user may reasonably want them different).

```
readiness.csv
   │  every cell kept as text (io_utils)          ← no type inference, no silent NaN
   ▼
coerce to numbers, map null tokens to missing
   │
   ▼
zero → missing for fields where 0 is impossible   ← per-field sentinel policy
   │
   ▼
merge duplicate rows per date                     ← most complete; ties to the latest
   │
   ▼
readiness_day  (one row per date, append-aware)
   │
   ▼
MOD-FAT-001 reads the configured fields           ← sleep, energy by default
   │
   ▼
if any configured field ≤ threshold → load × 0.95, cited with readiness:<date>
```

---

1. Input

Today readiness arrives only from the exported `readiness.csv`: one row per recording, and several rows can share a date. Every cell is read as text and stored verbatim in `raw_row` before anything is interpreted.

Manual entry (daily, occasional, or partial) feeds the same merge. A manual value wins **per field**; the imported row fills the fields it does not provide (`DEC-READINESS-03`). The old `squat_readiness` / `bench_readiness` / `deadlift_readiness` columns are retained only as calibration labels; they are not accepted as manual input (`DEC-READINESS-06`). Manual readings are appended to `readiness_entry`, never written into `readiness_day`, so a correction preserves the earlier value.

2. Missing values: the sentinel policy

The export uses `0` for "not measured" in several columns, and a physical zero is impossible there. Those fields become missing, never zero:

field	why 0 is missing
`Weight`	0 kg bodyweight is impossible
`HRV` , `RHR`	0 ms / 0 bpm are incompatible with life
`Fat_Percent`	0% body fat is impossible
`Squat_Readiness` , `Bench_Readiness` , `Deadlift_Readiness`	the 1-5 scale has no zero bucket

The 1–5 fields (`Sleep` , `Energy` , `Mood` , `Soreness` , `Stress` , `Nutrition` , `Hydration`) have **no** sentinel: a zero would be an invalid entry, not a missing one, and is treated as unparseable rather than silently converted.

This is `numeric.ZERO_IS_MISSING` and `numeric.ZERO_IS_VALID`; the split exists because the same `0` means different things in different columns (AGENTS.md §5).

3. Duplicate rows on one date

Several rows for a date usually **conflict** rather than repeat. The documented rule (`DEC-READINESS-01`, OPEN-QUESTIONS Q6):

1. count how many fields each row actually measured (after sentinel handling);
2. keep the **most complete** row;
3. on a tie, keep the **later** row in the file — the only ordering the export provides.

`readiness_day.merge_note` records how many candidates there were, which one won, and why. The assumption that completeness equals authority is `ASSUMPTION-04`; it is a choice, not a measurement.

4. Storage

One row per date in `readiness_day`, with the 1–5 fields, the device fields (HRV, RHR, bodyweight, body fat, wearable score), and a nullable `composite`. The legacy `grip_strength_kg` column remains for historical rows but is no longer accepted or scored; grip is tracked with `measure --metric grip_strength_kg` (`DEC-GRIP-01`).

`readiness_day.composite` is deliberately null. The computed score is a separate `readiness_score` assessment (`docs/READINESS-SCORE.md`), so the imported row stays a measurement record. Its HRV/RHR baseline and optional wearable blend are explicit and shown with the result; they are not written back into `readiness_day.composite`.

Manual readings live in `readiness_entry`: one row per field, append-only, with source `manual`, a timestamp and an optional note. Re-logging a date appends; the latest entry per (date, field) wins. `queries.readiness_day` overlays those entries on the imported row and returns a `sources` map and a `manual_merge_note` with both values, so a partial manual day keeps the imported fields around it and the override is auditable.

5. What actually moves a load

The implemented score reads the five core subjective fields (sleep, energy, mood, stress, soreness) and needs at least three of them plus sleep or energy. HRV and RHR enter only with a 28-day baseline (at least 7 readings) at half weight; an optional wearable score is a 10% blend. It produces a global score and per-category scores; overrides are 0.00–1.00. The old `MOD-FAT-001` configured-field rule remains as a **floor**:

```
modifier:
  fields: [sleep, energy]
  at_or_below: 2
```

Higher is better on every readiness scale, so "at or below" is the risk side. `Stress = 5` means no stress; a naive `stress >= 4` reading would be inverted.

When a field crosses the threshold the prescribed loads are multiplied by the configured fatigue factor (0.95), rounded down, and every touched item cites `MOD-FAT-001` and `readiness:<date>`.

There is also a conservative **set-reduction valve**: when the global score is below `set_reduction.score_below`, or the sleep/energy floor fires, each item loses `set_reduction.sets` (minimum one set). `include_essential` is now **true** by default, so main lifts lose one set as well as accessories; set it false to keep main-lift volume fixed. The warning names the affected sets, and the prescription notes record the setting. This is fatigue management, not a proven strength optimisation.

Missing is not "fine". A current/future workout is refused until readiness is "ready"; missing or insufficient data is reported as such. Historic dates remain reproducible. The old sleep/energy floor still applies when a score cannot be formed.

6. Changing it

Everything a user might reasonably want different is in `config/readiness.v1.yaml`:

key	effect
`modifier.fields`	which readiness fields can downgrade a session
`set_reduction`	whether poor readiness removes sets, at what score, how many, and whether essentials are included
`modifier.at_or_below`	the threshold on those fields
`merge.strategy` / `merge.tie_break`	which duplicate row wins; only the implemented values are accepted
`score.baseline_days` / `score.baseline_min_readings`	HRV/RHR baseline window and minimum
`score.hrv_weight` / `score.rhr_weight`	device-field weights (default 0.5)
`score.wearable_weight`	optional wearable blend (default 0.10)
`composite.enabled`	reserved; stays off until metrics are declared

An unimplemented value raises rather than being ignored — a config that silently did nothing would leave the athlete believing they had changed the behaviour.

Changing a modifier field or threshold **changes a prescribed load**, so it is a decision, not a preference of convenience: record it in `docs/DECISIONS.md` when you do.

7. What is deliberately not done

- The global/category score is implemented in `READINESS-SCORE.md`; `readiness_day.composite` itself stays null. The old sleep/energy trigger remains a floor.
- Grip strength stays a capacity trend, not a daily score input.
- The wearable score is not used to override constraints or the injury ceiling.
- No inference of readiness from training data, sleep reports or anything else: absent means absent.

REFERENCES — the scientific sources behind the rules

A consolidated bibliography of the papers retrieved while building the evidence notes. Every entry was fetched during this work (Europe PMC / publisher records) and has a DOI. Where a paper informed a decision, the "used for" column says which.

Dates are the publication dates in the retrieved record (some are 2026 issues). This is a reference list, not an endorsement: study quality varies, and the evidence notes say where a conclusion is weaker than it looks.

1. Proximity to failure, RIR and effort

#	Reference	DOI	Used for
1	Robinson ZP, Pelland JC, Remmert JF, Refalo MC, Jukic I, Steele J, Zourdos MC. Exploring the Dose-Response Relationship Between Estimated Resistance Training Proximity to Failure, Strength Gain, and Muscle Hypertrophy: A Series of Meta-Regressions. Sports Medicine, 2024. PMID 38970765.	10.1007/s40279-024-02069-2	Strength gains similar across a wide RIR range; hypertrophy improves closer to failure. Basis for widening the window to RIR 1-4.
2	Refalo MC, Helms ER, Trexler ET, Hamilton DL, Fyfe JJ. Influence of Resistance Training Proximity-to-Failure on Skeletal Muscle Hypertrophy: A Systematic Review with Meta-analysis. Sports Medicine, 2023. PMID 36334240.	10.1007/s40279-022-01784-y	No hypertrophy advantage for momentary failure over non-failure; no advantage of >25% over 20-25% velocity loss.
3	Wu S, Xie L, Zhao Z, Zheng X, Yin X, Hou J, Zhou K, Tang P. Effects of resistance training performed to repetition non-failure on exercise performance in healthy adults: a systematic review and meta-analysis. BMC Sports Science, Medicine and Rehabilitation, 2026. PMID 42410632.	10.1186/s13102-026-01861-z	Non-failure slightly better for dynamic strength; no difference for hypertrophy, endurance or power.
4	Vieira AF, Umpierre D, Teodoro JL, Lisboa SC, Baroni BM, Izquierdo M, Cadore EL. Effects of Resistance Training Performed to Failure or Not to Failure on Muscle Strength, Hypertrophy, and Power Output: A Systematic Review With Meta-Analysis. J Strength Cond Res, 2021;35(4):1165-1175. PMID 33555822.	10.1519/JSC.0000000000003936	No overall strength difference; hypertrophy advantage for failure disappeared with equalised volume.
5	Vasconcelos T, Alves AR, Refalo MC, Lucas GC, Vasconcelos PR, Paulo Brito J, Casanova F, Vilaça-Alves J, Oliveira R. Elbow Flexor Adaptations to Failure Versus Repetitions-in-Reserve Training: A Unilateral Within-Participant Study in Resistance-Trained Adults. Muscles, 2026. PMID 42784046.	10.3390/muscles5030061	1-3 RIR produced hypertrophy and strength comparable to failure.
6	Hermann T, Mohan AE, Enes A, Sapuppo M, Piñero A, Zamanzadeh A, Roberts M, Coleman M, Korakakis PA, Wolf M, Refalo M, Swinton PA, Schoenfeld BJ. Without Fail: Muscular Adaptations in Single-Set Resistance Training Performed to Failure or with Repetitions-in-Reserve. Med Sci Sports Exerc, 2025. PMID 40249908.	10.1249/MSS.0000000000003728	Failure may modestly favour some hypertrophy measures, but strength was similar at 2 RIR.

#	Reference	DOI	Used for
7	Wiedenmann T, Kunellis D, Rappelt L, Held S, Wicker P, Donath L. Trainability of repetitions-in-reserve estimation in younger and older adults: a parallel-group repeated-measures study. BMC Sports Science, Medicine and Rehabilitation, 2026. PMID 42632893.	10.1186/s13102-026-01997-y	RIR estimation error decreases with practice (up to 2.3 reps); supports a buffer rather than a tight veto.
8	Ferrari L, Pochetti M, Vialetto L, Bochicchio G. Estimating Repetitions in Reserve for Resistance Exercise: Effect of Exercise Intensity and Age on Prediction Accuracy. Preprint, 2026.	10.21203/rs.3.rs-8899476/v1	RIR 3 estimated well at 80% 1RM, poorly at 60%; error is load-dependent.
9	Qin X, Liu B, García-Ramos A. Gauging proximity to failure in the bench press: generalized velocity-based vs. %1RM-repetitions-to-failure approaches. BMC Sports Science, Medicine and Rehabilitation, 2025. PMID 40133968.	10.1186/s13102-025-01098-2	RIR-velocity least biased of the general approaches, but still ~±1.3 reps.
10	Chen W, Zhang H, Li R, Chen Z, Zheng J, Zhang X, Li Z. Velocity-Based Monitoring Lacks Both Accuracy and Applicability for Estimating Repetitions in Reserve During the Hexagonal Bar Deadlift. J Strength Cond Res, 2026. PMID 42328880.	10.1519/JSC.0000000000005549	No condition met the RIR-from-velocity accuracy criteria for the hex bar; basis for withholding trap-bar VBT targets.
11	Cowley N, Refalo M, Nicholson V, Timmins R, Munteanu G, Weakley J. Effects of different resistance training loads, exercises, and volume-regulated proximities to failure on acute and short-term neuromuscular, perceptual, and metabolic responses. Journal of Sports Sciences, 2026. PMID 42458937.	10.1080/02640414.2026.2703381	Fatigue rises as sets approach failure; supports keeping a buffer.
12	Varela-Olalla D, Del Campo-Vecino J, Balsalobre-Fernández C. Influence of Proximity to Failure, Relative Intensity, and Volume on Voluntary Performance and Fatigue Symptoms After Resistance Training: A Systematic Review. J Strength Cond Res, 2025;39(9):e1129-e1168. PMID 40644670.	10.1519/JSC.0000000000005194	Proximity to failure and volume are the main determinants of acute fatigue.

2. Velocity loss and VBT

#	Reference	DOI	Used for
13	Jukic I, Castilla AP, Ramos AG, Van Hooren B, McGuigan MR, Helms ER. The Acute and Chronic Effects of Implementing Velocity Loss Thresholds During Resistance Training: A Systematic Review, Meta-Analysis, and Critical Evaluation of the Literature. Sports Medicine, 2023.	10.1007/s40279-022-01754-4	Velocity-loss thresholds trade volume against fatigue; higher loss is not automatically better.
14	Zhang X, Feng S, Li H. The Effect of Velocity Loss on Strength Development and Related Training Efficiency: A Dose-Response Meta-Analysis. Healthcare, 2023;11(3):337.	10.3390/healthcare11030337	20-30% velocity loss associated with best strength outcomes; lower loss more efficient.
15	Claassen N, Siegel SD, Sproll M, Lebelt N, Bargende AV, Fasold AM, Warneke K. Reliability, Device Agreement and Validity of Load-Velocity Profiles: A Systematic Review with Meta-analysis. Sports Medicine - Open, 2026.	10.1186/s40798-026-01102-0	Velocity sensors are reliable (ICC ~0.90-0.91), but 1RM prediction from general profiles carries meaningful error.
16	Serrano-Cañadillas C, Jiménez-Lozano A, Alonso-Escribano A, Sánchez-Medina L, González-Badillo JJ, Rodríguez-Rosell D. Analysis of the Load-Velocity Relationship in the Full Squat Exercise in Female Soccer Players of Different Ages. J Strength Cond Res, 2026.	10.1519/JSC.0000000000005624	Mean velocity explains ~96% of variance in relative load in the squat.

#	Reference	DOI	Used for
17	Huang Y, García-Ramos A, Chen Z, Li Z, Yu J, Geng Y, Li D, Jia B, Lam WK. The Effect of Weightlifting Belt on the Relative Load-Velocity Relationship and One-Repetition Maximum Prediction Accuracy in the Free-Weight Back Squat. J Strength Cond Res, 2026.	10.1519/JSC.0000000000005523	Belts change the load-velocity relationship; comparisons need like-for-like setup.
18	Behm DG, Konrad A, Nakamura M, Alizadeh S, Culleton R, Hadjizadeh Anvar S, Pearson LT, Ramirez-Campillo R, Sale DG. A narrative review of velocity-based training best practice: the importance of contraction intent versus movement speed. Applied Physiology, Nutrition, and Metabolism, 2025.	10.1139/apnm-2024-0136	Intent matters; velocity is not only a load proxy.
19	Stevens LJ, Carey DL, Gastin PB, James LP. Validity and Reliability of a 3-Point Load-Velocity Profile for Predicting Hex Bar Deadlift 1 Repetition Maximum. J Strength Cond Res, 2026.	10.1519/JSC.0000000000005628	Individual load-velocity profiles predict hex-bar 1RM well; reviewed, not used in a decision.
20	Janicijevic D, Senturk D, Akyildiz Z, Gu Y, García-Ramos A. Hexagonal Barbell Deadlift One-repetition Maximum Estimation Using Velocity Recordings. International Journal of Sports Medicine, 2024.	10.1055/a-2330-3001	Individual profiles estimate hex-bar 1RM; reviewed, not used.

3. Deadlift biomechanics and spinal cost

#	Reference	DOI	Used for
21	Swinton PA, Stewart A, Agouris I, Keogh JW, Lloyd R. A biomechanical analysis of straight and hexagonal barbell deadlifts using submaximal loads. J Strength Cond Res, 2011.	10.1519/JSC.0b013e3181e73f87	Trap bar produced lower peak lumbar, hip and ankle moments; supports trap bar cost 2 vs conventional 3.
22	Hanen NC, Ben Mansour K, Ertel GN, Duchene Y, Gauchard GC. Biomechanical analysis of conventional and sumo deadlift. Frontiers in Bioengineering and Biotechnology, 2025.	10.3389/fbioe.2025.1597209	Sumo had lower erector spinae activation and more upright posture than conventional, but greater frontal/transverse moments. Supports sumo cost 2, not a sumo-vs-trap ranking.
23	Stahl CA, Regni G, Tanguay J, McElfresh M, Trihy E, Diggin D, King DL. A Biomechanical Comparison of the Back Squat and Hexagonal Barbell Deadlift. J Strength Cond Res, 2024;38(5):815-824.	10.1519/JSC.0000000000004729	Hex bar involves greater forward lean than a back squat; reviewed for context.
24	Zhang J, Huang YC, Liao YH, Chen CY, Mundel T, Lieu FK, Tsai SC. Effects of free-weight resistance training based on hexagonal barbell deadlift in older women: A 24-week randomized controlled trial. Experimental Gerontology, 2026.	10.1016/j.exger.2026.113246	Trap-bar training was safe and improved trunk function in older women; reviewed, not used.

4. Strength, testing and mortality

#	Reference	DOI	Used for
25	Jochem C, Leitzmann M, Volaklis K, Aune D, Strasser B. Association Between Muscular Strength and Mortality in Clinical Populations: A Systematic Review and Meta-Analysis. J Am Med Dir Assoc, 2019.	10.1016/j.jamda.2019.05.015	Low strength associated with higher all-cause mortality (HR 1.80); supports keeping strength measurable.
26	LaMonte MJ, Hyde ET, Nguyen S, Castro E, Seguin-Fowler RA, Eaton CB, Miller CR, Di C, Stefanick ML, LaCroix AZ. Muscular Strength and Mortality in Women Aged 63 to 99 Years. JAMA Network Open, 2026.	10.1001/jamanetworkopen.2025.59367	Grip strength and chair-stand performance inversely associated with mortality; reviewed for context.

#	Reference	DOI	Used for
27	Rosenberger F, Schneider J, Schlueter K, Paratte JL, Wiskemann J. Vertebral fracture during one repetition maximum testing in a breast cancer survivor: A case report. Medicine, 2021.	10.1097/MD.0000000000025705	1RM testing carries non-zero risk in older/osteoporotic populations; supports the 3RM/5RM default and `--max` gate.
28	Deboutte J, Alcazar J, Riesbeck M, Walker S, Delecluse C, Van Roie E. Validity of the individualized load-velocity profile to predict one-repetition maximum on a pneumatic leg press device in adults aged 55-81 years. Experimental Gerontology, 2026.	10.1016/j.exger.2026.113174	Velocity-based 1RM estimation can be equivalent with an individual profile; supports submaximal testing.
29	Hammert WB, Kataoka R, Yamada Y, Sallberg RW, Kang A, Buckner SL, Loenneke JP. Non-Specific Strength Changes Between High- and Low-Load Isotonic Resistance Training: A Systematic Review and Meta-Analysis. Sports Medicine, 2026.	10.1007/s40279-025-02370-8	Strength is load-specific; supports keeping heavy/low-rep schemes for strength.
30	Lu W, Du Z, Zhou A. Fast and Medium Tempo Resistance Training with a Low Number of Repetitions in Trained Men: Effects on Maximal Strength and Power Output. Journal of Human Kinetics, 2023.	10.5114/jhk/161472	Low-rep (3) training at 85% improved strength and power; reviewed for context.
31	Ruple BA, Plotkin DL, Smith MA, et al. The effects of resistance training to near failure on strength, hypertrophy, and motor unit adaptations in previously trained adults. Physiological Reports, 2023.	10.14814/phy2.15679	Low vs high RIR training in trained adults; reviewed for context.

5. Periodization and autoregulation

#	Reference	DOI	Used for
32	Zhang Z, Ya X, Zhao X, Liu Z, Luo J, Liu Y, Bu Y. Comparison of linear and undulating periodization resistance training on athletic capacities and health promotion: a systematic review and meta-analysis. Frontiers in Public Health, 2026.	10.3389/fpubh.2026.1707627	Linear and undulating periodization comparable; DUP framed as variety/fatigue management, not superior.
33	Vargas-Molina S, García-Sillero M, Romance R, Petro JL, Jiménez-García JD, Bonilla DA, Kreider RB, Benítez-Porres J. Traditional and Undulating Periodization on Body Composition, Strength Levels and Physical Fitness in Older Adults. International Journal of Environmental Research and Public Health, 2022;19(8):4522.	10.3390/ijerph19084522	Older adults improved similarly under traditional and undulating training.
34	Hickmott LM, Chilibeck PD, Shaw KA, Butcher SJ. The Effect of Load and Volume Autoregulation on Muscular Strength and Hypertrophy: A Systematic Review and Meta-Analysis. Sports Medicine - Open, 2022.	10.1186/s40798-021-00404-9	Autoregulated load vs standardized percentage showed no significant strength difference; supports readiness as fatigue management.

6. Detraining and return to training

#	Reference	DOI	Used for
35	Yamashita D, Hirata K, Yamazaki K, Mujika I, Miyamoto N. Effect of two weeks of training cessation on concentric and eccentric knee muscle strength in highly trained sprinters. PLOS ONE, 2023.	10.1371/journal.pone.0288344	Two weeks of cessation already reduced fast concentric and eccentric strength; supports a re-entry rule at ~14 days.

#	Reference	DOI	Used for
36	Rønnestad BR, Hansen J, Hollan I, Spencer M, Ellefsen S. Impairment of Performance Variables After In-Season Strength-Training Cessation in Elite Cyclists. Int J Sports Physiol Perform, 2016.	10.1123/ijspp.2015-0372	Eight weeks of cessation reduced strength/power adaptations.
37	Grgic J. Use It or Lose It? A Meta-Analysis on the Effects of Resistance Training Cessation (Detraining) on Muscle Size in Older Adults. Int J Environ Res Public Health, 2022;19(21):14048.	10.3390/ijerph192114048	Muscle size largely retained for 12-24 weeks of cessation, lost at 31-52 weeks; supports not panicking after a short break.
38	Huang Y, Li C, Chen J, Wang Z, Zhao D, Yang L, Zhang Z, Jiang Y, Zhang X, He B, Liu Z. A Multidimensional Regression Model for Predicting Recurrence in Chronic Low Back Pain. Eur J Pain, 2025.	10.1002/ejp.4793	Recurrence is common (38% in this cohort); a symptom-free streak is weak evidence, not clearance.
39	Wongwitwichote K, Yu CWG, Mansfield M, Deane J, Falla D. Can physical and psychological factors predict pain recurrence or an exacerbation of persistent non-specific low back pain? A protocol for a systematic review and meta-analysis. BMJ Open, 2025.	10.1136/bmjopen-2024-096594	Background figure only (about two-thirds relapse within a year). This is a **protocol**, not a result; the primary cohort used instead is ref 40.

7. Injury, loading tolerance and recurrence

#	Reference	DOI	Used for
40	da Silva T, Mills K, Brown BT, Pocovi N, de Campos T, Maher C, Hancock MJ. Recurrence of low back pain is common: a prospective inception cohort study. J Physiother, 2019;65(3):159-165. PMID 31208917.	10.1016/j.jphys.2019.04.010	69% (95% CI 62-74) recurrence within 12 months, median 139 days; **>2 previous episodes predict recurrence**. Basis for scaling the review window with prior flares (`DEC-INJURY-02`).
41	Hayden JA, Ellis J, Ogilvie R, Malmivaara A, van Tulder MW. Exercise therapy for chronic low back pain. Cochrane Database Syst Rev, 2021;9:CD009790. PMID 34580864.	10.1002/14651858.CD009790.pub2	Moderate-certainty evidence that exercise is probably effective for chronic low back pain vs no treatment (pain MD -15.2/100). Supports measuring the response to loading rather than avoiding it.
42	Lopes AD, Rizzo RR, Hespanhol L, Costa LO, Kamper SJ. Exercise for patellar tendinopathy. Cochrane Database Syst Rev, 2025;5:CD013078. PMID 40421598.	10.1002/14651858.CD013078.pub2	Very-low-certainty evidence for pain vs no treatment; **no trial measured adverse events**. The knee link stays advisory (`DEC-INJURY-01`).
43	Rathleff MS, Graven-Nielsen T, Hölmich P, Winiarski L, Krommes K, Holden S, Thorborg K. Activity Modification and Load Management of Adolescents With Patellofemoral Pain: A Prospective Intervention Study Including 151 Adolescents. Am J Sports Med, 2019;47(7):1629-1637. PMID 31095417.	10.1177/0363546519843915	Load management plus pain monitoring produced 86% success at 12 weeks and 81% at 12 months. Supports a load-tolerance model for the knee region.
44	Silbernagel KG, Thomeé R, Eriksson BI, Karlsson J. Continued sports activity, using a pain-monitoring model, during rehabilitation in patients with Achilles tendinopathy: a randomized controlled study. Am J Sports Med, 2007;35(6):897-906. PMID 17307888.	10.1177/0363546506298279	Continuing tendon-loading activity under pain monitoring was not worse than six weeks of active rest. Supports pain-guided progression over rest, in tendinopathy (not the low back).

#	Reference	DOI	Used for
45	McDevitt AW, Young JL, Cleland JA, Hiefield P, Snodgrass SJ. Physical therapy interventions used to treat individuals with biceps tendinopathy: a scoping review. Braz J Phys Ther, 2024;28(1):100586. PMID 38219522.	10.1016/j.bjpt.2023.100586	Only 14 studies, mostly of modalities; "a dearth of evidence" for conservative management. Why the biceps link is the weakest of the three.
46	Boufadel P, Daher M, Lopez R, Fares MY, Lawand J, Khan AZ, Abboud JA. Return to Sport After Distal Biceps Tendon Repair: A Systematic Review. Am J Sports Med, 2025;53(7):1769-1778. PMID 39836380.	10.1177/03635465241295618	91.5% return to sport at a mean 6.3 months after repair (42 studies, 1093 patients). A surgical timeline, deliberately not modelled by the signal.

8. Readiness scoring

#	Reference	DOI	Used for
47	Saw AE, Main LC, Gastin PB. Monitoring the athlete training response: subjective self-reported measures trump commonly used objective measures: a systematic review. Br J Sports Med, 2016;50(5):281-291.	10.1136/bjsports-2015-094758	Subjective wellness is more sensitive to training load than most objective measures; supports a subjective core for a readiness score.
48	Hooper SL, Mackinnon LT, Howard A, Gordon RD, Bachmann AW. Markers for monitoring overtraining and recovery. Med Sci Sports Exerc, 1995;27(1):106-112.	10.1249/00005768-199501000-00019	Daily multi-item wellness tracked staleness in swimmers; a precedent for a composite, but with no validated weights.
49	Govus AD, Coutts A, Duffield R, Murray A, Fullagar H. Relationship Between Pretraining Subjective Wellness Measures, Player Load, and Rating-of-Perceived-Exertion Training Load in American College Football. Int J Sports Physiol Perform, 2018;13(9): 1204-1210.	10.1123/ijspp.2016-0714	Pretraining wellness relates to subsequent load; observational support for asking before training.
50	Knowles OE, Drinkwater EJ, Urwin CS, Lamon S, Aisbett B. Inadequate sleep and muscle strength: Implications for resistance training. J Sci Med Sport, 2018;21(9):936-941.	10.1016/j.jsams.2018.01.012	Sleep restriction can reduce maximal strength in compound movements; supports sleep as a readiness input.
51	Manresa-Rocamora A, Sarabia JM, Javaloyes A, Flatt AA, Moya-Ramón M. Heart Rate Variability-Guided Training for Enhancing Cardiac-Vagal Modulation, Aerobic Fitness, and Endurance Performance: A Methodological Systematic Review with Meta-Analysis. Int J Environ Res Public Health, 2021;18(19):10299.	10.3390/ijerph181910299	HRV-guided training improved vagal indices but performance effects were small/non-significant; RHR was not improved. Supports trend/ context, not a hard gate.
52	Schaffarczyk M, Sperlich B. Heart rate variability-guided endurance training: evaluating strengths, weaknesses, opportunities, and threats for load prescription and adjustment. Front Sports Act Living, 2026.	10.3389/fspor.2026.1858271	No standard multimodal decision framework for HRV; supports transparent, configurable thresholds rather than a single formula.
53	Munteanu G, Halson SL, Huynh M, Jukic I, García-Ramos A, Fernandez F, Cowley N, Weakley J. The Effect of Menstrual Cycle Phase, Symptoms, Motivation, and Readiness to Perform on Resistance Training Performance. Sports Med, 2026.	10.1007/s40279-026-02459-8	Perceived readiness/motivation can relate to resistance-training performance, but the study is small and not a per-lift scale.

#	Reference	DOI	Used for
54	Bell L, Ruddock A, Boriel J, Maden-Wilkinson T, Thompson SW, Wright KJ, Burke K, Rogerson D. Effects of a 5-Day Back Squat Overreaching Protocol on Strength Performance, Perceived Recovery and Wellness Responses: A Pilot Trial. J Funct Morphol Kinesiol, 2025;10(2):227.	10.3390/jfmk10020227	Hooper Wellness Index and perceived recovery tracked a squat overreaching block; supports multi-item wellness as a trend marker, in a small pilot.
55	Silder A, Zifchock R, Brown L, Sessoms P, Jones D. The Association Between Grip Strength, Upper Body Power, and Limb Dominance in a Military Population. Mil Med, 2024.	10.1093/milmed/usae227	Grip strength relates to upper-body power; a capacity/ strength marker, not evidence of daily readiness.
56	Long H, Li L, Zhang L. Prioritizing next-day readiness decline in elite female soccer using readiness, workload, and wellness data: A temporally validated and calibrated monitoring framework. PLOS ONE, 2026.	10.1371/journal.pone.0358036	Readiness decline can be modelled from readiness, workload and wellness, but the outcome is next-day self-report and the sport is soccer; supports a monitoring score, not a lifting prescription.

| 57 | Toohey LA, Drew MK, Cook JL, Finch CF, Gaida JE. Is subsequent lower limb injury associated with previous injury? A systematic review and meta-analysis. Br J Sports Med, 2017;51(23):1670-1678. | 10.1136/bjsports-2017-097500 | Previous injury is a risk factor for subsequent lower-limb injury; ACL history was associated with later hamstring injury (RR 2.25). Supports an attention indicator based on history, not a probability. | | 58 | Lou S, Zhu H, He Z, Ye B, Sun G, Shen C. The application of machine learning in the prediction of sports injuries: systematic review and meta-analysis. Sci Rep, 2026. | 10.1038/ s41598-026-59571-y | Pooled sensitivity 0.79 and specificity 0.71 across 10 small models, with substantial methodological differences; not ready for individual prediction. | | 59 | Yuan J, Zeng Q, Wang A, Zhang Y, Li J. Machine Learning Applications in Non-Contact Lower Limb Sports Injury Prediction: A Systematic Review. J Sports Sci Med, 2026. | 10.52082/jssm.2026.172 | AUCs up to 0.91 in 15 heterogeneous studies; calls for large prospective validation. Supports caution, not a deployed predictor. | | 60 | Impellizzeri FM, Tenan MS, Kempton T, Novak A, Coutts AJ. Acute:Chronic Workload Ratio: Conceptual Issues and Fundamental Pitfalls. Int J Sports Physiol Perform, 2020;15(6):907-913. | 10.1123/ijspp.2019-0864 | ACWR has not been shown to be causal for injury; manipulating it is an overinterpretation. Why load-spike language stays descriptive. | | 61 | Herr NL, Bunn JA, Williams ML. Workload and Wellness as Risk Factors for Anterior Cruciate Ligament Injury in a Female Collegiate Lacrosse Athlete: A Level 3 Clinical CASE Report. J Athl Train, 2026. | 10.4085/1062-6050-0619.25 | A single case report linking a load-reload pattern and poor wellness to an ACL injury; hypothesis-generating only. |

| 62 | Hickmott LM, Butcher SJ, Chilibeck PD. Conceptualizing a load and volume autoregulation integrated velocity model to minimize neuromuscular fatigue and maximize neuromuscular adaptations in resistance training. Eur J Appl Physiol, 2025. | 10.1007/s00421-025-05709-1 | Autoregulation adjusts load/volume from readiness and performance; no numeric stacking cap is defined. Supports a guardrail, not a precise cap. | | 63 | Suchomel TJ, Nimphius S, Bellon CR, Hornsby WG, Stone MH. Training for Muscular Strength: Methods for Monitoring and Adjusting Training Intensity. Sports Med, 2021;51(10):2121-2143. | 10.1007/ s40279-021-01488-9 | Reviews RIR/RPE, VBT and autoregulatory methods for adjusting intensity; supports using RIR/load history as calibration features, but gives no universal cap. |

9. Fatigue modelling and cardio load

#	Reference	DOI	Used for
64	Hellard P, Avalos M, Lacoste L, Barale F, Chatard JC, Millet GP. Assessing the limitations of the Banister model in monitoring training. J Sports Sci, 2006;24(5):509-520.	10.1080/02640410500244697	The Banister model fits performance but its parameters are ill-conditioned/unstable; a good fit is not identification.
65	Kontro H, Mastracci A, Cheung SS, MacInnis MJ. The three-dimensional impulse-response model: Modeling the training process in accordance with energy system-specific adaptation. PLOS ONE, 2026.	10.1371/journal.pone.0341721	A single impulse assumes training types are interchangeable; proposes system-specific dimensions. Basis for not using one global fatigue scalar.
66	Piatrikova E, Willsmer NJ, Altini M, Jovanović M, Mitchell LJG, Gonzalez JT, Sousa AC, Williams S. Monitoring the Heart Rate Variability Responses to Training Loads in Competitive Swimmers Using a Smartphone Application and the Banister Impulse-Response Model. Int J Sports Physiol Perform, 2021;16(10):1471-1480.	10.1123/ijspp.2020-0201	HRV, sRPE and wellness can feed an impulse-response model; not a validated individual prescription.
67	Wang W, Zhao Y, Hou X, Yu W, Ren H, Ren H. Advancing training effectiveness prediction in mass sport through longitudinal data: A mathematical model approach based on the Fitness-Fatigue Model. PLOS ONE, 2025.	10.1371/journal.pone.0337824	Fitness-fatigue models need individual longitudinal data; generic parameters are a prior.

10. Wearable readiness composites

#	Reference	DOI	Used for
68	Spetz L, Rogestedt J, Nilsson R, Mattsson CM, Larsen FJ. Validating subjective ratings with wearable data for a nuanced understanding of load-recovery status in elite endurance athletes. Sports Med Open, 2025.	10.1186/s40798-025-00958-y	Wearable metrics are useful context but do not replace subjective load-recovery ratings; supports a low-weight blend.
69	Khodr R, Kamal L, Minerbi A, Gupta G. Accuracy, utility and applicability of the WHOOP wearable monitoring device in health, wellness and performance — a systematic review. medRxiv, 2024 (preprint).	10.1101/2024.01.04.24300784	Sleep/HR accuracy is acceptable, but the proprietary recovery score is not a validated readiness measure; supports bounded use.
70	Zhang P. Predicting athletic performance in track and field athletes based on wearable physiological and psychological indicators: an interpretable machine learning study. Front Physiol, 2026.	10.3389/fphys.2026.1912453	Wearable physiological/readiness indicators can relate to performance in groups, not an individual prescription.

11. Core and trunk training

#	Reference	DOI	Used for
71	Wang XQ, Zheng JJ, Yu ZW, Bi X, Lou SJ, Liu J, Cai B, Hua YH, Wu M, Wei ML, Shen HM, Chen Y, Pan YJ, Xu GH, Chen PJ. A meta-analysis of core stability exercise versus general exercise for chronic low back pain. PLoS One, 2012;7(12):e52082. PMID 23284879.	10.1371/journal.pone.0052082	Short-term benefit over general exercise, **no difference at 6 months**. No case for core as a specific treatment.

#	Reference	DOI	Used for
72	Saragiotto BT, Maher CG, Yamato TP, Costa LOP, Menezes Costa LC, Ostelo RW, Macedo LG. Motor control exercise for chronic non-specific low-back pain. Cochrane Database Syst Rev, 2016;1:CD012004. PMID 26742533.	10.1002/14651858.CD012004	MCE is **not superior to other forms of exercise** for chronic LBP; exercise choice is preference/cost/safety.
73	Ferreira GE, Lin CC, Stevens ML, Hancock MJ, Latimer J, Kelly P, Wisbey-Roth T, Maher CG. Exercise Is Medicine, But Perhaps Not for Preventing Low Back Pain: A Randomized Trial of Exercise and Education to Prevent Low Back Pain Recurrence. J Orthop Sports Phys Ther, 2021;51(4):188-195. PMID 33789433.	10.2519/jospt.2021.10187	12 weeks of exercise and education did **not** reduce recurrence vs an education booklet (63% vs 57%; HR 1.09). Core work must not be sold as back insurance.
74	Kayantaş İ. The effects of core training on athletic performance (speed and jumping) in different sports: a meta-analysis. Front Physiol, 2026;17:1946447. PMID 42890561.	10.3389/fphys.2026.1946447	49 studies; positive effects on speed (ES 0.60) and jumping (0.49). Supports some core work for performance, not a claim about barbell strength.
75	Lopes TJA, Simic M, Chia L, Terra BS, Alves DS, Bunn PDS, Rodrigues AI, Lima MDS, Ribeiro FM, Vilão P, Pappas E. Trunk endurance, posterior chain flexibility, and previous history of musculoskeletal pain predict overuse low back and lower extremity injury: a prospective cohort study of 545 Navy Cadets. J Sci Med Sport, 2021;24(6):555-560. PMID 33358586.	10.1016/j.jsams.2020.11.020	Failing a 60-s plank was associated with overuse injury (OR 3.3, 95% CI 1.2-8.8). Observational; does not validate plank training as prevention.
76	Ellestad SH, Holcomb TP, Swiergol AM, Holmstrup ME, Dicus JR. The Quantification of Muscle Activation During the Loaded Carry Movement Pattern. Int J Exerc Sci, 2024;17(4):480-490. PMID 38665162.	10.70252/nwue9985	Farmer's and suitcase carries produce high bilateral trunk EMG. Mechanistic support for a carry as core work.
77	Aleais DAS, Sullivan K, Ferreira P, Marchetti PN, Marchetti PH. Acute Dose-Response of Duration During the Isometric Forearm Plank Exercise on Muscle Thickness, Echo-Intensity, Peak Force, and Perception of Effort in Recreationally-Trained Participants. Int J Exerc Sci, 2022;15(4):676-685. PMID 35991937.	10.70252/rdkt2873	Only 3-minute planks produced acute swelling/ fatigue and a force drop; 1- and 2-minute holds did not. Argues for 30-60 s holds progressed by load/leverage.

12. Longevity, aerobic fitness and physical function

#	Reference	DOI	Used for
78	Momma H, Kawakami R, Honda T, Sawada SS. Muscle-strengthening activities are associated with lower risk and mortality in major non-communicable diseases: a systematic review and meta-analysis of cohort studies. Br J Sports Med, 2022;56(13):755-763. PMID 35228201.	10.1136/bjsports-2021-105061	16 cohorts; 10-17% lower all-cause mortality, largest reduction around 30-60 min/ week (J-shaped); combined aerobic + strengthening better than either.
79	Kodama S, Saito K, Tanaka S, Maki M, Yachi Y, Asumi M, Sugawara A, Totsuka K, Shimano H, Ohashi Y, Yamada N, Sone H. Cardiorespiratory fitness as a quantitative predictor of all-cause mortality and cardiovascular events in healthy men and women: a meta-analysis. JAMA, 2009;301(19):2024-2035. PMID 19454641.	10.1001/jama.2009.681	Each 1-MET higher fitness = RR 0.87 for all-cause mortality (13% lower); low vs high fitness RR 1.70.

#	Reference	DOI	Used for
80	Mandsager K, Harb S, Cremer P, Phelan D, Nissen SE, Jaber W. Association of Cardiorespiratory Fitness With Long-term Mortality Among Adults Undergoing Exercise Treadmill Testing. JAMA Netw Open, 2018;1(6):e183605. PMID 30646252.	10.1001/jamanetworkopen.2018.3605	122,007 patients; inverse, graded relation with no upper limit; elite vs low fitness HR 0.20.
81	Bull FC, Al-Ansari SS, Biddle S, et al. World Health Organization 2020 guidelines on physical activity and sedentary behaviour. Br J Sports Med, 2020;54(24):1451-1462. PMID 33239350.	10.1136/bjsports-2020-102955	150-300 min moderate or 75-150 min vigorous aerobic activity plus 2+ muscle-strengthening days per week.
82	Sherrington C, Fairhall NJ, Wallbank GK, Tiedemann A, Michaleff ZA, Howard K, Clemson L, Hopewell S, Lamb SE. Exercise for preventing falls in older people living in the community. Cochrane Database Syst Rev, 2019;1:CD012424. PMID 30703272.	10.1002/14651858.CD012424.pub2	108 RCTs; balance and functional exercise reduces the number of people who fall by ~22% (RR 0.78). Evidence is in adults 60+.
83	Balachandran AT, Steele J, Angielczyk D, Belio M, Schoenfeld BJ, Quiles N, Askin N, Abou-Setta AM. Comparison of Power Training vs Traditional Strength Training on Physical Function in Older Adults: A Systematic Review and Meta-analysis. JAMA Netw Open, 2022;5(5):e2211623. PMID 35544136.	10.1001/jamanetworkopen.2022.11623	Power training modestly improves physical function vs traditional strength training (SMD 0.30, low certainty) in adults 60+.
84	El Hadouchi M, Kiers H, de Vries R, Veenhof C, van Dieën J. Effectiveness of power training compared to strength training in older adults: a systematic review and meta-analysis. Eur Rev Aging Phys Act, 2022;19:18. PMID 35953775.	10.1186/s11556-022-00297-x	15 trials; power training better for muscle power (SMD 0.99) and activity tests.
85	Helgerud J, Høydal K, Wang E, Karlsen T, Berg P, Bjerkaas M, Simonsen T, Helgesen C, Hjorth N, Bach R, Hoff J. Aerobic high-intensity intervals improve VO2max more than moderate training. Med Sci Sports Exerc, 2007;39(4):665-671. PMID 17414804.	10.1249/mss.0b013e3180304570	4×4 min at 90-95% HRmax improved VO2max more than matched work at 70% or lactate-threshold intensity.
86	Ferraro-Farro D, Bandeira-Guimarães M, Blanco-Rambo E, Vieira AF, Cadore EL, Benítez-Flores S. Does Sprint Interval Training Cause Interference in Concurrent Training? A Meta-Analysis Study. Int J Sports Med, 2026;47(9):747-759. PMID 41734815.	10.1055/a-2820-4527	Nine RCTs; sprint intervals plus resistance training did not blunt strength/power and raised VO2max (SMD 0.78). Supports a small interval dose alongside lifting.
87	Morton RW, Murphy KT, McKellar SR, Schoenfeld BJ, Henselmans M, Helms E, Aragon AA, Devries MC, Banfield L, Krieger JW, Phillips SM. A systematic review, meta-analysis and meta-regression of the effect of protein supplementation on resistance training-induced gains in muscle mass and strength in healthy adults. Br J Sports Med, 2018;52(6):376-384. PMID 28698222.	10.1136/bjsports-2017-097608	49 trials; protein supplementation beyond ~1.62 g/kg/day produced no further fat-free-mass gain.

#	Reference	DOI	Used for
88	Celis-Morales CA, Welsh P, Lyall DM, Steell L, Petermann F, Anderson J, Iliodromiti S, Sillars A, Graham N, Mackay DF, Pell JP, Gill JMR, Sattar N, Gray SR. Associations of grip strength with cardiovascular, respiratory, and cancer outcomes and all cause mortality: prospective cohort study of half a million UK Biobank participants. BMJ, 2018;361:k1651. PMID 29739772.	10.1136/bmj.k1651	502,293 participants; higher grip strength associated with lower all-cause and cause-specific mortality. Prognostic marker, not a training target.

13. Vertical pull and power work

#	Reference	DOI	Used for
89	Doma K, Deakin GB, Ness KF. Kinematic and electromyographic comparisons between chin-ups and lat-pull down exercises. Sports Biomech, 2013;12(3):302-313. PMID 24245055.	10.1080/14763141.2012.760204	Direct comparison: chin-ups show greater biceps and erector activity, lat-pulldown greater rectus abdominis activity during the eccentric phase; chin-up is the more "functional" lift. Supports substitutable, not identical.
90	Johnson D, Lynch J, Nash K, Cygan J, Mayhew JL. Relationship of lat-pull repetitions and pull-ups to maximal lat-pull and pull-up strength in men and women. J Strength Cond Res, 2009;23(3): 1022-1028. PMID 19387371.	10.1519/JSC.0b013e3181a2d7f5	1RM lat-pull and 1RM pull-up correlate strongly (r = 0.78 in men) but are distinct strength expressions.
91	Li Q, Yan J, Qiao M, Quan J, Chen Y, Gong M, Niu W, Wang L. Eight-week lat pull-down resistance training with joint instability leads to superior pull-up endurance performance and reduced antagonist coactivation in recreationally active male college students. Eur J Sport Sci, 2025;25(1):e12243. PMID 39716392.	10.1002/ejsc.12243	Lat-pulldown training improved pull-up endurance and pulldown strength; supports transfer between the two.
92	Di Fonza D, Di Claudio G, Colantuono G, Persichini L, Cerulo N, Sangregorio B, Buonsenso A, Fiorilli G, Calcagno G, di Cagno A. Electromyographic Analysis of Latissimus Dorsi Activation During Common Resistance Training Exercises: A Narrative Review. J Funct Morphol Kinesiol, 2026;11(3):315. PMID 42647355.	10.3390/jfmk11030315	23 studies; pull-up variations show broadly similar latissimus activation across grips; lat-pulldown bar path matters. Acute EMG, not longitudinal outcomes.

14. Fatigue-fitness, recovery and e1RM error

#	Reference	DOI	Used for
93	Banister EW, Calvert TW, Savage MV, Bach TM. A systems model of training for athletic performance. Aust J Sports Med, 1975;7:57-61.	—	The original fitness-fatigue impulse-response model: performance = fitness − fatigue, with exponential decays.

#	Reference	DOI	Used for
94	Busso T, Häkkinen K, Pakarinen A, Carasso C, Lacour JR, Komi PV, Kauhanen H. A systems model of training responses and its relationship to hormonal responses in elite weight-lifters. Eur J Appl Physiol Occup Physiol, 1990;61:48-54. PMID 2289497.	10.1007/BF00236693	The direct strength application: six elite weight-lifters over one year; predicted and actual performance correlated, but the fatigue index was not clarified. Small sample.
95	Chiu LZ, Barnes JL. The fitness-fatigue model revisited: implications for planning short- and long-term training. Strength Cond J, 2003;25(6):42-51.	10.1519/1533-4295(2003)025<0042:TFMRIF>2.0.CO;2	Practical interpretation of the model for strength coaches; no new validation.
96	Vermeire KM, Van de Casteele F, Gosseries M, Bourgois JG, Ghijs M, Boone J. The Influence of Different Training Load Quantification Methods on the Fitness-Fatigue Model. Int J Sports Physiol Perform, 2021;16(9):1261-1269.	10.1123/ijspp.2020-0662	Different load metrics fit similarly but evolve differently; the day of peak positive effect moved by ~20 days. Load metric is not interchangeable.
97	Vermeire K, Ghijs M, Bourgois JG, Boone J. The Fitness-Fatigue Model: What's in the Numbers? Int J Sports Physiol Perform, 2022;17(5):810-813.	10.1123/ijspp.2021-0494	Parameters depend on starting values, fitting technique and load input; do not use general constants.
98	Marchal A, Benazieb O, Weldegebriel Y, Méline T, Imbach F. Statistical flaws of the fitness-fatigue sports performance prediction model. Sci Rep, 2025;15:3706.	10.1038/s41598-025-88153-7	Bayesian re-analysis of two datasets: non-identifiable parameters and the fatigue component did not improve out-of-sample prediction. Central reason not to let the model drive loads.
99	Morán-Navarro R, Pérez CE, Mora-Rodríguez R, de la Cruz-Sánchez E, González-Badillo JJ, Sánchez-Medina L, Pallarés JG. Time course of recovery following resistance training leading or not to failure. Eur J Appl Physiol, 2017;117:2387-2399.	10.1007/s00421-017-3725-7	Failure slowed recovery of jump height and bar velocity at 24-48 h (measured to 72 h) versus the same volume in sets of 5.
100	Sousa CA, Zourdos MC, Storey AG, Helms ER. The Importance of Recovery in Resistance Training Microcycle Construction. J Hum Kinet, 2024;91:205-223.	10.5114/jhk/186659	24-study review: failure, volume, lower-body/multi-joint, eccentric and long-length work increase recovery demands; wide individual variation.

#	Reference	DOI	Used for
101	Zourdos MC, Goldsmith JA, Helms ER, Trepeck C, Halle JL, Mendez KM, Cooke DM, Haischer MH, Sousa CA, Klemp A, Byrnes RK. Proximity to Failure and Total Repetitions Performed in a Set Influences Accuracy of Intraset Repetitions in Reserve-Based Rating of Perceived Exertion. J Strength Cond Res, 2021;35(2S):S158-S165.	10.1519/JSC.0000000000002995	RIR prediction error was ~2 reps at 1 RIR, ~3.7 at 3 RIR and ~5 at 5 RIR in the squat; e1RM from RIR carries that error.
102	Remmert JF, Laurson KR, Zourdos MC. Accuracy of Predicted Intraset Repetitions in Reserve (RIR) in Single- and Multi-Joint Resistance Exercises Among Trained and Untrained Men and Women. Percept Mot Skills, 2023;130(3):1239-1254.	10.1177/00315125231169868	RIR prediction error was not explained by sex, training experience or experience rating RIR; the error is a property of the method.
103	Greig L, Stephens Hemingway BH, Aspe RR, Cooper K, Comfort P, Swinton PA. Autoregulation in Resistance Training: Addressing the Inconsistencies. Sports Med, 2020;50(11):1873-1887.	10.1007/s40279-020-01330-8	Readiness, fatigue and autoregulation are defined inconsistently; performance- and perception-based adjustments are not interchangeable.

| 104 | Day ML, McGuigan MR, Brice G, Foster C. Monitoring exercise intensity during resistance training using the session RPE scale. J Strength Cond Res, 2004;18(2):353-358. PMID 15142026. | 10.1519/R-13113.1 | Session RPE was reliable across low, moderate and high resistance-training intensities (ICC 0.88); supports capturing one post-session number. | | 105 | Egan AD, Winchester JB, Foster C, McGuigan MR. Using Session RPE to Monitor Different Methods of Resistance Exercise. J Sports Sci Med, 2006;5(2):289-295. | — | Session RPE distinguished traditional, super-slow and power squat protocols; perceived effort is not a simple function of load. |

15. Per-lift readiness

#	Reference	DOI	Used for
106	Larsen S, Kristiansen E, van den Tillaar R. Effects of subjective and objective autoregulation methods for intensity and volume on enhancing maximal strength during resistance-training interventions: a systematic review. PeerJ, 2021;9:e10663.	10.7717/peerj.10663	14 studies: RIR/RPE-based autoregulation increased 1RM, but methods were heterogeneous. Supports effort as an input, not a per-lift readiness score.
107	Moura RE, Bezerra da Silva RF, Gomes LMS, Ramos da Silva JL, Henrique RDS, Sousa FAB, Fonseca FS. Monitoring Bar Velocity to Quantify Fatigue in Resistance Training. Int J Sports Med, 2024;45(8):624-632.	10.1055/a-2316-7966	Back-squat mean propulsive velocity was sensitive to acute fatigue, especially at 40% and 80% 1RM; supports a velocity component at a standardised load.
108	Ibrahim AH, Beaumont CT, Strohacker K. Implementing Meta-Session Autoregulation Strategies for Exercise - A Scoping Review. Int J Exerc Sci, 2024;17(4):382-404.	10.70252/dabq1369	11 studies: subjective readiness measures often lacked validity evidence and fidelity was rarely reported. Why the per-lift score is an assumption and downward-only.

16. Detraining and e1RM decay

#	Reference	DOI	Used for
109	Yang Y, Chen SC, Chen CN, Hsu CW, Zhou WS, Chien KY. Training Session and Detraining Duration Affect Lower Limb Muscle Strength Maintenance in Middle-Aged and Older Adults: A Systematic Review and Meta-Analysis. J Aging Phys Act, 2022;30(3):552-566.	10.1123/japa.2020-0493	Strength maintenance after detraining depends on training and detraining duration; not a per-week rate, which is why the decay is bounded and called an assumption.
110	Yılmaz U, Buzdağlı Y, Bilgin S, Bilgin S, Sepil Y, Ozan M, Bağaçlı UH, Şıktar E, Baygutalp N, Kılıç Baygutalp N. Multidimensional Physiologic Adaptations to a 14-Day Detraining Period in Elite Powerlifters: Neuromuscular, Morphological, and Hormonal Responses. J Strength Cond Res, 2026;40(9):1085-1094.	10.1519/JSC.0000000000005487	Two weeks of detraining altered neuromuscular performance while muscle size was preserved; supports a grace period before decaying an old e1RM.

17. Selection comparability and variant transfer

#	Reference	DOI	Used for
111	Morrissey MC, Harman EA, Johnson MJ. Resistance training modes: specificity and effectiveness. Med Sci Sports Exerc, 1995;27(5):648-660. PMID 7674868.	—	Classic specificity review: adaptations are specific to the training mode.
112	Saeterbakken AH, Stien N, Paulsen G, Behm DG, Andersen V, Solstad TEJ, Prieske O. Task Specificity of Dynamic Resistance Training and Its Transferability to Non-trained Isometric Muscle Strength: A Systematic Review with Meta-analysis. Sports Medicine, 2025.	10.1007/s40279-025-02225-2	Dynamic training transfers less to isometric tests than to dynamic tests; supports specificity.
113	Haugen ME, Vårvik FT, Larsen S, Haugen AS, van den Tillaar R, Bjørnsen T. Effect of free-weight vs. machine-based strength training on maximal strength, hypertrophy and jump performance — a systematic review and meta-analysis. BMC Sports Sci Med Rehabil, 2023.	10.1186/s13102-023-00713-4	Strength changes are specific to the tested modality; direct pooled differences are small.
114	Herrera-Bermudo JC, Puente-Alcaraz C, Díaz-Sánchez P, González-Badillo JJ, Rodríguez-Rosell D. Influence of Grip Width on the Load-Velocity Relationship and 1RM in the Bench Press. J Strength Cond Res, 2024;38(12):2013-2021.	10.1519/JSC.0000000000004920	Grip width changes the bench load-velocity relationship and 1RM.
115	Pérez-Castilla A, Jerez-Mayorga D, Martínez-García D, Rodríguez-Perea Á, Chirosa-Ríos LJ, García-Ramos A. Influence of Grip Width and Anthropometric Characteristics on the Bench-Press Load-Velocity Relationship. Int J Sports Physiol Perform, 2020.	10.1123/ijspp.2019-0549	The load-velocity relationship differs across bench grip widths.
116	Pérez-Castilla A, Jukic I, Haff GG, García-Ramos A. The Bench Press Grip Width Does Not Affect the Number of Repetitions Performed at Different Velocity Loss Thresholds. Int J Environ Res Public Health, 2021;18(3):1057.	10.3390/ijerph18031057	Counterpoint: fatigue/repetition response can be similar despite different load bases.
117	Spitz RW, Gonzalez AM, Ghigiarelli JJ, Sell KM, Mangine GT. Load-Velocity Relationships of the Back vs. Front Squat Exercises in Resistance-Trained Men. J Strength Cond Res, 2019;33(2):301-306.	10.1519/JSC.0000000000002962	Back and front squat do not share a load-velocity relationship.
118	Krzyszkowski J, Kipp K. Load-dependent mechanical demands of the lower extremity during the back and front squat. J Sports Sci, 2020.	10.1080/02640414.2020.1766738	Front/back squat mechanics differ by load; supports non-equivalence.

#	Reference	DOI	Used for
119	Johansson DG, Marchetti PH, Stecyk SD, Flanagan SP. A Biomechanical Comparison Between the Safety-Squat Bar and Traditional Barbell Back Squat. J Strength Cond Res, 2024;38(5):825-834.	10.1519/JSC.0000000000004719	SSB vs straight bar differs in kinematics, kinetics and load.
120	Vantrease WC, Townsend JR, Sapp PA, Henry RN, Johnson KD. Maximal Strength, Muscle Activation, and Bar Velocity Comparisons Between Squatting With a Traditional or Safety Squat Bar. J Strength Cond Res, 2021;35(2S):S1-S5.	10.1519/JSC.0000000000003541	SSB differs in strength, activation and velocity.
121	Hecker KA, Carlson LA, Lawrence MA. Effects of the Safety Squat Bar on Trunk and Lower-Body Mechanics During a Back Squat. J Strength Cond Res, 2019;33(7S):S45-S51.	10.1519/JSC.0000000000002912	SSB alters trunk/lower-body mechanics.
122	Heinecke ML, Mauldin ML, Hunter ML, Mann JB, Mayhew JL. Relationship of Barbell and Dumbbell Repetitions With One Repetition Maximum Bench Press in College Football Players. J Strength Cond Res, 2021;35(2S):S66-S71.	10.1519/JSC.0000000000003539	Barbell and dumbbell bench do not share a clean rep/1RM relationship.
123	Farias DA, Willardson JM, Paz GA, Bezerra ES, Miranda H. Maximal Strength Performance and Muscle Activation for the Bench Press and Triceps Extension Exercises Adopting Dumbbell, Barbell, and Machine Modalities Over Multiple Sets. J Strength Cond Res, 2017;31(7):1879-1887.	10.1519/JSC.0000000000001651	Bench modality changes performance, repetitions and activation.
124	Baz-Valle E, Schoenfeld BJ, Torres-Unda J, Santos-Concejero J, Balsalobre-Fernández C. The effects of exercise variation in muscle thickness, maximal strength and motivation in resistance trained men. PLOS ONE, 2019;14(12):e0226989.	10.1371/journal.pone.0226989	Random variation produced similar strength/hypertrophy but better motivation; no evidence for algorithmic selection.

18. Weakness/injury targeting and individual response

#	Reference	DOI	Used for
125	Robinson ZP, Helms ER, Trexler ET, Steele J, Hall ME, Huang CJ, Zourdos MC. N of 1: Optimizing Methodology for the Detection of Individual Response Variation in Resistance Training. Sports Medicine, 2024.	10.1007/s40279-024-02050-z	Individual response variation needs repeated within-person designs; group averages can hide different responses.
126	Rauch JT, Ugrinowitsch C, Barakat CI, et al. Auto-Regulated Exercise Selection Training Regimen Produces Small Increases in Lean Body Mass and Maximal Strength Adaptations in Strength-trained Individuals. J Strength Cond Res, 2020;34(4): 1133-1140.	10.1519/JSC. 0000000000002272	Small advantage for auto-regulated selection in one small study; not a weakness-targeting algorithm.
127	Parkinson AO, Apps CL, Morris JG, Barnett CT, Lewis MGC. The Calculation, Thresholds and Reporting of Inter-Limb Strength Asymmetry: A Systematic Review. J Sports Sci Med, 2021.	10.52082/jssm.2021.594	Asymmetry calculation, thresholds and reporting are inconsistent.
128	Helme M, Tee J, Emmonds S, Low C. The associations between unilateral leg strength, asymmetry and injury in sub-elite Rugby League players. Phys Ther Sport, 2023.	10.1016/j.ptsp.2023.05.003	No leg-strength measure was related to injury relative risk; only a weak association with time lost.

#	Reference	DOI	Used for
129	Tyler TF, Mullaney MJ, Mirabella MR, Nicholas SJ, McHugh MP. Risk Factors for Shoulder and Elbow Injuries in High School Baseball Pitchers: The Role of Preseason Strength and Range of Motion. Am J Sports Med, 2014.	10.1177/0363546514535070	Population-specific association between supraspinatus weakness and major injury; not a general selection rule.
130	Knapik JJ, Bauman CL, Jones BH, Harris JM, Vaughan L. Preseason strength and flexibility imbalances associated with athletic injuries in female collegiate athletes. Am J Sports Med, 1991.	10.1177/036354659101900113	Mixed historical evidence on preseason imbalance and injury.
131	Sprague AL, Couppé C, Pohlig RT, Snyder-Mackler L, Silbernagel KG. Pain-guided activity modification during treatment for patellar tendinopathy: a feasibility and pilot randomized clinical trial. Pilot Feasibility Stud, 2021.	10.1186/s40814-021-00792-5	Pain-guided modification is feasible in rehabilitation; not evidence for algorithm-selected exercise substitution.

19. Deadlift variants and feedback

#	Reference	DOI	Used for
132	Escamilla RF, Francisco AC, Kayes AV, Speer KP, Moorman CT. An electromyographic analysis of sumo and conventional style deadlifts. Med Sci Sports Exerc, 2002;34(4):682-688.	10.1097/00005768-200204000-00019	Sumo has more quadriceps/less gastrocnemius activity than conventional; both are legitimate variants.
133	Escamilla RF, Francisco AC, Fleisig GS, et al. A three-dimensional biomechanical analysis of sumo and conventional style deadlifts. Med Sci Sports Exerc, 2000;32(7):1265-1275.	10.1097/00005768-200007000-00013	Kinematic/kinetic differences; no injury-outcome ranking.
134	Hagerupsen K, Pedersen S, Giller NB, et al. Comparison of resistance training using barbell half squats and trap bar deadlifts on maximal strength, power performance, and lean mass in recreationally active females: an eight-week randomised trial. BMC Sports Sci Med Rehabil, 2024;16:124.	10.1186/s13102-024-00911-8	Similar adaptations when loads are heavy and target muscles are trained; exercise selection is secondary.
135	Weakley J, Cowley N, Schoenfeld BJ, et al. The Effect of Feedback on Resistance Training Performance and Adaptations: A Systematic Review and Meta-analysis. Sports Med, 2023;53(9):1789-1803.	10.1007/s40279-023-01877-2	Feedback improves acute performance and technical competency; does not mandate routine post-session review.

Not included

Search results that were unrelated to this project (e.g. cerebral autoregulation, blood-flow restriction, ACL rehabilitation, load carriage) were not retained. A reference here is not a claim that the study is definitive; the evidence notes state the limitations.

Where the references are used

- `docs/RIR-EVIDENCE.md` — effort window (1-4, target 2-3)
- `docs/TRAINING-DYNAMICS.md` — 1RM testing, rep ranges, DUP
- `docs/VBT-EVIDENCE.md` — velocity calibration and velocity loss
- `docs/SPINAL-COST.md` — ordinal spinal-cost ratings

- `docs/INJURY-SIGNAL.md` — recurrence, load tolerance and the injury-to-exercise links
- `docs/READINESS-SCORE.md` — global/per-lift readiness evidence and proposal
- `docs/CORE-TRAINING.md` — core/trunk training evidence and the programme gap
- `docs/LONGEVITY-REVIEW.md` — modality review against a longevity-first goal
- `ROADMAP.md` §4.13 — cardio load and impulse-fatigue model plan

RETROSPECTIVE WALK-FORWARD — method and result

Status: implemented 2026-10-20. This is retrospective, not prospective. **Command:** `strength-coach validate --max-dates 12` (also on the Prescribe page).

It changes no prescription, load or model parameter.

---

1. Method

For each evaluation date:

1. fit/model on data strictly before that date;
2. predict the next observed session outcome;
3. compare with a simple baseline;
4. move forward and repeat.

- **Continuity:** at each observed return, fit the continuity calibration on the prefix and compare its factor with the fixed rule and the actual e1RM ratio.
- **Fatigue:** at each cutoff, fit the strength fitness-fatigue model on the prefix and predict the next observed e1RM for a lift; baseline is persistence (last observed e1RM).
- **Volume:** no predictive volume model exists; only descriptive/causal limits are reported.
- **Selection:** no counterfactual exists (one exercise per slot), so it is excluded.

The report uses only earlier data per date, but the candidate models and features were chosen with knowledge of the full dataset. That is why it is retrospective.

2. Result (12 evaluation dates)

model	predictions	relative MAE	baseline MAE	skill
Strength fatigue	25 lift-sessions	0.158	0.074 persistence	**−112%**
Continuity calibration	12 returns	0.150	0.128 fixed rule	**−17%**

Both models are **worse than the simple baseline** in this retrospective check.

Volume has no predictive model; selection has no counterfactual.

3. Interpretation

- The fatigue model should **not** be adopted, and should not drive a load.
- The calibrated continuity factor should **not** replace the fixed fallback.
- No prospective shadow period is started for these candidates: they failed the retrospective screen first.
- This is a small, single-athlete, non-stationary record. The result is enough to reject adoption, not enough to prove the models can never work.
- If a future model beats these baselines retrospectively, the next step is a frozen prospective test, not immediate use.

4. What would count as a positive result

- a pre-declared improvement over the persistence/fixed baseline;
- a confidence interval excluding no difference;
- no safety or calibration failure;
- then a frozen prospective period with no retuning.

Until then: keep fixed continuity, keep the fatigue model shadow-only, and keep volume/selection decision rules conservative.

RIR EVIDENCE — why the effort window is 1-4 with a 2-3 target

The constitution used to hold primary compounds at RIR 2-3 and forbid RIR 0-1. The athlete asked for that window to be evidence-based rather than fixed. This note records the evidence and the resulting decision (`DEC-RIR-02`).

Evidence

Strength is similar across a wide range of RIR. A series of meta-regressions on estimated proximity to failure found that confidence intervals for the strength slope contained the null in every best-fit model: strength gains did not meaningfully depend on RIR (Robinson, Pelland, Remmert, Refalo, Jukic, Steele & Zourdos, Sports Medicine 2024, doi:10.1007/s40279-024-02069-2).

Failure is not required for strength, and may be slightly worse. A 2026 meta-analysis of 20 studies (556 participants) found non-failure training superior for dynamic strength (SMD 0.24, 95% CI 0.06-0.42) and equivalent for isometric strength, hypertrophy, endurance and power (Wu et al., BMC Sports Sci Med Rehabil 2026, doi:10.1186/s13102-026-01861-z). An earlier meta-analysis found no overall strength difference and a hypertrophy advantage for failure that disappeared once volume was equalized (Vieira et al., J Strength Cond Res 2021, doi:10.1519/JSC.0000000000003936).

Hypertrophy does respond to proximity to failure, but not enough to justify a hard floor. The 2024 meta-regression found hypertrophy increased as sets were taken closer to failure; a 2023 meta-analysis found no advantage for momentary failure over non-failure for hypertrophy (ES 0.12, 95% CI -0.13 to 0.37) (Refalo et al., Sports Medicine 2023, doi:10.1007/s40279-022-01784-y). Training at 1-3 RIR produced hypertrophy and strength comparable to failure in recent within-participant trials (Vasconcelos et al. 2026, doi:10.3390/muscles5030061).

RIR is self-reported and imprecise. Estimation error improves with practice — mean absolute error fell by up to 2.3 reps across six sessions — and the error is load-dependent: at 80% 1RM lifters estimate RIR 3 accurately, while at 60% 1RM they substantially underestimate how many reps remain (Wiedenmann et al. 2026, doi:10.1186/s13102-026-01997-y; Ferrari et al. 2026, doi:10.21203/rs.3.rs-8899476/v1). A tight 2-3 veto therefore punishes estimation error as much as effort.

Closer to failure costs more fatigue. Neuromuscular and perceptual fatigue rise as sets approach failure (Cowley et al. 2026, doi:10.1080/02640414.2026.2703381; Varela-Olalla et al., J Strength Cond Res 2025, doi:10.1519/JSC.0000000000005194). That matters for a longevity-first athlete with a back history.

No direct injury evidence. There is no good trial evidence that RIR 2-3 is safer than RIR 1-4. The fatigue and technique-degradation argument is plausible but is reasoning, not data.

Decision

`PRG-RIR-001` now:

- **allows RIR 1-4** on primary compounds;
- keeps **RIR 0 forbidden** (a grinding set is the failure mode the rule exists to prevent);
- keeps **max singles and 1RM testing inside a block forbidden**, except a declared test session (see `docs/TRAINING-DYNAMICS.md`);

- keeps **2-3 as the default target** — the prescription aims there, and the menu's "harder"/"easier" options move within 1-4.

Why not 0-5: RIR 0 is the fatigue/injury-adjacent case the athlete wants excluded, and RIR 5 is far enough from failure that it is hard to call a working set. Why not keep 2-3: the evidence does not support that floor, and RIR 1 is a legitimate strength exposure.

How to change it again

The window lives in `config/rules/programme.v1.yaml` (`params.default_rir`) and in `domain.rir.ALLOWED_RIR_WINDOW` (a consistency test keeps them equal). Changing it also changes:

- the proposal menu (wider window → more options),
- `validate_prescription`'s `RIR_OUTSIDE_WINDOW` check,
- the rubric's `RIR_ADHERENCE` score.

That is why a change needs a `DEC-` row and updated tests, not just a YAML edit.

Full bibliography with DOIs: `docs/REFERENCES.md`.

RULES DOCUMENT FORMAT

The constitution is the source of truth. It is stored as **versioned YAML** so that every claim the system makes can cite a stable, immutable rule ID.

1. Why a document rather than prose in a prompt

Three requirements force this:

1. **Citable.** The output must say "load reduced because `PRG-FAT-001` fired", and that ID must still mean the same thing in six months.
2. **Machine-checkable.** The policy engine must be able to veto an LLM suggestion. A rule that only exists as prompt text cannot veto anything.
3. **Versioned.** When the constitution changes, old prescriptions must still explain themselves against the version that was in force.

So each rule has a human `statement`, machine `params`, and a `check` that names a registered Python predicate. The rule text is never parsed as an expression — there is no little DSL to get wrong. YAML declares which predicate and with what arguments; Python implements the predicate once and tests it.

2. Anatomy of a rule

```
- id: PRG-RIR-001                # stable forever; never reused, never renumbered
  title: short human label
  severity: veto | downgrade | warn | advisory
  scope:                         # what the rule applies to
    exercises: [ssb_squat, ...]   # canonical ids
    sessions: [squat, hinge, bench_squat, bench_hinge]
  statement: >                   # the constitution, in words
    ...
  params:                        # every number the predicate needs
    main_lift_rir: {min: 2, max: 3}
  inputs: [set.rir, exercise.primary_compound]   # what must exist to evaluate
  check:
    fn: primary_compound_rir_window   # registry key
    args: {allow_rir_zero_when_unset: true}
  citations: ["constitution §2 effort bullet"]
  depends_on_data_decisions: [DQ-SENT-03]
  on_fail: {action: veto, code: RIR_FLOOR}
  notes: why this is worded the way it is
```

Severity semantics

severity	policy engine effect
`veto`	the proposed prescription is rejected; deterministic fallback used
`downgrade`	the offending item is reduced/modified; prescription survives
`warn`	recorded, surfaced in the report, no change
`advisory`	informational only (e.g. ACWR, which is contested)

The `inputs` field and the `unknown` verdict

A rule whose `inputs` are absent returns **`unknown`**, never `pass`. This is how the system stays honest about things it cannot measure: symptoms (no field exists), session duration (no timestamps), within-set velocity loss (no per-rep data). An `unknown` verdict is reported as "not assessable — " and never scored as compliant.

Modifiers are not rules

Fatigue handling is a **modifier**, not a pass/fail rule: it transforms a prescription rather than judging one. Modifiers live in the same document under `modifiers:` and are applied in order by the prescription engine before the policy engine evaluates the result. `MOD-FAT-001` now uses the computed readiness score (with the sleep/energy floor), global or category overrides, and a 20% combined readiness + continuity cap; the details are in `READINESS-SCORE.md`.

3. Citation scheme

Every rationale sentence carries one or both of:

- a **rule citation**: `PRG-RIR-001` — resolved through `rule_document.version`;
- a **data citation**: a stable fact id such as `set:<session>/<index>` or `feature:rolling_readiness/<date>`.

The critique and prescription store these as `rationale_refs`, so a citation cannot be hallucinated: the validator rejects any reference that does not resolve.

4. Five encoded rules

These are the constitution's own words, encoded. The full set lives in `config/rules/programme.v1.yaml`. Nothing here is invented — where the constitution is ambiguous, the ambiguity is recorded in `notes` and raised in `docs/OPEN-QUESTIONS.md`.

PRG-RIR-001 — Effort window on primary compounds (severity: `veto`)

> "Main lifts at RIR 2–3 (trap-bar RIR 3–4). No RIR 0–1 sets, no max singles, no 1RM testing inside a block. This is a hard rule, not a preference."

Amended: the window is RIR 1–4 with a 2–3 target (`DEC-RIR-02`), because the proximity-to-failure evidence does not support a 2–3 veto and RIR estimates carry roughly ±1–2 reps of error (`docs/RIR-EVIDENCE.md`). The trap-bar clause was removed earlier (`DEC-WINDOW-01`). One window now applies to every primary compound. The original wording is kept above so the change is visible rather than silently rewritten.

```
scope: {exercises: "$primary_compounds"}
params:
  default_rir: {min: 1, max: 4}
  per_exercise: {}
  forbid_max_single: true          # reps == 1 at >= 90% e1RM, except a declared test
check: {fn: primary_compound_rir_window}
on_fail: {action: veto, code: RIR_FLOOR}
notes: >
  Scoped to `primary_compound`, not `main_lift`, because the most-performed squat in
  an export may not be one of the constitution's named lifts. Scoping to main_lift
  would leave most heavy squatting unregulated. Depends on DQ-SENT-03: if the modal
  RIR value is a default bucket it must not be read as an easy set. The only exception
  to `forbid_max_single` is a declared test session (DEC-TEST-01), recorded through
  `strength-coach test` as an e1RM override rather than a training set.
```

PRG-DP-001 — Double progression (severity: `downgrade`)

> "Hold load until top of rep range on all sets at RIR ≥2, then add the smallest increment."

```
scope: {exercises: "$accessories_and_back_extension"}
params:
  top_of_range_all_sets: true
  min_rir_at_top: 2
  increment_kg: {default: 2.5, per_exercise: {Lateral Raise: 2.5, Ab Wheel: 2.5}}
check: {fn: double_progression_decision}
on_fail: {action: downgrade, code: HOLD_LOAD}
notes: >
  "Smallest increment" is per-exercise config: this log contains 0.5, 1.0 and 2.5 kg
  steps, so a global 2.5 kg constant would be wrong for small lifts. Also requires
  sets to be comparable: a set interrupted by a typo or a missing RIR does not count
  toward "all sets".
```

PRG-SCHED-001 — Erector work is exclusive to Days A and C (severity: `veto`)

> "Back extensions (Days A, C only — never on the hinge day)... do not stack a hinge plus direct erector work in the same session."

```
scope: {sessions: [squat, hinge, bench_squat, bench_hinge]}
params:
  allowed_days: [A, C]
  forbid_with_patterns: [hinge]
  direct_erector_flag: true
check: {fn: direct_erector_scheduling}
on_fail: {action: veto, code: ERECTOR_STACKING}
notes: >
  Implemented as a property check (`direct_erector` and `pattern == hinge`) rather
  than a name list, so a new erector exercise is covered automatically. This is what
  `substitutable_for_back_extension: false` protects: an RDL can never be counted as
  the erector dose, so it can never satisfy this rule by proxy.
```

PRG-EREC-001 — Erector dose ramp with a stability gate (severity: `warn`)

> "Dose-ramped over 8–12 weeks (weeks 1–3 bodyweight, 4–6 add 5–10 kg, 7–9 add a set, 10–12 consolidate). Progress only if two consecutive sessions are stable with no symptom flare."

```
scope: {exercises: [back_extension]}
params:
  weeks: [{weeks: [1,3], dose: bodyweight, added_kg: 0},
          {weeks: [4,6], dose: load, added_kg: [5,10]},
          {weeks: [7,9], dose: add_set, added_kg: [5,10]},
          {weeks: [10,12], dose: consolidate}]
  stability_sessions: 2
  symptom_gate: required
check: {fn: erector_dose_ramp}
inputs: [block.week_index, session.sets, symptom.flare]
on_fail: {action: warn, code: ERECTOR_RAMP_HOLD}
notes: >
  Two inputs, both surfaced rather than guessed: (1) the symptom signal is manual
  (a pre-session 0-3 back level in `symptom_log`), so until one is logged the rule
  returns `unknown` rather than assuming no flare; (2) since DEC-BACKEXT-02, historical
  `Hyperextensions` rows DO count toward this dose, so there is real history to ramp
  from — but week 1 still needs an explicit anchor date from the athlete.
```

PRG-SPINE-001 — Weekly high-fatigue spinal exposure cap (severity: `veto`)

> "Flag any week containing more than ~3 high-fatigue spinal-loading exposures (e.g. trap-bar + rack pull + RDL + good morning + hyperextension)."

```
scope: {unit: iso_week}
params:
  cap: 3
  counting: weighted            # sum of spinal_cost, not a raw count
  costs: {back_extension: 1, trap_bar_deadlift: 2, rdl: 2, jefferson_curl: 2,
          rack_pull: 3, good_morning: 3, conventional_deadlift: 3}
check: {fn: weekly_spinal_cap}
on_fail: {action: veto, code: SPINAL_CAP}
notes: >
  Ambiguous in the constitution: "exposures" could mean distinct exercises, sessions,
  or sets. Measured on distinct exercises the cap almost never fires, which would make
  it decorative; measured per set it fires every week. A weighted sum is the middle
  reading and is explicitly ASSUMPTION-02 — the cost values are an ordinal design
  choice, not a measurement.
```

5. What is deliberately not a rule

- `Trap-bar RIR 3–4` — **removed** by `DEC-WINDOW-01`. It was a per-exercise override inside `PRG-RIR-001`; the override map remains, empty, so re-adding a per-lift window stays a config change rather than a code change.
- Power work (3–5 reps, full recovery, stop on decline) — **prescribed since 2026-10-20** (`DEC-POWER-01`, Day A `box_jump`), but still **not a rule**: the log has no rep-level power data, so compliance cannot be scored (Q4). The slot is excluded from rep-scheme undulation and carries no rule citation; the prescription warns to stop when height or quality drops.
- Session time cap 60–80 min — cannot be measured (no timestamps). Encoded as an `advisory` placeholder that always returns `unknown` today, so the gap is visible instead of forgotten.

SANITY CHECKS — external verification of classification and formulas

The exercise taxonomy and the arithmetic in `strength_coach/domain/` were written from training-domain reasoning, then checked against published material. This file records what was checked, what the check changed, and what remains unverifiable.

Status legend: `CONFIRMED` (source agrees), `CORRECTED` (source contradicted the choice, config changed), `UNVERIFIABLE` (no source can settle it), `CONTESTED` (credible disagreement in the literature).

A. Exercise taxonomy

Claim	Verdict	Basis
A trap-bar pull is a **hinge** pattern	CONFIRMED	Universally classed as a hip-hinge/deadlift variant; handle position changes torso angle, not the movement pattern.
A trap bar has **lower lumbar demand** than a straight bar	CONFIRMED (direction)	The neutral-grip, higher-handle position reduces the trunk flexor moment. Exact ratio is not measurable from a log, hence an ordinal `spinal_cost` with an explicit ASSUMPTION tag.
Trap bar still counts as a high-fatigue spinal exposure	CONFIRMED by the programme	It stays in the high-fatigue set even though it is the mildest member.
"Hyperextension" and "back extension" name the same movement family	CONFIRMED	The 45-degree back-extension bench is commonly called a hyperextension; Starting Strength's own forum uses the terms interchangeably (startingstrength.com). This validates a forward-only synonym — it says nothing about whether historical rows used the same bench, so backward mapping stays disabled.
Reverse hyperextension is **not** direct erector work	CONTESTED	Usually described as glute/hamstring dominant with high erector involvement, not isolated spinal-erector loading. Kept `direct_erector: false` **and** added `substitutable_for_back_extension: false`, so it can never satisfy the erector dose even if the label is debated.
Good morning is high spinal cost	CONFIRMED	Long lever arm, barbell on the back, high lumbar moment.
Jefferson curl is direct erector work	CONFIRMED	Loaded spinal flexion/extension is its purpose.
Kettlebell swing is **not** high-fatigue spinal	CONFIRMED (spirit)	Ballistic hip hinge: high velocity, short spinal loading duration. Split into its own `hinge_ballistic` pattern so it never consumes a spinal-cap slot.
Ab wheel is core, not erector dose	CONFIRMED	Anti-extension isometric; erectors stabilise but are not the trained target.
Planks log **seconds** in a reps column	CONFIRMED by the data	Values in the tens-to-hundreds for a hold are durations, not repetitions. Modelled as `rep_semantics: seconds`.
Pause-bench variants are programme lifts	CORRECTED	They had been left as generic accessories; the brief's hinge day allows "close-grip **or pause** bench", so `bench_press_pause` is a first-class main lift with `vbt_family: bench`.
A non-main-lift squat may still be a primary compound	CORRECTED	Kept `main_lift: false` for programme role, but added `primary_compound: true` and `programme_slot_of`. Rationale: an effort rule scoped only to `main_lift` would leave most heavy squatting unregulated.

Ordering bugs these checks surfaced

Regex order is the fragile part of a taxonomy, and two real misclassifications were caught by running the profiler against a real export rather than by reading the config:

- `Lying Leg Curls` matched the biceps `curl` pattern → knee-flexion patterns now run first.
- Pause-bench variants fell through to the generic horizontal-push accessory bucket.

Both have regression tests in `tests/test_exercises.py`.

B. What the logging app itself is doing

The app's product page confirms three things that matter for design (sheikogold.com):

1. **Readiness capture matches the export** — "typical readiness questions... optional inputs like HRV, RHR, & BF%". The readiness columns are app-native, not user-invented.
2. **Workouts are generated on demand and valid for a bounded window** — so one workout is one session, but the export carries no session id. This independently confirms that session reconstruction is required and that a date is the only boundary available.
3. **The app deliberately rotates exercises**: it "will sometimes assign you new exercise variations to test out... It'll treat each exercise variation it assigns as a stress test."

Point 3 has a direct design consequence: **high exercise rotation may be prescribed, not chosen.** In a representative export the carry-over of exercise choice within three sessions is markedly higher for self-directed sessions than for app-assigned ones.

Consequence: the session-provenance label is the only signal available for this distinction, and the rotation rubric must be split by it. A critique that penalises rotation without conditioning on provenance penalises the athlete for following the prescription.

C. Formulas and feature logic

Choice	Verdict	Basis and consequence
Within-set **velocity loss** can be computed	**CORRECTED — impossible**	Velocity loss is defined from the series of individual rep velocities inside a set; the literature reports per-repetition mean propulsive velocity across a set (Wiley/ Physiological Reports). A per-set average has no within-set variance, so the rubric must use a clearly labelled **set-to-set decay proxy** or report the metric as not measurable.
Epley e1RM	CONFIRMED with a caveat	Epley and Brzycki are accurate at low rep counts and degrade as reps rise (sportRxiv). e1RM is therefore computed only for `reps <= 10` (configurable) and tagged `e1rm_quality: ok|degraded`; above a hard cap it is refused outright.
RIR-adjusted e1RM via `w * (1 + (reps + RIR)/30)`	**ASSUMPTION**	"Add RIR to reps" is a common practical convention (e.g. consumer RIR-aware calculators such as TrainerStudio) but it is an approximation, not a validated equation. Stored with `rir_adjustment: epley_plus_rir` so a series can be recomputed under another convention. Never presented as measured.
e1RM is meaningful for every exercise	**CORRECTED — no**	It is suppressed for bodyweight/erector work and for partial-ROM variants (rack pull, RDL, good morning), where the extrapolation produces an inflated index rather than a maximal estimate. A variant "e1RM" exceeding the competition lift's is the tell.
ACWR as a hard rule	**CONTESTED — demoted**	The "sweet spot" reading is actively disputed (Europe PMC critique; Gabbett). Advisory feature only; may never veto.

Choice	Verdict	Basis and consequence
Session duration can be measured	**CORRECTED — impossible**	No timestamps anywhere in the export. The time cap cannot be scored; only a set-count proxy, which must be labelled as such.
Symptoms / flare tracking	UNVERIFIABLE	No such field exists in either export. Symptom-dependent rules cannot run without a new manual input.

D. Assumptions this file creates or confirms

- **ASSUMPTION-05** — the switch date 2026-08-17 is the athlete's "about two months ago". The identity of "Competition Deadlift" is date-scoped (`DEC-TRAPBAR-02`), and the exact date may need correcting. Inferred from the log name; not verifiable from data.
- **ASSUMPTION-02** — `spinal_cost` values are an ordinal design choice, not a measurement. Sumo at 2.5 is a half-step the athlete chose: each of sumo and the trap bar has evidence of lower sagittal demand than a conventional pull (Swinton 2011 for the trap bar; Hanen 2025 for sumo), but no study ranks them against each other (`docs/SPINAL-COST.md`). Must be calibrated against symptom history.
- **ASSUMPTION-06** — a 4-week symptom-free streak with loaded sessions is treated as weak positive evidence for a suitability review. Relapse is common, so this is a prompt to look, not a clearance (`DEC-BACKSIGNAL-01`). Since `DEC-INJURY-02` , four weeks is the **base** of a scaled window, not the whole rule.
- **ASSUMPTION-08** — the injury window scales by 14 days and 2 tolerated sessions per prior flare, capped at 84 days and 12 sessions. The predictor (more than two previous episodes) is from da Silva 2019; the step sizes are a design choice (`DEC-INJURY-02`).
- **ASSUMPTION-09** — a loaded session is verified as tolerated by a symptom reading within 7 days and no flare in that window. The direction (pain-guided progression) is supported in tendinopathy; the 7-day window is not measured (`DEC-INJURY-03`).
- **ASSUMPTION-11** — the readiness composite uses equal weights over present subjective fields as a transparent starting point; the weights are not validated.
- **ASSUMPTION-12** — per-lift readiness is constructed from global readiness, an optional self-rating and an injury ceiling; no validated per-lift instrument exists.
- **ASSUMPTION-13** — the injury ceilings (active 0.50/0.60, monitoring 0.80/0.85) are design choices, not measurements.
- **ASSUMPTION-14** — readiness cut-points, load factors and the combined-reduction cap are design choices, not findings; the cut-points and injury caps are approved for the plan.
- **ASSUMPTION-15** — three core subjective fields are the minimum for a numeric score; a design choice, not a finding.
- **ASSUMPTION-16** — the injury attention tiers are descriptive, not probabilistic, and do not predict a new injury.
- **ASSUMPTION-17** — the attention indicator is grouped by movement/muscle category; regions remain the evidence unit, and the category list/links are config.
- **ASSUMPTION-18** — the old per-lift readiness values are a black-box reference, not ground truth; calibration measures mimicry, not validity.
- **ASSUMPTION-19** — overrides are bounded by constraints and injury ceilings; "override wins" applies to the computed score only.
- **ASSUMPTION-20** — calibration features and the date split are design choices until tested.
- **ASSUMPTION-21** — a 20% cap on combined readiness + continuity reduction is a guardrail, not an evidence-based threshold.
- **ASSUMPTION-22** — main-lift load features are used only in calibration and only from data available before the session.
- **ASSUMPTION-23** — the proposed core dose (2 sessions/week, 2-3 sets, 30-60 s holds, anti-extension + anti-rotation/lateral, progressed by load/leverage) is a synthesis of the evidence, not a validated minimum. The evidence gives no dose-response for this population (`docs/CORE-TRAINING.md`).

- **ASSUMPTION-34** — the readiness-drop gate for exercise substitution uses the prior 28 days, a 10-reading minimum, an absolute 0.70 guard, a 0.10 median drop, and a 0.65 fallback; these are policy choices, not validated cut-points (`DEC-SELECT-06`).
- **ASSUMPTION-35** — selection tax metadata (axial/systemic/skill/ROM) is ordinal and config-authoritative; unknown metadata blocks a “less taxing” claim (`DEC-SELECT-06`).
- **ASSUMPTION-33** — the under-MEV accessory progression uses a three-week window and one-set increments; the MEV landmarks remain coaching defaults, not measured thresholds (`DEC-VOLUME-03`).
- **ASSUMPTION-32** — the six-week scheduled-deload interval and its 10% load / one-set / one-RIR prescription are broad coaching defaults, not a measured optimum (`DEC-PROGRAMME-05`).
- **ASSUMPTION-31** — the MEV/MAV/MRV landmarks and the seven-day MRV window are broad coaching defaults, not measured individual thresholds (`DEC-VOLUME-02`).
- **ASSUMPTION-30** — the continuity calibration minimums (5 returns/band, 12 total), 0.15 bootstrap-width ceiling, 0.85 lower bound and 365-day cap are transparent defaults, not validated (`DEC-CONTINUITY-03`).
- **ASSUMPTION-29** — the exercise-selection weights (readiness 0.35, stress 0.25, rotation 0.20, preference 0.10, fatigue 0.10), 28-day rotation window and 0.15 injury penalty are transparent defaults, not validated (`DEC-SELECT-01`).
- **ASSUMPTION-28** — the fixed 42-day/4-day half-lives and the 8-observation/28-day minimums for the strength shadow model are assumptions, not fitted findings; the model never changes a load (`DEC-FFM-01`).
- **ASSUMPTION-27** — the 21-day grace, 1%/week decay and 10% cap for an old e1RM are transparent defaults, not a validated detraining rate (`DEC-E1RM-DECAY-01`).
- **ASSUMPTION-26** — the e1RM correction minimum (3 tests), 95% bootstrap CI, ±10% bound and median-ratio estimator are assumptions, not validated; the correction is disabled in config if the evidence is ambiguous (`DEC-E1RM-CAL-02`).
- **ASSUMPTION-25** — the per-lift readiness weights (e1RM 0.20, RIR 0.15, velocity 0.15) and bands are transparent defaults, not validated; the score can only lower a load (`DEC-LIFT-READY-01`, `docs/LIFT-READINESS.md`).
- **ASSUMPTION-24** — the recent-stress decay constants (48 h half-life, 6/3 decayed hard-set thresholds, 48 h warning window) are transparent defaults, not validated parameters. The indicator is advisory and never changes a load (`DEC-STRESS-01`).
- **ASSUMPTION-10** — the injury-to-exercise link is an **exposure** map (direct weight 2, indirect 1), not a causal claim. It says which lifts load a region, not which lift caused an injury or which one prevents recurrence.
- **ASSUMPTION-07** — the continuity thresholds (14/28/56 days and their load/RIR changes) are adapted from detraining studies but are not measured cut-points (`DEC-CONTINUITY-01`).
- **ASSUMPTION-03** — RIR-adjusted e1RM is an approximation.
- **ASSUMPTION-04** — plank/carry values in the reps column are seconds. Strongly implied by the magnitudes; no schema states it.
- **ASSUMPTION-05** — a session is one calendar date. Unverifiable without timestamps.

E. What these checks deliberately did not do

- They did not invent targets. No velocity range, RIR band or load increment here is sourced from the literature as "correct for this athlete"; all of it is calibrated from data or left as config awaiting the user.
- They did not use any source to override the programme. Where the literature and the programme disagree (e.g. ACWR), the programme's intent is kept as product behaviour and the metric is demoted to advisory.

SELECTION COMPARABILITY — evidence and proposed guard

Status: implemented 2026-10-20. This note documents the evidence and the guard that is now STRENGTH-PROGRAM-REVIEW.md and proposes a comparability model. The implemented classification is in `config/selection-comparability.v1.yaml`; the selector, engine, CLI and web UI enforce it.

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1. The problem

The selector can substitute one configured variant for another: SSB squat -> front squat, competition bench -> dumbbell bench, pull-up -> lat pulldown, and so on. It re-solves the load from the selected lift's own history, which is mechanically correct.

That still leaves two separate questions:

1. **Can the load be transferred?** Usually no; the lifts have different load bases.
2. **Is the stimulus still the intended one?** Not necessarily, even when the pattern matches.

The current safeguards are real but incomplete: pattern match, constraints, suitability, injury ceiling, one substitution per slot, and explicit opt-in. None of them classifies comparability.

2. What comparability means here

Three different claims are often conflated:

claim	meaning	evidence needed
same movement family	both are squats/benches/hinges	taxonomy pattern
same load basis	the same e1RM/history can be used without a conversion	direct load/1RM comparison
same stimulus	adaptation and fatigue cost are close enough to be interchangeable	longitudinal equivalence, not EMG alone

The selector currently checks only the first. The proposed guard is about the second and third.

3. Evidence

3.1 Specificity: a different mode is a different adaptation

The classic review of resistance-training modes concludes that adaptations are specific to the mode and exercise used (Morrissey, Harman & Johnson 1995). A recent meta-analysis reaches the same conclusion for dynamic versus isometric strength: dynamic resistance training transfers less well to isometric tests than to dynamic tests (Saeterbakken 2025).

Free-weight versus machine training shows the same pattern: direct pooled differences are small, but strength changes are **specific to the tested modality** (Haugen 2023).

Strength: strong for specificity; not evidence that every variant is useless, and not a claim that any particular substitution is harmful.

3.2 Bench grip width

Grip width changes the load-velocity relationship and the 1RM value:

- Herrera-Bermudo 2024: grip width influenced the bench-press load-velocity relationship and 1RM.
- Pérez-Castilla 2020: the load-velocity relationship differed across four grip widths.

A counterpoint matters:

- Pérez-Castilla 2021: grip width did **not** change the number of repetitions at fixed velocity-loss thresholds.

So the fatigue response can be similar while the load basis is not. A narrow-grip bench is not a neutral relabelling of a competition bench.

3.3 Front squat versus back squat

Front and back squats have different load-velocity relationships (Spitz 2019) and load-dependent lower-body mechanics (Krzyszkowski & Kipp 2020).

They share a pattern and a broad stimulus, but not a load basis. A front-squat e1RM cannot be used as a back-squat e1RM.

3.4 Safety squat bar versus straight bar

The SSB changes mechanics and performance:

- Johansson 2024: kinematic and kinetic differences; the traditional bar lifted more weight and produced greater hip extensor demand.
- Vantrease 2021: differences in maximal strength, muscle activation and bar velocity.
- Hecker 2019: altered trunk and lower-body mechanics.

SSB -> front squat is therefore not a like-for-like swap. Each can be a valid training choice; they are not one load basis.

3.5 Barbell versus dumbbell bench

Barbell and dumbbell bench do not share a clean rep/1RM relationship (Heinecke 2021). A direct comparison of barbell, dumbbell and Smith-machine bench also found differences in maximal performance, repetitions and muscle activation (Farias 2017).

The dumbbell version adds independent-arm stability and a different range/stability demand. The load must come from the selected implement's history.

3.6 Pull-up versus lat pulldown

The latissimus activation is broadly similar across pull-up variations (Di Fonza 2026), and lat-pulldown training improves pull-up endurance (Li 2025).

That is evidence of **carryover**, not evidence of a common load basis. One is bodyweight plus added load; the other is an external stack. They can substitute as a stimulus without sharing an e1RM.

3.7 Exercise variation is not automatically better

Random session-to-session exercise variation produced similar strength and hypertrophy to a fixed programme, with better motivation (Baz-Valle 2019).

This is useful: variation is not harmful by default. It does **not** show that an algorithmic selector improves outcomes, or that any two variants are load-comparable.

3.8 The selector itself has no outcome evidence

There is no trial of:

- readiness/stress/rotation-driven selection versus a fixed programme;
- automatic substitution versus human choice;
- a same-pattern filter preserving the intended stimulus;
- a parent-variant e1RM ratio improving prescriptions.

The strength-programme review therefore rates **specificity strong, algorithmic selection weak**. The guard is a safety and coherence measure, not a proven performance intervention.

4. What the evidence supports

1. **Do not transfer loads across different implements.** Barbell, dumbbell, machine, SSB, front squat and safety-bar variants have different load bases.
2. **Do not treat a shared pattern as proof of equivalence.** Bench, squat and hinge each contain variants with different mechanics.
3. **Do use the selected lift's own history.** Re-solving from the selected lift is necessary; it is not sufficient to make the substitution neutral.
4. **Similar EMG does not establish load equivalence.** Pull-up/lat-pulldown is the clearest example.
5. **Variation itself is defensible.** The guard should not ban variation; it should make the load and stimulus change visible.
6. **No parent-variant ratio is validated.** A ratio can be a future prior, not a conversion.

5. Implemented solution

5.1 Classify each variant, not just its pattern

Add explicit comparability metadata to each exercise or to each slot-variant pair:

field	examples	why
load_basis	barbell, dumbbell, machine, bodyweight, added-load, implement	decides whether a load can transfer
movement_family	squat, hinge, horizontal push, vertical pull	broad pattern only
rom_profile	full, partial, box, pause, deficit	changes load and velocity
stability	bilateral, unilateral, machine-supported	changes load and skill
bodyweight	yes/no	bodyweight vs external load
variant_class	exact, near, different	the guard's decision

The class must be configuration, not inferred from the display name.

5.2 Three classes

class	meaning	apply policy
A — load-comparable	same implement and load basis, same broad ROM/stability; e.g. a true duplicate/alias	auto-eligible, but still uses selected lift history

class	meaning	apply policy
B — near-comparable	same pattern but different grip, bar, pause, box or stability; e.g. competition <-> pause bench, SSB <-> high-bar	**confirmation required**; show the load-basis difference and selected lift history
C — different lift	different implement or load basis; e.g. barbell <-> dumbbell bench, pull-up <-> lat pulldown, trap bar <-> straight bar, SSB <-> front squat	**never automatic**; human must choose it as a programme change

The exact A/B boundary is a decision, not an evidence claim. Bench pause/narrow and SSB/high-bar are the hard cases; the evidence shows they are not identical, so the conservative default is **B**, not A.

5.3 Load handling

- Always re-solve from the **selected** lift's own e1RM/history.
- Never copy the old item's load or citation.
- If the selected lift has no usable history, show unknown; do not borrow a parent ratio.
- Show the selected lift, its e1RM source/age, the old load and the new load.
- Record the comparability class and confirmation in the selection note.
- A parent-variant ratio is allowed only after prospective validation, and even then as a display/prior, not a silent conversion.

5.4 Selection and UI

- The selector's default candidate set is class A/B; class C is hidden behind "show all" or an explicit programme-change action.
- Automatic apply is limited to class A. Class B requires a confirmation that states: "This changes the load basis; the load will be re-solved from the selected lift."
- Class C requires the same confirmation plus a visible "different lift" warning and a reason.
- The LLM sees only the filtered list and can never select class C.
- The shadow ranking can still show class C as information; it must not apply it.
- The existing one-substitution-per-slot and explicit opt-in remain.

5.5 Rollout

1. **Shadow audit** — classify the current candidate list and report pair classes/load deltas; change no prescriptions.
2. **Config** — add the comparability fields and a default class per configured pair.
3. **Opt-in confirmation** — enable class B confirmation in Prescribe/CLI.
4. **Only then** consider any automatic class A apply.

5.6 Acceptance criteria

- class C can never be applied automatically;
- class B requires a confirmation and records it;
- every applied substitution resolves load from the selected lift's id;
- no load, e1RM, or PRG-DP-001 citation is inherited from the old lift;
- no parent-ratio conversion without a validated, dated calibration;
- an unknown selected-lift history is shown as unknown, never filled from the old lift;
- tests use synthetic pairs for each class and an explicit different-implement case.

6. Open questions

1. Is bench pause/narrow a B (near) or can any pair ever be A? The load-velocity evidence says B is the safe default.

2. Is SSB <-> high-bar B, while SSB <-> front squat C? Both change mechanics; front squat also changes the load basis more substantially.
3. Should class be per exercise, per slot pair, or both? Pair-level is more precise but more configuration.
4. What counts as sufficient confirmation: one click, a typed reason, or both?
5. Are parent-variant e1RM ratios ever acceptable as a prior? If so, only after prospective validation and with a visible uncertainty interval.
6. Does the selector need a same-stimulus score, or is a hard class enough? A hard class is simpler and auditable.

7. Recommendation

Adopt the **three-class guard**, default all non-identical variants to **B**, and reserve **C** for different implements/load bases. Keep variation available, but make the load-basis change explicit and require confirmation before a prescription changes.

8. Implementation status

Implemented 2026-10-20:

- `config/selection-comparability.v1.yaml` defines load-basis groups, default B and explicit B/C pairs;
- `prescribe/selection.py` classifies candidates against the current lift and hides C unless `include_different` is requested;
- `prescribe/engine.py` applies only A automatically; B requires `selection_confirmed`; C is never applied by the selector;
- CLI `select` shows the class and `--include-different`; `run --selection-confirm` confirms B;
- the web Prescribe form has a near-comparable confirmation and a “show different lifts” checkbox; the selection panel shows class badges;
- tests cover default filtering, B confirmation, C blocking and selected-lift load resolution.

Still open: the exact A/B boundary for pause/narrow bench and SSB/high-bar, pair-vs-exercise metadata, and any future parent-ratio prior. Weakness/injury targeting is deliberately out of scope; see WEAKNESS-INJURY-SELECTION.md and `DEC-SELECT-04`.

SPINAL COST — what the ordinal actually rests on

`spinal_cost` is an **ordinal 1-3 weight** used to add up a week's high-fatigue spinal work against the cap, and to spot stacking. It is not a measurement in newtons, a moment, or an injury probability. This note records the evidence behind each rating and where the rating is an assumption.

What is measured in the literature

Straight bar vs hexagonal (trap) bar. A biomechanical comparison at submaximal loads found the hexagonal bar produced **lower peak moments at the lumbar spine, hip and ankle** and a higher peak knee moment, while allowing a heavier 1RM (Swinton et al., J Strength Cond Res 2011, doi:10.1519/jsc.0b013e3181e73f87). That supports a lower spinal cost for the trap bar than a straight-bar deadlift.

Conventional vs sumo. At 85% 1RM, sumo showed greater frontal- and transverse-plane joint moments and range of motion, while conventional showed greater hip flexion, ankle dorsiflexion and **greater erector spinae thoracis activation** (Hanen et al., Front Bioeng Biotechnol 2025, doi:10.3389/fbioe.2025.1597209). That supports a lower sagittal extensor demand for sumo than conventional.

No head-to-head sumo vs trap bar study. There is no direct comparison of lumbar moments or erector demand between those two lifts, so their relative rating is not measured.

The ratings

lift	cost	basis
conventional deadlift (straight bar)	**3**	highest peak lumbar moment of the compared barbells (Swinton 2011); most trunk flexion
sumo deadlift	**2.5**	lower erector activation and more upright than conventional (Hanen 2025); no sumo-vs-trap measurement, so placed between the trap bar (2) and conventional (3)
trap-bar deadlift	**2**	lower peak lumbar moment than a straight bar (Swinton 2011); no sumo-vs-trap measurement
back extension	1	direct erector work but low absolute load; ordinal judgement
RDL / rack pull / good morning	2-3	hinge variants; ordinal judgement, not measured

The **sumo rating of 2.5 is an assumption** (`ASSUMPTION-02`), not a finding. The evidence shows sumo has lower sagittal demand than a conventional pull, but it does not rank sumo against the trap bar. The half-step is the programme owner's decision: it records "between the trap bar and conventional" without pretending to a measured value. Whether a half-step is meaningful on an ordinal scale is an open question (`OPEN-QUESTIONS.md` Q11).

What would change a rating

- A direct biomechanical comparison of sumo and trap bar at matched relative loads (lumbar moment, erector activation, trunk inclination).
- Athlete-specific symptom data: a lift that repeatedly precedes a flare deserves a higher effective cost even if the group average says otherwise.
- A change to the cap or the exposure definition, which would require rethinking the bins.

Until then, the ratings are a **conservative ordinal prior**, and the corpus records that (`docs/SANITY-CHECKS.md`, ASSUMPTION-02).

Full bibliography with DOIs: `docs/REFERENCES.md`.

Strength programme review — 2026-10-20

Scope: the strength programme only. Cardio, video, the LLM narrative and the web UI are out of scope. This is the evidence snapshot; implementation status is recorded in the note and table below.

Review status: CLOSED 2026-10-20. No open programme-design items remain here. Future validation, parent-ratio and targeting work is tracked in ROADMAP.md §4.20.

Implementation note (2026-10-20): the P0/P1 fixes for power loading, hinge balance, rolling session types, upper-body row/press, readiness set reduction, volume management v1 and **selection comparability** were subsequently implemented; Plan A restored squat volume and raised trap-bar hinge volume while keeping the spinal cap. A scheduled deload is now implemented as an optional suggestion with automatic reactive triggers; a standardised warm-up, a 3RM/5RM cadence and accessory under-MEV progression are implemented. Individualized MRV is closed as not estimable from the available data; main-lift volume experimentation is deliberately not done. The retrospective walk-forward check is done; continuity and fatigue lost to their baselines, so neither is adopted; prospective validation is a future gate only for a candidate that first beats the retrospective baseline. Weakness/injury targeting is deliberately deferred (`DEC-SELECT-04/05`). There are no open programme-design items left in this review.

Method: read the templates, constitution, prescription engine, evidence notes and references; generated A/B/C prescriptions from the local database; checked the planned weekly dose and the main evidence claims against the cited literature. The local 12-week audit is in `private/docs/STRENGTH-PROGRAM-AUDIT.md` (git-ignored, personal data).

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1. The programme as implemented

Current weekly shape (Plan A, 2026-10-20)

Default prescriptions use rolling session types; the legacy A/B/C override has been removed. overrides.

session type	work	sets
squat	box jump 3×3–5; SSB squat **5×3–5**; bench 4×3–5; back extension 3×10–12; dumbbell row 3×8–12; ab wheel 3×6–10	21
hinge	trap-bar deadlift **5×3–5**; narrow/pause bench 4×3–5; pull-up/pulldown 3×6–10; dumbbell row 3×8–12; split squat 3×8–10	18
bench_squat	bench 4×3–5; SSB squat support **4×3–5**; dumbbell row 3×8–12; overhead press 3×6–10; ab wheel 3×6–10	17
bench_hinge	bench 4×3–5; single-leg RDL 3×8–10; pull-up/pulldown 3×6–10; dumbbell row 3×8–12; overhead press 3×6–10; hanging leg raise 3×8–12	19

Over six sessions (two weeks): squat, hinge, bench+squat, squat, hinge, bench+hinge. Planned average is **~57 working sets/week** across three sessions.

Current main-lift volume:

lift	sets/week
squat (SSB main + support)	**7**
bench (main + support)	12
trap-bar deadlift	**5**
single-leg RDL	1.5
back extension	3
horizontal row	9

lift	sets/week
overhead press	3
split squat	3
core	6
power	3

Effort and progression

- Main lifts: **RIR 2** target, allowed window **RIR 1–4**; RIR 0, max singles and 1RM testing inside a block are forbidden.
- Accessories: double progression, hold load until the top of the range on all sets at RIR ≥ 2, then add the smallest increment.
- Main-lift load: solved from the lift's own e1RM (Epley + RIR) with a bounded measured-test calibration and age decay.
- Undulation: 4-week cycle (strength / strength / maximal / strength; accessory / volume / accessory / volume), only when a block anchor is declared. A scheduled deload week is configured every 6 weeks; reactive triggers still override it.
- Readiness and continuity can reduce load and add RIR; poor readiness can also remove one accessory set. The combined readiness + continuity load reduction is capped at 20%. The fitness-fatigue model does **not** change loads.
- Exercise selection can apply at most one substitution per slot when explicitly enabled; the load is re-solved from the selected lift's history. A/B/C comparability classes gate automatic application, and the LLM may only choose from the Python-filtered list.

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2. Evidence check

Legend: **Strong** = consistent meta-analytic evidence; **Moderate** = a reasonable body of evidence with caveats; **Weak/Methodology** = plausible practice or product decision, not settled by trials.

#	Claim in the programme	Evidence status	What the evidence says	Verdict
1	3×/week full-body is a good strength frequency	Moderate	When weekly volume is equated, higher frequency appears to help strength somewhat; 3× full-body is within the supported range. Frequency is not a free extra dose. (frequency meta-analysis; 2025 dose-response)	Supported
2	Weekly volume around the current plan is enough	Moderate	Hypertrophy rises with weekly volume to roughly 10+ sets per muscle with diminishing returns; strength is less volume-dependent. The current plan has ~7 squat, 12 bench and ~5 heavy-hinge sets/ week, plus accessory volume. (Schoenfeld 2017; 2025 dose-response)	Supported after Plan A; hinge still below bench volume
3	Main lifts at 3–5 reps	Strong for strength	Heavy, lower-rep training is more specific for 1RM strength when effort is matched. (Hammert 2026)	Supported
4	RIR 2 target, RIR 1–4 allowed, RIR 0 forbidden	Moderate	Strength gains are similar across a wide RIR range; non-failure is at least as good for dynamic strength; hypertrophy improves closer to failure; RIR self-estimation carries meaningful error. (Robinson 2024; Wu 2026; RIR-EVIDENCE)	Supported and conservative

#	Claim in the programme	Evidence status	What the evidence says	Verdict
5	Accessories at RIR 2 with double progression	Moderate	The effort is defensible for a back-sensitive athlete, but hypertrophy-oriented accessory work may progress more slowly than a failure-inclusive programme. This is a deliberate trade-off, not an evidence claim that RIR 2 is optimal.	Coherent trade-off
6	Power work: box jumps 3×3–5, quality stop	Moderate for function, weak for this exact dose	Power training improves physical function in older adults, with low certainty; the exact 3–5 set prescription is practice, not a validated protocol. (power meta-analysis)	Keep, label as practice
7	4-week undulating cycle	Moderate	Linear and undulating periodization produce comparable strength and body-composition outcomes. DUP is variety/fatigue management, not a superiority claim. (Zhang 2026)	Supported as structure, not advantage
8	Scheduled and reactive deload policy	Weak/Methodology	Practical deload recommendations exist, but the comparative evidence for a fixed deload schedule is thin. The implemented policy is a low-volume week every six weeks plus reactive triggers, explicitly framed as fatigue management rather than a proven strength intervention. (practical deload review)	Implemented as a policy; evidence remains weak
9	Readiness reduces load, not volume	Moderate/Weak	Autoregulated load has not clearly beaten standard percentage prescription; readiness is a fatigue-management tool, not a proven strength optimiser. (Hickmott 2022)	Reasonable safety choice
10	Trap-bar hinge is the main hinge; direct erector work is not stacked with it	Moderate for trap-bar biomechanics; methodology for the stacking rule	Trap-bar deadlift has lower peak lumbar moments than a straight bar. The no-stacking rule is plausible fatigue management but not established by trials. (Swinton 2011; TRAINING-GUIDELINES)	Trap-bar supported; stacking is methodology
11	Fixed continuity bands, now calibrated from returns	Weak for fixed cut-points; improving with calibration	Detraining occurs within weeks, but exact −5%/−10% cut-points are assumptions. The calibration model estimates observed e1RM retention per band and falls back to fixed values when a band is sparse. (CONTINUITY; Yamashita 2023)	Honest and improving
12	Exercise selection can substitute variants	Weak for algorithmic selection; strong for specificity	Specificity matters; variation is not automatically beneficial. There is no trial showing algorithmic selection beats a fixed programme. Substituting a different load basis is a programme change, not a neutral swap. (specificity review; exercise variation review)	Useful shadow tool; needs guardrails
13	Strength is worth measuring for longevity	Moderate observational	Higher strength is associated with lower all-cause mortality; this is association, not proof that a particular programme causes longevity. (Jochem 2019)	Supports the goal, not the exact dose

Evidence limitations

- Most sources are short-term resistance-training trials in varied populations; this athlete's back history and adherence pattern are not represented.
- Several programme rules are explicitly methodology, not evidence: direct-erector scheduling, the spinal-exposure cap, the exact accessory RIR floor, and the 3×/week structure.
- The e1RM and load solver introduce measurement error; the programme's apparent precision is greater than the underlying RIR/e1RM data.
- No evidence supports the current exercise-selection algorithm as a strength intervention.

3. Critique

What is good

1. **The core strength dose is evidence-aligned.** Three full-body sessions, 3–5 reps on the main lifts, RIR 2 with a 1–4 window, and no failure is a coherent strength programme.
2. **It is safety-conscious without being fear-driven.** RIR 0 and max singles are excluded; direct erector work is separated from the hinge; trap bar is used; readiness can only reduce load. This fits a lifter with a back history.
3. **The deterministic layer is strong.** Loads are solved from lift-specific e1RM, every item cites a rule, constraints and injury signals are applied, and the LLM cannot compute or veto.
4. **Progression is explicit.** Double progression for accessories, e1RM-based loading for main lifts, and a configurable undulating cycle are all auditable.
5. **The system is honest about uncertainty.** The fatigue model is shadow-only, continuity now has a calibrated/fixed fallback, and the evidence notes distinguish assumptions from findings.
6. **The power and pulling additions are directionally right.** Power work and a vertical pull are better than a pure squat/bench/deadlift programme for a longevity-first athlete.

What is weak or risky

A. Main-lift volume — applied

Applied 2026-10-20 (Plan A + session types). The original plan gave bench 12 sets/week, squat 7 sets/week and hinge 4 sets/week once. The rolling programme initially restored squat but left heavy hinge at 4 and squat at ~5.5; Plan A corrected that to squat **~7** and trap-bar **~5** sets/week. The added hinge is a single-leg RDL support, not a second heavy hinge; heavy hinge frequency remains about once per week.

This is a **volume-first** resolution: the evidence favours added volume over added frequency when weekly volume is matched, and a second heavy hinge would exceed the spinal cap. The spinal cap remains the hard ceiling.

B. Power work loading — applied

Applied 2026-10-20. The power slot now uses `load_source: bodyweight`; it ignores stored external load and e1RM history, carries no double-progression citation, and stops on quality loss. `box_jump` remains bodyweight-only in the taxonomy. Loaded power work, if wanted, must be a separate variation with its own prescription, not this slot.

C. Exercise selection comparability — applied

Applied 2026-10-20. The selector now classifies every candidate against the current lift:

- **A** — load-comparable; auto-eligible;
- **B** — near-comparable (same pattern, changed grip/bar/ROM/stability); explicit confirmation required;
- **C** — different lift/load basis; hidden by default and never applied automatically.

The load is always re-solved from the selected lift's own history; the old load/citation is not transferred. The implementation is in `config/selection-comparability.v1.yaml`, `prescribe/selection.py`, `prescribe/engine.py`, the CLI and the Prescribe panel. Evidence and the class boundary are in SELECTION-COMPARABILITY.md (`DEC-SELECT-03`, Q21).

D. Scheduled deload — applied

Applied 2026-10-20 (`DEC-PROGRAMME-05`). A configurable low-volume week runs every six weeks from the declared block anchor. It removes one set from each item (minimum two), reduces load by 10% and adds one RIR under `MOD-DELOAD-001`; reactive triggers (two weeks over the spinal cap or two flares) still apply. The evidence for a fixed deload is weak, so this is described as fatigue management, not a proven strength intervention. The readiness set valve remains a separate, within-week response.

E. Upper-body balance — applied

Applied 2026-10-20. The session types now include a horizontal row on every session and a vertical press on the bench-heavy sessions; the vertical pull remains. This closes the original gap without changing the bench main-lift volume.

F. Rolling next-session policy — applied

Applied 2026-10-20. New prescriptions use session types and a rolling next-session rule; Canonical session types now replace the old labels. The forecast shows a calendar, but the day's session is selected from the rolling balance rather than a strict session-type sequence.

Applied above: the rolling rule preserves the main-lift balance without a strict weekly A/B/C calendar. This is a scheduling policy, not a new training principle.

G. Readiness set reduction — applied

Applied 2026-10-20. Poor readiness can now reduce accessory sets by a configured amount as well as reducing load and raising RIR. Main lifts keep their volume unless the config explicitly includes essential work. Exercise selection remains restricted by the A/B/C comparability guard; it is not a readiness-driven free swap.

H. Main-lift progression depends on e1RM estimates

The solver uses e1RM from recent sets, with calibration and decay. This is sensible, but it inherits RIR estimation error and formula error. There is no top set/AMRAP or regular submaximal test in the template; e1RM can drift without a deliberate calibration point.

Suggestion: keep the existing `test` command but schedule a 3RM/5RM check every 8–12 weeks, or use a repeatable top-set standard. Do not add routine 1RM testing.

Is it good?

Yes as a safe, evidence-aware and now reasonably balanced strength programme. No as a finished, validated or fully fatigue-managed programme.

The core — 3×/week full body, 3–5 reps, RIR 2, no failure, autoregulated load, conservative back rules — is defensible and better than most generic templates. The original weaknesses were programme-design imbalances and one concrete power-load bug, not a bad philosophy.

The current template includes the fixes that mattered:

1. rebalanced hinge and squat volumes — **done** (Plan A + session types);
2. bodyweight-only power slot — **done**;
3. horizontal pulling and a vertical press — **done**;
4. comparability-guarded selection — **done**;
5. fatigue-management valve — **done**: readiness set reduction, volume cap, reactive and scheduled deload.

The remaining gaps are the calibration/validation items in §4, not a missing deload policy.

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4. Suggested changes, prioritised

Status reflects the implementation, not only the original recommendation.

Implementation checklist

- [x] Power work bodyweight/implement-only.
- [x] Rolling session types instead of rigid A/B/C.
- [x] Upper-body row in every session.
- [x] Overhead press on bench-emphasis sessions.
- [x] Readiness set reduction (accessories + main lifts) and editable valve.
- [x] Continuity calibration from observed returns, with fixed-band fallback.
- [x] Dorsiflexion readiness category for the tib-bar lift.
- [x] Volume v1: ledger, MEV/MAV/MRV, conservative MRV cap.
- [x] Selection comparability A/B/C guard.
- [x] Plan A: squat ~7 sets/week and trap-bar ~5 sets/week.
- [x] Scheduled deload / broader fatigue-management policy (`DEC-PROGRAMME-05`).
- [x] Under-MEV accessory progression (`MOD-VOL-002`).
- [x] MRV question closed — not estimable without independent volume variation; no experiment.
- [x] 3RM/5RM calibration cadence (`test --status`).
- [x] Explicit warm-up prescription (50/70/85 ramp).
- [x] Review closed: prospective validation is moved to ROADMAP §4.20 as a gate for a future candidate. The retrospective check rejected the current continuity/fatigue models; no prospective period is active.
- [x] Parent-variant e1RM ratios moved to ROADMAP §4.20. Weakness/injury targeting is deferred there (`DEC-SELECT-04/05`); only the display-only own-history context is implemented.

priority	change	reason	effort	status
P0	Fix power items: bodyweight/ implement only, no e1RM/double progression	A loaded box jump is not the intended stimulus	Small	**Done**
P0	Rebalance main-lift volume: restore squat and raise hinge volume without breaking the spinal cap	Current 4 hinge vs 12 bench and reduced squat was not a balanced strength dose	Medium	**Done** — Plan A, volume-first; heavy hinge still ~1×/week
P0	Restrict automatic selection to load-comparable variants; require confirmation for different implements	Prevents a valid load solve from silently changing the stimulus	Medium	**Done** (A/B/C guard)
P1	Add a horizontal row and a vertical press (3 sets each), or explicitly trade bench accessory volume	Upper-body balance and shoulder health	Small	**Done**
P1	Add a scheduled low-volume week or a readiness set-reduction option	Set valve and scheduled/ reactive deload policy exist; evidence is weak but the gap is now addressed	Medium	**Done** — set valve + scheduled deload (`MOD-DELOAD-001`)
P1	Define a rolling next-session policy that preserves A/B/C balance across missed weeks	Current fixed sequence is brittle	Medium	**Done** (session types)
P2	Schedule a 3RM/5RM calibration every 8–12 weeks; no routine 1RM	Keeps e1RM honest without adding maximal testing	Small	**Done** — 10-week reminder

priority	change	reason	effort	status
P2	Add warm-up/ramp sets to the prescription or document the assumption explicitly	The template only describes working sets	Small	**Done** — 50/70/85 ramp
P2	Validate the calibrated continuity and selection models prospectively before letting them drive loads automatically	Avoid overfitting to a small, noisy log	Large	**Moved** to ROADMAP §4.20 — no candidate; current models failed the retrospective baseline
P3	Parent-variant e1RM ratios and weakness/injury selection	Useful later, but not needed to fix the current programme	Large	**Moved** to ROADMAP §4.20; targeting deferred (`DEC-SELECT-04/05`)

5. Bottom line

- **Evidence-aligned:** frequency, heavy main-lift reps, RIR buffer, no-failure, autoregulated load, trap-bar choice, measurable strength.
- **Methodology, not evidence:** erector scheduling, spinal cap, exact accessory RIR, 3×/week calendar.
- **Weakest links now:** data depth and future model validation; neither is a programme-design defect in the current template.
- **Verdict:** a good, safer and more balanced default for a back-sensitive lifter. The review is **closed**; future model/ratio/targeting work is in ROADMAP §4.20.

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6. Sources

Primary sources used in this review:

- Robinson et al. 2024, proximity to failure
- Wu et al. 2026, non-failure vs failure
- Refalo et al. 2023, failure and hypertrophy
- Hammert et al. 2026, load specificity
- 2025 resistance-training dose response
- Schoenfeld et al. 2017, weekly volume
- Frequency meta-analysis
- Zhang et al. 2026, linear vs undulating
- Hickmott et al. 2022, autoregulation
- Power training in older adults
- Practical deload review
- Swinton et al. 2011, trap-bar biomechanics
- Yamashita et al. 2023, detraining
- Jochem et al. 2019, strength and mortality
- Specificity review
- Exercise variation review

Internal evidence notes: `TRAINING-GUIDELINES.md`, `RIR-EVIDENCE.md`, `TRAINING-DYNAMICS.md`, `CONTINUITY.md`, `REFERENCES.md`.

TRAINING DYNAMICS — testing, rep ranges, and undulation

This note collects the evidence behind three questions the athlete asked: whether occasional 1RM testing is worth it, whether rep counts should be adjustable, and whether daily undulating periodization (DUP) and readiness-driven variation belong in the programme. It is a design note, not a decision record; decisions live in `docs/DECISIONS.md`.

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1. Occasional 1RM testing

What the evidence supports

- Maximal strength is a meaningful health marker. A meta-analysis of 39 studies (39,852 participants) found the lowest strength category had a hazard ratio of 1.80 for all-cause mortality versus the highest, and each 5 kg of strength was associated with HR 0.72 (Jochem et al., JAMDA 2019, doi:10.1016/j.jamda.2019.05.015). Association, not proof that testing causes longevity — but it supports keeping strength measurable.
- Actual 1RM testing is a standard assessment in trials and is generally considered safe in healthy, resistance-trained adults. Direct data on injury rates are thin.
- Risk is not zero. A vertebral fracture during 1RM leg-press testing has been reported in an older breast-cancer survivor with previously unknown osteoporosis (Rosenberger et al., Medicine 2021, doi:10.1097/MD.0000000000025705). Reviews of testing in older adults describe 1RM testing as time-consuming and carrying injury risk, and validate velocity-based alternatives instead (Deboutte et al., Exp Gerontol 2026, doi:10.1016/j.exger.2026.113174).
- Submaximal estimates are good enough for programming. A 3-5RM test plus the RIR-adjusted Epley formula gives an e1RM within the precision the load solver uses (it rounds down to 0.5 kg anyway). Velocity-based estimation can be equivalent to measured 1RM when the load-velocity profile is individual (Deboutte 2026), but general profiles are less reliable.

Recommendation for this athlete

- Test **3RM or 5RM**, not a true single: same information for load prescription, far less spinal load, and it fits the existing e1RM override mechanism (`source = measured`).
- Cadence: every **8-12 weeks**, or when the solver's e1RM and the athlete's perceived effort diverge for several sessions — not on a fixed calendar week.
- Always with safeties, never on the trap bar (the hinge day already carries the highest spinal cost), and never during a back-flare window.
- A true 1RM stays possible but should be an explicit, declared exception, not a routine.

Open issue: `PRG-RIR-001` forbids "max singles and 1RM testing inside a block". With no fixed block (DEC-PROGRAMME-01), "inside a block" is currently always true, which bans testing forever. A test-session exception needs defining before a `test` command can exist.

---

2. Rep ranges — fixed, or adjustable?

Today reps are fixed per template slot (e.g. 3-5 on main lifts, 8-12 on accessories), and the model may not change them. That is a design choice, not an evidence claim.

What the evidence supports

- **Strength** is specific to heavy loading: lower-rep, higher-load training produces larger 1RM gains than low-load training when effort is matched; high- and low-load training differ in non-specific strength too (Hammert et al., Sports Med 2026, doi:10.1007/s40279-025-02370-8).
- **Hypertrophy** is similar across a wide rep range when sets are taken close enough to failure and volume is matched; the practical band is roughly 5-30 reps, with low loads requiring more proximity to failure.
- **Longevity / health**: the benefit comes from being strong and trained, not from a specific rep bracket. Moderate loads (roughly 6-12) with RIR 1-3 capture most of the strength and mass benefit at lower joint and fatigue cost; some heavier work preserves the strength-specific stimulus.
- **Readiness** interacts with reps: at low readiness the same load is closer to failure, which is why the fatigue modifier cuts load rather than reps.

Recommendation

- Keep the current day character (A squat-led, B hinge, C bench-led) but make the rep range a **menu of validated schemes** rather than a single fixed pair, e.g. `strength` (3-5), `volume` (6-8), `accessory` (8-12), each with a solver-computed load.
- The model may **choose a scheme** (menu), never write a rep number; Python computes the matching load.
- Do not let a single session move reps arbitrarily: variation belongs at the day/week level, so the block still has a coherent stimulus.

---

3. DUP and readiness-driven variation

What the evidence supports

- Linear and undulating periodization produce **comparable** strength and body-composition outcomes in a 29-study meta-analysis (Zhang et al., Front Public Health 2026, doi:10.3389/fpubh.2026.1707627); undulating was better only for lean mass in obese participants in short-term subgroups. In older adults, traditional and undulating training also improved strength and fat-free mass similarly (Vargas-Molina et al., IJERPH 2022, doi:10.3390/ijerph19084522).
- Autoregulated load (RIR- or velocity-based) versus standardized percentage prescription showed **no significant difference** in 1RM strength (MD 2.07 kg, 95% CI -0.32 to 4.46, p = 0.09) in a 15-study meta-analysis (Hickmott et al., Sports Med Open 2022, doi:10.1186/s40798-021-00404-9); velocity-loss thresholds ≤25% were favourable for some outcomes.

What that means, honestly

- DUP is **not superior** to linear periodization in the literature. It is a legitimate way to organise variety and manage fatigue, and it may help adherence; it should not be sold as a strength advantage.
- Readiness-based adjustment is **not proven to beat** fixed prescription either. It is a sensible way to avoid bad-day grinding, and it fits the athlete's longevity goal, but its value is fatigue management rather than extra gains.
- Therefore: implement DUP and readiness variation as **configurable structure**, not as a claimed optimisation, and keep the deterministic rules (RIR window, spinal cap, deload triggers) above them.

---

4. Proposed shape (needs a decision before building)

1. **`test` session type — implemented (`DEC-TEST-01`).** `strength-coach test` records a 3RM/5RM (or a `--max` 1RM with confirmation) as a `measured` e1RM override, with a 28-day warning for submaximal tests and a 56-day limit for 1RM tests unless `--force`. The test set is not logged as training volume, and `--e1rm` lets the athlete override the computed value.

2. **Rep scheme menu — implemented (`DEC-REPS-01`).** Named schemes: `maximal` (1-3), `strength` (3-5), `volume` (6-8), `accessory` (8-12). The model may name a scheme, never a number; Python re-solves the load for it. `maximal` is excluded on hinge lifts: a near-maximal trap-bar pull is the highest-risk exposure for this athlete.
3. **Weekly undulation — implemented (`DEC-UNDUL-01`).** A 4-week config cycle keyed to the declared anchor: week 3 main lifts at `maximal`, weeks 2/4 accessories at `volume`. With no anchor there is no cycle. It is variety/fatigue management, not a claimed advantage.
4. **Readiness response.** Keep MOD-FAT-001 as the load cut; add an optional "reduce one set when readiness is poor" modifier, config-gated.

Each of these changes what is prescribed, so each needs a `DEC-` row and an acceptance test.

Readiness/injury and scheme selection: `WORKOUT-TYPE-READINESS-EVIDENCE.md`.

Full bibliography with DOIs: `docs/REFERENCES.md`.

TRAINING GUIDELINES — external evidence check on the programme

Purpose: check a fixed-methodology strength programme against published resistance-training evidence, and record where they agree, where they differ, and where the literature is too weak to settle anything.

This file is about the strength programme. The wider question — whether the programme covers the modalities that matter for a longevity-first goal (cardio, power, pulling, assessment) — is reviewed in `LONGEVITY-REVIEW.md`.

The programme's constitution wins every conflict — it is the product's methodology. This file only says where a rule is well supported, over-constrained, or unsupported.

Honesty note: several full texts were paywalled or blocked when checked. Where that happened the finding is cited at the level reported in the abstract or title and marked `[title/abstract only]`. No effect sizes are quoted that were not verified.

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1. Volume: sets per muscle per week

The best-known dose-response review finds hypertrophy increasing with weekly volume up to roughly 10+ sets per muscle group per week, with diminishing returns beyond (Schoenfeld, Ogborn & Krieger, J Sports Sci 2017 — tandfonline). A 2024 meta-analysis models dose-response across strength, hypertrophy and other outcomes and finds the shape differs by outcome (Sports Medicine 2024), so a single "sets per week" target cannot serve both goals.

Implication for a 3x/week full-body programme: it is fully compatible with this, because every session touches every muscle group. The meaningful audit is therefore not sessions per week but **sets per muscle group per week**, which most logging apps never compute. Recommendation: add `sets_per_muscle_group_per_week` to the feature store, keyed off the exercise taxonomy rather than the raw muscle label.

2. Frequency

When weekly volume is equated, training a muscle group more often does not clearly increase hypertrophy; for **strength**, higher frequency appears to help somewhat (weekly frequency meta-analysis, PubMed 30076500 `[title/abstract only]`).

Implication: 3x/week full-body sits at the high end of what the evidence supports for strength and is well within it. A rule like "missed sessions must not leave any muscle group without stimulus for a week" is a practical robustness rule, not a frequency-science claim — which is the right justification for it, and should be described that way.

3. Proximity to failure

A 2024 series of meta-regressions on estimated proximity to failure finds hypertrophy benefits from training closer to failure, while strength effects are less clear-cut (Sports Medicine 2024).

Implication: the window is now RIR 1-4 with a 2-3 target (`DEC-RIR-02`, evidence). It remains a conservative choice — failure is not required, and RIR estimation is imprecise — but RIR 1 is a legitimate strength exposure, so the rule no longer forbids it. The consequence worth stating plainly:

> Main lifts are still never taken to failure, and accessories only progress at RIR >= 2. That is a legitimate choice for a lifter managing symptoms — but it means **hypertrophy-oriented accessory work may progress more slowly than a failure-inclusive programme.** A critique must not score "too easy" as a failure when the constitution forbids the alternative.

It also couples to Q1: if the modal RIR value is a default rather than a real rating, apparent average effort is understated and the prescription is systematically too light.

4. VBT and velocity loss

Velocity loss is defined from the series of per-rep velocities inside a set; the literature reports mean propulsive velocity per repetition across a set (Physiological Reports). Many exports store only one **average** velocity per set, in which case within-set velocity loss is **not computable** — a single mean has no within-set variance.

Implication: any rubric line about "adherence to the velocity-loss cutoff" must either be dropped or explicitly relabelled as a set-to-set decay proxy. This is an arithmetic limitation, not a data-quality complaint.

5. Load monitoring (ACWR)

Acute:chronic workload ratio is **contested** — the "sweet spot" interpretation has been directly challenged (Europe PMC 7485291), and workload monitoring is broader than a single ratio (Gabbett, 2017). **Decision:** ACWR is computed as an advisory feature only and may never veto a prescription.

6. Where the literature is too weak to settle the programme

These are domain judgements in the constitution, not literature deficits. Do not over-claim support for them:

Rule	Evidence status
Direct erector work is its own progression, dose-ramped over 8-12 weeks	**No comparable prospectively-validated protocol found.** A clinical caution is a perfectly good reason; it is not an evidence-based protocol.
Never stack a hinge plus direct erector work in the same session	**Plausible fatigue-management logic, not established by trials.** Keep it; label it a methodology choice.
Use high handles on a trap bar if low handles irritate the back	Sound practical reasoning; not a trial result.
Power work: 3-5 reps, full recovery, stop on decline	Consistent with general power-training practice, but **cannot be scored** when the export has no per-rep power data.

7. How weakness-targeted exercise selection should work

"We add exercises for your weaknesses" is where a fixed-methodology coach most easily drifts into a chatbot inventing programmes. Two guardrails:

1. **A weakness must be evidenced** — a measured ratio outside a configured band, a stalled e1RM over N sessions, a movement pattern with insufficient weekly sets, or a reported symptom. "The model thinks the posterior chain is weak" is not evidence.

2. **A weakness may only select from a pre-approved exercise pool.** Adding a new exercise to the pool is a user decision, recorded in the taxonomy config, after which it appears in the baseline automatically.

A useful corollary: the **most-logged** lift is not automatically the weakest, and a programme that allocates time by logging frequency will over-serve whatever the athlete enjoys or tracks most. Allocation must follow evidence, not volume.

8. Anomaly detection that is evidence-based, not aesthetic

Programme audits should look for regime changes rather than judging individual sessions:

- a main lift's monthly volume collapsing to zero while a substitute rises (a common signature of an irritation being worked around),
- grinding counts rising alongside a volume peak,
- a lift's velocity or e1RM trending down while its volume trends up,
- near-empty months that are breaks rather than missed sessions.

These are questions to ask the athlete, not verdicts. A substitution pattern that looks like avoidance may be a correct clinical decision.

V0 SCOPE — the smallest end-to-end slice

1. The slice

One command:

```
python -m strength_coach.cli run --date 2025-06-10 --llm off   # any date in your export
```

that:

1. ingests both CSVs,
2. cleans them under the documented sentinel policy,
3. reconstructs sessions,
4. ~~calibrates VBT from history **as of that date** (no look-ahead)~~ — **deferred to v0.1** by `DEC-VBT-01`: the mapping from a velocity band to a prescribed target is not specified precisely enough to implement without inventing it. The step is kept visible here rather than deleted.
5. produces a **critique** of the most recent session with rubric scores,
6. produces a **prescription** for the next Day B session,
7. writes an append-only record of both,
8. and prints a rationale where every load cites a rule id or a data point.

Why Day B for v0: it is the cleanest single-day template (trap-bar deadlift + close-grip/pause bench), it is the only day with a per-exercise RIR override (trap bar 3–4), and it contains **no direct erector work**, which sidesteps the unanchored erector ramp (PRG-EREC-001) that is blocked on Q5. Day A/C would drag in unresolved-day-identity and erector-ramp questions on day one.

2. In scope

Area	v0 includes
Ingest	both CSVs, raw preserved, quarantine of null-date rows, `source_row` traceability
Cleaning	per-field zero sentinels, unit normalisation, `reps_unit` (reps vs seconds), typo flags, **no deduplication** (DQ-DUP-01)
Sessions	one per date, workout label retained, `POSSIBLE_DOUBLE_SESSION` flag
Load	`load_basis` resolution (DQ-BW-01), total vs external load
Features	tonnage (external only), e1RM with method+quality tags, per-lift baseline, weekly spinal cost, session frequency
VBT	calibration with the L0–L4 fallback ladder, `calibration_id` versioning; velocity targets **or an explicit `uncalibrated` — deferred to v0.1** (`DEC-VBT-01`); prescriptions carry `calibration_ids = []` until then
Rules	load `config/rules/programme.v1.yaml` ; evaluate PRG-RIR-001, PRG-DP-001, PRG-SCHED-001, PRG-SPINE-001; PRG-EREC-001 returns `unknown`
Prescribe	deterministic Day B prescription: per item sets/reps/load/RIR/rest/essential/rationale refs; fatigue modifier MOD-FAT-001
Critique	rubric items that are measurable now: RIR adherence, progression adherence, weekly spinal cost, missing essential work, redundancy. Each with score + evidence refs
Policy	veto/downgrade with fallback prescription
Store	SQLite, append-only, migrations
Trace	run + run_event, replayable

Area	v0 includes
LLM	**interface only**, plus `NullLLM`. One optional narrative call behind `--llm on`; the run must be identical in structure with it off
Tests	unit + integration + one golden prescription per day type present in the window

3. Explicitly out of scope for v0

Each of these is deferred with a reason, not forgotten:

Deferred	Why
RAG / vector store	rules are cited by ID; exact lookup is better than semantic search here. RAG is only for programme prose, later
Critique/prescription agents (real LLM)	the deterministic core must be proven first; `NullLLM` exercises the failure path from day one. **Scheduled for v0.1** (`ROADMAP.md` §3.4): the model may propose a modified session, but only inside the rules and only with policy approval
Days A and C	blocked by the SSB question (H1) and the erector ramp anchor (Q5)
Erector dose ramp (PRG-EREC-001)	returns `unknown` — no symptom input, no anchored week 1
Velocity-loss scoring	not computable from set means (SANITY-CHECKS §C); v0 reports `velocity_decay_proxy` with `measurable: false`
Session-duration scoring	no timestamps exist (DQ-SESSION-01)
Power-work module	`Avg_power` absent on 66% of sets; the power rule cannot be scored
ACWR	contested; advisory only, and not needed for a single-day prescription
Bevel / HRV / RHR / grip strength	normalisation layer is designed but there is no data yet
Rotation scoring	needs the app-vs-self provenance decision (see SANITY-CHECKS §B)
Labelled 10-session eval set	v0.1; v0 ships the harness skeleton and 3 hand-marked sessions
VBT velocity targets in prescriptions	deferred by `DEC-VBT-01`: the step from a calibrated velocity band to "do this set at 0.62 m/s" is underspecified, and prescribing it would be fabricated precision. The calibration code and tests remain; only the prescription wiring is postponed
UI	explicitly deferred by the brief

4. Acceptance criteria

v0 is done when all of these are true:

1. `python -m pytest` passes; every domain function in `utils` and `domain` is unit-tested.
2. `python tools/profile_data.py` reproduces `docs/EVIDENCE.md` exactly.
3. `python tools/baseline.py --as-of <date>` reproduces the baseline for that date (the quantitative baseline lives in the private docs).
4. A prescription for the chosen date is **executable**: every load is a positive multiple of a plausible increment, every main-lift RIR is inside its window, no two erector exposures coexist, weekly spinal cost is inside the cap.
5. Deleting the LLM entirely (`--llm off` or a raising client) produces the same prescription, differing only in narrative fields.
6. Re-running the same `run` is idempotent in content and appends no duplicate artifacts.

7. The critique's rubric includes the three known failure patterns from this history: grinding sets (RIR 0–1 on primary compounds), hinge/erector stacking, and exercise rotation — with rotation split by workout label.
8. Every numeric rationale field references a resolvable rule ID or data-point ID.

5. Rough effort estimate

Given the utils/domain layer already exists and is tested:

Piece	estimate
ingest + clean + quarantine	1 day
session reconstruction	0.5 day
store + migrations + repositories	1 day
features (tonnage, e1RM, spinal, frequency)	0.5 day (mostly written in `features.baseline`)
VBT calibration + ladder	1 day
rules loader + registry + policy	1 day
prescription engine (Day B) + fallback	1 day
critique rubric (deterministic)	1 day
CLI + trace	0.5 day
tests + golden + fixes	1.5 day
total	**~9 working days**

The estimate assumes the 8 open questions are answered around the time ingest lands. Q1 (RIR semantics) and Q2 (load basis) block the prescription engine specifically, because both change the load a single rep-resolution away.

Variant pattern audit — 2026-10-20

Status: **approved and applied 2026-10-20** for the high-confidence pattern/muscle fixes. Targeted online check of the 115 generated identities, plus the pre-existing split-squat entry. Most names were high-confidence by movement semantics; the exceptions below were checked online.

Decisions / corrections

raw	id	current pattern	proposed	confidence	status	n
Dumbbell Pullover	`dumbbell_pullover`	`lat_isolation`	`lat_isolation`	medium	**confirmed**	K d
Face Pull	`face_pull`	`rear_delt_pull`	`rear_delt_pull`	high	**applied**	P M is t c
Half-Kneeling Single Arm Cable Row	`half_kneeling_single_arm_cable_row`	`horizontal_pull`	`horizontal_pull`	high	**applied**	P M w
Landmine Press	`landmine_press`	`vertical_push`	`vertical_push`	medium	**confirmed**	H k o
Rear Lateral Raise	`rear_lateral_raise`	`rear_delt_pull`	`rear_delt_pull`	high	**applied**	P M is
Reverse Nordic Curl	`reverse_nordic_curl`	`hamstring_eccentric`	`knee_extension`	high	**applied**	R o P E o h h w
Seated Dumbbell External Rotation	`seated_dumbbell_external_rotation`	`rotator_cuff`	`rotator_cuff`	high	**applied**	P M is
Speed Bench Press	`speed_bench_press`	`horizontal_push`	`horizontal_push`	medium	**applied**	P in M is
T Bar Row	`t_bar_row`	`horizontal_pull`	`horizontal_pull`	high	**applied**	P M is

raw	id	current pattern	proposed	confidence	status	r
Tib Bar Calves	`tib_bar_calves`	`plantar_flexion`	`dorsiflexion`	high	**applied**	T d p s (w w c
Split Squat / Bulgarian Split Squat	`split_squat (existing)`	`squat`	`squat_unilateral`	high	**applied**	B g S S t

Full table

raw	id	current	proposed
1 Arm Row	`1_arm_row`	`horizontal_pull`	`horizontal_pull`
1 Arm Tricep Pushdown	`1_arm_tricep_pushdown`	`triceps_extension`	`triceps_extension`
Barbell Calf Raises	`barbell_calf_raises`	`plantar_flexion`	`plantar_flexion`
Barbell Curl	`barbell_curl`	`elbow_flexion`	`elbow_flexion`
Belt Squat	`belt_squat`	`squat`	`squat`
Bench Dips	`bench_dips`	`horizontal_push_bw`	`horizontal_push_bw`
Bench Press (elevated legs)	`bench_press_elevated_legs`	`horizontal_push`	`horizontal_push`
Bench Press (explosive)	`bench_press_explosive`	`horizontal_push`	`horizontal_push`
Bench Press (from pins)	`bench_press_from_pins`	`horizontal_push`	`horizontal_push`
Bench Press (medium grip)	`bench_press_medium_grip`	`horizontal_push`	`horizontal_push`
Bench Press (negative)	`bench_press_negative`	`horizontal_push`	`horizontal_push`
Bench Press (reverse grip)	`bench_press_reverse_grip`	`horizontal_push`	`horizontal_push`
Bench Press (w/bands)	`bench_press_w_bands`	`horizontal_push`	`horizontal_push`
Bench Press (w/cambered bar)	`bench_press_w_cambered_bar`	`horizontal_push`	`horizontal_push`
Bench Press (w/chains)	`bench_press_w_chains`	`horizontal_push`	`horizontal_push`
Bench Press (wide grip)	`bench_press_wide_grip`	`horizontal_push`	`horizontal_push`

raw	id	current	proposed
Bent Over Row	bent_over_row	horizontal_pull	horizontal_pull
Bird dog	bird_dog	core_stability	core_stability
Board Press	board_press	horizontal_push	horizontal_push
Cable Bent-over Triceps Extension	cable_bent_over_triceps_extension	triceps_extension	triceps_extension
Cable Glute Kickback	cable_glute_kickback	hip_extension	hip_extension
Cable Pull Through	cable_pull_through	hip_extension	hip_extension
Close Grip Pulldown	close_grip_pulldown	vertical_pull	vertical_pull
Deadlift (above knees, off blocks)	deadlift_above_knees_off_blocks	hinge	hinge
Deadlift (below knees, off blocks)	deadlift_below_knees_off_blocks	hinge	hinge
Deadlift (below knees, off blocks, pause above)	deadlift_below_knees_off_blocks_pause_above	hinge	hinge
Deadlift (from center block)	deadlift_from_center_block	hinge	hinge
Deadlift (pause above knees)	deadlift_pause_above_knees	hinge	hinge
Deadlift (pause below knees)	deadlift_pause_below_knees	hinge	hinge
Deadlift (plus deadlift from above knees)	deadlift_plus_deadlift_from_above_knees	hinge	hinge
Deadlift (plus deadlift from below knees)	deadlift_plus_deadlift_from_below_knees	hinge	hinge
Deadlift (snatch grip, slow descent)	deadlift_snatch_grip_slow_descent	hinge	hinge
Deadlift (to knees)	deadlift_to_knees	hinge	hinge
Deadlift (to knees, plus full deadlift)	deadlift_to_knees_plus_full_deadlift	hinge	hinge
Deadlift (w/ bands)	deadlift_w_bands	hinge	hinge
Deadlift (w/ chains)	deadlift_w_chains	hinge	hinge

raw	id	current	proposed
Deadlift (w/ chains, off blocks)	deadlift_w_chains_off_blocks	hinge	hinge
Decline Bench Press	decline_bench_press	horizontal_push	horizontal_push
Deep Squat	deep_squat	squat	squat
Deficit Deadlift (conventional)	deficit_deadlift_conventional	hinge	hinge
Deficit Deadlift (pause above knees)	deficit_deadlift_pause_above_knees	hinge	hinge
Deficit Deadlift (pause below knees)	deficit_deadlift_pause_below_knees	hinge	hinge
Deficit Deadlift (sumo)	deficit_deadlift_sumo	hinge	hinge
Deficit Deadlift (w/ chains)	deficit_deadlift_w_chains	hinge	hinge
Depth Jumps	depth_jumps	power_jump	power_jump
Dips	dips	horizontal_push_bw	horizontal_push_bw
Dumbbell Bench Press	dumbbell_bench_press	horizontal_push	horizontal_push
Dumbbell Chest Supported Row	dumbbell_chest_supported_row	horizontal_pull	horizontal_pull
Dumbbell Kickbacks	dumbbell_kickbacks	triceps_extension	triceps_extension
Dumbbell Press (alternating)	dumbbell_press_alternating	vertical_push	vertical_push
Dumbbell Pullover	dumbbell_pullover	lat_isolation	lat_isolation
Dumbbell Row	dumbbell_row	horizontal_pull	horizontal_pull
Dumbbell Tricep Extension	dumbbell_tricep_extension	triceps_extension	triceps_extension
Dumbell Curl (alternating)	dumbell_curl_alternating	elbow_flexion	elbow_flexion
Face Pull	face_pull	rear_delt_pull	rear_delt_pull
Flat Dumbbell Flyes	flat_dumbbell_flyes	horizontal_push_isolation	horizontal_push_iso

raw	id	current	proposed
Flat-bench Sit-ups	flat_bench_sit_ups	core	core
Floor Press	floor_press	horizontal_push	horizontal_push
French Press	french_press	triceps_extension	triceps_extension
Front Raise	front_raise	shoulder_abduction	shoulder_abduction
Front Squat	front_squat	squat	squat
Front Squat (to box)	front_squat_to_box	squat	squat
Hack Squat	hack_squat	squat	squat
Half-Kneeling Single Arm Cable Row	half_kneeling_single_arm_cable_row	horizontal_pull	horizontal_pull
High Bar Squat	high_bar_squat	squat	squat
High To Low Cable Chop	high_to_low_cable_chop	core	core
Hip Thrust	hip_thrust	hip_extension	hip_extension
Incline DB Bench Press	incline_db_bench_press	horizontal_push	horizontal_push
Incline Press	incline_press	horizontal_push	horizontal_push
Incline Sit-ups	incline_sit_ups	core	core
Jump Squats	jump_squats	power_jump	power_jump
Kettlebell Swing	kettlebell_swing	hinge_ballistic	hinge_ballistic
Landmine Press	landmine_press	vertical_push	vertical_push
Landmine Squat	landmine_squat	squat	squat
Leg Extension	leg_extension	knee_extension	knee_extension
Lying Leg Curls	lying_leg_curls	knee_flexion	knee_flexion
Nordic Curl	nordic_curl	hamstring_eccentric	hamstring_eccentric
Overhead Press (behind the neck)	overhead_press_behind_the_neck	vertical_push	vertical_push
Overhead Press (seated)	overhead_press_seated	vertical_push	vertical_push
Overhead Tricep Extension	overhead_tricep_extension	triceps_extension	triceps_extension
Pallof Press	pallof_press	core_stability	core_stability
Preacher Curl	preacher_curl	elbow_flexion	elbow_flexion
Pushups (chest emphasis)	pushups_chest_emphasis	horizontal_push_bw	horizontal_push_bw

raw	id	current	proposed
Pushups (tricep emphasis)	pushups_tricep_emphasis	horizontal_push_bw	horizontal_push_bw
Rear Lateral Raise	rear_lateral_raise	rear_delt_pull	rear_delt_pull
Reverse Nordic Curl	reverse_nordic_curl	hamstring_eccentric	knee_extension
SL Glute Bridge	sl_glute_bridge	hip_extension	hip_extension
Seal Row	seal_row	horizontal_pull	horizontal_pull
Seated Cable Row	seated_cable_row	horizontal_pull	horizontal_pull
Seated Calf Raises	seated_calf_raises	plantar_flexion	plantar_flexion
Seated Dumbbell External Rotation	seated_dumbbell_external_rotation	rotator_cuff	rotator_cuff
Shoulder Shrugs	shoulder_shrugs	upper_trap	upper_trap
Side Lunges	side_lunges	squat_unilateral	squat_unilateral
Single Arm Cable Triceps Kickback	single_arm_cable_triceps_kickback	triceps_extension	triceps_extension
Single Leg DB RDL	single_leg_db_rdl	hinge_unilateral	hinge_unilateral
Speed Bench Press	speed_bench_press	horizontal_push	horizontal_push
Squat (2s pause halfway down)	squat_2s_pause_halfway_down	squat	squat
Squat (2s pause halfway up)	squat_2s_pause_halfway_up	squat	squat
Squat (2s pause)	squat_2s_pause	squat	squat
Squat (down slow, up fast)	squat_down_slow_up_fast	squat	squat
Squat (from pins)	squat_from_pins	squat	squat
Squat (narrow stance)	squat_narrow_stance	squat	squat
Squat (negatives)	squat_negatives	squat	squat
Squat (partial range of motion)	squat_partial_range_of_motion	squat	squat

raw	id	current	proposed
Squat (slow downwards & upwards)	`squat_slow_downwards_and_upwards`	`squat`	`squat`
Squat (to box)	`squat_to_box`	`squat`	`squat`
Squat (w/ bands)	`squat_w_bands`	`squat`	`squat`
Squat (w/ chains)	`squat_w_chains`	`squat`	`squat`
Standing Single-Arm Cable Curls (Shoulder Flexed)	`standing_single_arm_cable_curls_shoulder_flexed`	`elbow_flexion`	`elbow_flexion`
Straight Arm Pulldown	`straight_arm_pulldown`	`lat_isolation`	`lat_isolation`
Straight Bar Cable Curl	`straight_bar_cable_curl`	`elbow_flexion`	`elbow_flexion`
T Bar Row	`t_bar_row`	`horizontal_pull`	`horizontal_pull`
Tib Bar Calves	`tib_bar_calves`	`plantar_flexion`	`dorsiflexion`
Tricep Pushdown	`tricep_pushdown`	`triceps_extension`	`triceps_extension`
Upright Row	`upright_row`	`shoulder_abduction`	`shoulder_abduction`
Split Squat / Bulgarian Split Squat	`split_squat (existing)`	`squat`	`squat_unilateral`

Sources used

- Reverse Nordic: PubMed 39214520
- Tibialis raise / dorsiflexion: tibialis raise
- Face pull: Wikipedia
- T-bar row: PureGym, BarBend
- Dumbbell pullover: Legion
- Landmine press: Onnit
- Split squat: Training & Conditioning

VBT CALIBRATION

Goal: turn historical velocity data into **per-lift, per-rep-count velocity targets** the prescription engine can use, without inventing numbers when the data is thin.

This is a deterministic, versioned code function. It is never an LLM decision.

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1. What counts as a clean calibration set

A set may enter calibration only if **all** of these hold. Every exclusion is counted and reported per lift, so a target with `n = 4` is visibly weak rather than silently weak.

#	Criterion	Reason
1	`Avg_velocity > 0`	`0.000000` is "device not used" (DQ-SENT-01), not slow
2	`RIR` inside the **2-3 target band** (`rir_mod.target_for`)	calibration deliberately uses the narrower target band, not the 1-4 allowed window (`DEC-RIR-02`); RIR 6 excluded as suspect (DQ-SENT-03)
3	`1 ≤ reps ≤ 8`	velocity is rep-count dependent; long sets are not VBT sets
4	`load_basis ≠ unknown`	DQ-BW-01 could not classify the load
5	not quarantined, no `REPS_TYPO` flag	corrupt rows cannot set a target
6	`exercise.vbt_family` not null	non-VBT movements (back extension, planks) are excluded structurally
7	`load ≥ 0.60 × e1RM`	a 40%-e1RM warm-up has no informative velocity target; configurable
8	velocity within `median ± 4·MAD` of its (lift, rep) group	single-session device glitches; iterative, max 2 passes

Expected magnitudes in a real export (reproduce with `tools/profile_data.py`):

- only about a third of all sets carry any velocity at all, and
- after the filters above, a high-volume main lift typically retains on the order of 100–200 clean sets, a moderately-used one a few dozen, and a rarely-logged variant fewer than 30 — which is precisely why the fallback ladder below is mandatory rather than a nicety.

2. What a target is

A target is **conditional on rep count**, not a single number per lift:

```
target(lift, reps, rir_band) = (median_velocity, low, high)
```

where `low/high = median ∓ k·MAD`, with `k = 2.0` and a **relative floor** so a zero-dispersion sample cannot collapse the band onto the median (see `utils.stats.robust_band`). Storing the band instead of a point is what makes the check "did velocity fall outside the expected range" honest at small n.

Why not a single "target velocity" per lift: velocity at 3 reps and at 6 reps are different quantities, and a target that ignores rep count will systematically flag one of them.

3. Sparse data: a strict fallback ladder

Never interpolate silently. Resolve in order and record which level was used:

level	rule	recorded as
L0	lift + rep count directly, if `n ≥ 12` and `≥ 3` distinct sessions	`scope: lift`
L1	pool by `vbt_family` at the same rep count, if `n_family ≥ 24`. Only genuine variants pool: bench grips, squat types. Named hinges (trap bar, sumo, RDL, conventional) each have their own family, because their load-velocity curves differ	`scope: family`
L2	not implemented. Pooled rep-count fits were designed but the shipped calibration uses lift and family only, because a pooled line can hide a lift-specific curve	—
L3	not implemented (same reason)	—
L4	no target	`scope: none`, `vbt_status: uncalibrated`

Consequence at L4: **the prescription omits a velocity target entirely** and says so. It does not guess. A lift with a few dozen clean sets will frequently land on L1 (its family), and the prescription must label the target as family-derived rather than lift-derived.

4. Manual override (per lift)

`config/calibration/overrides.v1.yaml`:

```
overrides:
  - exercise_id: ssb_squat
    rep_count: 3
    rir_band: [2, 3]
    velocity_low: 0.42
    velocity_high: 0.52
    reason: "SSB sets feel slower than comp squat at the same RIR; coach call."
    author: user
    recorded_on: <date>
    expires_on: null      # or a date, after which the computed target resumes
```

Rules: - an override **always wins** over a computed target and is tagged `source: override`; - a prescription records the `calibration_id` it used, so overriding later does not rewrite history; - an expired override is ignored and the computed target resumes — no silent rot.

5. Versioning and immutability

Every computation creates a new `vbt_calibration` row:

`calibration_id`, `exercise_id`, `scope`, `rep_count`, `rir_band`, `velocity_median`, `velocity_mad`, `velocity_low`, `velocity_high`, `n_sets`, `n_sessions`, `window_start`, `window_end`, `method_version`, `params_hash`, `computed_at`, `supersedes_id`.

Nothing is updated in place. Recalibrating after new data produces a new row; a past prescription still points at the old one and still explains itself. `params_hash` covers `k`, the relative floor, the warm-up threshold, the RIR band and the rep cap, so a change to any of those is visible as a different calibration even if the numbers coincide.

6. What cannot be computed — velocity loss

See `VBT-REP-LEVEL.md` for the post-hoc cost/benefit and the decision not to add manual rep-level entry.

Within-set velocity loss requires a **per-rep velocity series**. The export has one `Avg_velocity` per set, so within-set loss is arithmetically unavailable (`docs/SANITY-CHECKS.md` §C). It remains deferred (`DEC-VBT-02`, `DEC-VBT-03`), and the prescription does not pretend otherwise.

Two options, and the choice is Q4:

1. **Report it as not measurable** and score no rubric line for it. Honest.
2. **Set-to-set decay proxy**: within one session, at the same load and rep count, the drop from the first working set's velocity to the last, expressed as a percentage. This captures some of the same fatigue signal and is cheap. It must be labelled `velocity_decay_proxy`, never "velocity loss".

Recommendation: implement both, with `velocity_loss_measurable: false` carried in the feature record so the critique cannot accidentally print a real-looking velocity-loss score.

7. Test strategy for this module

- Private integration test (git-ignored): the clean-set counts in §1 are asserted against the real CSV, so a change in exclusion logic cannot silently change the calibration base. The expected numbers are personal data and are not committed.
- Unit tests with synthetic frames for each exclusion criterion (one row each) and for each fallback level boundary (`n = 11` vs `12`, `23` vs `24`).
- Determinism test: same inputs + same `params_hash` ⇒ byte-identical output.
- Immutability test: recomputation inserts a new row and leaves the old one untouched.

8. What is wired now (v0.1)

`build_prescription` calibrates from the stored clean sets on every run and attaches a target only where the ladder produces one:

- the target is the calibrated **median mean velocity** for that lift, rep count and RIR band;
- the item records `velocity_target` and `vbt_basis` (scope + sample size);
- the prescription cites the stored calibration rows in `calibration_ids`;
- **no target** is attached when the session was already modified by a safety rule (the load no longer matches the calibration), or when no lift has 12 clean sets across 3 sessions;
- **hex-bar deadlift targets are withheld** regardless of sample size, because the RIR-velocity relationship for that lift failed accuracy criteria (`docs/VBT-EVIDENCE.md`).

The prescription notes say which case applied, and the CLI prints a `vel` column when any item has a target. A missing target is visible, not silent.

9. Upgrade path (placeholder)

The current input is one mean velocity per set. A real velocity-loss controller needs:

- per-rep velocity samples (or first/last rep), for within-set loss and stop rules;
- a progressive loading test per lift, for an individual load-velocity line and a minimal velocity threshold;
- device/belt/setup metadata, so curves are compared like-for-like;
- a method-version bump, with old calibration rows kept and superseded, never edited.

The calibration already carries `METHOD_VERSION` and a typed `CalibrationParams`, so the upgrade is additive: a v2 method can be written beside the current one and selected by config. Until then, the app uses what the export provides and says when that is not enough.

VBT EVIDENCE — what velocity can and cannot tell us

A knowledge base for the velocity-target wiring. It records the evidence, the best practices that follow from it, and which parts of VBT this app does **not** do with the data currently available. Decisions live in `docs/DECISIONS.md`; the mechanics live in `docs/VBT-CALIBRATION.md`.

---

1. What the evidence supports

Load-velocity profiles are reliable when measured with a valid device. A 2026 systematic review of 63 sensor studies and 38 1RM-prediction studies found good-to-excellent reliability for commercial velocity sensors (pooled ICC 0.90-0.91), while 1RM prediction from a general profile carries meaningful error (Claassen et al., Sports Med Open 2026, doi:10.1186/s40798-026-01102-0). Individual profiles are better than group ones.

The relationship is strong within a lift. In the full squat, mean velocity explained about 96% of the variance in relative load ($R^2 = 0.958$) across age groups (Serrano-Cañadillas et al., J Strength Cond Res 2026, doi:10.1519/JSC.0000000000005624). A belt changes the relationship: it raises 1RM and shifts velocities at high relative loads, so measurements must be compared like-for-like (Huang et al. 2026, doi:10.1519/JSC.0000000000005523).

Velocity loss is a dose variable, not a pass/fail. A dose-response meta-analysis found a reverse-U relationship with 1RM gain: **20-30% velocity loss** was associated with the best strength outcomes, while lower loss was more efficient per repetition (Zhang et al., Healthcare 2023, doi:10.3390/healthcare11030337). A systematic review of acute and chronic responses reached a compatible conclusion: higher velocity loss means more fatigue, and the adaptation depends on the threshold chosen (Jukic et al., Sports Medicine 2023, doi:10.1007/s40279-022-01754-4).

Hypertrophy does not need a high velocity loss. Pooling the velocity-loss trials found no advantage for >25% over 20-25% loss for muscle growth (ES 0.08, 95% CI -0.16 to 0.32) (Refalo et al., Sports Medicine 2023, doi:10.1007/s40279-022-01784-y).

General velocity-to-RIR rules are weak. A bench-press study found the RIR-velocity relationship least biased of three general approaches, but errors were still about ±1.3 reps (Qin et al. 2025, doi:10.1186/s13102-025-01098-2). For the **hexagonal-bar deadlift**, no condition met the accuracy criteria for estimating RIR from mean velocity, and the authors concluded the relationship is not fit for real-time monitoring (Chen et al., J Strength Cond Res 2026, doi:10.1519/JSC.0000000000005549).

Intent matters as much as the number. A narrative review of best practice stresses explosive intent versus merely moving slowly; a set's velocity depends on how the athlete intends to move, not only on the load (Behm et al., Appl Physiol Nutr Metab 2025, doi:10.1139/apnm-2024-0136).

2. Best practices that follow

practice	why
Build an **individual** load-velocity profile (3-5 loads) when precision matters	group profiles carry more 1RM error (Claassen 2026)
Compare **like-for-like**: same lift, setup, belt, device, rep count	belt and setup shift the curve (Huang 2026)
Use **mean velocity**, not peak, for set-to-set comparison	mean is the more stable construct

practice	why
Stop sets at a **target velocity loss**: 20-25% for hypertrophy, 10-20% for strength/power	Zhang 2023; Jukic 2023
Treat **RIR-from-velocity** as a weak prior, not a measurement	Qin 2025; Chen 2026 (hex bar fails)
Do not prescribe velocity thresholds for the **hex-bar deadlift**	Chen 2026
Track **trend within an athlete**, not against a population table	individual curves differ

3. What our data can support today

The export gives **one mean velocity per set**, plus load, reps, RIR and date. That is enough to:

- estimate a per-lift, per-rep-count median velocity inside an RIR band;
- attach a target to a prescription where at least 12 clean sets across 3 sessions exist (or a family pool of 24, and only for genuine variants such as bench grips or squat types — named hinges are never pooled with each other);
- withhold a target where the evidence says velocity is not trustworthy (hex bar).

It is **not** enough to:

- compute within-set velocity loss (needs per-rep samples or start/end velocities);
- build an individual load-velocity profile (needs a progressive loading test);
- estimate a minimal velocity threshold at 1RM (needs near-maximal sets);
- control for belt, device or technique changes.

4. Consequences in the app

- A target is `target(lift, reps, rir band)`, never a single number per lift.
- No target is better than a guessed one; the prescription says "unavailable" and why.
- The target is a **median with a robust band**, and the basis records scope and sample size.
- A session already modified by a safety rule does not get a velocity target: the load it was calibrated against no longer holds.
- Hex-bar targets are withheld regardless of sample size.

5. Upgrade path (placeholder)

To move from "mean-velocity target" to real velocity-based autoregulation, the input needs:

- per-rep velocity samples, or the first/last rep velocity of each set, so **velocity loss** can be computed and a set stopped at 20-25%;
- a **progressive loading test** per lift to fit an individual load-velocity line and a minimal velocity threshold;
- device and setup metadata (belt, machine, sensor position) so curves are comparable;
- a calibration method version bump, with the old targets kept as superseded rows.

None of that is faked now. The calibration parameters already live in one dataclass (`CalibrationParams`) and the method has a version string, so the upgrade is additive.

Full bibliography with DOIs: `docs/REFERENCES.md`.

Rep-level VBT — post-hoc value, cost and decision

Status: not adopted. Rep-level velocity logging is not implemented, and manual per-rep entry after every workout is not justified. If a device already exports per-rep samples, keep the raw data for a possible retrospective analysis.

This note records why the feature is not being built now.

1. The distinction that matters

form	when available	what it can do
set-level mean velocity	after the set/workout	calibrate load–velocity bands; provide readiness context; existing use
rep-level velocity	after the set/workout, unless streamed live	compute within-set velocity loss; describe fatigue/technique retrospectively
live rep-level velocity	during the set	stop the set at a velocity-loss target; adjust load/volume in real time

The programme does not have a live VBT feed. Adding rep-level data after the workout therefore does **not** enable live autoregulation.

2. What post-hoc rep-level data can still do

1. **Retrospective fatigue monitoring** - compute within-set velocity loss for each set; - compare the same lift and load across sessions; - flag unusually grindy sessions.
2. **Consistency check on RIR** - a set logged at RIR 2 with a very large velocity loss may have been closer to failure; - this is a check, not a replacement for RIR.
3. **Calibrate future load–velocity relationships** - with enough per-rep data, estimate how velocity loss relates to load, reps and performance; - could improve future readiness or set-termination rules.
4. **Associate with other outcomes** - velocity loss versus sRPE, next-session performance, readiness and back symptoms; - descriptive covariate only.
5. **Data-quality audit** - implausible velocity curves, device-placement issues, rep-count errors.

3. What it cannot do

- stop a set at a velocity-loss threshold;
- adjust load or volume during the session;
- provide live autoregulation;
- reliably infer RIR from velocity, especially for the trap bar;
- establish an individual velocity-loss threshold from one athlete’s retrospective data;
- justify an automatic programme change.

The main evidence-supported use of velocity loss is **live set termination** — roughly 10–20% for strength/power and 20–30% for hypertrophy. Captured after the workout, the same number is descriptive, not actionable.

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4. Cost versus benefit

For **manual per-rep entry after every workout**:

- high logging burden;
- transcription and device-placement error;
- no in-session action;
- marginal information beyond sRPE, actual-vs-expected reps, readiness and back symptoms;
- one athlete cannot support a validated individual threshold.

Not worth building.

For **automatic device export that already contains per-rep samples**:

- low marginal cost;
- worth archiving raw files;
- useful for a focused retrospective analysis later.

The difference is whether the data arrives automatically or has to be typed.

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5. Decision

1. **No rep-level VBT logging UI, schema or import path now.**
2. **Keep set-level VBT** as the usable input; existing calibration and targets remain.
3. **Do not add manual per-rep entry.**
4. If a device already exports per-rep velocity, **preserve the raw export** outside the normal ingestion path. Do not contort the current schema for data with no current use.
5. Revisit only if: - per-rep data can be captured automatically; - there is a specific question it can answer better than sRPE/performance/readiness; - a retrospective baseline shows velocity loss adds information; - any future use is descriptive first, not a live stop rule.

This is consistent with:

- `DEC-VELOCITY-01` — within-set velocity loss is not measurable from one mean per set;
- `DEC-VBT-02` — per-rep loss and MVT wait for richer input;
- `docs/VBT-EVIDENCE.md` — velocity loss evidence and its live-use context;
- `docs/VBT-CALIBRATION.md` — what the current calibration can and cannot compute.

---

6. What would change the decision

A future phase could revisit rep-level VBT if:

- a device provides per-rep samples automatically;
- the data include load, reps and RIR;
- the athlete wants a specific question answered, for example:
- does velocity loss predict next-session performance better than sRPE?
- does velocity loss flag a back-flare prodrome?
- the analysis is pre-specified and descriptive, with no automatic programme change.

Until then, the honest position is: **no rep-level VBT feature.**

Volume management — evidence and implementation

Status: v1 done (2026-10-20). “Done” means the evidence check, ledger, configured landmarks, conservative MRV cap, readiness valve and UI/CLI/API are implemented. It does **not** mean individualized MRV (closed as not estimable), per-muscle volume or prospective validation; the under-MEV accessory progression is a separate deterministic rule (`MOD-VOL-002`). A separate scheduled/reactive deload policy is described in `DEC-PROGRAMME-05` .

Scope: weekly hard-set volume for the strength programme. This note records the evidence used for the volume ledger and the conservative MRV cap; it is not a claim that MEV/MRV are measured individual constants.

---

1. What the evidence supports

Volume dose-response

- Weekly volume is associated with more hypertrophy up to a point of diminishing returns; the shape differs for strength, hypertrophy and other outcomes. A 2025 meta-regression of weekly volume and frequency is the main current source (Sports Medicine 2025).
- An earlier dose-response review found hypertrophy increasing to roughly 10+ sets per muscle per week with diminishing returns (Schoenfeld et al. 2017).
- Strength is generally less volume-dependent than hypertrophy; very high volumes are not required for strength and can add fatigue. Examine's summary of the strength-volume question: Examine.

MRV, MEV and the "volume landmarks"

- **MEV** (minimum effective volume), **MAV** (maximum adaptive volume) and **MRV** (maximum recoverable volume) are coaching constructs popularised by RP/Israetel, not validated individual measurements: RP volume landmarks.
- There is no validated test that measures an individual's MRV. The landmarks are useful as **configurable guardrails and vocabulary**, not as measured thresholds.
- The programme therefore uses broad pattern-level ranges, not a precise per-muscle MRV.

Deloads and autoregulation

- A 2026 randomised within-subject deload trial in untrained young men examined hypertrophy and strength endurance after a deload; this is a narrow population and does not establish a general deload schedule (Scientific Reports 2026).
- Autoregulation has not consistently beaten a well-designed fixed programme. A trial in trained men found no additional benefit from autoregulation within a mixed periodised plan (autoregulation trial; Examine summary).
- Therefore the implementation is **conservative**:
- a ledger reports under/within/high/over;
- a cap trims **non-essential** sets only when the recent week plus the proposed session is above the configured MRV;
- under-MEV is reported; a separate accessory-only rule adds one set after three weeks below MEV (`MOD-VOL-002`);
- the readiness set valve remains a separate recovery rule.

---

2. Landmarks in this programme

Configured in `config/volume.v1.yaml`. They are broad, pattern-level defaults and are `ASSUMPTION-31`.

pattern	MEV	MAV	MRV
squat	6	8–12	14
hinge	6	6–9	10
hinge unilateral	0	0–6	8
horizontal push	8	10–16	18
vertical push	3	4–8	10
horizontal pull	4	6–10	12
vertical pull	3	4–8	10
core	3	4–9	12

Why these values:

- they sit around the broad dose-response ranges, with a lower hinge MRV because the hinge is the highest spinal-cost pattern and the athlete has a back history;
- unilateral hinge and core have lower MEV because they are accessory patterns;
- horizontal push has a higher MRV because it is the largest current strength emphasis;
- none of these is an individual measurement.

The ledger reports the **average per week over the window**, not one session's set count.

Programme adjustment (2026-10-20): the squat-emphasis SSB is 5×3–5, bench+squat squat support is 4×3–5, and trap-bar hinge is 5×3–5. Planned weekly squat volume is ~7 sets and heavy hinge ~5 sets; the back-extension and single-leg RDL doses are unchanged. This came from the review decision `DEC-PROGRAMME-04`, not from a new MRV estimate.

3. Implemented behaviour

Ledger

`features/volume.py` counts working sets by pattern over the configured window and returns:

- weekly counts, total, average;
- MEV, MAV, MRV;
- status: `under`, `within`, `high`, `over`, `untargeted`.

Exposed as:

```
strength-coach volume --weeks 4 --date 2026-10-20
```

and the **Weekly volume** panel on Prescribe / `GET /api/volume`.

MRV cap

Before validation, the engine adds the prior seven days' working sets for each pattern to the proposed session. If the total exceeds MRV:

- non-essential items in that pattern lose one set at a time until the total is at or below MRV;

- essential items are never cut automatically;
- if only essential work remains above MRV, the prescription warns and leaves the decision to the athlete;
- changed items cite `MOD-VOL-001`.

This is not a per-muscle measurement and not a deload. It is a guardrail against accidental volume stacking.

Under-MEV

When a pattern is below MEV across the configured three-week window, the engine adds **one accessory set** to the first non-essential item in that pattern (`MOD-VOL-002`), up to the configured per-item maximum. Main lifts are never increased by this rule. The MRV cap runs afterwards, so the result still cannot exceed the weekly ceiling.

This is a small deterministic progression, not an individual MRV or an experiment.

Readiness valve

The readiness set valve is separate: poor readiness removes one set from accessories and main lifts (config `include_essential`), regardless of the weekly landmark. It is editable on the Readiness page and recorded in the `MOD-FAT-001` warning.

---

4. Limitations

- MEV/MAV/MRV are not measured for this athlete; the values are config assumptions.
- The ledger counts hard sets, not effective reps, proximity to failure, load or tonnage.
- The seven-day MRV window is a practical approximation of a week, not an individual recovery measurement.
- The cap is asymmetric: it never adds work below MEV and never cuts an essential item.
- The evidence does not support a fixed automatic deload schedule; the scheduled deload is a configurable fatigue-management policy (`DEC-PROGRAMME-05`), not a claim of proven superiority.
- Volume management does not itself trigger a deload; the deload policy is separate.

---

5. What would strengthen this

1. Record a per-pattern RPE/soreness/fatigue response and test whether MRV predicts next-session performance.
2. Compare the ledger's under/over states with actual e1RM changes prospectively.
3. Calibrate the landmarks from this athlete's returns before widening them.
4. If deloads are added, test them against a no-deload control rather than assuming benefit.

---

6. Sources

- 2025 volume/frequency meta-regression
- Schoenfeld 2017 volume dose-response
- RP volume landmarks — coaching construct
- 2026 deload RCT
- Autoregulation trial
- Examine strength-volume summary

WEAKNESS / INJURY-TARGETED SELECTION — evidence and decision

Status: targeting deferred; display-only own-history context implemented. No exercise is selected because it targets an injury or a weak muscle. Injury remains a constraint/ceiling/penalty. The selector shows an own-history e1RM ratio, which never enters the score.

---

1. Why this is not obvious

Two different claims are often mixed:

1. **Weakness targeting:** choose exercises because a lift or muscle is lagging.
2. **Injury targeting:** choose exercises because a region is injured or at risk.

Both need a validated definition and a tested intervention. Neither currently exists for this single-athlete programme.

2. Evidence

Individual variation is real, but not predictable from one programme

True individual response variation requires repeated within-person designs; group averages can hide different responses (Robinson 2024).

Auto-regulated exercise selection produced small lean-mass and bench-press advantages over fixed selection in one small study, but no squat advantage; it was not a weakness-targeting algorithm (Rauch 2020).

A shared pattern is not a validated weakness measure

Exercise specificity means training a lift improves that lift. It does **not** establish that choosing the lowest e1RM is the right programme adjustment. A low e1RM can reflect technique, practice, fatigue or measurement error.

Asymmetry and injury prediction are inconsistent

Inter-limb strength asymmetry has inconsistent calculation methods and thresholds (Parkinson 2021).

In sub-elite rugby players, no unilateral leg-strength measure was related to injury relative risk; relative strength had only a weak association with time lost (Helme 2023).

Some population-specific exceptions exist: preseason supraspinatus weakness was associated with major shoulder/elbow injury in high-school pitchers (Tyler 2014), and older preseason imbalance work is mixed (Knapik 1991). These do not support a general automatic rule.

Injury-targeted exercise is not the same as rehabilitation

Pain-guided activity modification is feasible in patellar tendinopathy (Sprague 2021), but it is a rehabilitation strategy, not evidence that a selector should choose exercises from an injury map.

Exercise therapy is effective for chronic low back pain (Hayden 2021) and tendinopathy (Lopes 2025), but these trials test treatment programmes, not algorithm-selected substitutions.

Prior injury is a strong recurrence marker; that supports **caution and exposure management**, not "train the weak area to prevent recurrence."

3. Decision

1. **No weakness-targeted selection now.**
2. **No injury-targeted positive selection now.**
3. Injury status may only: - exclude a candidate, - penalise a candidate, - cap readiness. It may never boost a candidate because it "targets" an injury.
4. If a relative-weakness display is added later, it is **display-only**, based on the athlete's own trend, with uncertainty; it must not change the ranking.
5. Any future targeting must still obey the A/B/C comparability guard; C remains never automatic.
6. Revisit only after: - a validated weakness definition, - enough per-lift history, - a retrospective baseline result, - and a prospective or N-of-1 test showing benefit without harming the main lifts.

3a. What was implemented now

- Each selection candidate gets a **display-only** `history_context`: latest best e1RM versus the median of that lift's own earlier 84 days, with n and a plain label.
- It is **not** part of the score, ranking, class or prescription.
- It uses no parent-lift ratio and no population norm.
- No weakness target and no injury-positive selection were added.

4. Deferral gates — when this can be revisited

Recommended: not before v0.5 is complete, and not before six months of stable current-programme data (~April 2027). That is an earliest review point, not a commitment.

Weakness targeting may be reconsidered only when all of these exist:

1. **Data:** at least 12 sessions per targeted lift, across at least 12 weeks, under the current programme and with manual logging/import quality fixed.
2. **Definition:** a pre-registered weakness metric (own-trend e1RM, not a population ratio) and a declared target lift.
3. **Retrospective baseline:** the proposed targeting rule beats fixed selection on a walk-forward check, with uncertainty.
4. **Counterfactual design:** an opt-in N-of-1 or alternating-block test, not a single-path observational comparison.
5. **Safety:** the test shows no harm to the other main lifts or the back, and no increase in flare frequency.

Injury targeting is **not scheduled at all**. It may only be revisited if a clinician defines a specific rehabilitation goal and the athlete wants it; it is not a general selection feature. Until then, injury remains exclusion/penalty/ceiling only.

4a. Plan and earliest date

phase	deliverable	status
0	evidence, guardrails, display-only own-history context	**done**
1	12 sessions per target lift over 12 weeks; stable logging	waiting for data

phase	deliverable	status
2	pre-registered own-trend weakness metric + retrospective walk-forward baseline	not started
3	opt-in N-of-1 / alternating-block comparison, with safety outcomes	not started
4	decision: keep fixed, or add a bounded targeting rule	not reached

Earliest review: after v0.5 and roughly **April 2027**. This is a gate, not a scheduled feature. Injury targeting remains unscheduled at every phase.

5. What would be tested if this is revisited

- primary outcome: the targeted lift’s e1RM/performance, not “feels better”;
- guardrail outcomes: other main lifts, pain/injury, session completion;
- comparison: fixed programme vs targeted selection, with a pre-declared analysis;
- no automatic application until the comparison is positive.

No targeting implementation in this change.

WEB UI

A local browser interface over the same Python core. It is an additional surface, not a replacement for the CLI.

Run

```
./run-web.sh                      # launcher: finds the venv, opens the browser
./run-web.sh --no-open --port 8001
./run-web.sh --db other.db

strength-coach web                # http://127.0.0.1:8000
strength-coach web --open
strength-coach web --port 8001
```

`run-web.sh` refuses a non-loopback `--host` and moves `--db` before the subcommand (it is a global option in the CLI). It is convenience only: it ends in the same `strength-coach web`.

Local only, no authentication. The default bind is 127.0.0.1. Do not pass a public host.

Pages

- **Today** — goal, block week, readiness, injury signal, unreviewed count, recent sessions, and a **Descriptive progress** panel (sRPE coverage, performance ratio, weekly volume status, latest back symptom; monitoring only).
- **Import** — upload the training and readiness CSVs; parsed in memory, hashed and stored in SQLite, never written to the repository. New cumulative exports are overlap-matched against stored rows, so old history is not duplicated. An **After the import** panel states the procedure: check the report, review recent sessions to create marks, then calibrate later — importing alone does not require a calibration.
- **Review** — unreviewed/recent session sheet; engine scores shown for context; human marks, optional session RPE and session length, plus a **Run rubric calibration** button (same report as `strength-coach calibrate`), confidence and note; calibration report after saving.
- **Readiness** — any subset of 1-5 and device fields, including optional wearable readiness; manual values win per field; the computed global/category score is shown; per-athlete baseline settings and the shadow baseline report are available; overrides take 0.00–1.00 with a reason; the **Volume valve** sets the poor-readiness accessory set reduction; the volume ledger and MRV cap are on Prescribe. A current workout is refused until readiness is ready.
- **Prescribe** — date + session type (`auto` rolling, squat, hinge, bench+squat, bench+hinge); deterministic prescription, warnings and notes; every item cites rule ids. A `recent stress` column, a `readiness` column, the **strength fatigue model** panel (shadow: per-lift + overall fitness/fatigue/state with approximate 95% intervals, R^2/status; no load effect), the **weekly volume** panel (hard sets by pattern against MEV/MAV/MRV landmarks; the MRV cap trims accessory sets), the **continuity calibration** panel (per-band returns/fixed/applied and the `allow_relax_fixed` setting) and the **exercise selection** panel (preview rankings with A/B/C comparison classes and a display-only own-history e1RM ratio; optional LLM chooser). The session is chosen by the rolling emphasis rule; **Apply selection** makes A candidates replace the slot exercises, **Confirm near-comparable** is required for B, and C is hidden unless **show different lifts** is ticked (and can never be applied automatically). **Use LLM** appears only when Apply is ticked. Without the toggle the prescription is unchanged. The **Retrospective validation** panel runs the walk-forward check. A scheduled deload is shown as a suggestion with an **Apply deload**

button; it is not applied until confirmed. Reactive deloads are automatic. Applied deloads cite `MOD-DELOAD-001`. Main lifts show their warm-up ramp; the strength-test cadence is shown in the notes. The volume ledger now applies `MOD-VOL-002` (one accessory set after three weeks below MEV).

- **Workout** — start from a prescription, confirm the seven required readiness fields (device fields optional), log sets manually, remove a set, end with measured/manual duration and sRPE or mark sRPE not recorded, or abandon. A prescription snapshot is frozen at start. During the workout the plan is editable: adjust planned sets, swap/add/remove exercises; each change is an append-only event and the original prescription stays frozen. On completion the workout materialises, the end screen shows actual vs modified plan and actual vs original prescription, and the deterministic rubric is persisted. **Review workout** opens the rubric sheet; saving marks runs calibration and closes the workout. In-workout CSV import is available by paste or file with columns `Exercise,Set,RPE,Value,Weight,Reps`; preview first, then apply. The file has no date, so backfill is not supported.
- **Programme** — rolling session-type forecast; choose weeks, sessions/week, optional start date and first session type (`auto` or a specific type). every session is rebuilt on the day and nothing is persisted.
- **Cardio** — log a session (modality, minutes, HR, distance, sRPE) choose the window (4/12/26/52 weeks) and an optional as-of date: sessions, total/average minutes, weekly breakdown and the WHO reference as context. Capture only; no load changes.
- **Measures** — periodic outcomes (VO2max, grip, 5x sit-to-stand, steps, sitting), with latest value, change and entries, filtered by metric. Metrics and directions are config; nothing changes a load.
- **Exercises** — full canonical catalogue with search/filter, all properties per row, and an add/edit form. Edits are written to `config/exercises/overrides.v1.yaml`; ids are immutable.
- **Athlete** — append-only editing of profile, constraints, suitability, e1RM overrides, strength tests, block start and symptoms; shows the current tables, injury signal and the **e1RM calibration** report (current formula e1RM and last set for every lift, plus measured tests vs formula; a bounded correction is applied only when the confidence interval excludes 1.0).
- **Terminal** — restricted `strength-coach` command runner bound to the UI database.
- **Docs** — local Markdown rendered without external assets.

API

route	purpose
`GET /api/overview`	dashboard data
`POST /api/import`	`{training: {name, text}, readiness: {name, text}}` -> import report
`POST /api/prescribe`	`{date, session_type?, apply_selection?, selection_confirmed?, selection_llm?}` -> prescription; `session_type=auto` uses the rolling rule
`POST /api/workout/start`	`{date, plan, readiness, note?}` -> live workout; seven readiness fields required
`GET /api/workout/active`	active workout, or `{workout: null}`
`GET /api/workout/<id>`	workout detail
`POST /api/workout/<id>/sets`	`{exercise_id, load_kg, reps, rir, velocity?}` -> add manual set
`POST /api/workout/<id>/remove-set`	`{set_id}` -> append a removal event
`POST /api/workout/<id>/end`	`{session_rpe?, not_recorded?, duration_min?, notes?}` -> completed
`POST /api/workout/<id>/update-set`	`{set_id, load_kg, reps, rir, velocity}` -> append correction event
`POST /api/workout/<id>/import`	`{text, apply}` -> parse/preview/apply workout CSV; no date column

route	purpose
`POST /api/workout/<id>/import/undo`	undo the latest import batch
`POST /api/workout/<id>/abandon`	abandon the active workout
`GET /api/review/<session>`	session sheet
`POST /api/review/<session>`	`{scores, confidence, note, session_rpe, session_minutes}` -> marks + optional session detail
`GET /api/readiness/<date>`	merged readiness
`POST /api/readiness`	`{date, values, note}` -> merged readiness + score
`POST /api/readiness/override`	`{date, scope, value, reason}` -> merged readiness + score
`GET /api/readiness/calibrate`	calibration report text (no auto-apply)
`GET /api/state`	profile, constraints, suitability, e1RM, symptoms, injury signal
`GET /api/exercises`	canonical exercise catalogue for the pickers
`POST /api/programme`	rolling forecast; returns dated entries and prescriptions
`GET /api/cardio?weeks=N`	cardio summary for the window
`POST /api/cardio`	`{date, modality, minutes, avg_hr, max_hr, distance_km, srpe, note}` -> summary
`GET /api/fatigue?date=N`	strength fitness-fatigue shadow model (per-lift state and status)
`GET /api/validation?max_dates=N`	retrospective walk-forward report (not prospective)
`GET /api/continuity?date=N`	return-to-training calibration: per-band n/median/CI/fixed/applied and `allow_relax_fixed`
`GET /api/volume?date=N&weeks=4`	weekly hard-set ledger by movement pattern (advisory)
`GET /api/progress?date=N`	descriptive progress: sRPE coverage, performance ratio, volume status, latest back symptom
`POST /api/continuity/settings`	`{allow_relax_fixed}` -> update the calibration policy
`POST /api/readiness/volume`	update the readiness volume valve (enabled, score below, sets, include essentials)
`GET /api/selection?date=N&session_type=X&llm=0|1&include_different=0|1`	exercise-selection rankings with A/B/C comparability classes
`GET /api/exercises/catalog`	full exercise catalogue with source/properties
`POST /api/exercises`	create/update an entity in `config/exercises/overrides.v1.yaml`
`GET /api/measurements`	metric catalogue, per-metric summary and recent entries
`POST /api/measurements`	`{metric, value, date, method, note}` -> summary
`POST /api/terminal`	one restricted `strength-coach` command
`POST /api/state/profile` / `constraint` / `suitability` / `e1rm` / `test` / `symptom` / `block`	append-only state writes
`GET /api/calibrate`	calibration report
`GET /api/docs` / `GET /api/docs/<name>`	doc list / rendered doc

Design

The web layer is serialisation and routing only. It calls the same functions as the CLI: `build_prescription`, `build_facts` / `score_session`, `record_marks`, `record_manual`, and `backtest`. A missing value is still `unknown`; the UI does not fill it in.

Limitations

- No LLM proposal/chat UI; this is closed as CLI-only (`run --propose`), not pending.
- Workout mode is UI-first for start; CLI covers status/add/update/remove/end/import. Completed workouts materialise into `session` / `clean_set`; review/rubric is built.
- State editing is available; the taxonomy itself and programme rules remain config/CLI.
- No authentication; never expose beyond localhost.

Readiness, injury and workout type — evidence and decision

Status: decision recorded. No automatic readiness- or injury-driven change of rep scheme or session type.
Date: 2026-10-20

Question: should readiness, injuries and other metrics change the type of workout — strength, maximal, volume, hypertrophy — rather than only load/RIR/sets?

---

1. Current determinants of the rep scheme

The scheme is chosen by the programme, not by daily readiness:

- base template: main lifts 3–5, accessories 8–12;
- weekly undulation when a block anchor exists:
- week 1: main `strength`, accessory `accessory`;
- week 2: main `strength`, accessory `volume`;
- week 3: main `maximal`, accessory `accessory`;
- week 4: main `strength`, accessory `volume`;
- `maximal` (1–3) is excluded on hinge lifts;
- no anchor → template ranges are used.

Readiness currently changes load, RIR and sets. Injury can exclude/penalise/cap. Neither changes the scheme.

Evidence for the scheme groups: `TRAINING-DYNAMICS.md`.

---

2. Evidence

Periodisation / undulation

- Linear and undulating periodisation produce comparable strength and body-composition outcomes in a 29-study meta-analysis (Zhang et al. 2026, doi:10.3389/fpubh.2026.1707627).
- In older adults, traditional and undulating training also improved strength and fat-free mass similarly (Vargas-Molina et al. 2022, doi:10.3390/ijerph19084522).
- DUP is a variety/fatigue-management structure, not a proven superiority.

Autoregulation

- Autoregulated load (RIR/velocity) versus standardised percentage prescription showed no significant 1RM difference (MD 2.07 kg, 95% CI −0.32 to 4.46, p = 0.09) in a 15-study meta-analysis (Hickmott et al. 2022, doi:10.1186/s40798-021-00404-9).
- Autoregulation is defensible for avoiding bad-day grinding; it is not evidence for switching the training goal on a bad day.

Readiness monitoring

- Subjective self-reports are sensitive to training load, but heterogeneous and not a validated prescription rule (Saw et al. 2016, doi:10.1136/bjsports-2015-094758).

- Wellness indices track recovery/staleness, but no validated weights or cut-points.
- Perceived readiness can relate to performance in a small study (Munteanu et al. 2026), not a validated per-lift scale.
- HRV-guided training has small, inconsistent performance effects; HRV is context, not a hard daily gate.

Full readiness evidence: `READINESS-SCORE.md` and `READINESS-WEIGHT-EVIDENCE.md`.

Strength and hypertrophy specificity

- Strength is specific to heavy loading: lower-rep, higher-load training produces larger 1RM gains than low-load training when effort is matched (Hammert et al. 2026, doi:10.1007/s40279-025-02370-8).
- Hypertrophy is similar across a wide rep range when effort and volume are matched; roughly 5–30 reps, with low loads requiring more proximity to failure.
- Switching from a planned strength/maximal session to a volume session on a bad day therefore changes the stimulus, not merely the fatigue.

Injury and pain

- Pain-guided activity modification is feasible in patellar tendinopathy (Sprague et al. 2021, doi:10.1186/s40814-021-00792-5).
- Exercise therapy helps chronic low back pain (Hayden et al. 2021, doi:10.1002/14651858.CD009790.pub2), but that is treatment, not evidence for an automatic session-type switch.
- Prior injury is a recurrence marker; it supports caution, exposure management and excluding provocative movements.

---

3. Interpretation

Readiness and injury affect **capacity and tolerance**, not the training goal.

- A bad day reduces the load you can express at a given RIR. It does not mean hypertrophy training is now the right stimulus.
- Switching to a different rep scheme changes specificity, volume and fatigue. The evidence does not show that this improves outcomes.
- Injury is different: it can make a movement unsafe. The correct response is to exclude or modify the movement, not to silently change the whole session's goal.
- A maximal attempt on a very poor day may be unwise, but the evidence supports **not attempting it**, not automatically redefining the block.

The supported hierarchy is:

```
injury/constraint veto or exercise modification
  > load/RIR reduction
  > set reduction
  > reduce/stop the session
  > change the planned rep scheme (last resort, explicit)
```

---

4. Decision

1. **Do not change the rep scheme automatically from daily readiness.**
2. Keep scheme selection planned by the block/DUP cycle.
3. Readiness adjusts load, RIR and sets.

4. Injury may exclude/penalise/cap exercises; it may not silently turn a strength day into a hypertrophy day.
5. If readiness is so poor that a planned maximal exposure is unsafe, the safe action is to **skip or downgrade that exposure explicitly**, not to invent a new session type.
6. Any readiness-driven scheme change should be a visible suggestion with confirmation, and should be evaluated prospectively before becoming a rule.
7. Do not use sRPE, HRV or recent stress to choose strength vs hypertrophy; use them for load, volume and recovery context.

---

5. What would change this decision

- prospective evidence that readiness-driven scheme switching improves strength or adherence without increasing symptoms;
- a validated readiness threshold for maximal-attempt safety;
- a clinician-defined rehabilitation plan that specifies a different stimulus.

None exists now. The default stays: planned scheme, adjusted load/RIR/sets.

No implementation was done.